

FRM Part I Exam

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Questions with Answers - Valuation and Risk Models

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Reading 47: Measures of Financial Risk

Q.939 Billy Marquette has recently joined a small company that provides private commercial jets to royal families, government officials, and directors of big firms. Marquette is a retired commercial pilot with a very basic understanding of finance. On his first day, he is handed a report on risk management measures. The excerpt from the report says “due to volatility in oil prices, the company has a weekly 90% VaR of €20,000”. Which of the following is the most appropriate explanation of the excerpt?

- A. There is a 90% probability that the company will experience a loss of €2,000 on a weekly basis.
- B. There is a 10% probability that the company will experience a loss of €20,000 in any given week.
- C. There is a 90% probability, in any given week, that the company will experience a loss of more than €20,000.
- D. There is a 10% probability, in any given week, that the company will experience a loss in excess of €20,000.

The correct answer is **D**.

Value at Risk (VaR) is a statistical technique used to measure and quantify the level of financial risk within a firm or investment portfolio over a specific time frame. This metric is most commonly used by investment and commercial banks to determine the extent and occurrence ratio of potential losses in their institutional portfolios. Risk managers use VaR to measure and control the level of risk exposure. One can look at VaR as providing an estimate of the potential loss that could be made on an investment portfolio over a certain period of time, given normal market conditions and a certain level of confidence. In this case, a 90% VaR of €20,000 means that there is a 10% chance that the company will experience a loss exceeding €20,000 in any given week. This is because the 90% confidence level implies that losses will not exceed the VaR estimate 90% of the time. Therefore, the remaining 10% represents the probability that losses could exceed this amount, which is exactly what choice D states.

Choice A is incorrect. The statement does not imply a 90% probability of a weekly loss of €2,000. The VaR figure given is €20,000, not €2,000.

Choice B is incorrect. While it correctly identifies the 10% probability associated with VaR, it incorrectly states that this represents the likelihood of a loss exactly equal to €20,000 in any given week. In reality, VaR represents the minimum loss that could occur with a certain probability.

Choice C is incorrect. This choice incorrectly interprets the 90% confidence level as implying a 90% chance of losses exceeding €20,000 in any given week. In fact, there's only a 10% chance (100%-90%) that losses will exceed this amount.

Things to Remember

- Value at Risk (VaR) is a measure of the potential loss that could occur on an investment portfolio over a specific time frame, given normal market conditions and a certain level of confidence.
 - VaR is commonly used by banks and financial institutions to assess and manage risk exposure.
 - A 90% VaR of €20,000 means that there is a 10% chance that losses will exceed €20,000 in any given week.
 - VaR provides an estimate of the minimum loss that could occur with a certain probability, not an exact loss amount.
 - Risk managers use VaR to control and monitor the level of risk within a firm or investment portfolio.
-

Q.3328 An investment company has a portfolio which has the following ordered performance by historical data. Calculate the expected shortfall $ES_{0.95}$.

Probability	1%	5%	10%	12%	15%
Profit/Loss	-500	-300	-100	-90	-50

- A. 300
- B. 340
- C. 400
- D. 425

The correct answer is **B**.

Given a discrete distribution, the ES is the equivalent of:

$$ES_{\alpha} = \frac{1}{1-\alpha} \sum_{p=0}^{1-\alpha} [p^{\text{th}} \text{ highest loss}] \times [\text{probability of } p^{\text{th}} \text{ highest loss}]$$

At $\alpha = 0.95$,

$$ES_{0.95} = \frac{[(0.01 \times 500) + (0.04 \times 300)]}{0.05} = 340$$

Note: The sum of probabilities in the numerator must sum to $(1 - \alpha)$

Alternative Approach

To calculate the expected shortfall, we must ask ourselves, "If we are in the worst 5% of the loss distribution, what is the expected loss?" The first column of the given table makes it clear that the 5% tail of the distribution is composed of a 1% probability that the loss is 500 and a 4% probability that the loss is 300. Conditional on being in the tail of the distribution, there is, therefore, a 1/5 chance that the loss is 500 million and a 4/5 chance that it is 300. The expected shortfall (in millions of dollars) is, therefore:

$$\left(\frac{1}{5}\right) \times 500 + \left(\frac{4}{5}\right) \times 300 = 340$$

Q.3329 An investment company has a portfolio which has the following ordered performance by historical data. Calculate the expected shortfall, $ES_{0.99}$, i.e., at 99% level of confidence

Probability	1%	5%	10%	12%	15%
Profit/Loss	-500	-300	-100	-90	-50

- A. 168
- B. 400
- C. 460
- D. 500

The correct answer is **D**.

There is only one number 500 beyond 1%, therefore, the average is 500.

Further Explanation

The expected shortfall (also called conditional VaR) is the expected tail loss. It is the average of the worst $100 \times (1 - \alpha)\%$ of losses. For a discrete distribution, ES is derived as:

$$ES_{\alpha} = \frac{1}{1 - \alpha} \sum_{p=0}^{\alpha} (p^{\text{th}} \text{ loss} \times \text{probability of } p^{\text{th}} \text{ loss})$$

In words, to determine the expected shortfall at a level of confidence α , we must find the average of all the outcomes whose probability is less than or equal to $1 - \alpha$.

At a 99% confidence level, the significance level is 1%. To establish the expected shortfall at 1%, we must find the average of all the outcomes whose probability is less than or equal to 1%. in this case,

$$\text{Expected shortfall} = \frac{(0.01 * 500)}{0.01} = 500$$

Q.3397 A hypothetical portfolio of securities exhibits the following expected losses shown:

Name	Loss (million dollar)	Probability (%)
1	10	40%
2	20	35%
3	50	15%
4	100	5%
5	200	2.5%
6	225	2%
7	250	0.5%

Calculate the expected shortfall at the 95% and 99% confidence level?

- A. ES (95%) = \$225 million; ES (99%) = \$237.5 million
- B. ES (95%) = \$215 million; ES (99%) = \$237.5 million
- C. ES (95%) = \$217.5 million; ES (99%) = \$250 million
- D. ES (95%) = \$225 million; ES (99%) = \$250 million

The correct answer is **B**.

Note that the given data are in expected losses. The expected shortfall (also called conditional VaR) is the expected tail loss. It is the average of the worst $100 \times (1 - \alpha)\%$ of losses. For discrete distribution, ES is derived as:

$$ES_{\alpha} = \frac{1}{1 - \alpha} \sum_{p=0}^{\alpha} (\text{pth loss} \times \text{probability of pth loss})$$

In other words, to determine the expected shortfall at a level of confidence α , we must find the average of all the outcomes whose probability is less than or equal to $1 - \alpha$.

At a 95% confidence level, the significance level is 5%. To establish the expected shortfall at 5%, we must find the average of all the outcomes whose probability is less than or equal to 5%

$$ES_{0.95} = \frac{200 \times 2.5\% + 225 \times 2\% + 250 \times 0.5\%}{5\%} = 215 \text{ million dollars}$$

At a 99% confidence level, the significance level is 1%. To establish the expected shortfall at 1%, we must find the average of all the outcomes whose probability is less than or equal to 1%

$$ES_{0.99} = \frac{225 \times 0.5\% + 250 \times 0.5\%}{1\%} = 237.5 \text{ million dollars}$$

Q.3398 The VaR of a loan portfolio is computed at various confidence levels:

Confidence Level	VaR
95.0%	2%
95.5%	5%
96.0%	6%
96.5%	7%
97.0%	9%
97.5%	10%
98.0%	13%
98.5%	15%
99.0%	20%
99.5%	30%

What is the expected shortfall at the 97.5% confidence level?

- A. 0.1
- B. 0.15
- C. 0.195
- D. 0.2

The correct answer is **C**.

The expected shortfall at the 97.5% confidence level is computed by averaging all value of risk greater than the 97.5% confidence level.

Expected shortfall at the 97.5% confidence level is therefore,

$$\frac{13\% + 15\% + 20\% + 30\%}{4} = 19.5\%$$

Q.4639 A hypothetical portfolio has an annual 1% VaR of \$45,000. Which of the following statements is the **most likely** correct about the portfolio?

- A. The loss over the next year is expected to be at most \$45,000 in 1% of the cases.
- B. There is only a 1% chance that we will gain more than \$45,000 over the next year.
- C. The likelihood of losing more than \$45,000 over the next year is 1%.
- D. The likelihood of losing no more than \$45,000 over the next year is 1%.

The correct answer is **C**.

The Value at Risk (VaR) metric is a commonly used risk management tool that quantifies the potential loss that could happen in an investment portfolio over a specified period of time at a given confidence level. In this case, an annual 1% VaR of \$45,000 implies that there is a 1% chance that the portfolio could experience a loss exceeding \$45,000 over the next year. This is based on the statistical analysis of historical market trends and volatilities. The VaR metric is often used by financial institutions and investment firms to measure the extent of potential losses that could be incurred in extreme market conditions. Therefore, the statement 'The likelihood of losing more than \$45,000 over the next year is 1%' accurately reflects the interpretation of the given VaR value.

Choice A is incorrect. The statement is misleading as it suggests that the maximum loss in 1% of the cases will be \$45,000. However, VaR does not provide a maximum loss figure but rather a minimum loss figure at a certain confidence level.

Choice B is incorrect. This choice incorrectly interprets VaR as a measure of potential gains rather than potential losses. VaR measures the risk of potential losses, not gains.

Choice D is incorrect. This statement misinterprets VaR by suggesting that there's only 1% chance of losing no more than \$45,000 over the next year which contradicts with the concept of VaR which states that there's 1% chance we could lose more than \$45,000 over next year.

Things to Remember

- Value at Risk (VaR) is a statistical measure used to quantify the level of financial risk within a portfolio.
 - VaR provides an estimate of the maximum potential loss that a portfolio may experience over a specified time horizon at a given confidence level.
 - The confidence level in VaR represents the probability that the actual loss will not exceed the VaR estimate.
 - VaR is widely used in risk management to assess and manage the potential downside risk of investment portfolios.
 - Historical simulation, Monte Carlo simulation, and parametric methods are common approaches to calculate VaR.
-

Q.4640 The investment returns and the corresponding probabilities are given in the following table:

Returns	Probability
20%	0.1
30%	0.3
-10%	0.2
15%	0.3
7%	0.1

What is the standard deviation of the investment returns?

- A. 0.142
- B. 0.154
- C. 0.132
- D. 0.138

The correct answer is **A**.

The variance of the return R is given by:

$$\text{Var}(R) = E(R^2) - [E(R)]^2$$

Now,

$$E(R^2) = 0.1 \times (20\%)^2 + 0.3 \times (30\%)^2 + 0.2 \times (-10\%)^2 + 0.3 \times (15\%)^2 + 0.1 \times (7\%)^2 = 0.04024$$

$$E(R) = 0.1 \times 20\% + 0.3 \times 30\% + 0.2 \times -10\% + 0.3 \times 15\% + 0.1 \times 7\% = 0.142$$

$$\therefore \text{Var}(R) = 0.04024 - (0.142)^2 = 0.020076$$

Thus the standard deviation is given by:

$$\sqrt{0.020076} = 0.1417 = 14.17\%$$

Q.4641 An investor invests his funds in two correlated assets, A and B. The standard deviation of asset A is 20%, and that of B is 15%. The portfolio variance is 2.84%. Given that the investor has three times as much money in asset A than he has in asset B, what is the correlation coefficient between assets A and B?

A. 0.0962

B. 0.2133

C. 0.3994

D. 0.8078

The correct answer is **C**.

The variance of a portfolio is given by:

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2\rho w_A w_B \sigma_A \sigma_B$$

Where

w_A : the weight of asset A

w_B : the weight of asset B

σ_A : standard deviation of asset A

σ_B : standard deviation of asset B

ρ : correlation coefficient between asset A and B

Now let the amount invested in asset be B be w_B and thus:

$$\begin{aligned} w_A &= 3w_B \\ w_B + 3w_B &= 1 \\ \therefore w_B &= \frac{1}{4} \Rightarrow w_A = \frac{3}{4} \end{aligned}$$

Now,

$$\begin{aligned} w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2\rho w_A w_B \sigma_A \sigma_B &= 0.0284 \\ \left(\frac{3}{4}\right)^2 \times 0.2^2 + \left(\frac{1}{4}\right)^2 \times 0.15^2 + 2 \times \frac{3}{4} \times \frac{1}{4} \times 0.2 \times 0.15 \times \rho &= 0.0284 \\ \Rightarrow \rho &= \frac{0.0284 - \left(\frac{3}{4}\right)^2 \times 0.2^2 - \left(\frac{1}{4}\right)^2 \times 0.15^2}{2 \times 0.15 \times 0.2 \times \frac{1}{4} \times \frac{3}{4}} = 0.39944 \end{aligned}$$

Q.4643 The losses from a portfolio for one year are normally distributed with mean -10 and standard deviation 20. What is the value of the 99% expected shortfall?

- A. 52.85
- B. 37.4
- C. 42.85
- D. 26.43

The correct answer is **C**.

We start with the formula for the expected shortfall (ES) at a certain confidence level:

$$ES = \mu + \sigma \left(\frac{e^{-\frac{U^2}{2}}}{(1 - X)\sqrt{2\pi}} \right)$$

Here, μ is the mean of the distribution, σ is the standard deviation, U is the z-score that corresponds to our desired confidence level X .

For a confidence level of 99%, we find the z-score by inverting the cumulative distribution function (CDF), Φ^{-1} , of the standard normal distribution:

$$U = \Phi^{-1}(0.99) = 2.33$$

This z-score represents the point at which 99% of the distribution's values fall below it.

With our z-score U and the known values for the mean $\mu = -10$ and standard deviation $\sigma = 20$, we can now plug these into the ES formula:

$$ES = -10 + 20 \left(\frac{e^{-\frac{2.33^2}{2}}}{(1 - 0.99)\sqrt{2\pi}} \right) = 42.85$$

This means that, at the 99% confidence level, the expected shortfall of the portfolio—the average loss in the worst 1% of cases—is approximately \$42.85 million.

Q.6218 An investor is evaluating two portfolios to determine which lies on the efficient frontier. For each point on the efficient frontier, at least one portfolio can be constructed from all

available investments with the expected risk and return corresponding to that point. How does the inclusion of a risk-free asset impact the efficient frontier?

- A. The efficient frontier becomes a straight line, representing a linear relationship between expected return and standard deviation.
- B. The efficient frontier bends upwards, with higher risk for each level of expected return.
- C. The efficient frontier remains a curve but shifts upwards, providing higher returns for each level of risk.
- D. The efficient frontier shifts downwards, providing lower returns for each level of risk.

The correct answer is **A**.

By including a risk-free asset, the efficient frontier becomes a straight line because it creates a linear relationship between the expected return and standard deviation. This is due to the ability to mix the risk-free asset with a risky market portfolio, offering the highest Sharpe ratio.

B is incorrect because the efficient frontier does not bend upwards; instead, it becomes a straight line when a risk-free asset is available. This change allows for leverage, which increases the slope without changing the curvature.

C is incorrect because the efficient frontier does not remain a curve. The addition of risk-free assets transforms the efficient frontier into a straight line.

D is incorrect because the efficient frontier does not shift downwards. A downward shift would imply lower returns for each level of risk, which is contrary to the inclusion of a risk-free asset that enhances investment opportunities.

Things to Remember

- The introduction of a risk-free asset allows for the construction of a Capital Market Line (CML), which is a straight line on the risk-return graph, extending from the risk-free rate to the tangent point on the efficient frontier.
- The CML represents portfolios that combine the risk-free asset with the market portfolio, resulting in a linear risk-return trade-off.
- The slope of the CML is the Sharpe ratio, which measures the excess return per unit of risk.
- The inclusion of a risk-free asset enables investors to achieve any desired combination of risk and return along the CML by adjusting their proportion of investment in the

risk-free asset and the market portfolio.

Q.6219 Financial analysts often assume that returns of risky financial assets such as equities follow a normal distribution. However, empirical evidence shows that actual returns exhibit characteristics different from the normal distribution. Which of the following is a significant difference between the normal distribution and the actual return distribution of equities?

- A. The returns of equities usually exhibit lighter tails compared to a normal distribution.
- B. The returns of equities often exhibit lower kurtosis compared to a normal distribution.
- C. The returns of equities generally display skewness and heavier tails than those of a normal distribution.
- D. The returns of equities usually show fewer extreme values compared to what is predicted by a normal distribution.

The correct answer is **C**.

Empirical evidence shows that the returns of equities do not follow a normal distribution because they generally display skewness and heavier tails, which means a higher probability of extreme values compared to a normal distribution. This phenomenon is often referred to as "fat tails."

A is incorrect because equities exhibit heavier, not lighter, tails compared to a normal distribution. This implies a higher likelihood of extreme losses or gains.

B is incorrect because equity returns typically exhibit higher kurtosis, meaning they have more pronounced peaks and heavier tails than the normal distribution. Lower kurtosis would suggest less extreme values and a more flat-topped distribution.

D is incorrect because equities show more, not fewer, extreme values than predicted by a normal distribution, which is indicative of the heavy-tailed nature of actual return distributions.

Things to Remember

- Skewness refers to the asymmetry of the return distribution. Positive skewness indicates a distribution with a longer right tail, while negative skewness indicates a longer left tail.
- Kurtosis measures the "tailedness" of the distribution. Higher kurtosis means more frequent extreme returns (fat tails) compared to the normal distribution.
- Equity returns often exhibit **leptokurtosis**, characterized by a high peak and fat

tails, leading to more frequent extreme outcomes than predicted by the normal distribution.

- Assuming that financial returns are normally distributed can lead to underestimating the probability of distributive tail events, which can result in inadequate risk management strategies.
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Q.6220 A financial risk manager is using the VaR measure to evaluate the risk of a portfolio over a specific time period and confidence level. VaR is commonly used due to its simplicity and the ease of understanding the associated risk under normal market conditions. However, what are the major limitations of VaR despite its widespread use?

- A. VaR can be subadditive, which violates one of the desirable properties of a coherent risk measure.
- B. VaR does not capture the potential losses in the tail beyond the confidence level.
- C. VaR assumes normal distribution of returns, which may not accurately reflect the actual distribution of asset returns.
- D. VaR does not consider the correlations between different assets during periods of market stress.

The correct answer is **B**.

VaR does not account for or capture the potential losses beyond the specified confidence level. This can be crucial as significant losses beyond the VaR threshold can still occur, which VaR doesn't predict, thus leaving a part of the risk unaddressed.

A is incorrect because VaR can sometimes be non-subadditive, meaning the VaR of a combined portfolio can be greater than the sum of the VaRs of its individual components. This violates the subadditivity property of coherent risk measures. Therefore, the claim that VaR can be subadditive is incorrect; rather, the correct limitation is that it can be non-subadditive.

C is incorrect because VaR can be calculated using various methods, including historical simulation, which does not assume normal distribution. The assumption of normally distributed returns is a limitation specific to the parametric (delta-normal) approach, not VaR as a whole.

D is incorrect because VaR does account for correlations between different assets when calculated using the historical simulation or Monte Carlo simulation methods. These methods use actual or simulated return distributions that inherently include asset correlations.

Things to Remember

- VaR is a measure that estimates the maximum potential loss of a portfolio at a given confidence level over a specific time period.
 - VaR is affected by arbitrary parameters, like the confidence level and holding period, which can both influence its accuracy.
 - VaR can give a false sense of security as it assumes normal market conditions and may fail to predict extreme market events and crises, where correlations between assets can change drastically.
 - Alternative risk measures, such as Conditional VaR (CVaR), address some of VaR's limitations by estimating the expected loss beyond the VaR threshold, providing a fuller picture of tail risk.
-

Q.6221 Which of the following statements correctly describes the property of subadditivity in the context of a coherent risk measure?

- A. A risk measure is subadditive if adding a risk-free asset to a portfolio does not change its risk measure.
- B. A risk measure is subadditive if the risk of a combined portfolio does not exceed the sum of the risks of the individual portfolios.
- C. A risk measure is subadditive if increasing the size of a portfolio proportionally increases its risk measure.
- D. A risk measure is subadditive if the risk of a portfolio decreases linearly with additional diversification.

The correct answer is **B**.

A coherent risk measure must exhibit subadditivity, meaning that the risk measure of a combined portfolio should not exceed the sum of the risk measures of its individual components. This property encourages diversification because merging two portfolios should not increase the overall risk more than the sum of the individual risks.

A is incorrect because subadditivity does not relate to adding a risk-free asset; that property is more aligned with translation invariance.

C is incorrect because subadditivity is not about proportional increases in risk measure; this describes homogeneity.

D is incorrect because the property of subadditivity does not specify that risk decreases linearly with diversification; it simply states that the combined risk should not exceed the sum of individual risks.

Things to Remember

- The subadditivity property captures the benefits of diversification in risk management. Combined portfolios should have a risk measure that reflects these benefits.
- Subadditivity is crucial for risk managers when evaluating the cumulative risk of different asset classes to benefit from diversification effects.

Q.6222 Value at Risk (VaR) is a widely used risk management tool that estimates the potential loss in value of a portfolio over a defined period for a given confidence interval. Despite its popularity, there are significant limitations to VaR that question its effectiveness as a comprehensive risk measure. Why is VaR not considered a coherent risk measure?

- A. VaR fails to account for the current size of the portfolio.
- B. VaR fails the monotonicity property of coherent risk measures.
- C. VaR does not account for the skewness and kurtosis of return distributions.
- D. VaR fails the subadditivity test required for coherent risk measures.

The correct answer is **D**.

VaR is not considered a coherent risk measure because it fails the subadditivity test. Subadditivity would require that the risk measure of a combined portfolio should not exceed the sum of the risk measures of the individual portfolios, which VaR does not always satisfy. This failure undermines the principle of diversification in risk assessment.

A is incorrect because VaR does account for the size of the portfolio, as it measures potential losses in monetary terms based on given parameters.

B is incorrect because VaR generally satisfies monotonicity; if one portfolio is riskier than another, its VaR will generally be higher.

C is incorrect because VaR can be calculated using methods that take into account the skewness and kurtosis of return distributions, such as historical simulation or Monte Carlo simulation. These methods use actual or simulated return distributions, which include skewness and kurtosis.

Things to Remember

- A coherent risk measure satisfies four properties: subadditivity, translation invariance, positive homogeneity, and monotonicity.
 - Risk managers often complement VaR with measures such as ES, which is a coherent risk measure, to capture the tail risk better.
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Reading 48: Calculating and Applying VaR

Q.972 After using of the historical simulation method, you have been provided with the following 30 ordered percentage returns for an asset:

[-18, -16, -14, -12, -10, -9, -7, -7, -6, -6, -6, -5, -5, -4, -4, -4, -2, -1, 0, 0, 2, 3, 6, 12, 12, 13, 15, 15, 18, 28]

The value-at-risk (VaR) and expected shortfall (ES), at 90% confidence level, respectively, are closest to:

A. Var: 14; ES: 17

B. Var: 14; ES: 16

C. Var: 12; ES: 16

D. Var: 12; ES: 24

The correct answer is **A**.

VaR can be calculated as the $[(1 - 0.9) \times 30] = 3^{\text{rd}}$ worst observation, which is -14 and hence the VaR is 14.

The ES is the arithmetic average of losses that are **worse** than the VaR. Thus,

$$ES = \frac{(18 + 16)}{2} = 17$$

Q.973 Stuart Broad, a risk analyst working with Macquarie Bank compiles data of 100 simulated percentage returns of the bank's assets:

[-13.33, -12.25, -11.75, -10.66, -8.45, -7.13, -6.48, -5.29 ... 2.89, 3.56, 4.29, 5.38, 6.65, 7.89, 8.54, 9.64, 10.27, 12.28, 13.25]

Using the data, he calculates the expected shortfall (ES) and the value at risk (VAR) of the bank's assets at the 95 percent confidence level using the historical simulation method. What is the expected shortfall and the value at risk computed by Stuart Broad?

- A. Expected shortfall: 12; Value at Risk: 9.45
- B. Expected shortfall: 11.49; Value at Risk: 9.95
- C. Expected shortfall: 12; Value at Risk: 8.45
- D. Expected shortfall: 11.49; Value at Risk: 8.45

The correct answer is **C**.

VaR can be calculated as the $[(1 - 0.95) \times 100] = 5^{\text{th}}$ worst observation, which is -8.45 and hence the VaR is 8.45.

The ES is the arithmetic average of losses that are **worse** than the VaR. Thus,

$$ES = \frac{13.33 + 12.25 + 11.75 + 10.66}{4} = 12$$

Q.982 ANC National Bank handles a portfolio of assets amounting to USD 150 million. Antony Smith, a risk analyst, analyzes the portfolio and observes that the returns are independently identically normally distributed. The annual standard deviation of the portfolio is 0.55. What is the 20 day-Value at Risk at 95 percent confidence assuming 250 trading days in a year? Click [here](#) to see the standard normal table.

- A. USD 51.92 million
- B. USD 38.50 million
- C. USD 53.67 million
- D. USD 71.50 million

The correct answer is **B**.

$$\begin{aligned}\text{VaR} &= Z \times \text{standard deviation} \times \text{portfolio amount} \\ \text{Annual VaR} &= 1.65 \times 0.55 \times 150,000,000 = 136,125,000 \\ \text{Var (T days)} &= \text{1-day VaR} \times \sqrt{T} \dots\dots \text{formula} \\ \text{1-day VaR} &= \frac{\text{Annual VaR}}{\sqrt{250}} = \$8,609,300 \\ \text{20-day VaR} &= (\text{1-day VaR}) \times \sqrt{20} = \$38,501,694\end{aligned}$$

It's the standard deviation, σ , that determines where you start. If the σ given is annual, you'll start by calculating the annual VaR and then use the square root of the time rule to come up with VaR for shorter periods. Similarly, if given the daily σ , compute the 1-day VaR and then convert that appropriately.

Q.2616 Arthur Bell is the portfolio manager at FFF Investments. Recently, he bought 5,000 call options on stocks of one of the local growth-oriented oil refining companies that have never paid dividends. The strike price of the options was \$50. The underlying stock is trading at \$58 and has an annual volatility of return of 33%. Bell estimated the delta of these options to be 0.55. What is the approximate weekly (delta normal) 99% VaR of the position assuming 52 trading weeks in the year?

- A. \$7,725.57
- B. \$17,007
- C. \$6,659.97
- D. \$14,661.22

The correct answer is **B**.

In the delta-normal approach, we first calculate the VaR of the underlying. Then, we use the equation below to revalue the derivative by linear approximation (as the delta multiplied by the VaR of the underlying):

$$\begin{aligned}\text{VaR}_{\text{derivative}} &= \Delta \times \text{VaR}_{\text{underlying}} \\ \text{VaR of the underlying (stock)} &= \sigma \times z_{\alpha\%} \times \text{market value}\end{aligned}$$

But first, we have to convert the annual volatility, i.e., σ , into a weekly value:

$$\text{Annual volatility} = \text{weekly volatility} \times \sqrt{52}$$

$$\text{Thus, weekly volatility} = \frac{(33\%)}{\sqrt{52}} = 0.045763$$

$$\text{VaR of the underlying} = 0.045763 \times 2.33 \times \$58 = \$6.18438$$

$$\text{VaR of call option} = 0.55 \times \$6.18 = \$3.40$$

$$\text{VaR of 5,000 call options} = \$3.40 \times 5,000 = \$17,007$$

Note: 2.33 is the standard normal deviate at a 99% confidence level.

Q.2619 After a significant increase in the volatility of shares of USY Pharma, Ross Grand, senior portfolio manager, decides to hedge the existing position by buying 300 at-the-money call options. Shares of USY are trading at \$457 and the daily VaR of the underlying at 99% confidence is \$42.59. What is the daily 99% VaR of the options position using the delta-normal method?

- A. \$3,457.99
- B. \$5,791.77
- C. \$4,524.05
- D. \$6,388.50

The correct answer is **D**.

The key point here is that the call options are at the money, so we can assume $\text{delta} = 0.5$. (See the note at the end of the explanation if this is unclear.)

According to the delta-normal method,

$$\begin{aligned}\text{VaR of an option} &= \text{Delta} \times \text{VaR of underlying} \\ \text{VaR of the position} &= 0.5 \times \$42.59 \times 300 \text{ options} = \$6,388.50\end{aligned}$$

Note: At-the-money call options typically have a delta of 0.5, and the delta of out-of-the-money call options approaches 0 as expiration nears. The deeper in-the-money the call option, the closer the delta will be to 1, and the more the option will behave like the underlying asset. Since the delta of an option ranges between 0 and 1, an at-the-money option is right in between those two numbers.

For put options, at-the-money put options typically have a delta of -0.5, and the delta of out-of-the-money call options approaches 0 as expiration nears. The deeper in-the-money put option, the closer the delta will be to -1.

Q.2623 Ashli More prepares a presentation to the management board on the application of derivatives for hedging risk. She struggles with the classification of linear and non-linear derivatives. Which of the following is an example of linear derivatives?

- A. Futures on stocks, forwards on broad market indices, and plain vanilla European options on bonds.
- B. Interest rate swaps, interest rate caps, and plain vanilla American options on stocks.
- C. Futures on broad market indices.
- D. Futures on stocks and swaptions.

The correct answer is **C**.

Futures on broad market indices are considered linear derivatives. This is because the payoff of a futures contract is linear in nature, meaning it changes proportionally with the changes in the underlying asset. In other words, if the price of the underlying asset increases, the value of the futures contract also increases proportionally, and vice versa. This characteristic makes futures contracts linear derivatives. Broad market indices are a common underlying asset for futures contracts, making futures on broad market indices a typical example of linear derivatives.

Choice A is incorrect. While futures on stocks and forwards on broad market indices are examples of linear derivatives, plain vanilla European options on bonds are not. Options, whether they are European or American, call or put, are non-linear derivatives because their payoff is not directly proportional to the underlying asset's price movement.

Choice B is incorrect. Interest rate swaps can be considered as linear derivatives as their value changes in proportion with the underlying interest rate. However, interest rate caps and plain vanilla American options on stocks are non-linear derivatives due to their asymmetrical payoff structure.

Choice D is incorrect. Futures on stocks can be classified as linear derivatives but swaptions cannot be classified under the same category. Swaptions give the holder the right but not obligation to enter into a swap agreement in future which makes them a type of option and hence a non-linear derivative.

Things to Remember

- Linear derivatives have payoffs that change proportionally with the changes in the underlying asset.
- Non-linear derivatives have payoffs that are not directly proportional to the underlying asset's price movement.
- Futures contracts are considered linear derivatives because of their linear payoff structure.
- Options, such as plain vanilla European or American options, are examples of non-linear derivatives due to their asymmetrical payoff structures.
- Interest rate swaps can be classified as linear derivatives as their value changes in proportion with the underlying interest rate.
- Swaptions are a type of option and hence classified as non-linear derivatives.

Q.2625 The investment division of a bank is considering an investment of \$100,000,000 in one of the following:

- stocks of ALPHA Plc
- stocks of APPA Corporation
- 20-year bonds of APPA Corporation

Although the department seeks the investment with the highest expected return, to comply with bank's risk policies, the department cannot open a position with a daily 99% VaR higher than \$6,000,000.

The investment opportunities presented above have the following features:

	Expected Annual Return	Expected annual volatility of returns	Modified Duration	Market Price
Stocks of ALPHA	20%	50%	–	\$5 per stock
Stocks of APPA	13%	36%	–	\$30 per stock
Bonds of APPA	8%	10%	3.5	99% per \$100 nominal

Assuming zero daily returns and 252 trading days per year, which investment should the bank choose?

- A. Stocks of ALPHA
- B. Stocks of APPA
- C. Bonds of APPA
- D. It cannot be determined

The correct answer is **B**.

$$\text{VaR of ALPHA's stocks} = 50\% \times \frac{1}{\sqrt{252}} \times 2.33 \times \$100,000,000 = \$7,338,810.18$$

$$\text{VaR of APPA's stocks} = 36\% \times \frac{1}{\sqrt{252}} \times 2.33 \times \$100,000,000 = \$5,283,943.33$$

$$\text{VaR of APPA's bond} = 10\% \times \frac{1}{\sqrt{252}} \times 2.33 \times \$100,000,000 = \$1,467,762.036$$

Both the APPA's bond and APPA's stock have a VaR lower than \$6,000,000, but the bank should choose to invest in APPA's stock since it has a higher return than APPA's bond.

Q.2626 Two managers - X and Y - are looking to establish the 1-day VaR for a long position in an at-the-money call option on a non-dividend-paying stock with the following information: Current stock price: USD 100

Estimated annual stock return volatility: 15%

Current Black-Scholes-Merton call option value: USD 4.80,

Call option delta: 0.5

To compute VaR, manager X uses the delta-normal model, while manager Y opts for the Monte Carlo simulation method for full revaluation. Which manager will estimate a higher value for the 1-day 99% VaR?

- A. Manager Y.
- B. Manager X.
- C. Both managers will have the same VaR estimate.
- D. Insufficient information to determine.

The correct answer is **B**.

Manager X, who uses the delta-normal model, will estimate a higher value for the 1-day 99% VaR. This is because options are nonlinear derivatives, meaning their value is related to the market price of the underlying variable in a convex, non-linear relationship. The payoff of such products varies not only with the value of the underlying, but also with other elements such as interest rates, volatility, and dividends. The option's price function is convex with respect to the value of the underlying. For such a non-linear portfolio, the delta-normal model provides only a linear approximation which does not capture the positive effect of this curvature on the portfolio value. It understates the probability of high option values and overstates the probability of low option values. Therefore, for a long position in the call option, VaR and the expected shortfall under the delta normal model will be extremely high. On the other hand, for a short position in the call option, the VaR and the expected shortfall under the delta-normal model will be extremely low.

Choice A is incorrect. Manager Y, who uses the Monte Carlo simulation method for full revaluation, is not likely to estimate a higher value for the 1-day 99% VaR. This method takes into account all possible price paths and their associated probabilities, which includes both large and small changes in the option's value. Therefore, it tends to produce a more conservative (lower) VaR estimate compared to the delta-normal model.

Choice C is incorrect. Both managers will not have the same VaR estimate because they are using different methods of calculation. The delta-normal model used by Manager X assumes that returns are normally distributed and only considers linear risks such as delta risk. On the other hand, Monte Carlo simulation used by Manager Y considers all types of risks including non-linear risks such as gamma and vega risk.

Choice D is incorrect. There is sufficient information to determine which manager will likely estimate a higher value for 1-day 99% VaR based on their chosen methods of calculation.

Things to Remember

- Delta-normal model is a method used to estimate Value at Risk (VaR) by assuming that asset returns are normally distributed and only considering linear risks such as delta risk.
 - Monte Carlo simulation is a method used to estimate VaR by simulating all possible price paths and their associated probabilities, taking into account all types of risks including non-linear risks such as gamma and vega risk.
 - Options are nonlinear derivatives, meaning their value is related to the market price of the underlying variable in a convex, non-linear relationship.
 - The delta of an option measures the sensitivity of the option price to changes in the price of the underlying asset.
 - Expected shortfall is a risk measure that quantifies the expected loss in the tail of the distribution beyond the VaR level.
-

Q.3305 Bank A manages interest rate risk by monitoring the VaR using historical data. Bank A collects interest rate returns for 300 days and the data is sorted ascendingly. The lowest 10 interest rate returns are -4.2%, -4.0%, -3.8%, -3.2%, -3.0%, -2.5%, -2.3%, -2.2%, -2.0%, -1.7%. After 30 days, Bank A collects 30 more data points. However, none of these returns is less than -1.7%. What is the change in the 98% VaR as compared to the prior 30 days, assuming that all of the lowest 10 observations are still within the 300-day long historical window?

- A. Unchanged.
- B. VaR has increased by 0.01%.
- C. VaR has increased by 0.08%.
- D. VaR has increased by 0.11%.

The correct answer is **A**.

In general, if there are n ordered observations, and a confidence level $cl\%$, the $cl\%$ VaR is given by the $[(1 - cl\%) n]$ th highest observation. This is the observation that separates the tail from the body of the distribution. For instance, if we have 1,000 observations and a confidence level of 95%, the 95% VaR is given by the $(1 - 0.95)1,000 = 50$ st observation. There are 50 observations in the tail.

Given this information, the 98% VaR, given that $n = 300$ is:

98% VaR = $[(1 - 0.98)300]$ th value = 6th highest observation
Thus, VaR = -2.5%

One important note about this method is that the length of the historical window is fixed, i.e., the oldest observations exit the window as new observations are made. In this case, however, 30 more days have now elapsed (meaning that we have 30 new observations) but there hasn't been a loss big enough to dislodge any of the worst 10 observations made, which we assume are still within the historical window of 300 observations. As such, the 98% VaR will **still** be the 6th highest observation which happens to be -2.5%. In short, the VaR remains unchanged.

Things to Remember

- VaR (Value at Risk) is a measure used to assess the potential loss in value of a risky asset or portfolio over a specified time period under normal market conditions.
 - Historical VaR uses historical data to estimate potential losses, assuming that past performance is indicative of future performance.
 - The confidence level in VaR represents the probability that the actual loss will not exceed the estimated VaR.
 - When calculating VaR, the observations are sorted in ascending order, and the VaR is determined by the nth highest observation based on the confidence level.
 - Changes in the historical window can impact the VaR calculation, as new data may replace older data in the calculation.
-

Q.3307 Bank A manages interest rate risk by monitoring the VaR calculated using historical data. Bank A collects interest rate returns for 300 days and the data is sorted ascendingly. The lowest 10 interest rate returns are -4.2%, -4.0%, -3.8%, -3.2%, -3.0%, -2.5%, -2.3%, -2.2%, -2.0%, -1.7%. Calculate the 98% VaR.

- A. -2.0%
- B. -2.2%
- C. -2.5%
- D. -2.4%

The correct answer is C.

In general, if there are n ordered observations, and a confidence level $cl\%$, the $cl\%$ VaR is given by the $[(1 - cl\%)n]$ th highest observation. This is the observation that separates the tail from the body of the distribution. In this case, we have 300 observations and a confidence level of 98%; the 98% VaR is given by the $(1 - 0.98)300 = 6$ th observation, that's 2.5%

Things to Remember

- VaR (Value at Risk) is a measure used to assess the potential loss in value of a risky asset or portfolio over a specified time period with a certain confidence level.
- Historical VaR is calculated using historical data on asset returns and assumes that the past behavior of the asset will continue into the future.
- The confidence level in VaR represents the probability that the actual loss will not exceed the VaR estimate.
- Sorting the data in ascending order is a common step in calculating VaR as it helps in identifying the n th highest or lowest observation based on the confidence level.
- VaR is a key risk management tool used by financial institutions to quantify and manage their exposure to market risk, credit risk, and operational risk.

Q.3313 An option on the INMEX (Mexican) stock index is struck on 2,522 pesos. The delta of the option is 0.6, and the annual volatility of the index is 25%. Using delta-normal assumptions, what is the 10-day VaR of the option at the 95% confidence level? Assume 260 days per year.

- A. 204 pesos
- B. 61.0 pesos
- C. 115.8 pesos
- D. 122.4 pesos

The correct answer is **D**.

As per the delta-normal method, the VaR of a derivative position is given by:

$$\text{VaR}_{\text{Derivative}} = \text{Delta} \times \text{VaR}_{\text{Underlying risk factor}}$$

Thus, the first step is to determine the VaR of the underlying risk factor. i.e., the index. But before that, notice that the question asks for the 10-day VaR, which means it is important that we work out the 1-day VaR and then use the square root of time rule:

$$\begin{aligned}\text{T-day volatility} &= \text{1-day Volatility} \times \sqrt{T} \\ \text{260-day volatility} &= \text{1-day volatility} \times \sqrt{260}\end{aligned}$$

$$\text{Thus, 1-day volatility} = \frac{\text{260-day volatility}}{\sqrt{260}} = \frac{0.25}{\sqrt{260}} = 0.0155$$

Hence,

$$\begin{aligned}\text{1-day VaR of the underlying} &= 2,522 \times 0.0155 \times 1.65 = 64.5 \text{ pesos} \\ \text{10-day VaR} &= \text{1-day VaR} \times \sqrt{10} = 64.5 \times \sqrt{10} = 204 \text{ pesos}\end{aligned}$$

Finally,

$$\begin{aligned}\text{VaR}_{\text{Derivative}} &= \text{Delta} \times \text{VaR}_{\text{Underlying risk factor}} \\ &= 0.6 \times 204 = 122.4 \text{ pesos}\end{aligned}$$

Note that this questions mixes concepts from the chapters Calculating and Applying VaR and Measuring and Monitoring Volatility.

Q.3316 Consider a non-linear portfolio that depends on a share price. Given that delta and gamma of the portfolio are 20 and 2.5, respectively. All else constant, what is the corresponding portfolio change to a stock price increase of \$ 0.5?

- A. \$ 9.45
- B. \$ 10.63
- C. \$ 9.38
- D. \$ 10.31

The correct answer is **D**.

Given the delta and gamma of a portfolio, the change in portfolio is given by :

$$\Delta P = \delta \Delta S + \frac{1}{2} \gamma (\Delta S)^2$$

Where

δ = Delta

ΔS = Change in the share price

γ = Gamma

ΔP = Change in portfolio value. Thus,

$$\Delta P = 20 \times 0.5 + \frac{1}{2} \times 2.5 \times 0.5^2 = 10.3125$$

Q.3318 You have been asked to estimate the VaR of GreenWood Corp. The company's stock is currently trading at USD 308 and the stock has a daily volatility of 1.25%. Using the delta-normal method, the VaR at the 95% confidence level of a long position in an at-the-money put on this stock with a delta of -0.5 over a 1-day holding period is closest to:

- A. USD 3.12
- B. USD 2.15
- C. USD 3.167
- D. USD 4.52

The correct answer is **C**.

In the case of a linear derivative, VaR scales directly with the underlying risk factor, i.e.,

$$\begin{aligned}\text{VaR}_{\text{linear derivative}} &= \Delta \times \text{VaR}_{\text{underlying risk factor}} \\ \text{VaR} &= \Delta \times 1.645 \times \sigma \times S_0 = 0.5 \times 1.645 \times 0.0125 \times 308 \\ &= 3.1666\end{aligned}$$

Note: Just as the question dictates, this is an estimate: the accurate relationship is non-linear and we are actually omitting the curvature (option gamma). Also, we ignore the negative sign in the solution since a negative amount is implied.

Note 2: This question mixes concepts from the chapters Calculating and Applying VaR and Measuring and Monitoring Volatility.

Q.3320 A market risk manager has gathered historical P&L data for his financial institution over the last 100 days. He intends to determine the VaR and the conditional VaR (CVaR) at 90% level of confidence using the historical simulation method. The worst 15 observation gathered (in million CAD) are:

[-25, -27, -27, -28, -30, -32, -36, -38, -40, -43, -45, -52, -56, -58, -60]

Calculate the VaR and the conditional VaR (CVaR).

- A. VaR = 30; Conditional VaR = 46.
- B. VaR = 32; Conditional VaR = 47.6.
- C. VaR = 36; Conditional VaR = 47.6.
- D. VaR = 32; Conditional VaR = 46.

The correct answer is **B**.

We arrange the data in ascending order first:

-60, -58, -56, -52, -45, -43, -40, -38, -36, -32, -30, -28, -27, -25

VaR can be calculated as the $[(1 - 0.9) \times 100] = 10^{\text{th}}$ worst observation, which is -32 and hence the VaR is 32.

Conditional VaR is simply another name for the expected shortfall. The ES is the arithmetic average of losses that are worse than the VaR. Thus,

$$ES = \frac{(60 + 58 + 56 + 52 + 45 + 43 + 40 + 38 + 36)}{9} = 47.6$$

Note: Refer to table 2.3 of your chapter for proof.

Q.3394 A liquid asset K has a profit/loss distribution that's independent and identically distributed. The position has a one-day VaR of \$50,000 at the 95% level of confidence. Estimate the 10-day VaR of the same position at the 99% level of confidence.

- A. \$115,114
- B. \$70,000
- C. \$223,570
- D. \$200,000

The correct answer is C.

The first step should be to convert the one-day VaR at 95% to a one-day VaR at 99%. To do this, recall that:

$\text{VaR}_{95\%}$ uses the upper 5% point of the normal distribution, i.e., 1.645

And the $\text{VaR}_{99\%}$ uses the upper 1% point of the normal distribution, i.e., 2.326

Therefore, to convert the one-day VaR at 95% to a one-day VaR at 99%, we must multiply the former by a scale of $\frac{2.326}{1.645}$

$$\text{One-day VaR}_{99\%} = \frac{2.326}{1.645} \times \$50,000 = \$70,699$$

Next, recall that:

$$\text{VaR}(T \text{ days}) = \text{VaR}(1 \text{ day}) \times \sqrt{T}$$

Therefore,

$$\text{VaR}_{10 \text{ days}} = \$70,699 \times \sqrt{10} = \$223,569.87$$

Q.3395 Peter McLeish is a risk analyst at Quantum Bank. After estimating the 99%, one-day VaR of the bank's portfolio using historical simulation with 900 past days, he is concerned that the VaR is providing too little information on tail losses. He embarks on a deeper examination of simulation results. Sorting the simulated daily P&L from worst to best, he constructs the following table:

P & L Rank	1	2	3	4	5
	-2,000	-1,860	-1,800	-1,720	-1,630
P & L Rank	6	7	8	9	10
	-1,500	-1,400	-1,310	-1,260	-1,190
P & L Rank	11	12	13	13	15
	-1,110	-1,050	-990	-820	-750

Determine the 99%, one-day expected shortfall of the portfolio:

- A. 1,260
- B. 1,653
- C. 1,190
- D. 1,609

The correct answer is **B**.

When dealing with discrete data, let's first find the VaR. VaR can be calculated as the $[(1 - 0.99) \times 900] = 9^{\text{th}}$ worst observation, which is -1,260 and hence the VaR is 1,260.

The ES is the arithmetic average of losses that are worse than the VaR. Thus,

$$ES = (2,000 + 1,860 + 1,800 + 1,720 + 1,630 + 1,500 + 1,400 + 1,310)/8 = 1,652.50$$

Q.4667 The following are hypothetical ten worst returns for stock TGB from 120 days of data. Find the 1-day 95% VaR and expected shortfall, respectively. -15.72%, -10.92%, -6.50%, -3.56%, -6.90%, -2.50%, -5.30%, -4.31%, -12.12%, -3.45%,

- A. 0.053, 0.1043
- B. 0.0431, 0.0958
- C. 0.0431, 0.1149
- D. 0.0431, 0.053

The correct answer is **A**.

First, we rearrange starting with the worst day to the least bad day, as shown below:

-15.72%, -12.12%, -10.92%, -6.90%, -6.50%, -5.30%, -4.31%, -3.56%, -3.45%, -2.50%.

In this case, the VaR corresponds to the $(5\% \times 120) = 6^{\text{th}}$ worst day: -5.30%.

This implies ES is equivalent to an average of the 5 worst returns:

$$\text{ES} = \frac{(-15.72\% + -12.12\% + -10.92\% + -6.90\% + -6.5\%)}{5} = -10.43\%$$

Q.4668 Consider a linear portfolio consisting of short positions in 50 shares, each worth USD 10, and a long position in 150 shares each worth USD 25. What is the relative portfolio change (in USD) of the portfolio if the price of all shares increases by 2%?

A. 85

B. -85

C. 65

D. -65

The correct answer is **C**.

The change in a linear portfolio is given by:

$$\Delta P = \sum_i n_i \Delta S_i$$

Where

n_i : number of shares of stock i in the portfolio (negative for a short position)

S_i : Price of stock i

So,

$$\begin{aligned}\Delta P &= [-50 \times (0.02 \times 10)] + [150 \times (0.02 \times 25)] \\ &= -10 + 75 = 65\end{aligned}$$

So the change in the portfolio is an increase of USD 65.

Q.4670 Given that the one-day standard deviation of a portfolio is 90, what is the 10-day VaR with a 99% confidence level according to the delta-normal model?

- A. 604
- B. 505
- C. 662
- D. 645

The correct answer is **C**.

According to the delta-normal model, we can assume that the mean change in risk factor is zero for a short period of time so that VaR is given by:

$$\text{VaR} = \sigma_P U$$

Where

σ_P : standard deviation of the portfolio

U: point on the normal distribution where X is exceeded

In this case, $\sigma_P = 90$ and $U = -2.326$ and thus VaR is given by:

$$90 \times -2.326 = -209.340$$

We know that:

$$\text{VaR}(T, X) = \sqrt{T} \times \text{VaR}(1, X)$$

Thus the 10-day VaR is given by:

$$\sqrt{10} \times -209.340 = -661.99$$

Q.4671 Given that the daily standard deviation of portfolio change is 90, what is the 10-day expected shortfall with a 99% confidence level according to the delta-normal model? Please click [here](#) to view the standard normal table

- A. 640
- B. 759
- C. 650
- D. 749

The correct answer is **B**.

According to the delta-normal model, for a short period of time, we can assume that the mean change in risk factor is zero so that VaR is given by:

$$ES = \sigma_P \left(\frac{e^{-\frac{U^2}{2}}}{(1 - X) \sqrt{2\pi}} \right)$$

Where

σ_P : standard deviation of the portfolio

U: point on the normal distribution where X is exceeded

In this case, $\sigma_P=90$ and $U=2.326$ and thus ES is given by:

$$ES = 90 \left(\frac{e^{-\frac{(2.326)^2}{2}}}{(1 - 0.99) \sqrt{2\pi}} \right) = 240.0635$$

But we know that:

$$ES(T, X) = \sqrt{T} \times ES(1, X)$$

Thus the 10-day ES is given by:

$$\sqrt{10} \times 240.06 = 759.1473 \approx 759$$

Q.4675 A risk manager conducts 500 historical simulations to calculate one day, 99% VaR. Which of the following describes the value of VaR in this case?

- A. VaR would be the fifth-worst loss.
- B. VaR would be the average of first, second, third, and fourth-worst losses.
- C. VaR would be the sixth-worst loss.
- D. VaR would be the average of the fifth and sixth loss.

The correct answer is **A**.

Value at Risk (VaR) is a statistical technique used to measure and quantify the level of financial risk within a firm or investment portfolio over a specific time frame. In this case, the risk manager is calculating the one-day, 99% VaR using 500 historical simulations. This means that the VaR would be the fifth-worst loss. This is because when the losses are ordered in descending order, the fifth-worst loss (which is equivalent to 1% of 500) would represent the VaR. This value represents the maximum loss that the financial institution could experience with a 99% confidence level over a one-day period.

Choice B is incorrect. The Value at Risk (VaR) is not calculated as the average of the first, second, third, and fourth-worst losses. VaR represents a measure of risk that quantifies potential losses within a certain confidence level over a specific time horizon. In this case, with 500 simulations and a 99% confidence level, the VaR would be represented by the fifth-worst loss out of all simulations.

Choice C is incorrect. The VaR would not be represented by the sixth-worst loss in this scenario. As mentioned above, given 500 simulations and a 99% confidence level, it's the fifth-worst loss that represents the one-day VaR for this financial institution.

Choice D is incorrect. The calculation of VaR does not involve averaging any set of worst losses - in this case, it's not an average of fifth and sixth worst losses. It's simply represented by one single data point - here being the fifth-worst loss from all simulations.

Things to Remember

- VaR (Value at Risk) is a measure of the potential loss on an investment or portfolio over a specific time period for a given confidence interval.
- Historical simulation is a method of calculating VaR by using historical data to simulate potential future outcomes.
- The confidence level in VaR represents the probability that actual losses will not exceed the VaR estimate.
- When calculating VaR using historical simulations, the VaR is typically based on the nth-worst loss, where n is determined by the confidence level and number of simulations.

- VaR is a key tool used by risk managers to assess and manage the level of risk in a financial institution or investment portfolio.
-

Q.4676 The 100-day 99% VaR for a portfolio is 50. What is the corresponding 250-day 99% VaR?

- A. 80.54
- B. 65.25
- C. 79.06
- D. 78.54

The correct answer is **C**.

Using the formula

$$\begin{aligned} \text{VaR}(T, X) &= \sqrt{T} \times \text{VaR}(1, X) \\ \Rightarrow \text{VaR}(1, X) &= \frac{\text{VaR}(T, X)}{\sqrt{T}} \end{aligned}$$

We, therefore, need first to calculate the one-day 99% VaR and then translate it to 250-day VaR

$$\text{VaR}(250, 99\%) = \sqrt{250} \times \left[\frac{50}{\sqrt{100}} \right] = 79.06$$

Q.4677 A risk manager realizes over a period of one month, a portfolio he manages increased by USD 5 when the stock price increased by USD 0.1. What is the value of portfolio delta?

- A. 30
- B. 50
- C. 40
- D. 20

The correct answer is **B**.

Greek letter deltas (δ) is defined as:

$$\delta = \frac{\Delta P}{\Delta S}$$

Where ΔS is a small change in risk factors such as stock price and ΔP the corresponding change in the portfolio value.

So,

$$\delta = \frac{5}{0.1} = 50$$

Q.4678 A risk manager estimates 10-day 95% VaR using the delta-model to be USD 50. What is the corresponding one-day expected shortfall at a 95% confidence level? Please click here to view the standard normal table

A. 66.78

B. 56.78

C. 19.82

D. 94.56

The correct answer is **C**.

Since we are dealing with a short time period, then:

$$\text{VaR} = \sigma_P U$$

$$\text{ES} = \sigma_P \left(\frac{e^{-\frac{U^2}{2}}}{(1 - X) \sqrt{2\pi}} \right)$$

X: confidence level

U: point on the normal distribution where X is exceeded

σ_P : standard deviation of the portfolio change

From the question, we are given that the 10-day 95% VaR is 50, and thus one-day VaR is given by:

$$\frac{50}{\sqrt{10}} = 15.81$$

We have X=95%, then U = 1.645.

$$\begin{aligned} &= 15.81 = \sigma_P \times 1.645 \\ \sigma_P &= \frac{15.81}{1.645} = 9.612 \end{aligned}$$

Therefore, the ES is given by:

$$\text{ES} = \sigma_P \left(\frac{e^{-\frac{U^2}{2}}}{(1 - X) \sqrt{2\pi}} \right) = 9.612 \left(\frac{e^{-\frac{1.645^2}{2}}}{(1 - 0.95) \sqrt{2\pi}} \right) = 19.8220$$

Q.4679 In the context of financial risk management, the phenomenon of correlation breakdown is observed during periods of high market volatility, where correlations deviate from their usual patterns seen under normal market conditions. Given this scenario, what should be the primary focus when calculating Value at Risk (VaR) and Expected Shortfall (ES), two key measures used in risk management?

- A. Calculation of VaR and ES should concentrate on what happens in extreme market conditions.
- B. Calculation of VaR and ES should concentrate on what happens in normal market conditions.
- C. Calculation of VaR and ES should concentrate on what happens in both normal and extreme market conditions.
- D. Calculation of VaR and ES should ignore changes in correlation during high market volatility.

The correct answer is **A**.

Correlations in a high volatility period are quite different from those of normal market conditions. When calculating VaR or ES, risk managers should determine what will happen in extreme market conditions. Therefore risk managers should try to determine correlations in extreme conditions rather than those in normal market conditions.

Options B and C are incorrect since correlation breakdown implies that extreme market conditions should be considered rather than normal market conditions.

Options D is incorrect because ignoring changes in correlation during periods of high volatility would lead to inaccurate risk estimates. Correlation breakdowns can significantly impact the joint distribution of asset returns and, consequently, the calculation of risk measures like VaR and ES.

Things to Remember

- Correlation breakdown can lead to underestimation of portfolio risk during periods of high market volatility.
- During times of crisis, assets that were previously thought to be uncorrelated may become highly correlated.
- Calculating VaR and ES using historical data may not capture the true risk during periods of correlation breakdown.

- Stress testing is a useful tool to assess the impact of correlation breakdown on portfolio risk.
 - Using copulas can help model the joint distribution of asset returns more accurately, especially during times of correlation breakdown.
-

Q.4681 You have been given the following 30 ordered percentage returns of an asset: [-18,-16,-14,-12,-10,-9,-7,-7,-6,-6,-6,-5,-5,-4,-4,-4,-2,-1,0,0,2,3,6,12,12,13,15,15,18,28] What is the expected shortfall at a 90% confidence level?

- A. 16
- B. 17
- C. 18
- D. 17.5

The correct answer is **B**.

To locate the 10%, we take the third-worst return $(=(1-0.90)*30)$, which is -14. However, recall that VaR need not be represented as a negative.

The ES is the arithmetic average of losses beyond 90%. Thus,

$$ES = \frac{18 + 16}{2} = 17$$

Q.6223 A portfolio manager is analyzing the risk of a portfolio consisting of equities. The manager wants to determine the VaR for the portfolio over a one-year period at a 95% confidence level. The loss distribution is normal with a mean of 10 and a standard deviation of 20. What is the one-year VaR for the portfolio using the delta normal approach?

- A. 42.9
- B. 32.9
- C. 33
- D. 22.9

The correct answer is **A**.

The VaR at a 95% confidence level can be calculated using the formula:

$$\text{VaR} = \mu + Z\sigma$$

where Z is the critical value from the standard normal distribution ($Z=1.645$ for 95% confidence), μ is the mean, and σ is the standard deviation.

For this portfolio,

$$\text{VaR} = 10 + 1.645 \times 20 = 10 + 32.9 = 42.9$$

Q.6228 A portfolio manager wants to classify a portfolio containing various financial instruments to understand its risk behavior better. The portfolio consists of a combination of straightforward equity positions and complicated derivative structures such as options on those equities. Which of the following correctly describes this portfolio?

- A. The portfolio is linear because equity positions dominate.
- B. The portfolio is nonlinear because it includes derivative structures.
- C. The portfolio is linear concerning the equity positions but nonlinear concerning the derivatives.
- D. The portfolio is neither linear nor nonlinear, due to its mixed nature.

The correct answer is **C**.

The portfolio is linear concerning the equity positions because the change in value of the equity shares is directly proportional to changes in the share prices. However, it is nonlinear concerning the derivatives such as options, whose payoff structures vary non-linearly with changes in the underlying asset prices.

A is incorrect because while the equity positions are linear, the inclusion of derivative structures, which are nonlinear, means the portfolio as a whole cannot be classified as entirely linear.

B is incorrect because it overlooks the linear behavior of the equity positions, focusing only on the complex derivatives.

D is incorrect because we can classify the portfolio by breaking it down into its constituent components: linear for equities and nonlinear for derivatives.

Things to Remember

Linear instruments, like equities, have a direct, proportional relationship between changes in the instrument's value and changes in the underlying asset price.

Nonlinear instruments, such as options, have payoffs that change in a non-proportional manner relative to changes in the underlying asset's price. Their value can change at an accelerating or decelerating rate.

Q.6229 A portfolio manager is evaluating the risk measures Value at Risk (VaR) and Expected Shortfall (ES) for a portfolio with a normally distributed return. Which of the following statements most accurately describes why ES might be preferred over VaR for comprehensive risk management?

- A. VaR is better because it provides a clear threshold for potential losses.
- B. VaR provides a more conservative estimate of risk compared to ES.
- C. ES provides a better measure of extreme losses beyond the VaR threshold.
- D. ES is less reliable because it averages losses within the confidence interval.

The correct answer is **C**.

ES is preferred because it considers the average of the worst losses beyond the VaR threshold.

A is incorrect. VaR provides a threshold but does not account for the magnitude of losses beyond this threshold.

B is incorrect. ES considers the average of the worst losses beyond the VaR threshold, which provides a more comprehensive and often more conservative estimate of extreme risk.

D is incorrect. ES focuses on tail losses beyond the VaR threshold, not just within the confidence interval.

Things to Remember

Expected Shortfall (ES), also known as Conditional Value at Risk (CVaR), measures the average loss given that the loss exceeds the VaR threshold, providing a more comprehensive view of extreme risks.

VaR provides a threshold for potential losses at a specific confidence level but does not give information about the severity of losses beyond this threshold.

Q.6231 A portfolio manager is trying to understand the benefits and limitations of the historical simulation approach to computing Value at Risk (VaR). Which of the following statements correctly identifies a limitation of the historical simulation approach?

- A. It cannot capture nonlinear relationships between risk factors.
- B. It assumes past market behaviors will repeat themselves in the future.
- C. It assumes that asset returns are normally distributed.
- D. It is not suitable for portfolios with a large number of assets.

The correct answer is **B**.

The historical simulation approach assumes that past market behaviors are indicative of future risks, which may not always hold true, especially during periods of market stress or structural changes. This reliance on historical data can sometimes lead to inaccurate risk predictions if market conditions change significantly.

A is incorrect. The historical simulation approach can capture nonlinear relationships because it uses actual historical returns.

C is incorrect. The historical simulation does not assume that asset returns are normally distributed; it uses the actual distribution of historical returns.

D is incorrect. The approach can be applied to portfolios with a large number of assets, though it might become computationally intensive.

Things to Remember

- The historical simulation approach is non-parametric, meaning it does not assume any specific distribution for asset returns. It directly uses historical data to compute VaR, making it flexible in terms of distribution assumptions.
 - Method is most effective when historical data encompasses a range of market conditions similar to what might be expected in the future.
-

Q.6232 When using the historical simulation approach to estimate the Expected Shortfall (ES) for a portfolio at a 99% confidence level, how is the ES calculated from the historical loss distribution?

- A. By taking the average of the top 1% best-performing days.
- B. By identifying the 99th percentile loss and using it as the ES.
- C. By taking the average of the losses beyond the 99th percentile VaR.
- D. By using the Monte Carlo method to simulate the tail losses.

The correct answer is **C**.

Expected Shortfall (ES) at a 99% confidence level using the historical simulation approach involves calculating the average of the losses that occur beyond the 99th percentile VaR (i.e., the tail losses). This provides a more comprehensive measure of tail risk as compared to VaR.

A is incorrect because ES focuses on the worst-performing days, not the best-performing ones.

B is incorrect because while the 99th percentile loss is related to VaR, ES requires averaging the losses beyond this point.

D is incorrect because the historical simulation approach directly uses historical data, not the Monte Carlo simulation for tail losses.

Things to Remember

- ES is calculated as the average of losses beyond the VaR threshold, particularly at extreme percentiles like 99%.
 - Historical simulation provides a direct method to estimate ES using actual historical loss data, offering good insights into tail risk.
 - Proper ranking and accurate identification of losses beyond the desired confidence level are crucial for calculating ES using historical simulation.
-

Q.6233 A financial analyst uses the delta-normal approach to calculate the 1-day VaR of a portfolio composed of non-linear derivatives like options. The portfolio's delta is 0.5, the underlying's price volatility is 20%, and the initial value of the underlying is \$100. Assuming a 95% confidence level, what is the 1-day VaR of this portfolio?

- A. \$10
- B. \$16.45
- C. \$23.29
- D. \$7.23

The correct answer is **B**.

The delta-normal approach uses the delta of the portfolio to approximate the change in portfolio value. The 1-day VaR is calculated using the formula: $\text{VaR} = \Delta P = \delta \Delta S$.

Given $\delta = 0.5$, the underlying's standard deviation (σ) = $0.2 \times \$100 = \20 . At a 95% confidence level, the z-score equals -1.645. Thus, $\text{VaR} = 0.5 \times 20 \times 1.645 = \16.45 .

Things to Remember

- The delta-normal approach relies on approximating changes in portfolio value using the portfolio's delta and underlying asset's volatility.
 - VaR at a given confidence level often uses the corresponding z-score from the normal distribution.
 - This approach is straightforward but works best for smaller changes in value and less effectively for non-linear portfolios.
-

Q.6234 A financial analyst is assessing the sensitivity of a portfolio containing several options to changes in the price of the underlying asset. The analyst is particularly interested in understanding how the portfolio's value will change with small movements in the underlying asset price. Which of the following correctly describes the role of delta in this context?

- A. Delta measures the time decay of an option's value.
- B. Delta indicates the curvature of the portfolio's value.
- C. Delta represents the rate of change of the portfolio's value.
- D. Delta measures the sensitivity of the option's price to volatility.

The correct answer is **C**.

Delta quantifies how much the value of an option or portfolio of options is expected to change with a one-unit change in the price of the underlying asset. It is a first-order measure of sensitivity to price movements.

A is incorrect because delta measures price sensitivity, not time decay. Time decay is measured by theta.

B is incorrect because delta measures linear sensitivity, while curvature (second-order sensitivity) is measured by gamma.

D is incorrect because delta does not measure sensitivity to volatility changes; that measure is called vega.

Things to Remember

- Delta represents the rate of change of an option's or portfolio's value with respect to changes in the price of the underlying asset. It measures the sensitivity of the option's value to small movements in the underlying asset's price.
- Delta is a first-order risk measure, indicating the linear sensitivity of the portfolio's value. For example, a delta of 0.5 means that for every \$1 increase in the underlying asset's price, the option's value will increase by \$0.50.
- Time decay of an option's value is measured by theta, not delta. Theta quantifies how much the value of an option decreases as it approaches its expiration date.
- Curvature or the rate of change of delta itself is measured by gamma, which is a second-order risk measure. Sensitivity to changes in volatility is measured by vega, not delta.

Q.6236 A risk manager is tasked with calculating the 1-day 99% VaR for a portfolio consisting solely of linear derivatives using a sensitivity parameter Δ derived from underlying asset prices. The sensitivity parameter Δ is 1.2, and the 1-day 99% underlying factor VaR is \$400,000. What is the 1-day 99% VaR for the portfolio?

- A. \$480,000
- B. \$333,333
- C. \$500,000
- D. \$600,000

The correct answer is **A**.

For linear derivatives, the VaR can be calculated using the formula:

$$\text{VaR}_{\text{linear derivative}} = \Delta \times \text{VaR}_{\text{underlying factor}}$$

Given $\Delta = 1.2$ and the underlying factor $\text{VaR} = \$400,000$, the portfolio $\text{VaR} = 1.2 \times 400,000 = \$480,000$.

Things to Remember

- For linear derivatives, the VaR is calculated using the sensitivity parameter Δ multiplied by the underlying factor VaR.
 - The formula effectively measures the potential change in value of a linear derivative with proportional changes in the underlying factor.
-

Q.6237 A trader managing a portfolio with several linear-term government bonds seeks to calculate the 1-day 95% VaR. The portfolio has a sensitivity Δ of 0.8 to interest rate changes, and the 1-day 95% underlying interest rate VaR is estimated to be \$250,000. What is the portfolio VaR under these conditions?

- A. \$200,000
- B. \$300,000
- C. \$187,500
- D. \$125,000

The correct answer is **A**.

The VaR for a portfolio of linear derivatives is calculated using the formula

$$\text{VaR for linear derivative} = \Delta \times \text{VaR of the underlying factor}$$

Given that $\Delta = 0.8$ and the interest rate VaR = \$250,000, the portfolio VaR = $0.8 \times 250,000 = \$200,000$.

Things to Remember

- The sensitivity parameter Δ reflects how much the portfolio value changes with small changes in the underlying factor.
 - For interest rate sensitive portfolios, understanding the correct application of Δ ensures accurate risk assessments.
-

Q.6238 Why might the delta-normal approach underestimate the risk for portfolios with significant non-linear derivative positions?

- A. It overestimates the impact of skewness and kurtosis in the underlying asset returns.
- B. It ignores normal distribution assumptions.
- C. It inadequately models the higher-order risks associated with non-linear derivatives.
- D. It assumes perfectly correlated risk factors.

The correct answer is **C**.

The delta-normal approach may underestimate risk for significant non-linear derivatives because it inadequately models higher-order risks like gamma and vega, which are crucial for capturing the curvature and volatility sensitivity in derivatives' values.

A is incorrect because the delta-normal approach does not overestimate impact-associated higher moments like skewness and kurtosis.

B is incorrect because it does not ignore normal distribution assumptions; rather, it relies on them, which is part of its limitation.

D is incorrect because the assumption of perfectly correlated risk factors typically reduces diversification benefits, not generally increasing the overall risk estimate.

Things to Remember

- Non-linear derivatives like options have higher-order risks, such as gamma and vega; the delta-normal method inadequately accounts for these.
- Assuming normally distributed returns and linear risk factors often leads to underestimation of extreme and tail risks.
- Diversification impacts and risk factor correlations are treated simplistically in the basic delta-normal approach.

Q.6239 A risk manager is evaluating the effectiveness of different risk measures for a portfolio that includes equities, bonds, and derivatives. The manager wants to ensure that the chosen risk measure can adequately handle non-linear instruments and capture extreme market movements. Which of the following risk measures would be most appropriate in this scenario?

- A. Historical VaR

- B. Parametric VaR
- C. Monte Carlo VaR
- D. Standard Deviation

The correct answer is **C**.

Monte Carlo VaR is particularly effective for portfolios that include non-linear instruments like derivatives and for capturing extreme market movements. The method uses simulations to model a wide range of possible outcomes, allowing it to handle the complexities and non-linearities inherent in such portfolios.

A is incorrect because while Historical VaR uses real historical data and can capture some extreme movements, it may not fully account for non-linear instruments as effectively as Monte Carlo simulations.

B is incorrect because Parametric VaR assumes a normal distribution of returns and may not adequately capture the non-linearities and tail risks associated with derivatives.

D is incorrect because Standard Deviation is a measure of volatility, not risk, and does not capture the tail risks or non-linear characteristics of the portfolio's instruments.

Things to Remember

- Monte Carlo VaR is a simulation-based method that models a wide range of possible outcomes, making it highly effective for portfolios with non-linear instruments like derivatives.
- By generating numerous random price paths for the portfolio's assets, Monte Carlo VaR can capture extreme market movements and provide a more accurate risk assessment under various scenarios.
- This approach does not rely on the assumption of normally distributed returns, which is crucial for accurately measuring the risks associated with derivatives and other complex financial instruments.
- Monte Carlo VaR can be computationally intensive but offers a flexible and robust framework for evaluating the risk of portfolios with diverse asset classes and non-linear exposures.

Q.6240 An investment firm uses Monte Carlo simulation to estimate the 1-day 99% VaR and ES for their asset portfolio. What is a notable weakness of this method?

- A. It cannot handle the complexity of assets with non-linear payoff structures.
- B. The results can be very sensitive to the input assumptions and require extensive computational resources.
- C. It assumes normally distributed returns, which can lead to underestimated risks.
- D. It fails to provide insights into the underlying risk factors affecting the portfolio.

The correct answer is **B**.

A notable weakness of the Monte Carlo simulation method is its sensitivity to input assumptions and the extensive computational resources required. The accuracy of the output depends strongly on the quality of the input assumptions, and simulations can be time-consuming and resource-intensive.

A is incorrect because Monte Carlo simulation is well-suited to handling non-linear payoff structures.

C is incorrect because Monte Carlo simulations do not necessarily assume normally distributed returns; they can accommodate various distribution types.

D is incorrect because the method can provide insights into risk factors, though this depends on the design of the simulations.

Things to Remember

- The accuracy of Monte Carlo simulation results hinges on the quality of the input assumptions used in the model.
 - The method requires significant computational power, making it less practical for quick calculations but essential for detailed risk analysis.
 - Model risk and implementation risk can affect the reliability of the simulation outputs, necessitating careful scenario planning and validation.
-

Q.6241 When applying the Monte Carlo simulation method, a risk manager runs 10,000 simulations for a portfolio consisting of equity and option positions. How does the Monte Carlo simulation approach help in calculating the Expected Shortfall (ES) for such a complex portfolio?

- A. By providing a single-point estimate of the median loss in the portfolio.
- B. By generating a distribution of potential portfolio values and calculating the average of the losses beyond the VaR threshold.
- C. By relying on historical data to estimate the worst possible portfolio loss.
- D. By assuming that the portfolio returns are normally distributed and calculating ES directly from this assumption.

The correct answer is **B**.

The Monte Carlo simulation approach generates a distribution of potential portfolio values and calculates the Expected Shortfall (ES) by taking the average of the losses that occur beyond the VaR threshold. This method provides a detailed view of tail risk for complex portfolios that include non-linear payoff structures.

A is incorrect because ES is not a single-point estimate of the median loss; it is an average of tail losses.

C is incorrect because Monte Carlo simulations rely on hypothetical scenarios rather than just historical data to estimate portfolio losses.

D is incorrect because Monte Carlo simulations can be used with various assumed distributions and are not limited to normally distributed returns.

Things to Remember

- Monte Carlo simulation helps calculate Expected Shortfall (ES) by generating a distribution of potential portfolio values through numerous random scenarios. For example, running 10,000 simulations provides a comprehensive set of possible outcomes for the portfolio.
- To calculate ES, identify the portfolio's Value at Risk (VaR) at the desired confidence level (e.g., 95%). Then, focus on the worst losses beyond this VaR threshold.
- ES is computed by averaging the losses that occur beyond the VaR threshold, providing a more detailed understanding of tail risk and potential extreme losses for complex portfolios with non-linear payoff structures.

Q.6242 During the 2007-2008 financial crisis, many financial institutions observed a significant change in the correlations between asset returns under stressed market conditions. What is this phenomenon known as, and what implication does it have for VaR or ES analysis?

- A. It is known as systematic risk increase and implies higher diversification benefits.
- B. It is known as correlation breakdown and implies that standard risk models may underestimate risk during stress periods.
- C. It is known as volatility clustering and implies a steady risk profile across different periods.
- D. It is known as market normalization and implies that correlations become negligible.

The correct answer is **B**.

This phenomenon is known as correlation breakdown. During stressed market conditions, asset return correlations often increase. This implies that standard risk models, which typically assume stable correlations, may underestimate risk during stress periods, making VaR and ES calculations less reliable.

A is incorrect because correlation breakdown relates to increased correlations, which reduces diversification benefits, not increases them.

C is incorrect because volatility clustering pertains to periods of high volatility following one another but does not directly address changing correlations.

D is incorrect because correlation breakdown implies increased correlations, not negligible correlations, during market stress.

Things to Remember

- Correlation breakdown refers to the phenomenon where asset correlations increase during stressed market conditions, affecting risk models.
 - This breakdown leads to underestimating risk in VaR and ES models if not accounted for correctly.
 - Adjustments for time-varying correlations are necessary to improve risk estimation accuracy under stress scenarios.
-

Q.6243 An asset manager observes a significant increase in the correlations between various assets during a period of market stress. What challenges does this correlation breakdown present when using traditional VaR or ES models?

- A. Traditional models may overestimate diversification benefits and underestimate potential losses.
- B. Traditional models become impossible to apply as correlations stabilize.
- C. VaR will underestimate potential gains because it assumes that asset returns will move in the same direction.
- D. ES becomes less relevant because tail risk becomes of less importance.

The correct answer is **A**.

During a period of market stress, correlation breakdown means that correlations between various assets increase, reducing diversification benefits. Traditional VaR and ES models that assume stable correlations tend to overestimate diversification and underestimate potential losses, making them less reliable in such scenarios.

B is incorrect because increased correlations make model application more complex, not simpler.

C is incorrect because VaR models are designed to estimate potential losses, not gains. Increased correlations usually lead to underestimation of losses due to reduced diversification benefits.

D is incorrect because, in reality, tail risk becomes more critical, and ES becomes even more relevant during periods of increased asset correlations and market stress.

Things to Remember

- Correlation breakdown reduces diversification benefits, making traditional VaR and ES models underestimate risk during market stress.
 - Accurate risk modeling under stress conditions requires dynamically adjusting for increased asset correlations.
-

Q.6244 How should a financial institution adjust its risk modeling approach when dealing with correlation breakdown during periods of high volatility?

- A. Continue using the same correlation assumptions to keep the model simple and efficient.
- B. Incorporate stressed correlations estimated from past periods of high volatility.
- C. Eliminate correlations from the model as they become less relevant.
- D. Assume zero correlations to cover the worst-case scenario.

The correct answer is **B**.

To adjust for correlation breakdown during periods of high volatility, a financial institution should incorporate stressed correlations estimated from historical periods of high volatility. This helps in better capturing the risk dynamics and providing more accurate VaR and ES estimates under stress scenarios.

A is incorrect because continuing with the same correlation assumptions makes the model inaccurate under stress conditions.

C is incorrect because eliminating correlations altogether disregards the relationship between assets, oversimplifying the model and losing information.

D is incorrect because assuming zero correlations oversimplifies the risk profile and misrepresents potential diversification benefits.

Things to Remember

- Incorporating stressed correlations from past periods of high volatility improves the accuracy of risk models during periods of correlation breakdown.
 - Dynamic adjustment of correlation assumptions during stress conditions is crucial for robust VaR and ES estimation.
-

Q.6245 During a period of financial stress, a risk analyst observes that the correlations between the asset classes in their portfolio have significantly increased. What should be the analyst's primary concern regarding the calculation of VaR and ES?

- A. The increased correlations will lead to overestimating possible losses.
- B. The increased correlations will reduce the diversification benefits, potentially leading to underestimating the true risk.
- C. The calculation will remain unaffected, as standard models are robust to correlation changes.
- D. The increased correlations will necessitate the use of a historical simulation approach exclusively.

The correct answer is **B**.

Increased correlations during financial stress periods reduce diversification benefits, potentially leading to an underestimation of the true risk if the VaR and ES models don't account for changing correlations adequately.

A is incorrect because increased correlations during stress periods typically lead to underestimating rather than overestimating risk.

C is incorrect because standard VaR and ES models assuming constant correlations are affected significantly by changes in correlations.

D is incorrect because while historical simulation can be used, it's not the exclusive method; adjustments for correlations can be made in other models too.

Things to Remember

- High correlations reduce diversification effects, leading to potential underestimation of risk in VaR and ES models.
 - Risk models should dynamically adjust correlations, especially under stress scenarios, to accurately reflect risk.
-

Q.6246 An investment firm notices that the correlations between its portfolio's asset returns have risen significantly due to a market shock. What impact does this have on the firm's calculated VaR and ES, and how should the firm address this issue?

- A. VaR and ES will likely decrease, and the firm should continue using the same correlation matrix.
- B. VaR and ES will remain unchanged, and the firm should focus on other risk measures.
- C. VaR and ES will likely increase due to higher correlations, and the firm should adjust the models to account for this change.
- D. VaR and ES will become irrelevant, and the firm should stop using them temporarily.

The correct answer is **C**.

When correlations between asset returns rise significantly due to a market shock, VaR and ES are likely to increase because higher correlations reduce diversification benefits. The firm should address this issue by adjusting its models to account for the higher correlations, ensuring that risk measures accurately reflect the new risk environment.

A is incorrect because VaR and ES are more likely to increase, not decrease, with rising correlations.

B is incorrect because VaR and ES are significantly impacted by changes in correlations and should not be ignored.

D is incorrect because VaR and ES remain relevant even under changed conditions; they need adjustment to remain accurate.

Things to Remember

- Rising correlations during market shocks increase VaR and ES by reducing diversification benefits, reflecting higher overall risk.
 - Investment firms should adjust risk models to dynamically reflect changing correlations to maintain accurate risk assessments.
-

Q.6247 Which of the following highlights a key difference between worst-case scenario analysis and Value at Risk (VaR) analysis in the context of risk management?

- A. VaR identifies the maximum loss at a certain confidence level, while WCS details the extreme loss if the worst-case event occurs.
- B. VaR provides a detailed loss distribution, while WCS only looks at probable losses within a confidence interval.
- C. WCS measures expected losses accurately within a standard deviation, while VaR focuses on tail risks.
- D. Both methods provide the same level of detail regarding the distribution of losses in the tail.

The correct answer is **A**.

VaR is focused on predicting the maximum loss at a certain confidence level, typically neglecting how severe losses can get beyond this level. On the other hand, worst-case scenario (WCS) analysis goes beyond to detail the potential extremes of losses, assuming an unfavorable event causing worst outcomes.

B is incorrect because VaR does not provide a full loss distribution, and WCS doesn't limit itself to typical confidence intervals but provides insights into extreme scenarios.

C is incorrect because WCS focuses on tail-end losses beyond standard deviations, while VaR provides probable losses within confidence intervals.

D is incorrect because WCS and VaR do not provide an equivalent level of detail concerning tail loss distributions; WCS offers more granularity in the extreme loss spectrum.

Things to Remember

- VaR specifies maximum losses at a given confidence level but doesn't provide insights into the extremeness of potential further losses.
 - Worst-case scenario (WCS) offers detailed insights into extreme losses beyond what VaR provides, useful for robust risk management.
-

Reading 49: Measuring and Monitoring Volatility

Q.538 Given that σ_t^2 is the estimated variance at time t and μ_t is the realized return at time t , select the GARCH(1,1) model that will take the longest time to revert to its mean.

A. $\sigma_t^2 = 0.05 + 0.03\mu_{(t-1)}^2 + 0.91\sigma_{(t-1)}^2$

B. $\sigma_t^2 = 0.03 + 0.03\mu_{(t-1)}^2 + 0.92\sigma_{(t-1)}^2$

C. $\sigma_t^2 = 0.07 + 0.02\mu_{(t-1)}^2 + 0.94\sigma_{(t-1)}^2$

D. $\sigma_t^2 = 0.03 + 0.04\mu_{(t-1)}^2 + 0.91\sigma_{(t-1)}^2$

The correct answer is **C**.

To determine which GARCH(1,1) model will take the longest time to revert to its mean (or unconditional variance), we need to look at the persistence of the model, which is given by the sum of the coefficients on the lagged squared return (α_1) and the lagged variance (β_1). The closer this sum is to 1, the slower the model reverts to its mean.

Here are the sums for each choice:

- **Choice A:** $\alpha_1 + \beta_1 = 0.03 + 0.91 = 0.94$
- **Choice B:** $\alpha_1 + \beta_1 = 0.03 + 0.92 = 0.95$
- **Choice C:** $\alpha_1 + \beta_1 = 0.02 + 0.94 = 0.96$
- **Choice D:** $\alpha_1 + \beta_1 = 0.04 + 0.91 = 0.95$

Among these, the sum closest to 1, indicating the longest time to revert to the mean, is:

- **Choice C** with a sum of 0.96.

Therefore, the GARCH(1,1) model from **Choice C** will take the longest time to revert to its mean.

Choice A is incorrect. The sum of the ARCH term (0.03) and the GARCH term (0.91) in this model is 0.94, which is less than that in option C (0.96). This means that shocks to the variance will decay more quickly, and hence, the model will revert to its mean faster.

Choice B is incorrect. Similar to choice A, the sum of ARCH and GARCH terms here is 0.95 which again indicates a quicker reversion to mean compared to option C.

Choice D is incorrect. Despite having a higher ARCH term (0.04), this model's total persistence (sum of ARCH and GARCH terms = 0.95) remains lower than that in option C, implying a faster mean reversion time.

Note. In a GARCH(1,1) model, higher values of $\mu^2_{(t-1)}$ or $\sigma^2_{(t-1)}$ coefficients indicate slower decay of shocks or longer persistence - thus taking longer for variance to revert back to its long-term average or 'mean'. Therefore, among all options provided here, choice C with highest combined value for these coefficients would take longest time for mean reversion.

Things to Remember

- GARCH(1,1) model stands for Generalized Autoregressive Conditional Heteroskedasticity model with 1 lag each for the autoregressive and moving average terms.
 - Persistence in a GARCH model is determined by the sum of the coefficients of the lagged squared return and the lagged variance.
 - A higher persistence value indicates a longer time for the model to revert to its mean.
 - The ARCH term in a GARCH model captures the impact of past squared returns on current volatility, while the GARCH term captures the impact of past volatility on current volatility.
 - Mean reversion refers to the tendency of a variable to return to its long-term average over time.
-

Q.539 Martin Scholes, FRM, estimates daily variance h_t using the following GARCH model on daily returns r_t :

$$H_t = \alpha_0 + \alpha_1 r_{(t-1)}^2 + \beta h_{(t-1)}$$

Where $\alpha_0 = 0.004$, $\alpha_1 = 0.05$, $\beta = 0.93$ Approximate the long-run annualized volatility (Assume there are 252 trading days in a year).

- A. 0.2
- B. 7.144
- C. 0.45
- D. 0.5

The correct answer is **B**.

In the GARCH(1,1) model, the long-run average variance rate is given by:

$$\begin{aligned} h &= \frac{\alpha_0}{(1 - \alpha_1 - \beta)} \\ &= \frac{0.004}{(1 - 0.05 - 0.93)} = 0.20 \end{aligned}$$

Taking the square root, this gives 0.45 for daily volatility.

To determine the annualized volatility, we have to multiply 0.45 by $\sqrt{252}$, where 252 is the average number of trading days in a year excluding weekends and holidays.

Thus, annualized volatility = $0.45 * \sqrt{252} = 7.144$

Q.540 Consider the GARCH(1,1) model:

$$\sigma_t^2 = \omega + \alpha \mu_{(t-1)}^2 + \beta \sigma_{(t-1)}^2$$

Where $\alpha + \beta < 1$ Which of the following statements is INCORRECT regarding the volatility term structure predicted by the model above?

- A. If we assume that the long-run estimated variance remains unchanged as α and β increase, the volatility term structure predicted by the model reverts to the long-run estimated variance faster.
- B. If we assume that the long-run estimated variance remains unchanged as α and β increase, the volatility term structure predicted by the model reverts to the long-run estimated variance more slowly.
- C. If the current volatility estimate is above the long-run average volatility, we would expect the estimated volatility term structure to be downward-sloping.
- D. If the current volatility estimate is below the long-run average volatility, we would expect the estimated volatility term structure to be upward-sloping.

The correct answer is **A**.

In the GARCH(1,1) model, the sum of α and β represents the persistence of shocks to the variance. If this sum is close to 1, shocks to the variance will be highly persistent, meaning that the volatility term structure will revert to the long-run estimated variance more slowly, not faster. Therefore, increasing α and β would not result in a faster reversion to the long-run estimated variance, contrary to what the statement suggests.

Choice B is incorrect. The statement contradicts the behavior of the GARCH(1,1) model. As α and β increase, assuming that the long-run estimated variance remains unchanged, the volatility term structure predicted by this model reverts to the long-run estimated variance faster, not slower. This is because higher values of α and β imply a greater weight being placed on recent volatility estimates in determining future volatility.

Choice C is incorrect. While it's true that if current volatility estimate is above the long-run average volatility, we would expect a downward-sloping estimated volatility term structure; however this statement does not reflect any specific behavior related to changes in α and β parameters of GARCH(1,1) model which was asked in question.

Choice D is incorrect. Similar to choice C, although it's correct that if current volatility estimate is below the long-run average we would expect an upward-sloping estimated volatility term structure; but again this statement does not reflect any specific behavior related to changes in α and β parameters of GARCH(1,1) model which was asked in question.

Q.541 A risk manager at Meridian Bank uses the exponentially weighted moving average technique to model the daily volatility of a security, with λ equal to 0.95. The current daily volatility estimate stands at 1.8%. On a certain day, the security registers a closing price of \$10 and then \$8 the following day. Determine the updated estimate of volatility:

A. 0.04803

B. 0.0529

C. 0.1521

D. 0.1812

The correct answer is **A**.

The new variance forecast will be given by:

$$\lambda \sigma_{(t-1)}^2 + (1 - \lambda) u_{(t-1)}^2$$

where

σ_n^2 = variance estimate for day n

$\sigma_{(t-1)}^2$ = volatility estimate made at the end of day (t-2) for day (t-1)

$U_{(t-1)}^2$ = The most recent observation of the squared return (on day n – 1)
 $= \left(\frac{8-10}{10}\right)^2 = (-0.2)^2 = 0.04$

λ = exponential constant

Thus, the new variance forecast is

$$0.95 * 0.018^2 + (1 - 0.95) * 0.04 = 0.0023078$$

Hence

$$\text{Volatility} = \sqrt{0.0023078} = 0.04803$$

Q.542 Consider a portfolio with 60% invested in asset Y and 40% invested in asset Z. The mean and variance of the return on Y are 0 and 49 respectively. The mean and variance of the return on Z are 1 and 84 respectively. Given that the correlation coefficient between Y and Z is 0.4, determine the portfolio volatility.

- A. 43.4
- B. 6.59
- C. 8.5
- D. 23.1

The correct answer is **B**.

$$\begin{aligned}\sigma_p^2 &= w_Y^2 \sigma_Y^2 + w_Z^2 \sigma_Z^2 + 2w_Y \cdot w_Z \cdot \text{Corr}(y, z) \cdot \sigma_Y \cdot \sigma_Z \\ &= 0.6^2 * 49 + 0.4^2 * 84 + 2 * 0.6 * 0.4 * 0.4 \sqrt{(49 * 84)} \\ &= 17.64 + 13.44 + 12.32 = 43.40\end{aligned}$$

$$\text{Volatility} = \sqrt{43.40} = 6.59$$

Q.544 The decay factor used in an EWMA model is approximated to be 0.97. In addition, daily volatility is estimated to be 1%. Given that today's stock market return is 3%, determine the new estimate of volatility using the EWMA model.

- A. 0.00027
- B. 0.000124
- C. 0.01114
- D. 0.00567

The correct answer is **C**.

$$\sigma_n^2 = \lambda \sigma_{(n-1)}^2 + (1 - \lambda) u_{(n-1)}^2$$

Where $\lambda = 0.97$ and $u_{(n-1)} = 3\%$

Thus, $\sigma_n^2 = 0.97 * 0.01^2 + (1 - 0.97) * 0.03^2 = 0.000097 + 0.000027 = 0.000124$

$$\sigma_n = 0.01114 \text{ or } 1.11\%$$

Q.545 A generalized autoregressive conditional heteroskedastic (GARCH)(1,1) model has the following parameters: $\omega = 0.000003$; $\alpha = 0.05$; $\beta = 0.94$ Determine the implied long-run volatility level.

- A. 0.0003
- B. 0.07132
- C. 0.02732
- D. 0.01732

The correct answer is **D**.

$$\text{The long-run average variance} = \frac{\omega}{(1 - \alpha - \beta)} = \frac{0.000003}{(1 - 0.05 - 0.94)} = 0.0003$$

$$\text{The long-run volatility} = \sqrt{\frac{\omega}{(1 - \alpha - \beta)}} = \sqrt{0.0003} = 0.01732$$

Q.546 The dollar/sterling exchange rate at 5 P.M. yesterday was 0.78 and the most recent estimate of the daily volatility stands at 0.8%. The EWMA model used in the analysis has $\lambda = 0.9$. If the exchange rate at 5 P.M. today proves to be 0.775, find an estimate of the new daily volatility.

A. 0.007857

B. 7.855e-05

C. 6.4e-05

D. 6.17e-05

The correct answer is **A**.

The proportional daily change is $\ln(0.775/0.78) = -0.00643$.

The current daily variance estimate = $0.008^2 = 0.000064$.

$$\begin{aligned}\text{New daily variance estimate} &= 0.9 * 0.000064 + 0.1 * 0.00643^2 \\ &= 0.0000576 + 0.000004109 = 0.0000617\end{aligned}$$

$$\text{New daily volatility} = \sqrt{0.0000617} = 0.007857 \text{ or } 0.7857\%$$

Q.547 An analyst has interest in two assets, A and B. At the close of business yesterday, these assets had daily volatilities of 1.3% and 2.0% respectively. In addition, the assets were priced at \$40 for A and \$80 for B as at the close of business yesterday, and the estimated correlation coefficient between the two assets stood at 0.25. The EWMA model used by the analyst had $\lambda = 0.95$.

Compute an estimate of the covariance between A and B.

- A. 6.5e-05
- B. 0.0006
- C. 2.60005
- D. 0.05

The correct answer is **A**.

Recall that:

$$\text{Correlation coefficient} = \frac{\text{Cov}(A, B)}{(\text{Volatility A} * \text{Volatility B})}$$

This means that if we make the covariance the subject of the formula, we get:

$$\begin{aligned}\text{Cov}(A, B) &= \text{Correlation coefficient} * \text{Volatility A} * \text{Volatility B} \\ &= 0.25 * 0.013 * 0.02 = 0.000065\end{aligned}$$

Q.548 An analyst has interest in two assets, A and B. At the close of business yesterday, these assets had daily volatilities of 1.3% and 2.0% respectively. In addition, the assets were priced at \$40 for A and \$80 for B as at the close of business yesterday, and the estimated correlation coefficient between the two assets stood at 0.25. The EWMA model used by the analyst had $\lambda = 0.95$.

Assuming that the prices of the assets today are \$40.5 and \$80.5, update the correlation coefficient.

- A. 0.00625
- B. 0.01954
- C. 6.56e-05
- D. 0.2589

The correct answer is **D**.

Recall that:

$$\text{Correlation coefficient} = \frac{\text{Cov}(A, B)}{\text{Volatility A} * \text{Volatility B}}$$

This means that if we make the covariance the subject of the formula, we get:

$$\begin{aligned}\text{Cov}(A, B) &= \text{Correlation coefficient} * \text{Volatility A} * \text{Volatility B} \\ &= 0.25 * 0.013 * 0.02 = 0.000065\end{aligned}$$

The return for A is $0.5/40 = 0.0125$

Similarly, the return for B is $0.5/80 = 0.00625$

The new covariance estimate $0.95 * 0.000065 + 0.05 * 0.0125 * 0.00625 = 0.00006566$

For asset A, the new variance estimate $= 0.95 * 0.013^2 + 0.05 * 0.0125^2 = 0.0001684$

Therefore, new volatility $= \sqrt{0.0001684} = 0.01298$

For asset B, the new variance estimate $= 0.95 * 0.02^2 + 0.05 * 0.00625^2 = 0.000382$

Therefore, new volatility $= \sqrt{0.000382} = 0.01954$

The new correlation estimate $= \frac{0.00006566}{(0.01298 * 0.01954)} = 0.2589$

Q.556 Robert Kelly, FRM, uses the EWMA model to carry out daily updates of correlation and covariance rates between two random variables X and Y. The weight for the most recent covariance on day $n - 1$ is 0.80. The correlation estimate between X and Y on day $n - 1$ is 0.6. In addition, on day $n - 1$, X and Y have estimated standard deviations of 0.013 and 0.019 respectively. Also, the percentage change on day $n - 1$ for variables X and Y are 2% and 1% respectively. Calculate the updated covariance between X and Y on day n.

- A. 0.0125
- B. 0.41
- C. 0.0001586
- D. 0.0001482

The correct answer is C.

The exponentially weighted moving average (EWMA) model uses the following formula to compute the updated covariance between 2 random variables X and Y at time n:

$$\text{Cov}_n = \lambda \times \text{cov}_{n-1} + (1 - \lambda)X_{n-1} \times Y_{n-1}$$

Where:

λ = weight of the most recent covariance on day $n - 1$

X_{n-1} = percentage change for variable X on day $n - 1$

Y_{n-1} = percentage change for variable Y on day $n - 1$

We know that $\text{Cov}_{xy} = \rho_{xy} * \sigma_x \sigma_y$

$$\text{Cov}_{n-1} = 0.6 * 0.013 * 0.019 = 0.0001482$$

$$\text{Which gives: } \text{Cov}_n = 0.8 * 0.0001482 + 0.2 * 0.02 * 0.01 = 0.0001586$$

Q.557 Robert Kelly, FRM, uses the EWMA model to carry out daily updates of correlation and covariance rates between two random variables X and Y. The weight for the most recent covariance on day $n - 1$ is 0.80. The correlation estimate between X and Y on day $n - 1$ is 0.6. In addition, on day $n - 1$, X and Y have estimated standard deviations of 0.013 and 0.019 respectively. Also, the percentage change on day $n - 1$ for variables X and Y are 2% and 1% respectively. Compute the updated correlation between X and Y

- A. 0.1586
- B. 0.2152

C. 0.3088

D. 0.6152

The correct answer is **D**.

We know that $\rho_{xy} = \frac{\text{Cov}_{xy}}{\sigma_x \sigma_y}$

Updated variance of X = $\lambda \sigma_{X,n-1}^2 + (1 - \lambda) X_{n-1}^2 = 0.8 * 0.013^2 + 0.2 * 0.02^2 = 0.0002152$

Updated variance of Y = $\lambda \sigma_{Y,n-1}^2 + (1 - \lambda) Y_{n-1}^2 = 0.8 * 0.019^2 + 0.2 * 0.01^2 = 0.0003088$

The updated covariance between 2 random variables X and Y at time n:

$$\text{Cov}_n = \lambda \times \text{cov}_{n-1} + (1 - \lambda) X_{n-1} \times Y_{n-1}$$

Where:

λ = weight of the most recent covariance on day n – 1

X_{n-1} = percentage change for variable X on day n – 1

Y_{n-1} = percentage change for variable Y on day n – 1

We know that $\text{Cov}_{xy} = \rho_{xy} * \sigma_x \sigma_y$

$$\text{Cov}_{n-1} = 0.6 * 0.013 * 0.019 = 0.0001482$$

Which gives:

$$\text{Cov}_n = 0.8 * 0.0001482 + 0.2 * 0.02 * 0.01 = 0.0001586$$

Hence the updated correlation coefficient is,

$$\rho_n = \frac{0.0001586}{(\sqrt{0.0002152} * \sqrt{0.0003088})} = 0.615238$$

Q.558 A financial analyst uses daily data to estimate a GARCH (1,1) model as follows:

$$\sigma_n^2 = 0.000002 + 0.16r_{n-1}^2 + 0.74\sigma_{n-1}^2$$

She also has established that the most recent return and variance are 0.04 and 0.02, respectively.

Calculate the updated volatility.

- A. 1.51%
- B. 11.11%
- C. 3.33%
- D. 12.27%

The correct answer is **D**.

The Generalized Autoregressive Conditional Heteroskedasticity GARCH(1,1) model is defined as:

$$\sigma_n^2 = \omega + \alpha r_{n-1}^2 + \beta \sigma_{n-1}^2$$

Where:

σ_n^2 = updated variance at time n

ω = long-term average variance rate

α = weight of the most recent squared return

r_{n-1}^2 = most recent squared return

β = weight of the most recent variance rate estimate

σ_{n-1}^2 = most recent variance rate estimate

From the information provided in the question:

$$\sigma_n^2 = 0.000002 + 0.16 \times 0.04^2 + 0.74 \times 0.02 = 0.015058$$

Volatility is given by:

$$\text{Volatility} = \sqrt{\sigma_n^2} = \sqrt{0.015058} = 0.12271 \approx 12.27\%$$

Q.565 A certain analyst uses the EWMA model with $\lambda = 0.9$ to carry out an update of correlation and covariance rates. On day $n - 1$, the observed percentage changes for variables X and Y are 3% and 2% respectively. Historical data puts the correlation estimate between X and Y at 0.54 on day $n - 1$. Furthermore, the estimated standard deviations on day $n - 1$ are 1.2% and 1.4% for X and Y respectively.

Compute the new estimate of the covariance between X and Y on day n.

- A. 9.07205
- B. 0.1234
- C. 0.1416
- D. 0.0001416

The correct answer is **D**.

$$\text{Cov}_n = \lambda \times \text{cov}_{n-1} + (1 - \lambda)X_{n-1} \times Y_{n-1}$$

But first, we must determine Cov_{n-1}

We know that $\text{Cov}_{xy} = \rho_{xy} * \sigma_x \sigma_y$

$$\text{Cov}_{n-1} = 0.54 * 0.012 * 0.014 = 0.00009072$$

And

$$\text{Cov}_n = 0.9 * 0.00009072 + 0.10 * 0.03 * 0.02 = 0.000141648$$

Q.567 The following is a variance-covariance matrix:

$$\begin{bmatrix} 1 & 0 & 0.7 \\ 0 & 1 & 0.7 \\ 0.7 & 0.7 & 1 \end{bmatrix}$$

Determine the correlation rate between variables 2 and 3.

- A. 0.7
- B. 1
- C. 0.49
- D. 0

The correct answer is **A**.

Generally, a 3×3 covariance matrix is given by:

$$\begin{bmatrix} \text{Var}(1) & \text{Cov}(1, 2) & \text{Cov}(1, 3) \\ \text{Cov}(1, 2) & \text{Var}(2) & \text{Cov}(2, 3) \\ \text{Cov}(1, 3) & \text{Cov}(2, 3) & \text{Var}(3) \end{bmatrix}$$

Now,

$$\text{Corr}(2, 3) = \frac{\text{Cov}(2, 3)}{\text{Sd}(1) \times \text{Sd}(2)} = \frac{0.7}{1 \times 1} = 0.7$$

Q.940 A number of risk measures are based on the parametric approach, which assumes that the asset returns are normally distributed. However, mathematicians and statisticians have discovered that in reality, the asset returns deviate from normality. Which of the following options is least likely consistent with the assumption that the asset returns deviate from normality?

- A. Asset returns have fat-tailed distributions which means assets have a higher probability weight in their tails relative to the normal distribution.
- B. Asset returns have skewed distribution, which means that the declines in asset prices are more severe than increases in prices.
- C. Asset returns have unstable parameter values due to varying market conditions.
- D. Asset returns have symmetrical distributions which means they are evenly distributed around the mean returns.

The correct answer is **D**.

The statement that 'Asset returns have symmetrical distributions which means they are evenly distributed around the mean returns' is least consistent with the assumption that asset returns deviate from a normal distribution. In a normal distribution, the data is symmetrically distributed with no skewness, meaning that the mean, median, and mode of the distribution are all the same. The data points are evenly distributed around the mean, creating a bell-shaped curve. However, in reality, asset returns often exhibit skewness and kurtosis, deviating from this symmetrical, bell-shaped curve of a normal distribution. Skewness refers to the asymmetry in the distribution, while kurtosis refers to the 'tailedness' of the distribution. Therefore, the assertion that asset returns have symmetrical distributions is least likely to be consistent with the real-world observations of asset return distributions.

Choice A is incorrect. Empirical evidence and statistical analysis have shown that asset returns

often exhibit fat-tailed distributions, which means they have a higher probability weight in their tails relative to the normal distribution. This is a characteristic of many financial market returns and aligns with the observed reality.

Choice B is incorrect. Asset returns are often skewed, meaning that declines in asset prices can be more severe than increases in prices. This skewness is another common feature of financial market returns and aligns with the observed reality.

Choice C is incorrect. It's true that asset return parameters can be unstable due to varying market conditions. This instability can lead to changes in volatility, skewness, kurtosis etc., which again aligns with the observed reality of asset return distributions.

Things to Remember

- **Skewness in asset returns:** Skewness refers to the asymmetry in the distribution of asset returns, where one tail is longer or fatter than the other. Positive skewness indicates a longer right tail, while negative skewness indicates a longer left tail.
- **Kurtosis in asset returns:** Kurtosis measures the 'tailedness' of the distribution of asset returns. High kurtosis indicates heavy tails and a sharp peak, while low kurtosis indicates light tails and a flat peak.
- **Normal distribution assumptions:** The normal distribution assumes that data is symmetrically distributed with no skewness or kurtosis. It is characterized by a bell-shaped curve where data points are evenly distributed around the mean.
- **Fat-tailed distributions:** Fat-tailed distributions have a higher probability weight in the tails compared to a normal distribution. This implies that extreme events are more likely to occur than what would be expected under a normal distribution.
- **Market conditions impact on parameter values:** Market conditions such as economic events, geopolitical factors, and investor sentiment can lead to unstable parameter values in asset returns, causing changes in volatility, skewness, and kurtosis.

Q.941 Donald York is a quantitative analyst at Brooklyn Investments Hub, a tech investment company based in New York. York brings 5 years of experience in quantitative and statistical analysis. In one of his explanatory articles, he mentioned that when the mean and standard deviation of asset returns are the same for any given time period, the distribution of returns is said to be an unconditional distribution. In contrast, if the mean is the same at any time while the standard deviation of the return change over time, the return distribution is referred to as a

conditional distribution. Identify the correct option from the following.

- A. York's explanation regarding the unconditional distribution is incorrect.
- B. York's explanation regarding the conditional distribution is incorrect.
- C. York's explanation regarding the conditional and unconditional distribution is incorrect.
- D. York's explanation regarding the conditional and unconditional distribution is correct.

The correct answer is **D**.

Donald York's explanation regarding the conditional and unconditional distribution is correct. In the field of finance and investment, the distribution of asset returns plays a crucial role in risk management and investment decision-making. An unconditional distribution, as explained by York, is a type of probability distribution where the mean and standard deviation of asset returns remain constant for any given time period. This implies that the probability of achieving a certain level of return does not change over time. On the other hand, a conditional distribution is one where the mean remains constant, but the standard deviation of returns changes over time. This suggests that the volatility or risk associated with the asset returns can vary, depending on different conditions or factors. Both these concepts are fundamental to understanding the behavior of financial markets and making informed investment decisions. Therefore, York's explanation of both these concepts is accurate and aligns with the standard definitions used in finance and investment analysis.

Choice A is incorrect. York's explanation of an unconditional distribution is correct. An unconditional distribution does indeed have a constant mean and standard deviation for any given time period, as it does not take into account any conditions or changes over time.

Choice B is incorrect. York's explanation of a conditional distribution is also correct. In a conditional distribution, the mean remains constant while the standard deviation can vary over time, reflecting changes in volatility based on different conditions or events.

Choice C is incorrect. As explained above, both explanations provided by York regarding unconditional and conditional distributions are accurate.

Things to Remember

- Unconditional distribution: Mean and standard deviation of asset returns remain constant for any given time period
- Conditional distribution: Mean remains constant, but the standard deviation of returns changes over time
- Understanding the distribution of asset returns is crucial for risk management and investment decision-making

- Volatility or risk associated with asset returns can vary in a conditional distribution
 - Financial markets behavior and investment decisions are influenced by these distribution concepts
-

Q.942 An analyst is conducting fundamental and technical analysis on Pak-China Trading Co (PCTC) stocks. The analysis takes into account the stock's daily returns based on the mean and volatility of returns. If PCTC returns have a mean of 9.56 bps/day, a high volatility of 14.25 bps/day and a low volatility of 6.18 bps/day, determine which of the following distributions best fits the characteristics of PCTC's return distribution?

- A. Unconditional distribution.
- B. Regime-switching distribution.
- C. Unconditionally lognormal distribution.
- D. Conditional distribution.

The correct answer is **B**.

Regime-switching models allow for the possibility of more than one regime or state, with each regime having its own distinct statistical properties. In the context given, the two regimes are the high and low volatility states. As such, the best distribution that fits the characteristics of PCTC's return distribution is a regime-switching distribution. The high volatility of 14.25 bps/day suggests that the stock is risky in certain periods, while the low volatility of 6.18 bps/day suggests that it is relatively safe in other periods.

A is incorrect. The unconditional distribution assumes that the distribution of stock returns remains constant over time. Given the stock's varying volatility (two distinct regimes), an unconditional distribution wouldn't capture this variability.

C is incorrect. The unconditionally lognormal distribution assumes that stock returns are lognormally distributed at all times, without any variation in distribution parameters like volatility. Given the data, this doesn't fit since we have two distinct volatilities.

D is incorrect. Conditional distributions relate to models where today's distribution is conditional on past information, such as in GARCH models. They focus on time-varying volatility, but not necessarily on distinct regimes like the scenario presented.

Things to Remember

- Regime-switching models allow for multiple regimes or states, each with its own statistical properties.

- Unconditional distribution assumes a constant distribution of stock returns over time.
 - Unconditionally lognormal distribution assumes lognormal distribution of stock returns at all times.
 - Conditional distributions are based on past information and focus on time-varying volatility.
-

Q.3301 An investment company uses RiskMetrics to calculate the volatility. The volatility for the previous day is 0.02, and yesterday's return is 10%. What is the updated volatility if $\lambda = 0.97$ is used?

- A. 0.0028
- B. 0.0194
- C. 0.0262
- D. 0.053

The correct answer is **C**.

Adaptive volatility using the RiskMetrics model can be found as:

$$\begin{aligned}\sigma_t^2 &= \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1,t}^2 \\ &= 0.97 * 0.02^2 + 0.03 * 0.1^2 = 0.000688\end{aligned}$$

However, remember that variance is volatility squared.

$$\sigma_t = \sqrt{0.000688} = 0.0262$$

Q.3302 A hedge fund manages risk by calculating future volatility using RiskMetrics™ to calculate the volatility. The volatility of the portfolio today is 3% per day and today's observed return is 1%. The conditional volatility estimate, assuming that $\lambda = 0.9$, is closest to:

- A. 2.86%
- B. 3.49%
- C. 2.50%
- D. 3.21%

The correct answer is **A**.

The volatility according to RiskMetrics is given by:

$$\sigma_n^2 = (1 - \lambda) r_{n-1}^2 + \lambda \sigma_{n-1}^2$$

Where

λ = The weight given to the most recent variance estimate.

r_{n-1}^2 = estimate of most recent squared return

σ_{n-1}^2 = Most recent estimate of variance

From the information given,

$$\sigma_n^2 = (1 - 0.9) 0.01^2 + 0.9 \times 0.03^2 = 0.00082$$

Thus the volatility is

$$\sqrt{0.00082} = 0.02863 \approx 2.86\%$$

Q.3303 A hedge fund manages risk by calculating future volatility using historical standard deviation. The portfolio performance in the past 5 days are 2% (n-1), 4% (n-2), 6% (n-3), 2% (n-4), and 10% (n-5), respectively. The hedge fund uses the historical standard deviation (moving average) method to calculate volatility. What is the volatility estimate?

- A. 4.00%
- B. 5.66%
- C. 3.40%
- D. 3.35%

The correct answer is **D**.

$$\sigma_n = \sqrt{\frac{1}{m-1} \sum_{i=1}^m (r_{n-i} - \bar{r})^2}$$

Where:

$$\begin{aligned} \bar{r} &= \frac{1}{m} \sum_{i=1}^m r_{n-i} \\ &= \frac{1}{5} (0.02 + 0.04 + 0.06 + 0.02 + 0.1) = 0.048 \end{aligned}$$

Now,

$$\sigma_n = \sqrt{\frac{1}{4} [(0.02 - 0.048)^2 + (0.04 - 0.048)^2 + (0.06 - 0.048)^2 + (0.02 - 0.048)^2 + (0.1 - 0.048)^2]} \approx 3$$

Q.3377 The parameters of a generalized autoregressive conditional heteroskedastic (GARCH) (1,1) model are $\alpha = 0.05$, $\beta = 0.91$. Long-run volatility is 0.80%. If estimated daily volatility is 2% and recent stock return is 3%, compute the new estimated volatility using the GARCH (1,1) model.

- A. 0.04%
- B. 2.50%
- C. 1.50%
- D. 2.03%

The correct answer is **D**.

$$\gamma = 1 - \alpha - \beta = 1 - 0.05 - 0.91 = 1 - 0.96 = 0.04$$

$$\text{Weighted long-run variance} = \omega = \gamma V_L = 0.04 * (0.008)^2 = 0.00000256$$

New variance is given by:

$$\sigma_n^2 = \omega + \alpha u_{n-1}^2 + \beta \sigma_{n-1}^2$$

$$\text{Variance} = 0.00000256 + 0.05 * 0.03^2 + 0.91 * 0.02^2 = 0.00041156$$

$$\text{Volatility} = \sqrt{0.00041156} = 0.0203 = 2.03\%$$

Q.3378 The decay factor of exponentially weighted moving average (EWMA) model is estimated to be 0.95. If the estimated daily volatility is 2% and the recent stock return is 3%, compute the new estimated volatility using the EWMA model.

- A. 2.06%
- B. 2.03%
- C. 0.04%
- D. 2.50%

The correct answer is **A**.

New variance is given by

$$\sigma_n^2 = (1 - \lambda)u_{n-1}^2 + \lambda\sigma_{n-1}^2$$

$$\text{Variance} = 0.05 * 0.03^2 + 0.95 * 0.02^2 = 0.000425$$

$$\text{Volatility} = \sqrt{0.000425} = 2.06\%$$

Q.3383 An analyst is trying to update the estimated covariance by using the exponentially weighted moving average (EWMA) model with $\lambda = 0.91$. The analyst has gathered the following relevant data.

- Estimated standard deviation on day $n - 1$ for variables X : 2%
- Estimated standard deviation on day $n - 1$ for variables Y : 3%
- The correlation between them: 0.8
- The percentage change on day $n - 1$ for variable X : 2.5%
- The percentage change on day $n - 1$ for variable Y : 3.5%

What is the updated estimated covariance between them?

- A. cannot be estimated
- B. 0.00051
- C. 0.00048
- D. 0.00051555

The correct answer is **D**.

Covariance between X and Y on day $n-1$:

$$\text{Cov}_{n-1} = 0.8 * 0.02 * 0.03 = 0.00048$$

Updated covariance using EWMA:

$$\begin{aligned}\text{Cov}_n &= \lambda \text{Cov}_{n-1} + (1 - \lambda) X_{n-1} Y_{n-1} \\ &= 0.91 * 0.00048 + (1 - 0.91) * 0.025 * 0.035 \\ &= 0.00051555\end{aligned}$$

Q.3384 An analyst is trying to update the estimated covariance by using the exponentially weighted moving average (EWMA) model with $\lambda = 0.91$. The analyst has gathered the following relevant data.

- Estimated standard deviation on day $n - 1$ for variables X : 2%
- Estimated standard deviation on day $n - 1$ for variables Y : 3%
- The correlation between them: 0.8
- The observed return on day $n - 1$ for variable X : 2.5%
- The observed return on day $n - 1$ for variable Y : 3.5%

Given that the new Covariance between X and Y is 0.00051555, what is the updated estimated correlation between them?

- A. 0.825
- B. 0.8
- C. 0.91
- D. 0.2

The correct answer is **A**.

Compute updated standard deviation using EWMA.

$$\sigma_n^2 = \lambda \sigma_{n-1}^2 + (1 - \lambda) u_{n-1}^2$$

Updated variance for X = $0.91 * 0.02 * 0.02 + 0.09 * 0.025 * 0.025 = 0.00042025$

Updated standard deviation for X = $\sqrt{0.00042025} = 0.0205$

Updated variance for Y = $0.91 * 0.03 * 0.03 + 0.09 * 0.035 * 0.035 = 0.00092925$

Updated standard deviation for Y = $\sqrt{0.00092925} = 0.030483602$

Updated correlation = $\frac{\text{Cov}(X,Y)}{[\text{SD}(X) * \text{SD}(Y)]} = \frac{0.00051555}{0.0205 * 0.030483602} = 0.824993729 = 0.825$

Q.3385 Suppose that an analyst has gathered the following relevant data for estimating updated covariance using the EWMA model:

- Estimated standard deviation on day $n - 1$ for variables X : 2%
- Estimated standard deviation on day $n - 1$ for variables Y : 3%
- The correlation between them: 0.8
- The percentage change on day $n - 1$ for variable X : 25%
- The percentage change on day $n - 1$ for variable Y : 35%

Assuming that we choose $\lambda = 0.96$, what is the updated estimated covariance between variables X and Y using the EWMA model?

- A. 0.00041
- B. 0.00048
- C. 0.08402
- D. 0.003961

The correct answer is **D**.

$$\text{Cov}_{n-1} = 0.8 \times 0.02 \times 0.03 = 0.00048$$

Updated covariance using EWMA model is given by:

$$\begin{aligned}\text{Cov}_n &= (\lambda)\text{Cov}_{n-1} + (1 - \lambda)X_{n-1}Y_{n-1} \\ &= 0.96 \times 0.00048 + 0.04 \times 0.25 \times 0.35 \\ &= 0.0039608 \approx 0.003961\end{aligned}$$

Q.4661 A stock market investor records the stock price for five consecutive days as 20.20, 20.00, 21.20, 21.00, and 23.30. Estimate the daily volatility using the stock price returns.

- A. 10.25%
- B. 5.6%
- C. 11.45%
- D. 12.56%

The correct answer is **B**.

The variance rate is given by:

$$\sigma_n^2 = \frac{1}{m} \sum_{i=1}^m r_{n-i}^2$$

Recall that return are calculated by:

$$r_i = \frac{S_i - S_{i-1}}{S_{i-1}}$$

The following table gives the consecutive returns:

Day	Stock Price	r_i
1	20.2	–
2	20.0	–0.0099
3	21.2	0.06
4	21.0	–0.00943
5	23.3	0.10952

Now,

$$\sigma_n^2 = \frac{1}{m} \sum_{i=1}^m r_{n-i}^2$$

$$\frac{1}{5} [(-0.0099)^2 + (0.06)^2 + (-0.00943)^2 + (0.10952)^2] = 0.0031563$$

The volatility is the square root of the variance rate:

$$\therefore \sigma_n = \sqrt{0.0031563} = 0.056 = 5.6\%$$

Note: Your textbook in section 3.3 says the following: "The usual formula for calculating standard deviations from sample data would give the volatility estimated for day n from the return on the m previous days [using $m-1$]. In risk management, we usually simplify this formula [by replacing $m - 1$ by m]."

Also, it gives an example on which you can see that we use $1/m$ as the actual number of **days** and not the number of returns.

Q.4662 Consider a GARCH (1,1) model with $\omega=0.00005$, $\alpha=0.025$, and $\beta=0.90$, what is the value of long-run average volatility?

- A. 0.0245
- B. 0.0258
- C. 0.0051
- D. 0.00735

The correct answer is **B**.

The formula for the updated variance rate under GARCH(1,1) model is given by:

$$\sigma_n^2 = \alpha r_{n-1}^2 + \beta \sigma_{n-1}^2 + \gamma V_L$$

Where

V_L : long-run average variance rate

α : weight given to the most recent squared returns

β : weight given to the previous variance rate estimate

γ : weight given to long-run average variance rate

V_L can be defined as:

$$V_L = \frac{\omega}{\gamma} = \frac{\omega}{1 - \alpha - \beta} = \frac{0.00005}{1 - 0.925} = 0.0006667$$

Therefore, the long-run average volatility is given by:

$$\sqrt{0.0006667} = 2.58\%$$

Q.4663 The current volatility for stock prices is estimated to be 4% per day. Assuming the GARCH (1,1) model with $\omega=0.00005$, $\alpha=0.025$, and $\beta=0.90$, what is volatility estimate in 50 days?

- A. 0.0009
- B. 0.0044
- C. 0.0303
- D. 0.0262

The correct answer is **D**.

According to GARCH (1,1), the expected variance rate on day t is given by:

$$\sigma_{n+t}^2 = V_L + (\alpha + \beta)^t(\sigma_n^2 - V_L)$$

Where

V_L : long-run average variance rate

α : weight given to the most recent squared returns

β : weight given to the previous variance rate estimate

σ_n^2 : current volatility estimate

Now,

$$\begin{aligned} V_L &= \frac{\omega}{1 - \alpha - \beta} \\ &= \frac{0.00005}{1 - 0.025 - 0.90} = 0.00067 \end{aligned}$$

$$\text{Thus, } \sigma_{n+50}^2 = 0.00067 + (0.025 + 0.90)^{50}(0.04^2 - 0.00067) = 0.0006889$$

Expected volatility is the square root of the expected variance rate:

$$\sigma_{n+50} = \sqrt{0.0006889} \approx 0.00262$$

Q.4664 An investor notes that the closing stock price for asset A yesterday was USD 50, with a corresponding volatility of 2.5% per day. Similarly, the closing stock price for asset B was USD 30, with a corresponding volatility of 1.5% per day. Today, the stock price for asset A closed at

USD 45, and that of B closed at USD 35. The correlation coefficient between the stocks A and B on close of trading yesterday was 0.55.

Using the EWMA model with $\lambda=0.85$, what is the updated correlation coefficient between stocks A and B?

- A. 0.64
- B. -0.64
- C. 0.78
- D. -0.78

The correct answer is **D**.

According to the EWMA model,

$$s_n^2 = (1-\alpha)r_{n-1}^2 + \alpha s_{n-1}^2$$

For stock A, a return is -0.10 ($= \frac{45-50}{50}$) and that of B is 0.1667 ($= \frac{35-30}{30}$). Therefore, updated volatility for stock A is:

$$s_{An} = \sqrt{(1 - 0.85) \times (-0.10)^2 + 0.85 \times 0.025^2} = 0.04507$$

And that of B is:

$$s_{Bn} = \sqrt{(1 - 0.85) \times (0.1667)^2 + 0.85 \times 0.015^2} = 0.06603$$

Now using the formula:

$$\text{Corr}(A, B) = \frac{\text{Cov}(A, B)}{s_A s_B}$$

$$\alpha \text{Cov}(A, B) = \text{Corr}(A, B) \times s_A s_B$$

So the covariance yesterday was:

$$\text{Cov}(A_{n-1}, B_{n-1}) = 0.55 \times 0.025 \times 0.015 = 0.00020625$$

The covariance is updated using the formula:

$$\text{cov}_n = \alpha \text{cov}_{n-1} + (1-\alpha)x_{n-1}y_{n-1}$$

$$\begin{aligned}\text{Cov}(A_n, B_n) &= 0.85 \times 0.00020625 + (1 - 0.85) \times -0.10 \times 0.1667 \\ &= -0.002325\end{aligned}$$

Using the formula $\text{Corr}(A, B) = \frac{\text{Cov}(A, B)}{s_A s_B}$ once again, the new correlation is given by:

$$\text{Corr}(A_n, B_n) = \frac{-0.002325}{0.04507 \times 0.06603} = -0.7813$$

Q.6248 A financial analyst is reviewing the return distributions of various equity portfolios. It is commonly assumed that asset returns follow a normal distribution for simplicity and ease in modeling. However, empirical evidence often suggests otherwise. In what ways do actual asset return distributions typically deviate from the normal distribution?

- A. They exhibit lighter tails and higher mean returns compared to the normal distribution.
- B. They demonstrate negative skewness and fat tails, indicating a higher probability of extreme negative events.
- C. They show perfectly symmetrical distribution with consistent variance over time.
- D. They are less volatile than predicted by a normal distribution, with stable mean and variance parameters.

The correct answer is **B**.

Asset return distributions often deviate from the normal distribution by demonstrating negative skewness and fat tails. This means that there is a higher probability of extreme negative events than what is predicted by a normal distribution, making the returns more volatile and risky than a normal distribution would suggest.

A is incorrect because actual asset return distributions generally exhibit fatter tails rather than lighter tails and do not necessarily have higher mean returns compared to a normal distribution.

C is incorrect because asset return distributions are often not perfectly symmetrical and demonstrate volatility clustering, with periods of high and low volatility.

D is incorrect because actual asset returns are typically more volatile than predicted by a normal distribution, with unstable mean and variance parameters over time.

Things to Remember

- Asset returns frequently show negative skewness, indicating that extreme negative returns are more likely than extreme positive returns.
 - Fat tails in the distribution imply a higher probability of extreme events compared to a normal distribution, impacting risk measure calculations like VaR and ES.
-

Q.6249 What is one of the primary reasons for negative skewness and fat tails in asset return distributions, and what implications does this have for risk management?

- A. Stable market conditions leading to predictable returns.
- B. High-frequency trading causing frequent small fluctuations.
- C. Instability in market parameters and increased likelihood of extreme losses.
- D. Perfectly symmetrical distribution of asset returns.

The correct answer is **C**.

Negative skewness and fat tails in asset return distributions are often caused by the instability in market parameters, such as mean and standard deviation, resulting in a higher likelihood of extreme losses. This implies that traditional risk measures might underestimate the risk, making it critical for risk managers to account for these extreme events.

A is incorrect because stable market conditions do not lead to negative skewness and fat tails.

B is incorrect because high-frequency trading may lead to frequent small fluctuations but does not necessarily cause fat tails.

D is incorrect because asset return distributions are not perfectly symmetrical, as indicated by negative skewness and fat tails.

Things to Remember

- Instability in market parameters, such as mean and standard deviation, contributes to negative skewness and fat tails in asset return distributions.
 - Risk managers must acknowledge and model these extreme events to avoid underestimating potential risks.
 - Fat tails imply higher probabilities of extreme losses, necessitating adjustments in traditional risk models.
-

Q.6250 A financial analyst is analyzing a portfolio and needs to distinguish between conditional and unconditional distributions. The analyst knows that asset returns can follow these distributions under different market conditions. Which of the following best describes the difference between conditional and unconditional distributions?

- A. Conditional distributions assume that asset returns are constant over time, whereas unconditional distributions assume varying asset returns.
- B. Unconditional distributions assume constant mean and variance, while conditional distributions allow these to change with market conditions.
- C. Conditional distributions are based on historical averages, while unconditional distributions use forward-looking estimates.
- D. Conditional distributions use a static model, while unconditional distributions consider time-varying parameters.

The correct answer is **B**.

Unconditional distributions assume that the mean and variance of asset returns are the same for any given day, regardless of market and economic conditions. In contrast, conditional distributions account for changes in mean and variance depending on varying market conditions, offering a more realistic reflection of the financial environment.

A is incorrect because it reverses the assumption; conditional distributions account for varying returns while unconditional assume constancy.

C is incorrect because both conditional and unconditional distributions can use historical data; the difference lies in how they adjust for market conditions.

D is incorrect because it reverses the roles; conditional distributions consider time-varying parameters, whereas unconditional distributions use static parameters.

Things to Remember

- Unconditional distributions assume constant mean and variance for asset returns, ignoring market conditions.
- Conditional distributions recognize changing mean and variance, reflecting different market conditions and giving a more accurate risk assessment.

Q.6251 How does regime switching enhance the understanding of asset return distributions, and why is it important in quantifying volatility?

- A. It provides a constant mean for asset returns and ignores changes in volatility.
- B. It captures the occurrence of fat tails by considering both high and low volatility regimes with constant mean returns.
- C. It averages out the extreme highs and lows in asset prices to provide stable return estimates.
- D. It assumes that volatility is constantly high, thereby simplifying the analysis.

The correct answer is **B**.

Regime switching captures the occurrence of fat tails by considering multiple regimes that can have either high or low volatility while maintaining a constant mean for asset returns. This approach helps in resolving the fat tails problem and provides a more nuanced understanding of volatility in different market conditions.

A is incorrect because regime switching considers changes in volatility, rather than ignoring them.

C is incorrect because regime switching does not simply average out extremes but distinguishes between distinct regimes of volatility.

D is incorrect because regime switching does not simplify analysis by assuming constant high volatility but instead accounts for varying volatility regimes.

Things to Remember

- Regime switching accounts for multiple market conditions, capturing periods of both high and low volatility, which gives a better picture of return distributions.
 - By considering different volatility regimes, regime switching addresses the fat tails problem more effectively than models assuming constant volatility.
-

Q.6252 A portfolio manager is considering two approaches to estimate conditional volatility for a portfolio: the Exponentially Weighted Moving Average (EWMA) and the GARCH(1,1) model. What is a key advantage of the GARCH(1,1) model over the EWMA model?

- A. GARCH(1,1) assumes constant volatility over time, making calculations easier.
- B. GARCH(1,1) can capture the mean reversion property of volatility more effectively than EWMA.
- C. EWMA accounts for fat-tails in return distributions, making it superior to GARCH(1,1).
- D. EWMA is less sensitive to new data and provides smoother volatility estimates.

The correct answer is **B**.

The GARCH(1,1) model captures the mean reversion property of volatility more effectively than the EWMA model. This means that GARCH(1,1) can model how volatility tends to revert to a long-run mean over time, making it useful for predicting future volatility under changing market conditions.

A is incorrect because GARCH(1,1) does not assume constant volatility over time; it models changing volatility.

C is incorrect because EWMA does not specifically account for fat-tails in return distributions; GARCH(1,1) provides a more robust framework.

D is incorrect because EWMA is, in fact, more sensitive to new data with exponential weighting, and GARCH(1,1) provides better long-term volatility predictions.

Things to Remember

- The GARCH(1,1) model is effective in capturing mean reversion properties of volatility, useful for accurate long-term predictions.
- While EWMA is simple and responsive, GARCH(1,1) offers a more comprehensive approach by incorporating historical variance and return data.

Q.6253 A financial analyst is assessing two different models for estimating conditional volatility: EWMA and GARCH(1,1). Which of the following best describes a key difference between these two approaches?

- A. EWMA uses historical price data, while GARCH(1,1) does not.
- B. GARCH(1,1) can model long-run variance levels, while EWMA cannot.
- C. EWMA assumes normally distributed returns, whereas GARCH(1,1) does not.
- D. GARCH(1,1) provides more accurate volatility estimates than EWMA.

The correct answer is **B**.

The GARCH(1,1) model captures the mean reversion property of volatility more effectively than the EWMA model. This means that GARCH(1,1) can model how volatility tends to revert to a long-run mean over time. As a result, it provides more accurate predictions of future volatility under changing market conditions. This makes GARCH(1,1) useful for financial analysts and traders who need reliable volatility forecasts to manage risk and make informed investment decisions.

A is incorrect because both models primarily use historical price data.

C is incorrect because both models can be implemented under various distributional assumptions.

D is incorrect as the accuracy of each model can vary based on specific market conditions and the data used.

Things to Remember

- GARCH(1,1) (Generalized Autoregressive Conditional Heteroskedasticity) can model long-run variance levels, capturing the mean-reverting nature of volatility. It estimates current volatility based on past returns and past volatility, allowing for long-term volatility forecasting.
- EWMA (Exponentially Weighted Moving Average) assigns exponentially decreasing weights to past observations, emphasizing recent data more than older data. While it reacts quickly to changes in market conditions, it does not account for long-term mean reversion.

Q.6254 In the field of financial econometrics, accurately modeling and predicting volatility is crucial for risk management and investment decisions. Among the various models available, the GARCH(1,1) model and the EWMA model are widely used. What is a critical limitation of EWMA in estimating conditional volatility compared to the GARCH(1,1) model?

- A. EWMA fails to incorporate recent volatility data and changes slowly over time.
- B. EWMA cannot adjust for new information quickly, providing outdated volatility estimates.
- C. EWMA does not account for mean reversion in volatility, reducing its accuracy during changing market conditions.
- D. EWMA incorporates too many external variables, making the model overly complex.

The correct answer is **C**.

A critical limitation of the EWMA model is that it does not account for mean reversion in volatility. This oversight reduces its accuracy, particularly during periods of changing market conditions where volatility tends to revert to a long-run mean. The absence of this feature in the EWMA model means it cannot dynamically adjust to shifts in volatility levels, leading to less reliable forecasts. In contrast, the GARCH(1,1) model effectively captures mean reversion, allowing it to model how volatility returns to its long-run average over time. This capability makes GARCH(1,1) more adept at predicting future volatility under varying market conditions.

A is incorrect because EWMA does incorporate recent volatility data and responds quickly to new information.

B is incorrect because EWMA is generally sensitive to new information due to its exponential weighting mechanism.

D is incorrect because EWMA is actually simpler and does not incorporate too many external variables.

Things to Remember

EWMA (Exponentially Weighted Moving Average) assigns more weight to recent observations, allowing it to quickly reflect recent market changes. However, it does not account for mean reversion, a key behavior of financial volatility.

GARCH(1,1) (Generalized Autoregressive Conditional Heteroskedasticity) explicitly models mean reversion, providing a more accurate and reliable prediction of volatility over both short and long terms.

Q.6255 A financial analyst wants to use the exponentially weighted moving average (EWMA) approach to estimate the volatility of returns. What is a key characteristic of the EWMA model in estimating volatility?

- A. EWMA assigns equal weights to all historical return data.
- B. EWMA assigns declining weights to older historical return data.
- C. EWMA weights historical return data equally.
- D. EWMA assumes that volatility is constant over time.

The correct answer is **B**.

The EWMA model assigns declining weights to historical return data, giving more importance to recent returns. This means that recent return data has a larger impact on the current volatility estimate, making the EWMA model sensitive to recent changes in market conditions. However, a critical limitation of the EWMA model is that it does not account for mean reversion in volatility. This reduces its accuracy, particularly during periods of changing market conditions where volatility tends to revert to a long-run mean.

A is incorrect because EWMA does not assign equal weights; it assigns exponentially declining weights.

C is incorrect because the weights in EWMA are not equal and are not solely based on the length of the observation period.

D is incorrect because EWMA regularly updates volatility estimates with new data, reflecting changing market conditions.

Things to Remember

- The EWMA model gives greater weight to recent return data, making it more responsive to recent market changes.
 - The approach uses an exponential decay factor (λ) to determine the rate at which weights decline.
-

Q.6256 A risk manager needs to determine the updated volatility for a stock using the EWMA approach. The estimate of daily volatility over the previous day is 2% and the return on the asset over the previous day is -3%. Given a smoothing parameter (λ) of 0.94, what is the updated volatility?

- A. 2.5%
- B. 1.95%
- C. 2.25%
- D. 2.07%

The correct answer is **D**.

Using the EWMA formula for updating volatility:

$$\begin{aligned}\sigma_n^2 &= (1 - \lambda)r_{n-1}^2 + \lambda\sigma_{n-1}^2 \\ \text{New Volatility} &= \sqrt{(1 - 0.94)(-0.03)^2 + 0.94(0.02)^2} \\ &= \sqrt{(0.06)(0.0009) + (0.94)(0.0004)} \\ &\approx 0.02073 \text{ or } 2.07\%.\end{aligned}$$

- σ_n^2 : The updated variance for the next period (n).
- λ : The decay factor, which determines the weight given to past data.
- r_{n-1}^2 : The squared return for the previous day, (n - 1).
- σ_{n-1}^2 : The variance for the previous day, (n - 1).

Things to Remember

- The EWMA model updates volatility using the previous returns and previous volatility estimate, weighted by the smoothing parameter.
- The smoothing parameter (λ) determines the responsiveness of the volatility estimate to new data.
- A higher λ value implies slower responsiveness, while a lower λ value means faster responsiveness to return changes.

Q.6257 Despite having several disadvantages, the use of EWMA also comes with a number of positives. Which of the following is an advantage of using the EWMA model over simpler historical volatility methods?

- A. It assigns equal importance to all historical data points.
- B. It captures the mean reversion property of volatility better than other models.
- C. It provides a more responsive volatility estimate that quickly incorporates new information.
- D. It assumes that volatility remains constant over long periods.

The correct answer is **C**.

The EWMA model provides a more responsive volatility estimate that quickly incorporates new information. This feature makes it suitable for dynamic financial markets where recent data may be more indicative of current conditions. By assigning declining weights to historical return data, the EWMA model ensures that recent returns have a larger impact on the current volatility estimate. This responsiveness allows the EWMA model to adapt swiftly to market changes, making it particularly useful in environments where volatility is rapidly evolving.

A is incorrect because EWMA does not assign equal importance; it gives more weight to recent data.

B is incorrect because mean reversion is better modeled by GARCH rather than EWMA.

D is incorrect because EWMA does not assume constant volatility; it adapts to new data regularly.

Things to Remember

- The EWMA model adapts to new data quickly, making it preferable in volatile markets.
 - EWMA's weighting scheme ensures recent returns have a greater influence on volatility estimates.
-

Q.6258 Alternative approaches to weighting historical return data include methods like Historical Simulation and Multivariate Density Estimation (MDE). How does MDE differ from EWMA in weighting historical data?

- A. MDE assigns equal weights to all historical return data.
- B. MDE weights data based on the similarity between current and past conditions.
- C. MDE uses a linear weighting scheme where the weights decrease steadily over time.
- D. MDE assumes a constant volatility environment similar to EWMA.

The correct answer is **B**.

Multivariate Density Estimation (MDE) weights historical return data based on the similarity between current and past conditions, rather than using a fixed exponential decay factor as in EWMA. This allows MDE to better capture patterns that are contextually similar to the current market environment.

A is incorrect because MDE does not assign equal weights; it assigns weights based on similarity.

C is incorrect because MDE does not use a linear weighting scheme.

D is incorrect because MDE does not assume a constant volatility environment and differs fundamentally in its approach from EWMA.

Things to Remember

- MDE does not assume a constant volatility environment. Instead, it adapts to changing market conditions by adjusting weights based on the contextual similarity of past data.
- In contrast, EWMA uses a predefined decay factor that may not accurately reflect the complexities and nuances of market behavior, potentially limiting its effectiveness in certain scenarios.

Q.6259 A financial analyst is using the GARCH (1,1) model to estimate the volatility of a financial asset. If the parameters of the model are: $\omega = 0.01$, $\alpha = 0.05$, and $\beta = 0.90$, and the last observed variance is 0.04, and the last observed return is 0.03 what is the estimated volatility for the next period?

- A. 0.2146
- B. 0.0435
- C. 0.0545
- D. 0.046

The correct answer is **A**.

The GARCH (1,1) model estimates the next period's volatility using the formula:

$$\sigma_n^2 = \omega + \alpha \times r_{(n-1)}^2 + \beta \times \sigma_{(n-1)}^2$$

Where α and β are the weights given to the most recent squared return and the previous variance rate estimate, respectively.

Substituting the given values, we get:

$$\sigma_n^2 = 0.01 + 0.05 \times (0.03)^2 + 0.90 \times (0.04) = 0.046045$$

Therefore, the estimated volatility (standard deviation) is

$$\sqrt{0.046045} = 0.21458$$

Q.6260 A risk manager estimates the volatility of a stock using the GARCH (1,1) model and discovers that the sum of the coefficients α and β (Where α and β are the weights given to the most recent squared return and the previous variance rate estimate, respectively) is very close to 1. What implication does this have for the stock's volatility behavior?

- A. The stock's volatility will revert quickly to its long-term average.
- B. The stock's volatility will exhibit significant mean reversion and stabilize rapidly.
- C. The shocks to the stock's volatility will be persistent and decay slowly over time.
- D. The stock's volatility will remain unaffected by past returns.

The correct answer is **C**.

If the sum of the coefficients α and β is very close to 1 in a GARCH (1,1) model, it means that shocks to the stock's volatility will be persistent and decay slowly over time. Such a scenario implies that volatility shocks have long-lasting effects on the stock's volatility. This means that any significant change in volatility will have a prolonged impact, causing the volatility to remain elevated or subdued for an extended period. In such a scenario, volatility shocks have long-lasting effects on the stock's volatility, as the model suggests that the current volatility is heavily influenced by past volatility and past shocks. Consequently, the volatility process becomes highly dependent on its history, making it more challenging to return to a long-run mean. This persistence can be particularly important for risk management and forecasting, as it implies that the effects of market events or economic changes can endure much longer than in models with less persistent volatility.

A is incorrect because a sum close to 1 indicates slow, not rapid, reversion to the mean.

B is incorrect because a higher sum suggests less mean reversion and slower stabilization.

D is incorrect as it misinterprets the GARCH model's reflection of past return impacts on volatility.

Things to Remember

- In the GARCH(1,1) model, α represents the weight given to the most recent squared return (the immediate impact of new information or shocks on volatility).
 - β represents the weight given to the previous period's forecasted variance (how much past volatility influences current volatility).
-

Q.6261 A financial analyst needs to estimate long-horizon volatility using the GARCH (1,1) model. How can the analyst extend the 1-day volatility estimate to a 30-day time horizon, given a daily variance rate of 0.000025?

- A. Multiply the 1-day variance rate by 30 and take the square root.
- B. Multiply the 1-day volatility by 30 and take the square root.
- C. Square the 1-day variance rate and multiply by 900.
- D. Add up the daily volatility estimates for 30 days and average them.

The correct answer is **A**.

To estimate long-horizon volatility, the variance over T days is calculated by multiplying the 1-day variance rate by T. Therefore, the 30-day variance rate is 0.000025×30 . To obtain the 30-day volatility, take the square root of the variance: $\sqrt{(0.000025 \times 30)} \approx 0.0274$ or 2.74%.

B is incorrect because the multiplication and square-root process must be applied to the variance rate, not directly to volatility.

C is incorrect as this misinterprets the variance scaling law.

D is incorrect because adding up volatility estimates is not the correct approach; using variance is necessary for accurate calculations.

Things to Remember

- GARCH(1,1) Model Components:
 - α (Alpha): Measures the impact of recent returns on current volatility. A higher α indicates that recent price changes have a significant impact on volatility.
 - β (Beta): Represents the influence of past volatility on current volatility. A higher β suggests that past volatility has a strong influence on current volatility.
 - ω (Omega): A constant term that ensures the variance remains positive and defines the baseline level of volatility.
-

Q.6262 What is the implication of mean reversion in a GARCH (1,1) model for long-term volatility estimates?

- A. Volatility shocks have no lasting impact regardless of the shock size.
- B. Volatility will revert to a long-term average, reducing dependence on short-term shocks over time.
- C. Volatility will increase indefinitely with accumulating shocks.
- D. Volatility is assumed to be constant, with no fluctuations over time.

The correct answer is **B**.

In a GARCH(1,1) model, mean reversion implies that although short-term shocks may significantly impact volatility, the volatility will eventually revert to a long-term average. This characteristic indicates that even after substantial market events that cause spikes or drops in volatility, the effect of these shocks will diminish over time. The mean reversion property ensures that volatility does not perpetually deviate from its historical average, reducing the lasting impact of these shocks. Instead, volatility exhibits a tendency to return to stable average levels, providing a more predictable and consistent measure of market risk over the long term.

A is incorrect because short-term shocks do impact volatility temporarily.

C is incorrect as accumulating shocks will not increase volatility indefinitely; they revert to the mean.

D is incorrect because GARCH (1,1) explicitly models volatility fluctuation over time.

Things to Remember

- Mean reversion in a GARCH (1,1) model ensures that volatility will eventually return to a long-term average, regardless of short-term deviations.
 - This characteristic helps in maintaining stability in long-term volatility predictions.
-

Q.6263 A risk manager is using the GARCH (1,1) model to project long-term VaR. If the current estimated daily volatility is 3% and the model parameters are $\omega = 0.00005$, $\alpha = 0.15$, and $\beta = 0.80$, calculate the long-term variance rate.

- A. 0.25
- B. 0.0278
- C. 0.00025
- D. 0.001

The correct answer is **D**.

The long-term variance rate (VL) in the GARCH (1,1) model is calculated using the formula:

$$VL = \frac{\omega}{1 - \alpha - \beta}$$

Given $\omega = 0.00005$, $\alpha = 0.15$, and $\beta = 0.80$, we find:

$$VL = \frac{0.00005}{1 - 0.15 - 0.80} = 0.001$$

Things to Remember

- The long-term variance rate in GARCH (1,1) is driven by the constants ω , α , and β and their sum's relationship to 1.
 - Scaling daily GARCH estimates is crucial in projecting long-horizon VaR to maintain consistency with the desired time period.
-

Q.6264 An analyst wants to evaluate the effectiveness of implied volatility as a predictor of future volatility compared to historical volatility estimates. What is one significant advantage that implied volatility has over historical volatility?

- A. Implied volatility reflects the market's expectation of future volatility, while historical volatility relies solely on past data.
- B. Implied volatility assumes constant volatility over time, whereas historical volatility is more dynamic.
- C. Implied volatility does not depend on option prices for its calculation.
- D. Historical volatility is unaffected by market sentiments compared to implied volatility.

The correct answer is **A**.

Implied volatility reflects the market's expectation of future volatility, as it is derived from the prices of current options in the market. This measure captures the collective outlook of market participants on how volatile they believe the underlying asset will be over the life of the option. Because it incorporates traders' forecasts and sentiments, implied volatility is inherently forward-looking. In contrast, historical volatility is based solely on past price data of the asset, providing a backward-looking measure of how volatile the asset has been over a specified period. This reliance on historical data means that historical volatility does not account for current market conditions or future uncertainties.

B is incorrect because implied volatility does not assume constant volatility; it changes based on market conditions.

C is incorrect as implied volatility is indeed derived from option prices.

D is incorrect because historical volatility is not influenced by market sentiments, while implied volatility is.

Things to Remember

- Implied volatility is forward-looking, incorporating market expectations and sentiments into future volatility predictions.
 - Historical volatility, rooted in past return data, lacks the predictive insight provided by implied volatility.
-

Q.6265 Which of the following is a shortcoming of using implied volatility as a predictor of future volatility?

- A. Implied volatility can differ significantly across different options for the same underlying asset.
- B. Implied volatility is based on historical price data and may not reflect current market conditions.
- C. Implied volatility assumes that future volatility will be lower than historical volatility.
- D. Implied volatility is influenced by market supply and demand for options, which can distort the true expectation of future volatility.

The correct answer is **A**.

One shortcoming of using implied volatility is that it can differ significantly across different options for the same underlying asset. This variation, known as the "volatility smile" or "volatility skew," can lead to inconsistent volatility estimates depending on which option's prices are used for calculation. Consequently, relying on implied volatility from a single option might not provide a fully accurate or consistent measure of expected future volatility, making it challenging to obtain a unified view of market sentiment.

B is incorrect. Implied volatility heavily incorporates market sentiment, affecting its accuracy during unstable periods.

C is incorrect. Implied volatility does not inherently assume that future volatility will be lower (or higher) than historical volatility. It simply reflects the market's expectations of future volatility, which can be higher, lower, or similar to historical levels.

D is incorrect. While it is true that implied volatility can be influenced by market supply and demand for options, this is not considered a shortcoming specific to implied volatility.

Things to Remember

- Implied volatility can vary significantly between different options on the same underlying asset, creating inconsistencies.
- This variability introduces challenges in using implied volatility as a reliable predictor of future market movements.

Q.6266 A portfolio manager is considering using implied volatility from current options trading on a stock to predict future volatility. How does the assumption of constant volatility over time represent a limitation in this context?

- A. It assumes that implied volatility is a better predictor than historical volatility.
- B. It fails to account for the dynamic nature of market conditions, leading to potential inaccuracies.
- C. It leads to volatility estimates that are always lower than expected.
- D. It negates the influence of underlying asset prices on volatility prediction.

The correct answer is **B**.

Assuming constant volatility over time is a limitation because it fails to account for the dynamic nature of market conditions. Volatility often changes due to various economic, political, and market events, and the assumption of stability can lead to inaccurate predictions of future volatility.

A is incorrect because the assumption of constant volatility over time does not directly relate to the comparison between historical and implied volatility.

C is incorrect because the assumption results in volatility estimates that can be either higher or lower than expected, but not consistently lower.

D is incorrect because implied volatility does consider the influence of underlying asset prices, derived from option market data.

Things to Remember

- The assumption of constant volatility overlooks the inherent variability in market conditions, leading to potential inaccuracies in predictions.
 - Using implied volatility for predictions requires acknowledging its inherent limitations due to market fluctuations and sentiment variations.
-

Q.6267 A risk manager uses historical data to estimate the correlation between two assets in a portfolio. Given new market data, what is one method the manager can employ to update the correlation estimate?

- A. Use a fixed correlation value calculated years ago for consistency.
- B. Apply a rolling window approach where the correlation is recalculated using the most recent data periodically.
- C. Estimate the correlation once and assume it remains constant indefinitely.
- D. Assume zero correlation when new data is inconsistent with past trends.

The correct answer is **B**.

A rolling window approach allows the risk manager to update correlation estimates periodically using the most recent data. This method reflects current market conditions better and quickly incorporates new trends or changes in relationships between assets.

A is incorrect as using a fixed correlation value ignores recent market dynamics, leading to outdated estimates.

C is incorrect because correlations should be updated to reflect changing market conditions, not assumed constant indefinitely.

D is incorrect because assuming zero correlation disregards the actual relationship between assets, impacting risk assessment quality.

Things to Remember

- The rolling window approach recalculates correlation using the most recent data, updating estimates to reflect current market conditions.
 - This method helps in capturing dynamic changes in asset relationships, improving the accuracy of risk assessments.
-

Reading 50: External and Internal Credit Ratings

Q.1045 The rating of an issuer provided by a rating agency is a (an):

- A. Mere opinion on the issuer or securities issued by the issuer.
- B. Recommendation to purchase, sell, or hold any types of security.
- C. Indicator of the issuer's creditworthiness and also gives the price or relative value of specific securities.
- D. Recommendation based on quantitative analyses and business reviews.

The correct answer is **A**.

The rating provided by a credit rating agency is indeed a mere opinion on the issuer or securities issued by the issuer. It is important to understand that these ratings are not absolute or definitive assessments of an issuer's creditworthiness. Instead, they are subjective evaluations based on the agency's analysis and interpretation of the issuer's financial condition and future prospects. These ratings are intended to provide investors with a benchmark for comparing the relative credit risk of different securities. However, they do not guarantee the issuer's ability to meet its financial obligations, nor do they predict the future performance of the securities issued by the issuer. Therefore, while these ratings can be a useful tool for investors, they should not be the sole basis for investment decisions.

Choice B is incorrect. Credit ratings are not recommendations to purchase, sell, or hold any types of security. They are simply an assessment of the creditworthiness of the issuer or securities issued by the issuer.

Choice C is incorrect. While credit ratings do indicate an issuer's creditworthiness, they do not provide information on the price or relative value of specific securities. The pricing and valuation of securities involve other factors such as market conditions and investor sentiment which are beyond the scope of a credit rating agency's analysis.

Choice D is incorrect. Although quantitative analyses and business reviews may be part of a rating agency's process, a credit rating is not a recommendation based on these factors alone. It represents an overall opinion on the issuer's ability to meet its financial obligations.

Things to Remember

- **Credit Rating Agencies:** These agencies assess the creditworthiness of issuers and their securities by assigning ratings based on various factors such as financial condition, industry outlook, and economic conditions.
- **Credit Risk:** This is the risk that an issuer may not be able to meet its financial obligations, leading to potential losses for investors holding the issuer's securities.

- **Investment Decisions:** Investors should consider credit ratings as one of many factors when making investment decisions, along with other factors such as market conditions, risk tolerance, and investment objectives.
 - **Credit Rating Scale:** Ratings typically range from AAA (highest credit quality) to D (default), with intermediate ratings indicating varying levels of credit risk.
 - **Regulatory Considerations:** Regulators may require certain investors, such as pension funds or insurance companies, to consider credit ratings when making investment decisions to ensure prudence and risk management.
-

Q.1046 Which of the following statements is true with regard to the relationship between ratings and probabilities of default?

- A. Across all industries, the number of defaults monotonically increases as we move down the credit rank.
- B. A given rating is meant to be forward-looking; it is devised to pinpoint a precise probability of default.
- C. Ex-post information such as that provided in default tables or transition matrices does guarantee to provide ex-ante insights regarding future probabilities of default or migration.
- D. Credit ratings remain static and are not affected by changes in economic conditions or industry trends.

The correct answer is **A**.

Rating agencies publish on a regular basis table reporting observed default rates per rating category, per year, per industry, and per region. They show that ratings tend to have homogeneous default rates across industries. This implies that in every industry, the number of defaults increases as we move down the ratings order. However, it is important to bear in mind that for a given rating category, default rates can vary from industry to industry (e.g., a higher percentage of banks with a given rating, say BBB, will default when compared with firms in other industries with the same rating).

Option B is incorrect: Although a rating is meant to be forward-looking, it is not devised to pinpoint a precise probability of default, but rather to point to a broad risk bucket.

Option C is incorrect: Transition matrices serve as indicators of the likely path of a given credit at a given horizon. Ex-post information such as that provided in default tables or transition

matrices **does not** guarantee to provide ex-ante insights regarding future probabilities of default or migration.

Option D is incorrect: Credit ratings can and do change in response to shifts in economic conditions, industry trends, and the financial health of the rated entity. Ratings agencies regularly review and update ratings to reflect these changes.

Things to Remember

- Default rates can vary from industry to industry within the same rating category.
 - Ratings agencies use transition matrices to indicate the likely path of a credit at a given horizon.
 - Credit ratings are not static and can change in response to economic conditions, industry trends, and the financial health of the rated entity.
 - Ratings are meant to provide a broad indication of risk rather than a precise probability of default.
-

Q.1048 Which of the following statement is most likely to be true with regard to the impact of a rating downgrade/upgrade on the price of bonds/stocks?

- A. A rating downgrade is somewhat likely to increase the price of a bond.
- B. A rating downgrade is likely to decrease the price of a bond.
- C. A rating upgrade is unlikely to increase the price of the stock since the price only reflects the earnings expectations of investors.
- D. A rating downgrade is unlikely to decrease the price of a stock since the price largely reflects earnings expectations among consumers.

The correct answer is **B**.

A rating downgrade is likely to decrease the price of a bond. Credit ratings are a measure of a bond issuer's ability to meet its financial obligations. When a rating agency downgrades a bond's rating, it signals a higher risk of default. This increased risk makes the bond less attractive to investors, leading to a decrease in demand and, consequently, a decrease in the bond's price. For instance, if a rating agency downgrades a bond issue's rating from BBB to BB due to a deterioration in the firm's debt repayment ability (e.g., a fall in the current ratio), the bond's yield will increase substantially, and its price will fall. In this case, the bond will transition from investment-grade to below investment-grade. This transition may force many pension funds, which may only be allowed to hold investment-grade securities, to sell the bond. This selling

pressure will further depress the bond's already low price. Conversely, a rating upgrade, which signals a decrease in default risk, is likely to increase the bond's price by making it more attractive to investors.

Choice A is incorrect. A rating downgrade typically indicates a higher risk of default, which makes the bond less attractive to investors. As a result, the demand for the bond decreases and so does its price.

Choice C is incorrect. While it's true that stock prices reflect earnings expectations, they also take into account other factors such as risk level and market sentiment. A rating upgrade usually signals lower credit risk which can make the stock more appealing to investors, potentially leading to an increase in its price.

Choice D is incorrect. Similar to bonds, stocks are also affected by credit ratings. If a company's rating is downgraded, it suggests higher financial risk which can negatively impact investor sentiment and consequently lead to a decrease in the stock's price.

Things to Remember

- Credit ratings are important indicators of a bond issuer's ability to meet its financial obligations.
 - A rating downgrade signals a higher risk of default, making the bond less attractive to investors.
 - A rating upgrade signals a decrease in default risk, potentially increasing the bond's price.
 - Stock prices reflect not only earnings expectations but also factors like risk level and market sentiment.
 - A rating downgrade for a stock can negatively impact investor sentiment and lead to a decrease in its price.
-

Q.2810 The CRO of an investment bank is reviewing the internal rating assessment policies. He notices that the bank is using the through-the-cycle approach to rate the borrowers. He is concerned about the effectiveness of the current approach during recessions and asks to compare it with the at-the-point-in-time approach. Which of the following statements is correct?

- A. During recessions, the through-the-cycle approach tends to over-estimate risk during recessions.
- B. During recessions, the at-the-point-in-time approach tends to over-estimate risk during

recessions.

C. During recessions, both through-the-cycle and at-the-point-in-time approaches tend to over-estimate risk.

D. During recessions, both through-the-cycle and at-the-point-in-time approaches tend to under-estimate risk.

The correct answer is **B**.

The at-the-point-in-time approach assesses the credit quality over the near term and tends to amplify the effect of the business cycle. During recessions, the at-the-point-in-time approach tends to overestimate risk.

On the other hand, through-the-cycle internal ratings try to evaluate the permanent component of default risk. Unlike point-in-time ratings, they are said to be nearly independent of cyclical changes in the creditworthiness of the borrower. They are not affected by credit cycles, i.e. they are through the cycle. As a result, they are less volatile than at-the-point ratings and are valid for a much longer period (exceeding one year).

Option A is incorrect: The through-the-cycle approach does not over-estimate risk during recessions. Instead, it smooths out the cyclical variations in credit quality and provides a more stable assessment of default risk over the longer term. This approach is less sensitive to economic cycles, meaning it avoids the exaggerated risk estimates seen in the at-the-point-in-time approach during downturns.

Option C is incorrect: It is not accurate to say that both approaches over-estimate risk during recessions. The at-the-point-in-time approach tends to over-estimate risk because it closely follows the business cycle and reflects the deteriorated economic conditions during recessions. The through-the-cycle approach, on the other hand, aims to assess the borrower's credit risk over a full economic cycle, thereby avoiding the cyclical bias and not over-estimating risk during downturns.

Option D is incorrect: Neither the through-the-cycle nor the at-the-point-in-time approaches tend to under-estimate risk during recessions. The at-the-point-in-time approach tends to over-estimate risk by responding to immediate economic conditions. The through-the-cycle approach aims for a more balanced and stable risk assessment, neither significantly over-estimating nor under-estimating risk due to temporary economic fluctuations.

Things to Remember

- Point-in-time approach: Assesses credit quality over the near term and is more sensitive to economic cycles.
- Through-the-cycle approach: Evaluates the permanent component of default risk and is less sensitive to economic cycles.
- Cyclical variations: Through-the-cycle approach smooths out cyclical variations in

credit quality.

- **Stability:** Through-the-cycle approach provides a more stable assessment of default risk over the longer term.
 - **Volatility:** Through-the-cycle ratings are less volatile than at-the-point ratings.
 - **Validity period:** Through-the-cycle ratings are valid for a much longer period, exceeding one year.
-

Q.3436 Simon Bryan, FRM, is scrutinizing historical migration tables published by S&P's and Moody's. Which of the following statements would possibly appear under "additional information" below such tables?

- A. We should expect to see the highest level of rating stability in the intermediate term (five-year time frame). Risk ratings will tend to have changed more at both the one- year and ten-year horizons.
- B. We should expect to see the highest level of rating stability during the one-year timeframe. This stability will decline at both the five-year time frame and even more so at the ten-year horizon.
- C. We should expect to see the greatest amount of credit rating stability over long periods of time (e.g., ten years). Credit ratings will tend to change more during shorter periods of time.
- D. We should expect to see credit ratings change by about the same amount over time. The ratings transition matrix shows approximately the same figures for the one-year, five-year, and ten-year time horizons.

The correct answer is **B**.

The Rating Transition Matrix tables developed by renowned rating agencies show that credit ratings are their most stable over a one-year horizon and that stability decreases with longer horizons. This is because credit ratings are designed to be forward-looking assessments of credit risk, and the shorter the time horizon, the more accurate these assessments tend to be. Over longer time horizons, many factors can change, including the financial health of the issuer, economic conditions, and industry trends, which can all impact the issuer's ability to meet its debt obligations. Therefore, it is expected that the stability of credit ratings would decline over longer time horizons such as five years and ten years.

Choice A is incorrect. While it might seem logical to assume that rating stability would be highest in the intermediate term, this is not typically the case. Credit ratings are designed to be

forward-looking and incorporate expectations about future changes in a company's financial condition. Therefore, they are more likely to change over longer time horizons as new information becomes available and expectations adjust.

Choice C is incorrect. This statement contradicts the nature of credit ratings which are dynamic and subject to change over time due to various factors such as changes in economic conditions, industry trends, or company-specific events. Over long periods of time (e.g., ten years), it's more likely for these factors to cause significant changes in a company's creditworthiness leading to less stability in credit ratings.

Choice D is incorrect. Credit ratings do not tend to change by about the same amount over different time horizons. The likelihood of a rating change increases with the length of the horizon because there's more opportunity for new information that could affect a company's financial condition or risk profile.

Things to Remember

- Rating Transition Matrix tables show the stability of credit ratings over different time horizons.
 - Credit ratings are forward-looking assessments of credit risk.
 - Shorter time horizons tend to have more accurate credit rating assessments.
 - Factors such as financial health, economic conditions, and industry trends can impact credit ratings over time.
 - Longer time horizons provide more opportunity for new information to affect a company's financial condition or risk profile.
-

Q.3437 ANEX Financials (AF), a U.S. based firm, has just issued a two-year zero-coupon bond currently rated AA. Market analysts expect that one year from now:

- The probability that the rating of AF remains at AA is 90%
- The probability that the rating of AF is downgraded to A is 5%
- The probability that AF is upgraded to AAA is 5%

The risk-free rate is flat at 2%, and credit spreads for AAA-, AA-, and A-rated debt are flat at 40, 60, and 100 basis points, respectively. All rates are compounded annually. What is the best approximation of the expected value of the zero-coupon bond a year from today?

- A. 97.6
- B. 97.5
- C. 97.7
- D. 97.4

The correct answer is **B**.

In a year's time, the bond will be a 1-year zero-coupon bond. The bond has different expected values in each of the three scenarios outlined above. Precisely,

$$P_{AAA} = \frac{100}{(1 + 0.02 + 0.004)^1} = 97.66$$

$$P_{AA} = \frac{100}{(1 + 0.02 + 0.006)^1} = 97.47$$

$$P_A = \frac{100}{(1 + 0.02 + 0.01)^1} = 97.09$$

Note that as the ratings deteriorate, so does the value of the bond.
Expected value = $5\% \times 97.66 + 90\% \times 97.47 + 5\% \times 97.09 = 97.46$

Note: 1% = 100 basis points

Q.3438 Rating agencies make efforts to incorporate the effects associated with an economic cycle in their ratings. Although this practice is generally valid, it may lead to:

- A. Underestimation of the probability of default in an economic recession
- B. Overestimation of the probability of default in an economic recession
- C. Underestimation of the probability of default in an economic expansion
- D. Divergence of the interests of agency analysts and those of management

The correct answer is **A**.

Underestimation of the probability of default in an economic recession. Rating agencies aim to provide a long-term view of a firm's creditworthiness, which includes considering the potential impact of economic cycles. However, this approach can sometimes lead to inaccuracies. In particular, during an economic recession, the probability of default may be underestimated. This is because the severity of the recession may be greater than what was anticipated when the rating was assigned. As a result, firms may face more financial distress than what was initially expected, leading to a higher probability of default than what the rating suggests.

Choice B is incorrect. Overestimation of the probability of default in an economic recession is not a potential inaccuracy. In fact, during a recession, the risk of default generally increases due to deteriorating economic conditions and financial stress on entities. Therefore, it would be more accurate for rating agencies to anticipate higher probabilities of default during these periods.

Choice C is incorrect. Underestimation of the probability of default in an economic expansion is also not a potential inaccuracy that could arise from a rating agency's efforts to incorporate the effects of an economic cycle into their ratings. During periods of economic expansion, entities are generally more financially stable and less likely to default on their obligations; hence underestimating defaults during such times aligns with general expectations.

Choice D is incorrect. The divergence between the interests of agency analysts and those of management does not directly relate to inaccuracies arising from incorporating the effects of an economic cycle into credit ratings. This choice pertains more towards conflicts-of-interest issues within rating agencies rather than methodological inaccuracies related to accounting for cyclical effects.

Things to Remember

- Rating agencies use credit ratings to assess the creditworthiness of entities and their ability to meet financial obligations.
- Economic cycles refer to the fluctuations in economic activity that occur over time, including periods of expansion and recession.
- During economic recessions, entities may face higher risks of default due to factors such as reduced consumer spending, increased unemployment, and financial market instability.

- Rating agencies need to consider the potential impact of economic cycles on entities' creditworthiness when assigning ratings to provide a comprehensive view of their risk profile.
 - It is important for investors and stakeholders to understand the limitations of credit ratings, including the potential for inaccuracies during economic downturns.
-

Q.4683 Suppose a firm has a debt of \$20 million. If the recovery rate is 80%, and that there is a 0.03 chance that the loan will default, what is the expected loss?

- A. 0.12m
- B. 0.15m
- C. 0.09m
- D. 0.06m

The correct answer is **A**.

Since the recovery rate is 80%, the loss given default (LGD) is

$$\text{LGD} = 100\% - 80\% = 20\%.$$

The expected loss on the loan is, therefore, :

$$\text{EL} = \text{PD} \times \text{LGD} \times \text{EAD} = 0.03 \times 0.20 \times \$20 \text{ million} = \$0.12 \text{ million}$$

Q.4685 Suppose the hazard rate for the first three years is 0.01. Suppose further that the cumulative default probability over six years is 0.035. What is the probability of default between years 3 and 6?

- A. 0.0421
- B. 0.005446
- C. 0.09434
- D. 0.2701

The correct answer is **B**.

To calculate the probability of default between year 3 and year 6, we first calculate the average hazard rate over six years, which can be obtained as follows:

$$\begin{aligned}1 - \exp(-\bar{h} \times 6) &= 0.035 \\ \Rightarrow \bar{h} &= 0.005938\end{aligned}$$

The probability of default between years 3 and year 6 is therefore given by:

$$\begin{aligned}PD_{3,6} &= \text{Default during the first 3 years} - \text{Default during the first 6 years} \\ &= [\exp(-0.01 \times 3)] - [\exp(-0.005938 \times 6)] \\ &= 0.005446\end{aligned}$$

Q.6268 A credit rating agency is analyzing a corporate bond for potential investors. The agency uses its external rating scales to assign risk grades to the bond. In the process of assigning the rating, they follow specific steps as outlined in their methodology involving both qualitative and quantitative analyses. Given the relationship between ratings and default probabilities, what does a transition from an investment-grade rating (e.g., BBB) to a non-investment grade (e.g., BB) signify for a bond's default risk over time?

- A. The bond's default risk increases and becomes more sensitive to economic downturns.
- B. The bond's near-term prospects improve significantly despite being downgraded.
- C. There is no significant change in default risk as it is mostly determined by the issuer's external environment.
- D. The bond's default risk decreases, and the bond becomes less sensitive to market shocks.

The correct answer is **A**.

When a bond transitions from an investment-grade rating (e.g., BBB) to a non-investment grade (e.g., BB), it signifies an increase in the bond's default risk. This is because non-investment grade bonds are considered to be speculative and more vulnerable to economic downturns, reflecting higher risk to investors.

B is incorrect because a downgrade from investment-grade to non-investment grade indicates deteriorating financial health and higher default risk, not improved near-term prospects.

C is incorrect because the rating explicitly measures credit risk relative to other issuers and does change with an issuer's deteriorating financial condition, not just the external environment.

D is incorrect because a downgrade indicates worsening credit quality, thus increasing the bond's sensitivity to market shocks and elevating default risk.

Things to Remember

- External rating scales are ordinal measures of risk and the risk increases as we move down the scale. For instance, Moody's considers Baa and above as investment-grade, while those below as non-investment-grade.
- The rating process includes qualitative and quantitative assessments like management quality, competitiveness, economic cycles, and financial ratios to assign a credit rating.
- Externally issued ratings from agencies such as S&P or Moody's are vital for financial market participants as they signify the default risk related to the issuer or the specific instrument being rated.

Q.6269 Assume you are modeling the default probabilities for XYZ Corp's debt. Unconditional probabilities capture the likelihood of default by a given time from issuance, while conditional probabilities reflect the risk of default in a specific period, assuming XYZ Corp has survived up to that period. If the hazard rate remains constant over time, how would the conditional default probability for a given year, given survival up to the beginning of that year, be calculated?

- A. It would involve calculating 1 minus the exponential of the negative hazard rate times the number of years.
- B. It would be derived by multiplying the constant hazard rate by the survival probability up to the preceding period.
- C. It would require solving for the cumulative default using the hazard rate and subtracting the prior period's default probability from the cumulative probability.
- D. It would involve finding the survival probability at the start of the period and dividing it by the sum of survival probabilities across all periods from the start.

The correct answer is **B**.

If the hazard rate is constant, the conditional default probability for a given year, n given survival up to the beginning of that year, is computed using the multiplication of the constant hazard rate by the survival probability up to the end of the year ($n - 1$). The survival probability up to the preceding period ensures that we account for the fact that the entity has not defaulted until that point.

A is incorrect because this describes the computation for cumulative or unconditional default probability over multiple periods, not the conditional default probability for a single period.

C is incorrect because it does not utilize the survival probability directly, which is crucial for conditional probabilities.

D is incorrect as it misinterprets the use of survival probabilities across periods, an unnecessary complexity for conditional default probability calculations.

Things to Remember

- Constant hazard rate significantly simplifies the calculation of conditional default probabilities since it provides a steady estimate of default risk per period.
- Using real-life scenarios like constant hazard assumptions can be noted in practical applications, such as modeling default risk for corporate bonds under steady economic conditions.

Q.6270 In the context of a credit risk assessment, an analyst is examining the default probabilities for DEF Corp. The analyst notes the distinction between conditional and unconditional probabilities over a three-year horizon under a hazard rate of 0.04 per year. What can be inferred about the nature of these probabilities and their relevance to assessing DEF Corp's risk?

- A. Conditional probabilities will generally be lower than unconditional probabilities within the same timeframe.
- B. Unconditional probabilities will be exponentially greater than conditional probabilities within the same timeframe.
- C. Conditional probabilities reflect real-time default risk, while unconditional probabilities aggregate overall default risk up to a given time.
- D. Unconditional default probabilities will be zero if conditional probabilities are constant over time at 0.04.

The correct answer is **C**.

Conditional probabilities capture the real-time risk of default within a specific period given survival up to that period, while unconditional probabilities measure the overall cumulative risk of default up to a given time. With a constant hazard rate, understanding both types provides a comprehensive risk profile for DEF Corp's obligations.

A is incorrect because conditional probabilities could be higher for a short interval within a given period compared to the cumulative nature of unconditional defaults, depending on the hazard rate dynamics.

B is incorrect since unconditional probabilities grow cumulatively, not exponentially, relative to conditional probabilities.

D is incorrect because even with a constant hazard rate, unconditional probabilities grow cumulatively and will not be zero.

Things to Remember

- Conditional default probabilities are important for period-specific risk assessments, while unconditional default probabilities provide a total risk perspective over time.
- Useful in practical risk management strategies to prioritize actions based on immediate versus cumulative risks.

Q.6271 The hazard rate is a critical concept in credit risk modeling, representing the instantaneous default rate at any given time. Suppose you are analyzing the credit risk of a firm with a constant hazard rate of 0.06 per year. Using this hazard rate, what is the unconditional probability of default by the end of the third year?

- A. 0.0582
- B. 0.1647
- C. 0.9418
- D. 0.8353

The correct answer is **B**.

To calculate the unconditional probability of default by the end of the third year, use the formula: $[1 - \exp(-\text{hazard rate} \times \text{time})]$.

Given a constant hazard rate of 0.06 per year, the calculation is:

$$[1 - \exp(-0.06 \times 3) = 1 - \exp(-0.18) \approx 1 - 0.8353 = 0.1647]$$

Things to Remember

- Unconditional probability of default is a cumulative measure that considers the entire period from the beginning to the specific end point.
 - The exponential decay model helps in understanding how the risk of default accrues over time, which is essential for modeling long-term credit risk.
-

Q.6272 Assume a bond's hazard rate changes over time, initially at 0.04 but projected to increase by 0.01 each subsequent year. What will be the unconditional probability of default by the end of 3 years?

- A. 0.1618
- B. 0.0488
- C. 0.0582
- D. 0.1393

The correct answer is **D**.

The unconditional default probability is calculated in segments using the formula:

$$[1 - \exp(-(\sum \text{Hazard rates}))]$$

From the given information, the unconditional probability of default by the end of 3 years is:

$$[1 - \exp(-(0.04 + 0.05 + 0.06)) = 1 - \exp(-0.15) \approx 0.1393],$$

Things to Remember

- Variable hazard rates complicate the calculation but enhance modeling accuracy in real-world dynamic credit conditions.
 - Modeling flexibility provides encompassing, adaptable risk insights aiding thorough risk management assessment.
-

Q.6273 In the context of a loan, the recovery rate is defined as the proportion of the loan that is expected to be recovered in the event of default. If a loan has a recovery rate of 60%, what will be the expected loss on a loan with a notional amount of \$10 million and a default probability of 2%?

- A. \$200,000
- B. \$80,000
- C. \$40,000
- D. \$120,000

The correct answer is **B**.

The expected loss from a loan can be calculated using the formula:

$$[EL = EA \times PD \times LGD]$$

Where $LGD = 1 - \text{Recovery Rate}$.

Given a recovery rate of 60%, LGD is $1 - 0.60 = 0.40$. Therefore, expected loss is:

$$[0.02 \times 0.40 \times \$10 \text{ million} = \$80,000]$$

Things to Remember

- Recovery rate gives a percentage of the total loan amount recovered upon default.
 - Expected Loss (EL) accounts for both the probability of default (PD) and loss given default (LGD), and helps in understanding potential exposure.
-

Q.6274 Analysts often calculate the expected loss on a loan to gauge potential risk. If a corporate bond with a face value of \$50 million carries an unconditional default probability of 3% and a recovery rate of 65%, what is the expected loss?

- A. \$500,000
- B. \$750,000
- C. \$525,000
- D. \$975,000

The correct answer is **C**.

The expected loss (EL) can be calculated using the formula:

$$[EL = PD \times LGD \times \text{Notional Value}]$$

Given the recovery rate of 65%, LGD is $1 - 0.65 = 0.35$. Therefore, the expected loss is:

$$[0.03 \times 0.35 \times \$50 \text{ million} = \$525,000]$$

Things to Remember

- The interplay between notional value, recovery rate, and default probability ensures a cautious perspective on assessing potential financial risk.
-

Q.6275 A financial institution is evaluating a client's creditworthiness using both through-the-cycle (TTC) and point-in-time (PIT) ratings approaches. How do these approaches differ concerning sensitivity to economic cycles, and what are their respective benefits for the financial institution?

- A. TTC is more sensitive to economic cycles and offers valid short-term ratings, while PIT is less sensitive and provides stable long-term ratings.
- B. TTC is less sensitive to economic cycles, offering stable long-term ratings, while PIT is more sensitive and provides timely short-term ratings.
- C. TTC provides reactive short-term ratings sensitive to market conditions, while PIT offers longer-term stable ratings less affected by economic cycles.
- D. TTC offers annual cyclical adjustment ratings, while PIT applies static ratings unaffected by market changes.

The correct answer is **B**.

The through-the-cycle (TTC) approach is designed to be less sensitive to short-term economic cycles, providing stable and long-term ratings. This stability helps financial institutions manage customer relations without frequent rating fluctuations. On the other hand, the point-in-time (PIT) approach is more sensitive to current economic conditions and market dynamics, offering timely and short-term ratings based on the latest financial data and cyclical impacts.

A is incorrect because TTC actually offers long-term consistency, avoiding short-term economic cycle volatility, while PIT is short-term sensitive.

C is incorrect as it misinterprets the roles; PIT, not TTC, reacts to short-term financial conditions.

D is incorrect because it wrongly describes TTC as annually adjusting and PIT as static, both contrary to their dynamic rating characteristics.

Things to Remember

- TTC ratings provide a stable, long-term view of credit risk, beneficial for strategic planning and long-term investments.
- PIT ratings offer immediate responses to current financial conditions, useful for short-term risk assessments and dynamic market scenarios.

Q.6276 As a risk manager, you are tasked with implementing either through-the-cycle (TTC) or point-in-time (PIT) rating systems for loan evaluations. The choice impacts how the bank perceives borrower risk over time. Why might a bank prefer using TTC ratings over PIT ratings for its strategic loan portfolio management?

- A. TTC ratings offer stability by averaging out economic cycles, thereby minimizing frequent rating changes and ensuring consistent risk assessment.
- B. TTC ratings are more volatile, providing a real-time snapshot of borrower risk.
- C. PIT ratings focus on long-term trends, avoiding the short-term noise in borrower financials.
- D. TTC ratings are optimal for daily trading decisions due to their high frequency of updates.

The correct answer is **A**.

A bank may prefer using through-the-cycle (TTC) ratings for strategic loan portfolio management because these ratings average out the effects of economic cycles, providing stability and minimizing frequent rating changes. This consistency helps in better long-term decision-making and customer relationship management, as the ratings do not react excessively to short-term market volatility.

B is incorrect because TTC ratings are actually less volatile, designed to smooth out economic cycles, rather than providing real-time snapshots.

C is incorrect as it inaccurately attributes long-term analysis to PIT ratings, which are primarily short-term focused.

D is incorrect because TTC ratings are not intended for frequent updates; they are designed to provide consistent assessments over longer periods.

Things to Remember

- TTC ratings help banks maintain consistent credit assessments even through economic fluctuations, supporting strategic stability.
- This approach can be particularly beneficial during volatile economic periods, as it avoids overreacting to short-term financial shocks.

Q.6277 Credit risk assessment methods extend beyond traditional credit ratings provided by agencies like Moody's, S&P, and Fitch. Firms like KMV and Kamakura offer models to estimate default probabilities by analyzing various financial factors. What underlying principle do these models primarily utilize to determine a firm's likelihood of default?

- A. The firm defaults when its liquidity ratio falls below a certain threshold.
- B. The firm defaults if its market share declines significantly.
- C. The firm defaults when it experiences negative cash flows for three consecutive quarters.
- D. The firm defaults if the value of its debt exceeds the value of its assets.

The correct answer is **D**.

Models like those provided by KMV and Kamakura are based on the principle that a firm defaults if the value of its debt exceeds the value of its assets. These models use market value of equity, the amount of debt in the capital structure, and equity volatility to assess default risk.

A is incorrect as liquidity ratios focus on short-term financial health rather than overall default risk which is better captured by asset-debt relationships.

B is incorrect since market share decline does not directly equate to default risk without considering the financial structure elements covered in the models.

C is incorrect because while consecutive negative cash flows can indicate financial distress, the models prioritize the asset-debt comparison principle.

Things to Remember

- KMV and Kamakura methods utilize market-based indicators to assess credit risk, notably the distance to default framework.
 - In practice, these methods provide real-time, short, and medium-term perspectives on default probabilities.
-

Q.6278 Financial institutions often seek alternative methods to those provided by traditional credit rating agencies for assessing default risk. How do the models developed by KMV differ from traditional rating agency approaches concerning the time horizon and updating frequency of ratings?

- A. KMV models provide longer-term assessments and are updated infrequently.
- B. KMV models offer short-term and medium-term assessments and are updated continuously based on market data.
- C. KMV models focus exclusively on qualitative assessments and are updated annually.
- D. KMV models offer static assessments based on historical financial data.

The correct answer is **B**.

KMV models differ from traditional rating agency methods by offering more dynamic and frequently updated assessments of default risk. Unlike traditional methods, which often rely on periodic evaluations and qualitative assessments, KMV models provide short-term and medium-term risk evaluations that are continuously updated based on real-time market data. These models incorporate stock prices, volatility, and financial statements to generate a more immediate and responsive measure of a firm's credit risk. By leveraging market data, KMV models can quickly reflect changes in a firm's financial health and market conditions, offering a more timely and market-sensitive assessment of default risk compared to the more static and less frequently updated ratings from traditional agencies.

A is incorrect because KMV models emphasize shorter-term perspectives and frequent updates.

C is incorrect as KMV uses quantitative market-based data to dynamically update default probabilities, not mainly qualitative assessments.

D is incorrect because KMV models employ real-time data and do not rely solely on static historical data.

Things to Remember

- KMV models utilize financial market inputs, making them more adaptable to current economic conditions than traditional credit ratings.
- These models are pivotal in providing timely credit risk assessments, significantly benefiting risk management and dynamic financial decision-making.

Q.6279 During the evaluation of credit risk, some models like the one developed by Kamakura Corporation incorporate the volatility of a firm's equity along with other financial measures. How does the incorporation of equity volatility contribute to the assessment of default probabilities?

- A. Higher equity volatility usually indicates a lower risk of default as it reflects growth potential.
- B. Equity volatility has no significant impact on default probabilities in these models.
- C. Higher equity volatility signifies greater uncertainty and risk, thus increasing the probability of default.
- D. Equity volatility is used as a secondary measure for qualitative adjustment.

The correct answer is **C**.

Incorporating equity volatility into the assessment of default probabilities is crucial because higher volatility reflects increased risk and uncertainty about the firm's future financial performance. When a firm's stock price exhibits high volatility, it suggests that there is greater uncertainty and fluctuation in the market's perception of the firm's value and stability. This heightened uncertainty can lead to a higher likelihood of extreme negative outcomes, including the potential for default. Therefore, by accounting for equity volatility, models can more accurately gauge the level of risk associated with the firm, thereby providing a more precise estimate of the probability of default. This approach ensures that the assessment of default risk is sensitive to market conditions and changes in investor sentiment.

A is incorrect because higher equity volatility generally represents risk and instability rather than growth potential in default assessment frameworks.

B is incorrect as equity volatility is a significant factor in determining the likelihood of default, indicating the value fluctuations of the firm's assets.

D is incorrect because equity volatility is a core quantitative input in default prediction models, not merely a secondary qualitative adjustment.

Things to Remember

- Equity volatility reflects the risk perceptions of the firm's equity investors and provides valuable insights into the credibility and financial health of the firm.
- High volatility can suggest potential financial instability, making it a critical parameter in quantitative credit risk models.

Q.6280 External ratings provided by independent agencies such as Moody's, S&P, and Fitch are widely used to assess the risk of lending and borrowing. However, banks often use internal ratings to independently evaluate a borrower's creditworthiness. How do these internal ratings compare to external ratings in terms of methodology and applicability?

- A. External ratings tend to be less flexible compared to internal ratings, which can be customized based on an institution's specific risk factors.
- B. Internal ratings are generally less robust because they do not utilize quantitative analysis like external ratings.
- C. External ratings provide real-time risk assessments, whereas internal ratings are often too delayed to be useful.
- D. Internal ratings focus on qualitative assessments, making them less reliable than external ratings.

The correct answer is **A**.

External ratings by agencies such as S&P and Moody's provide standardized measures of risk and are less flexible to specific institutional needs. In contrast, internal ratings can be customized according to the institution's specific risk factors, allowing for more tailored and potentially more accurate creditworthiness evaluations.

B is incorrect because internal ratings use quantitative analyses, often adopting methodologies similar to those of external ratings, but with adjustments to suit specific institutional needs.

C is incorrect because external ratings are not necessarily real-time; internal ratings can be more frequently updated to reflect the current standing of specific borrowers.

D is incorrect since internal ratings include both qualitative and quantitative assessments, making them reliable tools for credit risk evaluation.

Things to Remember

- External ratings provide standardization and are beneficial for benchmarking across a wide range of entities but can lack the granularity needed for specific institutional risk assessments.
- Internal ratings offer flexibility and customization, which is advantageous for risk management processes tailored to institutional needs but require consistent validation and calibration.

Q.6281 A bank uses an internal rating system to ascertain the creditworthiness of its borrowers, relying on metrics such as Altman's Z-score and other discriminant analysis tools. How does the use of these customized internal rating tools impact ratings in terms of capital requirement modeling and regulatory compliance?

- A. Internal rating systems are often criticized for their lack of objectivity due to high dependence on qualitative assessments.
- B. Internal rating systems play a crucial role in determining the capital requirements by providing more accurate and specific inputs required under regulations such as Basel II.
- C. Internal ratings are typically outdated by the time regulatory compliance assessments are needed, making them less useful than external ratings.
- D. External ratings provide more comprehensive compliance data because they are universally accepted and standardized.

The correct answer is **B**.

Internal rating systems offer specific and accurate measures of risk, which are crucial for determining the capital requirements as per regulatory frameworks like Basel II. These precise measurements allow for better risk assessment and management, ensuring that banks meet the capital adequacy standards efficiently.

A is incorrect because, while qualitative components exist, internal ratings heavily incorporate quantitative measures, especially in regulated environments like banking.

C is incorrect as internal ratings can be updated regularly to reflect the most current financial condition and risk profile, aiding timely compliance.

D is incorrect because although external ratings are standardized, they can lack the specificity and real-time relevance needed for precise regulatory compliance assessments which internal ratings provide.

Things to Remember

- Internal ratings are essential for regulatory compliance, offering granular insights necessary for calculating capital requirements accurately under frameworks like Basel II.
- Proper validation and calibration of internal ratings ensure their effectiveness and reliability for ongoing risk management and compliance purposes.

Q.6282 A rating transition matrix is a powerful tool in credit risk management, showing the likelihood of transitioning from one credit rating to another over a specific period. Given the following one-year rating transition matrix for a firm originally rated BBB, interpret the significance of the values in context to the firm's credit risk:

Initial Rating at Year Start	AAA	AA	A	BBB	BB	B	CCC	Default
BBB	0.02%	0.33%	5.95%	86.93%	5.30%	1.17%	0.12%	0.18%

- A. The firm is most likely to retain its BBB rating with a moderate risk of downgrading to BB or lower.
- B. The firm has a high chance of upgrading to AA and minimal risk of default.
- C. There is a significant likelihood of the firm defaulting, as indicated by higher values in the default column.
- D. The firm has an equal probability of transitioning to any of the higher or lower ratings.

The correct answer is **A**.

The rating transition matrix for a firm initially rated BBB indicates that the firm has an 86.93% probability of maintaining its BBB rating. This suggests stability at the current rating level but also shows a moderate risk of downgrades, with a 5.30% chance of moving to BB, 1.17% to B, and a small but non-negligible 0.18% risk of defaulting.

B is incorrect because the probability of upgrading to AA is only 0.33%, much lower than the probability of retaining a BBB rating or downgrading.

C is incorrect as the probability of defaulting is relatively low at 0.18%, indicating minimal risk in the default column.

D is incorrect because the probabilities are not evenly distributed; the highest likelihood is maintaining the current BBB rating.

Things to Remember

- A rating transition matrix shows the likelihood that a financial instrument will migrate from one credit rating to another over time.
- The highest value in a row typically indicates the most likely outcome for the rating within the period described.

Q.6283 Consider the following ratings transition matrix for Evaluating Enterprises with an initial rating of BB over a one-year period. What insights can be derived about the credit migration risk and stability of BB-rated firms?

Initial Rating at Year Start	AAA	AA	A	BBB	BB	B	CCC	Default
BB	0.03%	0.14%	0.67%	7.73%	80.53%	8.84%	1.00%	1.06%

- A. BB-rated firms have a high probability of upgrading to AA or higher.
- B. BB-rated firms are stable, with the highest probability of maintaining their rating while facing a moderate risk of downgrade.
- C. BB-rated firms are most likely to reach default within a year.
- D. BB-rated firms have an equal chance of upgrading, downgrading, or defaulting.

The correct answer is **B**.

The matrix shows that BB-rated firms have an 80.53% probability of maintaining their BB rating, suggesting relative stability. There is also a significant risk of downgrade (8.84% to B), with smaller probabilities allocated to upgrades to BBB (7.73%), higher ratings, or default (1.06%). This distribution indicates that while BB firms are stable, they face moderate downward credit migration risk.

A is incorrect because the probabilities for upgrading to AA or higher are extremely low at 0.14% or lower.

C is incorrect since the probability of default is relatively low at 1.06%, indicating that default is not the most likely outcome.

D is incorrect as the matrix does not depict equal probabilities across different rating migration possibilities; retaining the BB rating is the highest probability.

Things to Remember

- The highest probability in a rating transition matrix reveals the most expected rating migration path.

Q.6284 Changes in credit ratings can have direct and pronounced impacts on various financial instruments. Suppose a firm's credit rating is downgraded from A to BBB. How is this likely to affect the firm's bond prices, and what can be the foreseen reaction in terms of investor behavior?

- A. Bond prices are likely to decrease as investors perceive higher credit risk, leading to increased selling pressure.
- B. Bond prices are likely to increase as investors anticipate higher future returns due to the rating change.
- C. There will be no significant impact on bond prices as credit ratings have little influence on market perception.
- D. Bond prices will remain stable as the downgrade reflects internal adjustments not immediately relevant to market dynamics.

The correct answer is **A**.

When a firm's credit rating is downgraded from A to BBB, it reflects an increased credit risk. Investors tend to sell bonds of downgraded firms, leading to decreased bond prices. The higher perceived risk typically results in decreased demand and increased yield requirements, contributing to the price drop.

B is incorrect because a downgrade usually implies higher risk, which does not attract investors unless adequately compensated by higher yields, thus lowering bond prices.

C is incorrect as credit ratings significantly influence market perception and investor behavior, especially concerning risk assessment.

D is incorrect because the market often reacts quickly to rating downgrades, reflecting in bond prices rather than remaining stable.

Things to Remember

- Bond prices generally decrease when a firm's credit rating is downgraded, reflecting higher risk and lower demand from investors.
 - Rating downgrades can lead to increased selling of the bonds as investors seek to minimize their exposure to higher risk.
-

Q.6285 A corporation's stock price is compared in relation to its credit rating changes. What typical impact does a credit rating downgrade have on the stock price and why?

- A. A downgrade typically leads to an increase in stock price due to perceived growth prospects.
- B. A downgrade leads to a decrease in stock price as it signals increased financial risk and potential weaknesses.
- C. There is negligible impact on stock prices following a downgrade because stock investors are more focused on earnings reports.
- D. Stock prices remain unchanged as credit rating changes are already factored into the market.

The correct answer is **B**.

A credit rating downgrade usually leads to a decrease in stock prices as it signals increased financial risk and potential weaknesses within the company. Investors tend to sell off stocks to avoid potential losses associated with increased danger of default or significant financial distress.

A is incorrect because a downgrade typically presents a risk perception rather than growth prospect, leading to reduced investor confidence.

C is incorrect as financial health indications through credit ratings critically influence stock prices beyond just earnings reports.

D is incorrect since market adjustments follow news of downgrades, demonstrating the perceived increased risk not being entirely pre-factored.

Things to Remember

- Stock prices often decline upon a credit rating downgrade due to increased perception of financial risk and possible operational weaknesses.
 - Investor sentiment reacts to credit rating changes as a credible reflection of a firm's financial condition and creditworthiness.
-

Q.6286 Credit Default Swap (CDS) spreads offer insight into the market's perception of credit risk. If an entity's credit rating is downgraded, how does this likely affect its CDS spread, and what does this indicate about market expectations?

- A. The CDS spread decreases, indicating reduced risks associated with the entity.
- B. The CDS spread remains unchanged, reflecting market stability irrespective of rating changes.
- C. The CDS spread increases, indicating higher perceived probability of default and rising credit risk.
- D. The CDS spread decreases, as the market has already anticipated and adjusted to the downgrade.

The correct answer is **C**.

When an entity's credit rating is downgraded, its CDS spread typically increases. This indicates that the market perceives a higher probability of default and rising credit risk. Investors demand higher premiums to insure against the increased risk associated with the entity.

A is incorrect because a downgrade implies higher risk, leading to increased spreads rather than decreased ones.

B is incorrect as rating changes significantly impact CDS spreads by altering the default risk perception.

D is incorrect since spreads reflect heightened risk rather than a reduction, with adjustments factoring in real-time rather than prior anticipation exclusively.

Things to Remember

- CDS spreads generally widen following a downgrade, reflecting increased perceived risk of default.
 - The relationship between CDS spreads and credit ratings helps investors gauge market sentiment regarding an entity's creditworthiness.
-

Q.6287 The financial crisis of 2007-2008 exposed significant failings in the use of credit ratings, particularly for structured products. Given these historical failures, what primary issues inherent in rating those products contributed to the crisis, and how did this affect the credibility of rating agencies?

- A. Rating agencies used overly conservative models, resulting in high accuracy but losing market trust.
- B. Rating agencies' refusal to rate structured products led to a vacuum of risk assessment in the market, facilitating unchecked risky behaviors
- C. Ratings were accurate but ignored by market participants, leading to defaults entirely unsurprising to rating agencies.
- D. The optimistic correlation assumptions and the ability of banks to game the rating models led to inaccurate ratings, contributing to widespread defaults and undermining the credibility of rating agencies.

The correct answer is **D**.

The financial crisis revealed that rating agencies used overly optimistic assumptions about correlations among mortgage defaults and that banks could manipulate rating models to achieve desired ratings. This led to inaccurate ratings and inflated assessments of creditworthiness of structured products, which defaulted at high rates during the crisis and severely damaged the credibility of the rating agencies.

A is incorrect because the models were not overly conservative; they were too optimistic, leading to overly favorable ratings.

B is incorrect because rating agencies did rate structured products, albeit inaccurately, driven by the desire to retain business from financial institutions.

C is incorrect as ratings were not accurate; market participants used those inaccurate ratings as bases for investment decisions.

Things to Remember

- The 2007-2008 financial crisis highlighted the critical issue of model manipulation and overly favorable assumptions in rating structured products.
- This led to increased regulatory scrutiny of rating agencies and diminished their credibility among market participants.

Reading 51: Country Risk

Q.2807 Aram Stone recently graduated from one of the most renowned German universities. During his time as an undergrad, Stone developed a unique algorithm that could completely change the development of Artificial Intelligence. Stone wants to patent his idea and initiate a startup, but he did not yet decide on the exact location. To evaluate legal risks, Stone found the rating presented in the table below:

Region	Overall Property Rights	Legal Property Rights	Physical Property Rights	Intellectual Property Rights
Central/Eastern Europe	4.78	4.64	5.47	4.22
Asia Oceania	4.77	4.42	5.44	4.44
Middle East & North Africa	4.76	4.61	5.42	4.26
Latin America	4.57	4.23	5.23	4.25
Africa	4.53	4.26	5.17	4.16
best protection <-> highest scores				

In which of the below regions should Stone register his company and patent if his main concern is the protection of his algorithm?

- A. Central/Eastern Europe
- B. Asia & Oceania
- C. Middle East & North Africa
- D. Latin America

The correct answer is **B**.

Since Stone developed a unique algorithm that could completely change the development of Artificial Intelligence, and that his main concern is the protection of his algorithm, then he should register his company in a region with the highest rating of Intellectual Property Rights protection. In this case, he should choose Asia & Oceania.

Things to Remember

1. Intellectual property rights are legal rights that provide creators protection for their inventions, literary and artistic works, symbols, names, and images. They enable creators to earn recognition or financial benefit from their inventions.
2. The protection of intellectual property rights varies from region to region. It's crucial for inventors and creators to understand the level of protection offered in different regions before deciding where to register their patents.
3. In this case, Aram Stone should consider not only the overall property rights score but also the specific score for intellectual property rights. Since his main concern is the protection of his algorithm, the region with the highest intellectual property rights score would be the most

suitable for him.

4. It's also important to consider other factors when choosing a location for a startup, such as the local business environment, access to resources, and potential market size.

Q.3433 Mendoza Valeria, FRM, works as a risk analyst at a Mexican conglomerate. She has been asked to evaluate USD-based bond investments issued by four different companies. According to her employer's guidelines, the firm has a strict policy of only investing in companies with investment-grade ratings on both the S&P rating scale and Moody's. In addition, the firm only invests in countries with favorable sovereign risk quality. Assuming the company is located in the paired country, which (country, company) pair, as outlined below, would be the most appropriate investment?

Country	Import Ratio	Debt Service Ratio	Company	S&P Rating	Moody's Rating
Jolly World	42%	300%	Brighter World Ltd.	A	Ba
Pluto	18%	30%	Green Leaf Corp	A	Aa
Northern Lights	8%	250%	Eastbrom Financial	BB	Baa
Norfork	30%	50%	Helsinki Inc.	BBB	Ba

- A. (Norfork, Helsinki Inc.)
- B. (Northern Lights, Eastbrom Financial)
- C. (Pluto, Green Leaf Corp)
- D. (Jolly World, Brighter World ltd.)

The correct answer is **C**.

The pair (Pluto, Green Leaf Corp) is the most suitable investment according to the conglomerate's guidelines. An investment-grade bond is defined as a bond that has received a rating of BBB or higher on the S&P rating scale or a rating of Baa or higher on Moody's rating scale. Green Leaf Corp has received an A rating on the S&P scale and an Aa rating on Moody's scale, both of which are considered investment-grade ratings. Furthermore, the country of Pluto has an import ratio of 18% and a debt service ratio of 30%. The import ratio is a measure of a country's total imports relative to its total foreign exchange reserves. A lower import ratio is generally indicative of a lower probability of default, which could potentially lead to a rescheduling of payments. The debt service ratio is a measure of a country's debt service payments (principal + interest) relative to its export earnings. A lower debt service ratio is generally indicative of a healthier (less risky) country. Therefore, given the favorable ratings of Green Leaf Corp and the positive economic indicators of Pluto, the pair (Pluto, Green Leaf Corp) is the most suitable investment according to the conglomerate's guidelines.

Choice A is incorrect. Norfork, Helsinki Inc. does not meet the conglomerate's investment

policy as it has a Moody's rating of Ba, which is below investment grade.

Choice B is incorrect. Northern Lights, Eastbrom Financial also fails to meet the criteria as its S&P rating is BB, which falls under speculative grade and not investment grade.

Choice D is incorrect. Jolly World, Brighter World Ltd., despite having an S&P rating of A (investment-grade), has a Moody's rating of Ba (below investment-grade). Therefore, it does not comply with the conglomerate's stringent policy that requires both ratings to be at least at the level of investment-grade.

Q.3434 Credit rating agencies like S&Ps and Moody's issue two different credit ratings for countries with an appetite for debt. These are the local currency debt rating and the foreign currency debt rating. Historically, it has been observed that defaults on local-currency-denominated debt are less frequent than foreign-currency-denominated debt. What's the main reason behind this observation?

- A. Local currency debt has a lower spread compared to that of foreign debt
- B. This is a statistical anomaly - ideally, defaults rates in the two categories of debt should be more or less equal
- C. Foreign currency-denominated debt has fewer political ramifications than local currency-denominated debt, making it easier to deal with the consequences from the perspective of a country's leadership
- D. Unlike local currency obligations, foreign currency obligations cannot be settled via monetary expansion

The correct answer is **D**.

The primary reason for the lower default rate on local-currency-denominated debt compared to foreign-currency-denominated debt is the ability of governments to use monetary expansion to settle their local currency obligations. Monetary expansion refers to the process by which a country's central bank increases the money supply, often by printing more money. This is a tool that governments can use to manage their local currency debt, as they have the authority to print their own currency. However, this option is not available for foreign currency obligations, as no government has the right to print foreign currencies. Therefore, when a country has foreign currency debt, it cannot use monetary expansion to manage its obligations, making defaults more likely. This explanation is consistent with the observation that credit ratings for local-currency-denominated debt are typically higher than those for foreign-currency-denominated debt.

Choice A is incorrect. While it's true that local currency debt often has a lower spread compared to foreign debt, this doesn't directly explain the trend of fewer defaults on local currency debt. The spread is a reflection of the perceived risk associated with the debt, not a cause for default rates.

Choice B is incorrect. This statement suggests that there should be no difference in default rates between local and foreign currency debts, which contradicts empirical evidence. It's not just a statistical anomaly; there are fundamental reasons why countries are less likely to default on their local currency debts.

Choice C is incorrect. The political ramifications of defaulting on any type of debt can be significant and complex, but this doesn't necessarily make defaults on foreign currency-denominated debt more common than those on local currency-denominated debt.

Things to Remember

- Local currency debt is issued in the country's own currency, while foreign currency debt is issued in a currency other than the country's own.
- Credit ratings provided by agencies like S&P and Moody's are crucial for investors to assess the creditworthiness of countries and their ability to repay debt.
- Monetary expansion can lead to inflation if not managed properly, impacting the value of the local currency.
- Defaulting on debt can have severe consequences for a country's economy, including reduced access to credit markets and increased borrowing costs.

Q.5393 In the hypothetical country of Eastportia, severe economic recession and political upheavals have resulted in the country defaulting on its sovereign debt in 2030. In the immediate aftermath of this event, which of the following consequences is Eastportia most likely to face?

- A. Military intervention from the countries to which it owes debt.
- B. Immediate liquidation of the country's assets to repay its debts.
- C. Significant impediments to its economic development and growth, difficulty raising funds, and a loss of investor confidence.
- D. An immediate improvement in its credit rating due to the recognition of its financial distress and default.

The correct answer is **C**.

In the modern era, defaulting on sovereign debt can have serious consequences for a country, including significant hindrances to its economic development and growth, an increase in difficulty raising funds, a loss of investor confidence, a deepening of economic downturn, and

political instability. These are likely to be the immediate aftermath of Eastportia's default.

A is incorrect. In the modern era, military intervention is not a usual consequence of a country defaulting on its debt, unlike in the past.

B is incorrect. Unlike a firm that defaults, a country cannot be liquidated. Instead, the debt is typically restructured or replaced.

D is incorrect. A default typically has a negative impact on a country's credit rating rather than an improvement, reflecting the increased risk to investors.

Things to Remember

- Sovereign debt default occurs when a country is unable to meet its debt obligations, leading to a range of economic and financial consequences.
 - Defaulting on sovereign debt can lead to a loss of investor confidence, making it harder for the country to borrow in the future.
 - Countries that default on their debt may face economic downturns, political instability, and challenges in attracting foreign investment.
 - Debt restructuring is a common approach taken by countries facing default, where the terms of the debt are renegotiated to make repayment more manageable.
 - Credit ratings agencies play a crucial role in assessing the creditworthiness of countries, and a default typically results in a downgrade of the country's credit rating.
-

Q.5394 Consider the case of Zeland, a fictional country with a high level of sovereign debt. Various factors influence the level of sovereign default risk in Zeland. Which of the following factors is most likely to influence the level of sovereign default risk?

- A. The country's economic diversification and export base.
- B. The level of political stability and effectiveness of governance.
- C. The currency in which the sovereign debt is issued.
- D. The historical track record of the country in servicing its debt.

The correct answer is **B**.

Political stability and the effectiveness of governance play a crucial role in determining the level

of sovereign default risk. A stable political environment and effective governance reduce the likelihood of policy disruptions, economic instability, and political upheavals that could hinder the country's ability to service its debt. A well-functioning government can implement sound fiscal and monetary policies, maintain investor confidence, and promote sustainable economic growth, reducing the risk of default.

A is incorrect. While economic diversification and a strong export base are important for a country's overall economic stability, they do not directly determine the level of sovereign default risk. A country could have a diverse economy and strong exports but still face default risk if other factors such as governance or debt sustainability are problematic.

C is incorrect. The currency in which the sovereign debt is issued does influence certain aspects of default risk, such as exchange rate risk. However, it is not the primary factor that determines the overall level of sovereign default risk. Other factors, such as economic and political stability, have a more significant impact on default risk.

D is incorrect. While a country's historical track record in servicing its debt can provide insights into its commitment to meeting obligations, it is not the most influential factor in determining the level of sovereign default risk. A country's circumstances and the current economic and political conditions are more critical indicators of default risk than historical performance alone.

Things to Remember

- Sovereign default risk refers to the likelihood that a country will fail to meet its debt obligations.
- Political stability and governance effectiveness are crucial factors in assessing sovereign default risk.
- Economic diversification and a strong export base can contribute to a country's overall economic stability.
- The currency in which sovereign debt is issued can impact exchange rate risk but is not the primary factor in determining default risk.
- A country's historical track record in servicing debt can provide insights but is not the most influential factor in assessing default risk.

Q.5395 When rating agencies measure sovereign default risks, which of the following factors is the most significant in their assessment?

A. Fiscal deficit and government debt levels

- B. Political stability and leadership
- C. Economic growth and GDP per capita
- D. Debt-to-GDP ratio and external debt exposure

The correct answer is **D**.

When rating agencies measure sovereign default risks, the most significant factor in their assessment is the debt-to-GDP ratio and external debt exposure. The debt-to-GDP ratio reflects the level of a country's government debt relative to its Gross Domestic Product (GDP), providing an indication of the country's ability to manage and service its debt obligations. A higher debt-to-GDP ratio increases the risk of default. Additionally, the external debt exposure measures the amount of debt owed to foreign creditors, which adds another layer of risk to a country's default potential.

A is incorrect. While fiscal deficit and government debt levels are important factors in assessing sovereign default risks, they are not the most significant in the rating agencies' assessment. These factors contribute to the debt burden of a country, but they do not capture the full picture of default risk, as they do not consider the country's economic capacity to manage its debt.

B is incorrect. Political stability and leadership are indeed relevant factors in evaluating a country's overall stability and the potential impact on its ability to meet debt obligations. However, they are not the most significant factor when rating agencies specifically measure sovereign default risks. Political stability is often considered alongside other factors in the assessment but may not have as direct an impact on default risk as the debt-related indicators.

C is incorrect. Economic growth and GDP per capita are important indicators of a country's economic health, but they are not the most significant factors in the rating agencies' assessment of sovereign default risks. While economic growth and high GDP per capita can contribute to debt sustainability, they are not the primary focus when evaluating default risks.

Things to Remember

- Sovereign default risk refers to the likelihood that a country will be unable to meet its debt obligations.
- Rating agencies assess sovereign default risk by analyzing various economic, financial, and political factors.
- Debt-to-GDP ratio is a key metric used to evaluate a country's debt sustainability and default risk.
- External debt exposure measures the amount of debt owed to foreign creditors, which can increase default risk.

- Political stability and leadership can impact a country's ability to manage its debt, but they are not the most significant factors in assessing sovereign default risk.
-

Q.5396 The Republic of Veritas, a highly industrialized nation with robust infrastructure, has an investment-grade rating by the leading credit rating agencies. However, recently the nation is experiencing high volatility in its credit spread due to global economic turmoil. An investor, Mr. Sterling, is examining Veritas's 10-year sovereign bond which has a yield of 6%, given the risk-free rate in the same currency is 3%. Given the country's investment-grade rating and the characteristics of sovereign credit spreads, Mr. Sterling is contemplating the factors that might have led to Veritas's higher credit spread. He is also aware of the relationship between the credit spread and the credit default swap (CDS) market. Based on the above scenario, which of the following statements best explains the reasons behind the high credit spread of Veritas's sovereign bond?

- A. The credit rating agencies have recently downgraded Veritas's credit rating.
- B. Speculators in the credit default swap market are driving up the cost of protection against Veritas's debt.
- C. Veritas has increased its public debt issuance significantly in the recent past.
- D. The 3% risk-free rate is causing artificial inflation of Veritas's credit spread.

The correct answer is **B**.

In this case, the credit spread is more volatile and has increased despite the country's investment-grade rating. One possible reason for this can be speculative activities in the CDS market, which can affect credit spreads. If speculators drive up the cost of protection on the debt (as in the case of Greece in 2010), bond holders might consider the yield to be too low, which can increase the credit spread.

A is incorrect. Veritas has an investment-grade rating. There is no information provided that suggests a recent downgrade of its credit rating.

C is incorrect. While an increased debt issuance can put pressure on the country's creditworthiness, it doesn't directly explain the high credit spread in the face of an investment-grade rating. More public debt could lead to higher credit spreads, but only if there is a corresponding increase in perceived default risk.

D is incorrect. The risk-free rate doesn't cause an artificial inflation of credit spreads. The credit spread is defined as the difference between the yield of the sovereign bond and the risk-free rate. While a low risk-free rate might contribute to a wider credit spread, it would not directly cause a higher credit spread independent of other factors.

Things to Remember

- **Credit Spread:** The credit spread is the difference in yield between a risk-free asset and a bond with credit risk. It reflects the compensation investors demand for taking on the credit risk of a particular issuer.
 - **Credit Default Swap (CDS):** A CDS is a financial derivative that allows an investor to "swap" or offset their credit risk with another party. It is often used as a form of insurance against the default of a bond or loan.
 - **Speculation in Financial Markets:** Speculators are individuals or entities that seek to profit from short-term fluctuations in asset prices. Their activities can impact market prices and volatility.
 - **Investment-Grade Rating:** An investment-grade rating indicates that a bond or issuer has a relatively low risk of default. It is a signal of creditworthiness and can influence the cost of borrowing for the issuer.
 - **Risk-Free Rate:** The risk-free rate is the theoretical rate of return of an investment with zero risk of financial loss. It serves as a benchmark for evaluating the return on other investments that carry risk.
-

Q.5397 Consider the hypothetical country of Norelia. Recently, economic turbulence has raised the risk of Norelia defaulting on its debts. Norelia's rating has not yet changed, but its credit spread has widened significantly, reflecting the heightened perceived risk among market participants. Simultaneously, the Credit Default Swap (CDS) market is experiencing an influx of speculators betting on Norelia's default, causing the price of protection to surge. Some financial experts argue that these speculative activities have indirectly pushed Norelia's bond yields up. However, others counter that Norelia's bonds themselves, rather than its CDS spreads, provide a more accurate representation of the country's credit spread. Which of the following best describes the factors contributing to the movements in Norelia's credit spreads and the potential implications?

- A. Norelia's widened credit spread primarily reflects speculators driving up the cost of CDS protection, exacerbating the country's financial difficulties by pushing up bond yields.
- B. Norelia's bond yields, rather than its CDS spreads, serve as the primary source of credit spread data. Therefore, the increase in bond yields directly suggests the country's deteriorating financial health, independent of speculator activity.

C. Norelia's credit spread is fundamentally tied to its unchanged rating, making the impact of speculator activity in the CDS market negligible.

D. Norelia's widened credit spread is due to the clustering effect in the CDS market, which has unfairly penalized Norelia due to its association with other troubled economies.

The correct answer is **B**.

The credit spread for sovereign debt in a specific currency is the excess interest paid over the risk-free rate in that currency. Norelia's bond yields, rather than its CDS spreads, should serve as the primary source of credit spread data. This is due to the possibility of the seller of protection defaulting in a CDS, as well as issues with market illiquidity and clustering. Therefore, if Norelia's bond yields have increased, it directly suggests the country's deteriorating financial health, regardless of speculator activity.

A is incorrect because it oversimplifies the role of speculators. While speculator activity might influence the CDS market, it doesn't necessarily directly drive up a country's bond yields or credit spreads. Moreover, many researchers argue that bond yields are not affected by CDS spreads.

C is incorrect as the credit spread can adjust more quickly to new information than ratings, and can therefore diverge from a country's rating. This is one of the characteristics that make credit spreads more granular and potentially more informative than ratings.

D is incorrect because while clustering (where the CDS spreads of groups of countries move together) is a phenomenon that can occur in the CDS market, it does not necessarily reflect the default risk of a specific country. Thus, it would be inaccurate to attribute Norelia's widened credit spread solely to a clustering effect.

Things to Remember

- Credit Spread: The excess interest paid over the risk-free rate in a specific currency for sovereign debt.
- Credit Default Swap (CDS): A financial derivative that allows an investor to "swap" or offset their credit risk with another party.
- Speculators: Market participants who make bets on the direction of asset prices, including CDS protection.
- Market Illiquidity: The condition where trading activity is low and it is difficult to buy or sell securities without causing a significant price change.
- Clustering Effect: The phenomenon where CDS spreads of groups of countries move

together, potentially impacting individual country credit spreads.

Q.5398 The Sovereign Republic of Midlandia, an emerging economy, has a recent credit rating that aligns with neighboring nations in its region. However, the Republic of Midlandia's bonds have exhibited lower credit spreads compared to its counterparts with similar ratings. Market participants perceive Midlandia as less risky, and there's an influx of foreign direct investments (FDI). Despite this favorable situation, the Midlandia government is worried about the potential for speculators to manipulate the credit spreads. Given the circumstances, which of the following measures should the government of Midlandia primarily consider to manage its credit spread?

- A. Diversifying its bondholder base.
- B. Strengthening the performance of its central bank.
- C. Reducing reliance on ratings, focusing instead on improving economic fundamentals.
- D. Seeking legislative measures to limit speculative trades as was done by the European Union in 2011 to curb bets on the euro zone crisis.

The correct answer is **C**.

Credit spreads, although correlated to credit ratings, provide more granular information on a country's ability to repay its debt. Thus, focusing on improving economic fundamentals to reassure bondholders about the country's ability to repay its debt would be more effective in managing credit spreads.

A is incorrect. While diversification could reduce the risk of sudden withdrawal of funds, it might not directly affect the credit spread or reduce the risk from speculators.

B is incorrect. While strengthening the performance of the central bank might improve the country's overall economic stability, it is not a direct method to control credit spreads or speculative activities.

D is incorrect. Seeking legislative measures to limit speculative trades similar to the European Union. While legislation could potentially limit speculators' impact on credit spreads, most researchers argue that bond yields are not significantly affected by credit default swap (CDS) spreads. Therefore, this might not be the most effective measure. Moreover, it would need to consider the possible negative effects of such regulation, like reducing market liquidity.

Things to Remember

- Credit spreads provide more granular information on a country's ability to repay its debt compared to credit ratings.
- Improving economic fundamentals can help reassure bondholders about a country's

ability to repay its debt, thus managing credit spreads.

- Diversifying bondholder base can reduce the risk of sudden fund withdrawals but may not directly impact credit spreads or speculative activities.
 - Strengthening the central bank's performance can enhance overall economic stability but may not directly control credit spreads or speculative activities.
 - Legislative measures to limit speculative trades may have potential negative effects like reducing market liquidity and may not be the most effective measure to manage credit spreads.
-

Q.5399 Consider the following scenario: Country X and Country Y both hold the same credit rating from a prominent credit rating agency. However, the credit spread for Country X's sovereign debt is significantly lower than that of Country Y, indicating that the market perceives Country X as less risky. Recently, new information has emerged that signals potential economic trouble for Country X. As an analyst, how do you expect this new information to impact the credit spreads for Country X and Country Y, and what implications might this have for their perceived risk?

- A. The credit spread for Country X will decrease, making it perceived as less risky than Country Y.
- B. The credit spread for Country X will increase, causing no change in the perceived risk of Country Y.
- C. The credit spread for both countries will increase due to their identical credit ratings.
- D. The credit spread for Country X will increase, possibly causing a reevaluation of the perceived risk for Country Y.

The correct answer is **D**.

The new information signaling economic trouble for Country X will likely cause its credit spread to increase, reflecting a higher risk of default. As credit spreads are more reactive to new information than credit ratings, this change can happen quickly. While Country X and Y share the same credit rating, they do not necessarily share the same credit spread, so an increase in Country X's credit spread won't automatically affect Country Y's. However, this new perceived risk for Country X might cause market participants to reevaluate the perceived risk for Country Y, even if Country Y's economic situation hasn't changed, due to their previous identical credit ratings.

A is incorrect because negative economic information would typically cause a credit spread to increase, reflecting higher risk.

B is incorrect. The new information could lead to a reevaluation of perceived risk across similarly rated countries, including Country Y.

C is incorrect. Credit spreads are more specific and responsive to individual country situations than ratings. Therefore, a change in one country's credit spread doesn't necessarily trigger a change in the other's, even if their credit ratings are the same.

Things to Remember

- Credit Spread: The difference in yield between a risk-free asset and a comparable asset with credit risk.
 - Credit Rating: An assessment of the creditworthiness of an entity, such as a government or corporation, based on its financial strength and ability to meet its financial obligations.
 - Market Perception: How investors and market participants view the risk associated with an investment or entity.
 - Default Risk: The risk that a borrower will be unable to meet its debt obligations.
 - Market Reactivity: How quickly and efficiently financial markets incorporate new information into asset prices.
-

Q.5400 The sovereign bond market is facing turbulent times. You are an analyst at a global investment firm tasked with assessing the risk of various countries. Given recent global events, there are concerns about the credibility and effectiveness of different metrics to evaluate sovereign credit risk. Your attention is drawn to the credit spreads and credit ratings assigned to countries X and Y. Both countries have similar credit ratings from the leading rating agencies. However, the credit spread for country X is notably lower than that of country Y. The speculator activity in the Credit Default Swap (CDS) market for these countries is under scrutiny, and the illiquidity of the CDS market has recently caused concerns. Which of the following explanations is most likely to account for the difference in the credit spreads between country X and country Y, despite their similar credit ratings?

- A. Country X and Y are rated by different credit rating agencies.
- B. Country Y may have a higher perceived default risk compared to Country X, reflected more quickly in the credit spread than in the credit rating.

C. The risk-free rate in Country Y is higher than that in Country X.

D. Speculators in the CDS market are manipulating the credit spreads of Country Y.

The correct answer is **B**.

Credit spreads and credit ratings aim to capture the same underlying risk: the risk of default by a country. However, they don't always move in lockstep. Credit spreads can adjust more quickly to new information than credit ratings, and they are more granular (i.e., there's a range of credit spreads associated with a given credit rating). As a result, it's possible for two countries to have similar credit ratings but different credit spreads if the market perceives one country to be riskier than the other.

A is incorrect. The same rating agencies typically rate most countries, and it's unlikely that a discrepancy between agencies would cause such a difference in credit spreads.

C is incorrect. A higher risk-free rate in Country Y could theoretically affect the credit spread, but it would not likely be the primary factor causing a difference in credit spreads between two countries with similar credit ratings.

D is incorrect. While it's possible for speculators to influence CDS spreads and, indirectly, credit spreads, this is not typically the main factor driving credit spreads, especially since regulatory measures have been taken to limit speculative activity in the CDS market.

Things to Remember

- **Credit Spread:** The difference in yield between a risk-free asset and a bond with credit risk. It reflects the market's perception of the risk of default by the issuer.
- **Credit Rating:** An assessment of the creditworthiness of an issuer, such as a country or a company, by a credit rating agency.
- **Credit Default Swap (CDS):** A financial derivative that allows an investor to "swap" or offset their credit risk with another party. It is often used as a measure of credit risk in the market.
- **Speculator Activity:** Speculators are investors who seek to profit from short-term price movements in the market. Their activity can sometimes impact asset prices, including credit spreads.
- **Illiquidity in Markets:** When a market lacks sufficient depth or trading volume, it can lead to challenges in buying or selling assets at fair prices, potentially causing concerns for investors.

Q.5401 As the head of risk management at a multinational corporation, you are tasked with evaluating the risk of expanding the company's operations into three countries: Alpha, Beta, and Gamma. You decide to use composite measures of risk to evaluate each country's risk profile. You are considering using Political Risk Services (PRS), Euromoney's country risk scores, and The Economist's country risk scores. You're aware that these services use different methodologies and that they all measure a mixture of political, financial, and economic risk. Your company has significant exposure to sovereign debt risk and political instability. Given your specific risk exposures and the information provided by these services, how should you proceed to effectively assess the risk of expanding into these countries?

- A. Assign equal weight to each service's country risk score for each country to come up with a composite risk measure.
- B. Choose the service whose scoring method aligns best with your company's specific risk exposures and use that service's country risk scores.
- C. Combine the country risk scores from all three services but assign higher weight to the service that best aligns with your company's specific risk exposures.
- D. Ignore the numerical scores and solely focus on the rankings of the countries provided by the services.

The correct answer is **C**.

Given the different methodologies and considerations of each service, it's most prudent to incorporate information from all three services. However, because your company has significant exposures to sovereign debt risk and political instability, you should assign a higher weight to the service that best assesses these specific risks. This approach ensures that the specific exposures of your company are well captured while not ignoring the other aspects of country risk that could become relevant in the future.

A is incorrect. Assigning equal weight to each service's country risk score doesn't account for the fact that these services may not equally reflect your company's specific risk exposures.

B is incorrect. Choosing one service that aligns best with your company's risk exposures might overlook valuable information provided by the other services. Country risk is multifaceted, and a composite risk measure should incorporate as many relevant factors as possible.

D is incorrect. Ignoring the numerical scores and solely focusing on the rankings of the countries might be oversimplifying the situation. The rankings can provide a good overview, but the numerical scores and accompanying narratives can give you much more granular and specific insights.

Things to Remember

- Country Risk Assessment: Evaluating the risk associated with investing or operating in

a particular country, taking into account political, economic, and financial factors.

- **Political Risk:** The risk of financial, market, or personnel losses due to political decisions or instability in a country.
 - **Sovereign Debt Risk:** The risk that a government may default on its sovereign debt obligations, leading to financial losses for investors.
 - **Composite Risk Measures:** Combining multiple risk assessments or scores to get a more comprehensive view of overall risk.
 - **Weighting:** Assigning different levels of importance or influence to different factors or assessments based on their relevance to the specific risk exposures of a company.
-

Q.6288 Analyzing a country's risk exposure involves examining its economic growth rates, political and legal risks, and economic structure. Suppose an investment is considered in two emerging markets: one with a high GDP growth rate but unstable political environment and another with moderate GDP growth but a stable legal and political framework. How might these factors influence the risk assessment of these countries?

- A. The country with the high GDP growth but unstable environment might present higher long-term investment risk due to potential political instability and weak legal enforcement.
- B. A high GDP growth rate automatically signifies lower risk regardless of political or legal instability, making it the preferable investment option.
- C. Political and legal risks are secondary to economic growth rates, so the rapidly growing economy is always less risky.
- D. Both countries are equally risky as economic growth and political stability do not significantly impact overall investment risk.

The correct answer is **A**.

The country with high GDP growth but political instability might present higher long-term investment risk due to potential political instability and weak legal enforcement. Economic growth is beneficial, but instability in political and legal systems can lead to unpredictable regulatory changes, expropriation risks, and challenges in contract enforcement, making investments more vulnerable.

B is incorrect because high GDP growth does not negate the risks posed by political and legal instability, which can significantly undermine investment security.

C is incorrect as political and legal risks are critically important and can outweigh the benefits of high economic growth by increasing uncertainty and potential losses.

D is incorrect because economic growth, political stability, and legal enforcement significantly influence the risk profile of an investment, making the assessment of overall risk complex and multi-faceted.

Things to Remember

- Countries with political instability and weak legal systems are often considered higher risk, even if they exhibit high economic growth rates.
- Comprehensive risk assessments should include evaluations of political risks, legal risks, and economic structure alongside economic growth metrics to gauge the full risk exposure.

Q.6289 Examining the role of composite risk measures, consider a country rated by Political Risk Services (PRS) and the World Bank. PRS uses political, financial, and economic dimensions while the World Bank uses six indicators to assess risk. What interpretative challenges might arise from utilizing these composite risk scores together?

- A. Both scores will accurately reflect the same country risk scenario since they focus on the same variables.
- B. Divergence in methodologies and indicators used may result in different risk interpretations, complicating comparative risk analysis.
- C. Composite scores from both sources will typically underrate the country's risk by not considering external factors.
- D. The interpretations are universally standardized, which allows for seamless comparison between the two scores.

The correct answer is **B**.

Interpreting composite risk measures from PRS and the World Bank can be challenging due to differences in their methodologies and the risk indicators used. PRS uses political, financial, and economic dimensions, while the World Bank considers six specific indicators such as corruption and political stability. This divergence can lead to varying risk interpretations, complicating direct comparisons.

A is incorrect because different dimensions and methodologies mean the scores may not reflect the same risk scenario accurately.

C is incorrect as underrating risk isn't typically due to missing out on external factors but rather due to methodological variances.

D is incorrect because the interpretation of these scores is not standardized, and each organization's approach can lead to different perspectives on risk.

Things to Remember

- Differences in methodologies between composite score providers can result in variations that make comparative risk analysis complex.
- Understanding the specific indicators and dimensions used by each rating service is essential for accurate risk interpretation.

Q.6290 When evaluating composite risk scores provided by services like the World Bank or PRS, what might be the potential implications of using these scores for portfolio management in emerging markets?

- A. Composite risk scores from these services are universally precise, ensuring reliable investment decisions without needing further due diligence.
- B. These scores often provide a generalized view of risk, and without understanding the specific variables included, there's a risk of misjudging the actual investment environment.
- C. Composite scores ensure rapid adjustments to market changes, accurately predicting foreseeable risks and negating the need for continuous portfolio reassessments.
- D. The scores are calibrated for mature markets only, making them less applicable to emerging market scenarios.

The correct answer is **B**.

Composite risk scores often provide a generalized view of risk encompassing various dimensions. For portfolio management in emerging markets, it is essential to understand the specific variables included in such scores. Otherwise, there is a risk of misjudging the actual investment climate due to the varying impact of each factor.

A is incorrect because while composite scores can be useful, they are not universally precise and should not replace comprehensive due diligence.

C is incorrect as composite scores do not always ensure rapid adjustments to market changes and may miss sudden developments.

D is incorrect because while there may be biases, composite scores are applicable to both mature and emerging markets, though their relevance depends on the variables considered.

Things to Remember

- Composite scores provide an overall risk perspective but do require careful interpretation, especially in the dynamic contexts of emerging markets.
- Detailed understanding of the components and methodologies ensures that investors apply these scores effectively within their comprehensive risk assessment frameworks.

Q.6291 Sovereign defaults on foreign currency debt have been more prevalent compared to local currency debt defaults in recent decades. This is primarily due to governments' inability to raise sufficient foreign currency reserves. How does this contrast with the dynamics of local currency debt defaults, and what are the main causes of each type?

- A. Foreign currency debt defaults often result from hyperinflation, whereas local currency debt defaults are generally due to currency depreciation.
- B. Both types of defaults occur predominantly due to external trade imbalances and poor fiscal management.
- C. Local currency debt defaults are frequently due to political instability, whereas foreign currency debt defaults stem from external economic conditions.
- D. Foreign currency debt defaults are driven by a lack of foreign reserves, while local currency debt defaults can arise from shared currencies, reluctance to print, or historical gold backups.

The correct answer is **D**.

Foreign currency debt defaults are largely attributable to a shortage of foreign currency reserves required for debt repayment. Governments cannot print foreign currency to meet these obligations, leading to defaults. In contrast, local currency debt defaults have been driven by issues such as mandatory gold backups, shared currencies limiting monetary policy flexibility, and a reluctance to print currency due to the associated economic and political risks.

A is incorrect because hyperinflation is not the primary driver of foreign currency debt defaults, and local currency defaults are not predominantly due to currency depreciation.

C is incorrect as local currency defaults can stem from structural financial reasons rather than primarily political instability.

B is incorrect since external trade imbalances and poor fiscal management are contributing factors but not the primary causes differentiating the two types of defaults.

Things to Remember

- Foreign currency debt defaults are often due to the inability to acquire sufficient foreign reserves.
- Local currency debt defaults can be caused by rigid monetary frameworks and reluctance to risk economic instability by printing more currency.

Q.6292 Sovereign credit spreads, or sovereign default spreads, and credit default swaps (CDS) are essential tools in evaluating country risk. How do sovereign credit spreads typically compare to credit ratings when assessing a country's credit risk?

- A. Credit ratings are more volatile and react immediately to market signals, while sovereign spreads tend to be more stable.
- B. Sovereign credit spreads provide real-time risk assessment and are more granular, while credit ratings are less volatile and offer long-term evaluations.
- C. Sovereign credit spreads are standardized and universally accepted, making them more reliable than credit ratings.
- D. Both sovereign spreads and credit ratings are equally used by markets, offering identical insights into country risk.

The correct answer is **B**.

Sovereign credit spreads provide real-time assessments of country risk, allowing for immediate reflection of market conditions and more granular risk differentiation. However, they can be more volatile due to market forces unrelated to default risk. Credit ratings offer more stable, long-term evaluations but may lag in reflecting real-time economic changes, providing less granularity in immediate risk shifts.

A is incorrect because credit ratings are generally less volatile compared to market-based spreads, which react to immediate market signals.

C is incorrect as credit spreads are not standardized universally and can be influenced by liquidity and market supply-demand factors.

D is incorrect because while both metrics are valuable, they complement each other rather than offering identical insights.

Things to Remember

- Sovereign credit spreads offer real-time risk assessments and provide granularity, making them effective for short-term evaluations.
- Credit ratings provide stable, long-term assessments, often lagging in real-time reflection but useful for long-term investment decisions.

Reading 52: Measuring Credit Risk

Q.562 Understanding the concept of a copula is essential in the field of finance, especially in risk management and modeling the dependency structure between different financial variables. Which of the following is the correct definition of a copula?

- A. A joint probability distribution between two uniformly distributed random variables.
- B. A joint probability distribution between two or more uniformly distributed random variables which still maintains their marginal distributions.
- C. The product of the marginal distributions of two or more random variables.
- D. A statistical tool that represents a multivariate distribution while still maintaining their individual marginal distributions.

The correct answer is **D**.

A copula is indeed a statistical tool that represents a multivariate distribution while still maintaining their individual marginal distributions. The term 'copula' is derived from the Latin word 'copulare', which means 'to connect'. In statistics, a copula is used to describe the dependence between random variables. The Gaussian copula model is a multivariate distribution function whose marginal probability distribution is uniform on the interval $[0, 1]$. The copula function connects or 'couples' the marginal distributions of each individual variable to form the joint distribution function. This allows the copula to capture the dependence structure between the variables, while the marginal distributions capture the individual characteristics of each variable. The use of copulas in financial markets is particularly important as they allow for the modeling of skewed and heavy-tailed distributions which are often observed in financial returns.

Choice A is incorrect. While a copula does involve the joint probability distribution of random variables, it is not limited to just two uniformly distributed random variables. It can be applied to two or more random variables with any type of distribution.

Choice B is incorrect. This choice partially describes a copula but it's incomplete. A copula does involve the joint probability distribution of two or more uniformly distributed random variables and maintains their marginal distributions, but this definition doesn't capture the full essence of a copula which also represents a multivariate distribution while maintaining individual marginal distributions.

Choice C is incorrect. The product of the marginal distributions of two or more random variables defines an independent joint distribution, not a copula. A copula goes beyond this by allowing for dependence structures between the variables.

Things to Remember

- Copulas are widely used in finance for risk management, portfolio optimization, and modeling financial variables.

- There are different types of copulas such as Gaussian copula, t-copula, Clayton copula, and Gumbel copula, each with its own characteristics and applications.
 - Copulas are particularly useful in capturing non-linear dependencies and tail dependencies between variables.
 - One important property of copulas is that they separate the marginal distributions from the dependence structure, allowing for more flexible modeling.
 - Copulas are also used in credit risk modeling, especially in the context of credit portfolio risk assessment and correlation modeling.
-

Q.1053 American International Bank sanctioned a loan to a corporate client. The following particulars are given in the credit note by the credit analyst of the client:

Exposure amount = 100 USD million

Loss rate = 10%

Probability of default = 20%

What is the expected loss of the loan?

- A. USD 2 million.
- B. USD 20 million.
- C. USD 10 million.
- D. USD 40 million.

The correct answer is **A**.

$$\begin{aligned}
 \text{Expected loss} &= \text{Probability of default at time H} * \text{Exposure amount at time H} \\
 &\quad * \text{Loss rate experienced at time H} \\
 &= PD * EA * LR \\
 &= 100 * 0.2 * 0.1 \\
 &= 2 \text{ million}
 \end{aligned}$$

Q.3074 BYJ commercial bank has \$100 million of retail exposures. The 1-year probability of default averages 2% and the recovery rate averages 60%. If the correlation parameter is estimated at 0.1, what will be the 1-year unexpected loss at 99.9% confidence? Please click here

to see the standard normal table.

- A. \$7.68 million
- B. \$8.01 million
- C. \$4.32 million
- D. \$12.8 million

The correct answer is **C**.

Assuming the loss distribution is lognormal, the α percentile of the distribution of the default rate logarithm is

$$\alpha\text{percentile for default rate} = N\left(\frac{N^{-1}(PD) - \sqrt{\rho}N^{-1}(1 - \alpha)}{\sqrt{1 - \rho}}\right)$$

Thus, the 99 default rate is given by;

$$\begin{aligned} V(0.999, 1) &= N\left[\frac{\{N^{-1}(0.02) + \sqrt{0.1}N^{-1}(0.999)\}}{\sqrt{1 - 0.1}}\right] \\ &= \frac{-2.05 + \sqrt{0.1} \times 3.09}{\sqrt{1 - 0.1}} = 0.128 \end{aligned}$$

This is showing that the 99.9% worst-case default rate is 12.8%

The 1-year unexpected loss at 99.9% confidence is given by:

$$\begin{aligned} &(WCDR - PD) \times LGD \times EAD \\ &= (0.128 - 0.02) \times (1 - 0.6) \times 100 \\ &= 4.32 \text{ million} \end{aligned}$$

Where:

α is the confidence level

ρ is the correlation between each pair of U_i distributions

N^{-1} is the inverse cumulative normal distribution

WCDR(Worst-Case Default Rate) is the 99.9 percentile of the default rate distribution

LGD is the Loss Given Default(equals one minus the recovery rate).

PD is the Probability of Default

EAD is the total exposure at default (i.e., the sum of the principals of all the loans).

Q.3440 An investor holds a portfolio of \$200 million. This portfolio consists of AA-rated bonds (\$120 million) and BB-rated bonds (\$80 million). Assume that the one-year probabilities of default for AA-rated and BB-rated bonds are 4% and 6%, respectively, and that they are independent. In the event of default, the recovery rate for AA-rated bonds is 65%, and the recovery rate for BB-rated bonds is 40%. Determine the one-year expected credit loss from this portfolio:

- A. \$1,680,000
- B. \$4,560,000
- C. \$4,500,000
- D. \$2,880,000

The correct answer is **B**.

The expected loss of the portfolio is the sum of the expected losses of individual assets.

$$EL = EA \times PD \times LR$$

For AA-rated bonds,
EA = \$120,000,000,
PD = 0.04, and
LR = 0.35
Thus,

$$EL_{AA} = 120,000,000 \times 0.04 \times 0.35 = \$1,680,000$$

For BB-rated bonds,
EA = \$80,000,000,
PD = 0.06, and
LR = 0.6
Thus,

$$EL_{BB} = 80,000,000 \times 0.06 \times 0.6 = \$2,880,000$$

$$\text{Portfolio Expected Loss} = \$1,680,000 + \$2,880,000 = \$4,560,000$$

Q.3441 A portfolio consists of two bonds. The credit VaR – as defined by the bondholder – is the maximum loss due to defaults at a confidence level of 99%, over a period of one year. The probability that the two bonds jointly default is 2%, with a default correlation of 25%. The bond value, default probability, and recovery rate are USD 500,000, 5%, and 50% for one bond, and USD 300,000, 3%, and 30% for the other. Determine the expected credit loss of the portfolio:

- A. USD 18,800
- B. USD 12,500
- C. USD 18,424
- D. USD 12,424

The correct answer is **A**.

The joint default probability and the default correlation are nugatory as far as the expected credit loss of the portfolio is concerned. In other words, they do no matter. The expected loss of the portfolio is simply the sum of the expected losses of individual assets.

$$EL = EA \times PD \times LR$$

For the first bond,

$$EA = \$500,000,$$

$$PD = 0.05, \text{ and}$$

$$LR = 0.5$$

Thus,

$$EL_{AA} = 500,000 \times 0.05 \times 0.5 = \$12,500$$

For the second bond,

$$EA = \$300,000,$$

$$PD = 0.03, \text{ and}$$

$$LR = 0.7$$

Thus,

$$EL_{BB} = 300,000 \times 0.03 \times 0.7 = \$6,300$$

$$\text{Portfolio credit loss} = \$12,500 + \$6,300 = \$18,800$$

Note: The joint probability of default and the default correlation would be important only in the calculation of the unexpected credit loss of the portfolio.

Q.4644 A bank has two assets outstanding, denominated in U.S. dollars. The correlation between the two assets is 0.4. Other details are as follows:

	Asset A	Asset B
EA	1,600,000	2,000,000
PD	1%	2%
LR	30%	40%

Calculate the expected loss (EL) of the portfolio.

- A. 22400
- B. 20800
- C. 18200
- D. 20200

The correct answer is **B**.

The expected loss of a portfolio is equal to the summation of expected losses of individual asset. That is,

$$\begin{aligned}
 EL_P &= \sum EA_i \times PD_i \times LGD_i \\
 &= [1,600,000 \times 0.01 \times 0.3] + [2,000,000 \times 0.02 \times 0.4] \\
 &= 20,800
 \end{aligned}$$

Q.4645 The amount of a loan issued by a bank is \$2 million, with a default probability of 0.1% over a period of one year. If the recovery rate is estimated to be 40%, what is the expected credit loss?

- A. \$800
- B. \$1,000
- C. \$1,200
- D. \$700

The correct answer is **C**.

The credit loss of a single asset is given by:

$$EL = EA \times PD \times LGD$$

But also

$$LGD = 1 - \text{Recovery Rate}$$

So, the expected credit loss is given by:

$$EL = \$2 \text{ million} \times 0.001 \times (1 - 0.4) = \$1,200$$

Q.4646 The amount of a loan issued by a bank is \$2 million, with a default probability of 0.1% over one year. If the recovery rate is estimated to be 40%, what is the standard deviation of expected credit loss?

- A. \$37,928.35
- B. \$30,567.65
- C. \$32,464.54
- D. \$35,890.75

The correct answer is **A**.

The standard deviation of the loss is given by:

$$\begin{aligned}\sigma_i &= \sqrt{p_i - p_i^2} [L_i (1 - R_i)] \\ &= \sqrt{0.001 - 0.001^2} [\$2 \text{ million} (1 - 0.4)] = 0.0379284 = 37,928.35\end{aligned}$$

Q.4647 The Bank of Africa has a portfolio of three \$2 million loans, each with a default rate of 0.5% over one year. If the correlation between the loans is 0.4 and the recovery rate is 40%, what is the mean of the portfolio credit loss?

- A. \$18,000
- B. \$12,000
- C. \$10,000
- D. \$9,000

The correct answer is **A**.

The mean of credit loss of a loan is given by:

$$p_i L_i (1 - R_i)$$

Nevertheless, we are given three loans with the same variables. So the mean of the portfolio is:

$$\begin{aligned}&= 3 \times p_i L_i (1 - R_i) \\ &= 3 \times 0.005 \times 2 \text{ million} \times (1 - 0.4) = \$18,000\end{aligned}$$

Q.4648 The Bank of Africa has a portfolio of three \$2 million loans, each with a default rate of 0.5% over one year. If the correlation between the loans is 0.4, and the recovery rate is 40%, what is the standard deviation of the portfolio credit loss?

- A. \$426,875
- B. \$38,685
- C. \$84,600
- D. \$196,593

The correct answer is **D**.

The variance of the credit loss of a loan portfolio is given by

$$\sigma_P^2 = n\sigma^2 + n(n-1)\rho\sigma^2$$

Since the loans have equal principal L , recovery rate R and default probabilities, we need to compute the common standard deviation as:

$$\begin{aligned}\sigma_i &= \sqrt{p - p^2} [L(1 - R)] \\ &= \sqrt{0.005 - 0.005^2} [2 \times (1 - 0.4)] = 0.0846\end{aligned}$$

Thus, the variance of the credit loss is given by

$$\begin{aligned}\sigma_P^2 &= n\sigma^2 + n(n-1)\rho\sigma^2 \\ \sigma_P^2 &= 3 \times 0.0846^2 + 3(3-1) \times 0.4 \times 0.0846^2\end{aligned}$$

Hence, the standard deviation of the credit loss is given by

$$\sigma_P = \sqrt{0.386852} = 0.196593 = \$196,593$$

Q.4650 Aiden Bank has a \$500 million loan portfolio with a PD of 0.5%. Assuming the Vasicek Model, what is the 99.9 percentile of the default rate if the correlation parameter is 0.25?

- A. 0.1305
- B. 0.0165
- C. 0.1169
- D. 0.0175

The correct answer is **C**.

According to the Vasicek model, the 99.9 percentile of the default rate is given by

$$\begin{aligned}
 \text{99.9 percentile for default rate} &= N\left(\frac{N^{-1}(\text{PD}) + \sqrt{\rho}N^{-1}(0.999)}{\sqrt{1-\rho}}\right) \\
 &= N\left(\frac{N^{-1}(0.005) + \sqrt{0.25}N^{-1}(0.999)}{\sqrt{1-0.25}}\right) \\
 &\Rightarrow N\left(\frac{-2.576 + 0.5 \times 3.09}{\sqrt{1-0.25}}\right) = N(-1.1905) = 0.1169 = 11.69\%
 \end{aligned}$$

Q.4652 A bank has a loan portfolio consisting of three loans A, B, and C with standard deviations of 1.25 each. The correlations matrix appears as follows:

	Loan A	Loan B	Loan C
Loan A	1	0	0.3
Loan B	0	1	0.6
Loan C	0.3	0.6	1

Suppose the size of loan A is increased by 1%. Using the Euler's theorem, calculate the contribution of loan A to the total standard deviation.

- A. 0.76
- B. 0.72
- C. 0.8
- D. 0.74

The correct answer is **D**.

According to Euler's theorem,

$$Q_i = x_i \frac{\Delta F_i}{\Delta x_i}$$

Where

Δx_i = small change in x_i

ΔF_i = small change in F_i

Q_i = ratio of ΔF_i to a proportional change $\frac{\Delta x_i}{x_i}$ in x_i

We need to first calculate the total loss on the portfolio.

Given a portfolio of three assets, the portfolio standard deviation is given by:

$$\sigma_p = \sqrt{\sigma_A^2 + \sigma_B^2 + \sigma_C^2 + 2\rho_{AB}\sigma_A\sigma_B + 2\rho_{AC}\sigma_A\sigma_C + 2\rho_{BC}\sigma_B\sigma_C}$$

The current portfolio standard deviation is

$$\sqrt{1.25^2 + 1.25^2 + 1.25^2 + 2 \times 0 \times 1.25 \times 1.25 + 2 \times 0.3 \times 1.25 \times 1.25 + 2 \times 0.6 \times 1.25 \times 1.25} \\ = 2.738613$$

Now if the size of loan A is increased by 1%, then the new standard deviation of loan A is given by

$$1.25 \times 1.01 = 1.2625$$

So, the new portfolio standard deviation is

$$\sqrt{1.2625^2 + 1.25^2 + 1.25^2 + 2 \times 0 \times 1.2625 \times 1.25 + 2 \times 0.3 \times 1.2625 \times 1.25 + 2 \times 0.6 \times 1.25 \times 1.25} \\ = 2.746048$$

The change in portfolio standard deviation = $2.746048 - 2.738613 = 0.007435$

So, that

$$Q_A = \frac{0.007435}{0.01} = 0.7435 \approx 0.74$$

The contributions of loans B and C can be calculated in a similar manner, i.e., increasing their

respective standard deviations by 1%, holding all other factors constant.

Q.4654 Barclays Bank has a \$600 million loan portfolio, and the recovery rate in the event of default is 40%. Assuming the Vasicek Model, the required regulatory capital is \$5 million. The 99.9 percentile for default rate is 0.0188. What is the probability of default for the loan portfolio?

A. 0.0051

B. 0.04

C. 0.0049

D. 0.0054

The correct answer is **C**.

The Basel II capital requirement for banks under the IRB approach is given by

$$(WCDR - PD) \times LGD \times EAD$$

Where WCDR is defined as the worst-case default rate, and it is 99.9 percentile of the default rate distribution defined as in Vasicek model, which in this case is 0.0188.

$$\begin{aligned} &\Rightarrow (WCDR - PD) \times LGD \times EAD = 5 \\ &(0.0188 - PD) \times (1 - 0.4) \times 600 = 5 \\ &\therefore PD = 0.004911 \end{aligned}$$

Q.4655 A bank issues a \$2 million loan, with a default probability of 0.5% over one year. The standard deviation of the expected credit loss is \$35,000. What is the recovery rate?

- A. 0.752
- B. 0.456
- C. 0.656
- D. 0.764

The correct answer is **A**.

The standard deviation of the loss is given by:

$$\begin{aligned}\sigma_i &= \sqrt{p_i - p_i^2} [L_i (1 - R_i)] \\ &= \sqrt{0.005 - 0.005^2} [2 (1 - R_i)] = 0.035 \\ &= 1 - \left[\frac{0.035}{2 \times \sqrt{0.005 - 0.005^2}} \right] = 0.7519\end{aligned}$$

Q.4656 A bank issues a \$7 million loan, with a default probability of 0.5% over a period of one year. If the recovery rate is estimated to be 35%, what is the expected credit loss on this loan?

- A. \$23,350
- B. \$22,750
- C. \$23,600
- D. \$22,850

The correct answer is **B**.

The credit loss of a single asset is given by:

$$\begin{aligned}EL &= EA \times PD \times LGD \\ &= \$7 \text{ million} \times 0.005 \times (1 - 0.35) = \$22,750\end{aligned}$$

Q.4657 An American bank recently issued a USD 5 million loan to a business entity, of which USD 2 million is currently outstanding. According to the bank's internal rating model, the business entity has a 0.5% chance of defaulting over the next year. In case that happens, the estimated loss rate is 25%. The probability of default and the loss rate have standard deviations of 7% and 17%, respectively. What is the value of unexpected loss?

- A. \$41,245.45
- B. \$42,461.75
- C. \$40,564.56
- D. \$45,563.45

The correct answer is **B**.

The unexpected loss is given by:

$$UL = EA \times \sqrt{PD \times \sigma_{LR}^2 + LR^2 \times \sigma_{PD}^2}$$

So in our case.

EA = USD 2,000,000

PD = 0.5%

LR = 25%

$\sigma_{LR} = 0.17$

$\sigma_{PD} = 0.07$

Thus,

$$\begin{aligned} &= 2,000,000 \times \sqrt{0.005 \times 0.17^2 + 0.25^2 \times 0.07^2} \\ &= \$42,461.75 \end{aligned}$$

Q.4658 The bank of Aides has a portfolio consisting of 100,000 loans, each amounting to \$1 million, and has a 1% probability of default in a year. The recovery rate is 40%, and the correlation coefficient is 0.3. Calculate α , the standard deviation of the loss from the loan portfolio as a percentage of its size.

A. 0.033

B. 0.045

C. 0.056

D. 0.045

The correct answer is **A**.

The α parameter is given by:

$$\alpha = \frac{\sigma\sqrt{1 + (n - 1)\rho}}{L\sqrt{n}}$$

For this case,

$L = \$ 1 \text{ million}$

$\rho = 0.3$

$n = 100,000$

$R = 0.4$

$$\begin{aligned}\sigma &= \sqrt{p - p^2} [L (1 - R)] \\ &= \sqrt{0.01 - 0.01^2} [1 (1 - 0.4)] = 0.05970\end{aligned}$$

Therefore,

$$\alpha = \frac{0.05970\sqrt{1 + (100,000 - 1)0.3}}{1 \times \sqrt{100,000}} = 0.03270$$

Reading 53: Operational Risk

Q.1071 Eurasia Bank Limited is following the basic indicator approach for calculating the operational risk amount for the year 2016. The financial details of the bank are given below:

	Income earned (In million USD)	Interest paid	Non-interest income
Year 2015	105	52	18
Year 2014	100	50	20
Year 2013	95	40	16

Based on the original Basel Accord, the bank must hold capital for operational risk for 2016 equal to:

- A. USD 10.60 million.
- B. USD 10.95 million.
- C. USD 11.05 million.
- D. USD 7.90 million.

The correct answer is **A**.

Based on the original Basel Accord, banks using the basic indicator approach must hold capital for operational risk equal to the average over the previous three years of a fixed percentage of positive annual gross income. Gross income is defined as net interest income plus non-interest income.

Net interest income plus non-interest income for the previous three years:

	Income earned (In million USD)	Interest paid	Net Income	Non-interest income	Gross Income
Year 2015	105	52	53	18	71
Year 2014	100	50	50	20	70
Year 2013	95	40	55	16	71

Average gross income over the previous three years: 70.667

Operational risk = 15% of average gross income over the previous three years: 10.60

Q.1074 A bank with annual revenues of \$4 billion has incurred a loss of \$200 million on account of operational risk. What would be the losses for a bank with a similar business profile but with revenues of \$12 billion? Assume the exponent for scaling losses is 0.23.

- A. USD 7.76 million
- B. USD 12.76 million
- C. USD 257.5 million
- D. USD 200.00 million

The correct answer is **C**.

$$\begin{aligned}\text{Loss of bank B} &= \left(\frac{\text{Revenue of bank B}}{\text{Revenue of bank A}} \right)^{0.23} * \text{Loss of bank A} \\ &= \left(\frac{12}{4} \right)^{0.23} * 200 \\ &= 3^{0.23} * 200 = 257.5 \text{ million}\end{aligned}$$

Q.1075 Your Canadian Bank has been using the standardized approach for the last three years. The board of directors has recently decided to use the advanced measurement approach for measuring operational risk for regulatory risk from the current year onwards. Considering the change in measurement approach, which of the following process has been adopted from the current year onwards in the risk management department of the bank?

- A. Regular reporting of operational risk losses throughout the bank.
- B. Well-documented operational risk management system.
- C. Regular independent review of operational risk management processes by internal auditors, external auditors, and supervisors.
- D. Estimation of unexpected losses based on an analysis of relevant internal and external data, and scenario analyses.

The correct answer is **D**.

The Advanced Measurement Approach (AMA) for operational risk, as per Basel II guidelines, requires banks to develop their own empirical model to quantify required capital for operational risk. Banks can use this approach only subject to approval from their local regulators. The AMA allows banks to use a wider range of operational risk measurement techniques, provided they can demonstrate to their supervisor that their techniques are sound, implemented with integrity, and that they meet a series of qualitative and quantitative standards. One of the key

requirements of the AMA is the estimation of unexpected losses based on an analysis of relevant internal and external data, and scenario analyses. This involves the use of internal loss data, relevant external loss data, scenario analysis, and factors reflecting the business environment and internal control systems. Therefore, with the shift from the standardized approach to the AMA, the bank would have adopted the process of estimating unexpected losses based on an analysis of relevant internal and external data, and scenario analyses, as stated in choice D.

Choice A is incorrect. Regular reporting of operational risk losses throughout the bank is a common practice in both the standardized and advanced measurement approaches. Therefore, this would not be a new procedure implemented due to the shift in risk measurement methodology.

Choice B is incorrect. Having a well-documented operational risk management system is also a requirement under both methodologies, hence it would not be unique to the advanced measurement approach.

Choice C is incorrect. Regular independent reviews of operational risk management processes by internal auditors, external auditors, and supervisors are part of good governance practices and are expected under both methodologies. Thus, this procedure would not have been newly implemented due to the transition from standardized approach to advanced measurement approach.

Things to Remember

- Basel II guidelines for operational risk management
 - Advanced Measurement Approach (AMA) for operational risk
 - Qualitative and quantitative standards for operational risk measurement
 - Use of internal loss data, external loss data, and scenario analysis in estimating unexpected losses
 - Approval process from local regulators for adopting AMA
-

Q.1076 Bank X is following the advanced measurement approach for measuring operational risk. Which of the following should be the operational risk capital computed for regulatory purposes?

- A. The bank must use 15% of net interest income over the previous three years.
- B. The bank's activities are divided into eight business lines. The average gross income over the last three years for each business line is multiplied by a "beta factor" for that business line, and the result is summed to determine the total capital.
- C. The bank must estimate one-year 99.9% VaRs for the seven categories of operational risks identified by the Basel Committee and then aggregate them to determine a single

one-year 99.9% operational risk VaR measure.

D. The bank must use 15% of net interest income plus non-interest income over the previous three years.

The correct answer is C.

The advanced measurement approach (AMA) is a method for operational risk management under the Basel II Accord. Under this approach, banks are allowed to develop their own empirical model to quantify required capital for operational risk. Banks can use this approach only subject to approval from their local regulators. The AMA allows a bank to use internal operational risk loss data, relevant external operational risk loss data, scenario analysis data and business environment and internal control factors in determining the bank's operational risk capital requirement. In this case, the bank must estimate one-year 99.9% Value at Risk (VaR) for each of the seven categories of operational risks identified by the Basel Committee. These categories include internal fraud, external fraud, employment practices and workplace safety, clients, products and business practices, damage to physical assets, business disruption and system failures, and execution, delivery and process management. The bank then aggregates these VaRs to determine a single one-year 99.9% operational risk VaR measure. This measure represents the capital that the bank needs to hold to cover unexpected losses from operational risks over a one-year period with a confidence level of 99.9%.

Choice A is incorrect. The advanced measurement approach (AMA) does not require the bank to use 15% of net interest income over the previous three years to calculate its operational risk capital. This method is too simplistic and does not take into account the specific risks associated with different types of operations.

Choice B is incorrect. While it's true that under Basel II, banks' activities are divided into eight business lines for operational risk assessment, this method described here refers to the Standardized Approach (TSA), not AMA. In TSA, average gross income over the last three years for each business line is multiplied by a "beta factor" for that business line, and then summed up to determine total capital requirement. However, in AMA, banks are allowed to use their own internal models to estimate operational risk.

Choice D is incorrect. Similar to choice A, using 15% of net interest income plus non-interest income over the previous three years as a measure of operational risk capital does not align with AMA guidelines set by Basel Committee on Banking Supervision (BCBS). The AMA requires more sophisticated modeling techniques that capture all significant aspects of operational risk faced by a bank.

Things to Remember

- Operational risk is the risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events.
- The Advanced Measurement Approach (AMA) allows banks to develop their own empirical model to quantify required capital for operational risk.

- AMA requires banks to use internal operational risk loss data, relevant external operational risk loss data, scenario analysis data, and business environment and internal control factors in determining operational risk capital.
 - Basel II identifies seven categories of operational risks: internal fraud, external fraud, employment practices and workplace safety, clients, products and business practices, damage to physical assets, business disruption and system failures, and execution, delivery and process management.
 - AMA aims to provide a more risk-sensitive approach to calculating operational risk capital compared to the Standardized Approach (TSA).
-

Q.1077 Loss severity and loss frequency are two distributions that are important in estimating potential operational risk losses for a risk type. With regard to these two distributions, which of the following is true?

- A. For loss frequency, the natural probability distribution to use is a Poisson distribution, and for the loss-severity probability distribution, a lognormal distribution is used.
- B. For loss frequency, the natural probability distribution to use is a lognormal distribution, and for the loss-severity probability distribution, a Poisson distribution is used.
- C. For loss frequency, the natural probability distribution to use is a Poisson distribution, and for the loss-severity probability distribution, a normal distribution is used.
- D. For loss frequency, the natural probability distribution to use is a normal distribution, and for the loss-severity probability distribution, a Poisson distribution is used.

The correct answer is **A**.

In the context of operational risk management, loss frequency and loss severity are two critical distributions used to estimate potential losses for a specific risk type. For loss frequency, the natural probability distribution to use is a Poisson distribution. The Poisson distribution is a discrete probability distribution that expresses the probability of a given number of events occurring in a fixed interval of time or space if these events occur with a known constant mean rate and independently of the time since the last event. This distribution assumes that losses happen randomly through time so that in any short period of time Δt , there is a probability $\lambda \Delta t$ of a loss occurring. For the loss severity probability distribution, a lognormal distribution is used. The lognormal distribution is a continuous probability distribution of a random variable whose logarithm is normally distributed. The parameters of this probability distribution are the mean

and standard deviation of the logarithm of the loss. Therefore, choice A accurately describes the natural probability distributions used for loss frequency and loss severity in operational risk management.

Choice B is incorrect. The lognormal distribution is not typically used for loss frequency in operational risk management. Loss frequency refers to the number of times a loss event occurs, which can only take on non-negative integer values (0, 1, 2,...). The Poisson distribution is more suitable for this purpose as it models events that occur at a constant mean rate with independent occurrences. On the other hand, the lognormal distribution is continuous and can take any positive real value, making it unsuitable for modeling count data like loss frequency.

Choice C is incorrect. While the Poisson distribution correctly represents loss frequency as explained above, a normal distribution does not accurately represent loss severity in operational risk management. Loss severity refers to the magnitude of losses when an event occurs and can take any non-negative real value (including zero). A normal distribution includes negative values which are not applicable in this context since losses cannot be negative. Instead, a lognormal distribution which only takes positive values would be more appropriate.

Choice D is incorrect. Neither part of this choice accurately represents natural probability distributions used in operational risk management. As explained earlier, a normal distribution does not suitably model loss frequency due to its ability to take negative values while Poisson distributions are inappropriate for modeling loss severity because they only represent count data.

Things to Remember

- Loss frequency refers to the number of times a loss event occurs
 - Loss severity refers to the magnitude of losses when an event occurs
 - Poisson distribution is used for modeling events that occur at a constant mean rate with independent occurrences
 - Lognormal distribution is used for continuous variables with positive values
 - Operational risk management involves estimating potential losses for specific risk types
-

Q.1085 The existence of an insurance contract causes the bank to behave differently than it otherwise would and increases the risks to the insurance company. This risk is known as:

- A. Adverse selection.
- B. Moral hazard.
- C. Wrong-way risk.
- D. Operational risk.

The correct answer is **B**.

Moral hazard is a term used in economics, finance, and insurance to describe the risk that arises when a party behaves differently because it is protected by an insurance contract. In the context of banking and insurance, moral hazard refers to the risk that the existence of an insurance contract will cause the bank to behave differently than it otherwise would. This changed behavior increases the risks to the insurance company. The bank, knowing that it is protected by the insurance, might take on more risk than it would without the insurance. This could lead to reckless behavior, such as lending to high-risk borrowers or investing in risky assets, which could potentially lead to significant losses for the insurance company. Therefore, moral hazard is a significant concern for insurance companies and is a critical factor that they need to consider and manage when underwriting insurance policies.

Choice A is incorrect. Adverse selection refers to a situation where an individual's demand for insurance (either the propensity to buy insurance, or the quantity purchased, or both) is positively correlated with the individual's risk of loss. It does not refer to a change in behavior due to the presence of an insurance contract.

Choice C is incorrect. Wrong-way risk occurs when exposure to counterparty credit risk increases as the credit quality of that counterparty deteriorates. This concept is typically applied in derivatives markets and does not relate directly to changes in behavior due to an insurance contract.

Choice D is incorrect. Operational risk refers to the prospect of loss resulting from inadequate or failed procedures, systems or policies; employee errors; system failures; fraud or other criminal activity; any event that disrupts business processes. While operational risks may be exacerbated by poor decisions influenced by moral hazard, they are not directly caused by changes in behavior due to an insurance contract.

Things to Remember

- Adverse selection refers to a situation where an individual's demand for insurance is positively correlated with the individual's risk of loss.
- Wrong-way risk occurs when exposure to counterparty credit risk increases as the credit quality of that counterparty deteriorates.
- Operational risk refers to the prospect of loss resulting from inadequate or failed procedures, systems or policies; employee errors; system failures; fraud or other criminal activity.
- Moral hazard is a significant concern for insurance companies and is a critical factor that they need to consider and manage when underwriting insurance policies.

Q.3444 A risk manager has established that there's a 90% probability that losses over the next year will not exceed \$30 million. Given that the power law parameter is 0.8, what is the probability of the loss exceeding \$10 million?

- A. 22%
- B. 24%
- C. 20%
- D. 23%

The correct answer is **B**.

The power law states that the probability of a random variable x exceeding a value V is given by:

$$p(v > x) = KV^{-\alpha}$$

where:

K is constant,

α is the power law parameter.

$$p(v > x) = KV^{-\alpha}$$

$$0.1 = K(30)^{-(0.8)}$$

$K = 1.5195$ Thus,

$$p(v > x) = 1.5195V^{-0.8}$$

when $x = 10$,

$$\text{Probability} = 1.5195 \times 10^{-0.8} = 0.24$$

Q.4598 The average loss frequency of Bank of Africa is estimated to be once every 18 months. What is the probability of three losses in a year for this bank?

- A. 0.0234
- B. 0.057
- C. 0.0254
- D. 0.0507

The correct answer is **C**.

We need to find λ which from the question we have:

$$\lambda = \frac{1}{1.5} = 0.6667 \text{ losses per year}$$

Now using the Poisson distribution

$$\Pr(n) = \frac{e^{-\lambda} \lambda^n}{n!}$$
$$\Pr(3) = \frac{e^{-0.6667} 0.6667^3}{3!} = 0.0254$$

Q.4599 Given the mean and the standard deviation of lognormal loss of a ban is 200 and 50 respectively, what is the variance of the logarithm of the loss?

A. 0.0606

B. 0.527

C. 0.069

D. 0.0629

The correct answer is **A**.

We know the mean of the logarithm of the loss is given by:

$$\ln\left(\frac{\mu}{\sqrt{1+w}}\right)$$

Furthermore, the variance is given by:

$$\ln(1+w)$$

Where

$$w = \left(\frac{\sigma}{\mu}\right)^2 = \left(\frac{50}{200}\right)^2 = 0.0625$$
$$\Rightarrow \text{Variance} = \ln(1.0625) = 0.0606$$

Q.4600 Given the mean and the standard deviation of the lognormal loss of a bank is 200 and 50, what is the standard deviation of the logarithm of the loss?

A. 0.2462

B. 1.0308

C. 0.0606

D. 0.4724

The correct answer is **A**.

We know the mean of the logarithm of the loss is given by:

$$\ln\left(\frac{\mu}{\sqrt{1+w}}\right)$$

Furthermore, the variance is given by:

$$\ln(1+w)$$

Where

$$\begin{aligned}w &= \left(\frac{\sigma}{\mu}\right)^2 = \left(\frac{50}{200}\right)^2 = 0.0625 \\&\Rightarrow \text{Variance} = \ln(1+w) = \ln(1.0625) = 0.0606 \\&\therefore \text{Standard deviation} = \sqrt{0.0606} \approx 0.2462\end{aligned}$$

Q.4601 Over the past ten years, the Bank of Yemen has had losses (in million euros) of 3, 6, 10, 50, 72, 101, and 200. What is the approximate amount of loss component of the bank under the SMA approach?

A. €656 million

B. €678 million

C. €756 million

D. €442 million

The correct answer is **C**.

Using the formula

$$7X + 7Y + 5Z$$

Where X, Y, and Z are the approximations of the average losses from the operational risk over the past ten years defined as:

- X - an average of all losses
- Y - an average of losses greater than 10 million
- Z - an average of losses greater than 100 million

From the question, the average total loss for the ten years is 442:

$$X = \frac{3 + 6 + 10 + 50 + 72 + 101 + 200}{10} = \frac{442}{10} = 44.2$$

The average losses greater than 10 million is:

$$Y = \frac{50 + 72 + 101 + 200}{10} = \frac{423}{10} = 42.3$$

And the average losses greater than 100 million is:

$$101 + 200 = \frac{301}{10} = 30.1$$

So the loss component is given by:

$$7 \times 44.2 + 7 \times 42.3 + 5 \times 30.1 = 756$$

Q.4604 The 90-percentile of a loss distribution is 30. Using the power law with $\alpha = 4$, what is the value of 95-percentile of the loss distribution?

- A. 35.68
- B. 30.45
- C. 25.56
- D. 27.89

The correct answer is **A**.

Recall that the power law is given by:

$$\Pr(v > x) = Kx^{-\alpha}$$

Therefore for the 90-percentile loss distribution, we have,

$$\begin{aligned} 0.10 &= K \cdot 30^{-4} \\ \Rightarrow K &= \frac{0.10}{30^{-4}} = 81,000 \end{aligned}$$

Thus, for the 95-percentile loss, we have to solve the equation:

$$\begin{aligned} 0.05 &= 81,000x^{-4} \\ \Rightarrow x &= \left(\frac{0.05}{81,000} \right)^{-\frac{1}{4}} = 35.68 \end{aligned}$$

Q.4606 Bank A has revenues of USD 50 billion and incurs a loss of USD 300 million. Another bank B has revenues of USD 40 billion. Given that the estimated loss for bank A is 250 million, which of the following is closest to the observed loss for Bank (Assume the scale adjustment is 0.23) B?

- A. USD 263.2 million
- B. USD 200 million
- C. USD 312.5 million
- D. USD 237.5 million

The correct answer is **D**.

Using the scale adjustment:

$$\begin{aligned}
 \text{Estimated Loss for Bank A} &= \text{Observed Loss for Bank B} \times \left(\frac{\text{Bank A Revenue}^{0.23}}{\text{Bank B Revenue}} \right) \\
 \Rightarrow 250 &= \text{Observed Loss for Bank B} \times \left(\frac{50^{0.23}}{40} \right) \\
 \text{Observed Loss for Bank B} &= 250 / \left(\frac{50^{0.23}}{40} \right) = 237.4929 \approx 237.5
 \end{aligned}$$

Q.4607 Over the last three years, an American bank earned an interest of USD 300 million and paid interest of USD 150 million on average. The bank's noninterest income over the last three years is USD 600 million on average. Under the indicator method, what is the value of the operational risk capital?

- A. \$75 million
- B. \$250.40 million
- C. \$112.50 million
- D. \$140.50 million

The correct answer is C.

Under the indicator approach, the required operational risk is equivalent to 15% of annual gross income over the previous three years. Also, recall that,

$$\begin{aligned}\text{Gross income} &= \text{Interest earned} - \text{Interest paid} + \text{Noninterest income} \\ &= 300 - 150 + 600 = 750 \\ \therefore \text{Gross Income} &= \text{USD 750 Million}\end{aligned}$$

Thus, the required operational risk capital is

$$15\% \times 750 = 112.5$$

Reading 54: Stress Testing

Q.1091 The senior management of the African Industrial Development Bank is reviewing the stress program post a severe financial crisis in Africa. With regard to the stress testing program, which of the following is most accurate?

- A. Senior management is ultimately responsible for the overall stress testing program, whereas the risk department is accountable for the program's implementation, management, and oversight.
- B. The Chief Risk Officer is ultimately responsible for the overall stress testing program, whereas the risk department is accountable for the program's implementation, management, and oversight.
- C. The risk committee is ultimately responsible for the overall stress testing program, whereas the risk department is accountable for the program's implementation, management, and oversight.
- D. The board of directors is ultimately responsible for the overall stress testing program, whereas the senior management is accountable for the program's implementation, management, and oversight.

The correct answer is **D**.

The board of directors is ultimately responsible for the overall stress testing program, whereas the senior management is accountable for the program's implementation, management, and oversight. This is because the board of directors, being the highest governing authority within an organization, is responsible for the overall strategic direction and governance of the organization, including its risk management practices. The senior management, on the other hand, is responsible for the day-to-day operations of the organization, which includes the implementation, management, and oversight of the stress testing program. Recognizing that many practical aspects of a stress testing program will be delegated, the involvement of the board in the overall stress testing program and of senior management in the program's design is essential. This ensures that the stress testing program aligns with the organization's strategic objectives and risk appetite, and that it is effectively managed and overseen to ensure its ongoing effectiveness and relevance.

Choice A is incorrect. While senior management plays a crucial role in the stress testing program, they are not ultimately responsible for it. The ultimate responsibility lies with the board of directors who oversee the overall strategic direction and risk appetite of the bank.

Choice B is incorrect. The Chief Risk Officer (CRO) has a significant role in managing and overseeing risk, but they are not ultimately responsible for the overall stress testing program. This responsibility rests with the board of directors.

Choice C is incorrect. Although the risk committee plays an important part in reviewing and recommending policies related to risk management, it does not hold ultimate responsibility for the overall stress testing program. This duty falls on the board of directors as per FRM guidelines.

Things to Remember

- Stress testing is a risk management technique used to evaluate the potential impact of financial or economic shocks on a bank or financial institution.
 - The board of directors is responsible for setting the overall risk appetite and strategic direction of the organization.
 - Senior management is responsible for the day-to-day operations and implementation of risk management practices within the organization.
 - The Chief Risk Officer (CRO) typically oversees the risk management function and ensures that risk management practices are in line with regulatory requirements and best practices.
 - The risk committee, if present, assists the board of directors in overseeing and reviewing risk management practices within the organization.
-

Q.1151 A bank has recently launched a fund for retail investors. The risk management team carries out stress testing of the newly launched fund to determine the impact of the fund on the bank's overall capital. Jason Bloomberg, a newly recruited risk manager, observes that the bank has an independent risk management team. He notes that the entire risk assessment and identification process is carried exclusively through stress testing. While examining the stress testing result, Jason observes that the test produces multiple potential losses under various scenarios. He also observes that the inputs from the bank's top economists were taken while modeling the stress testing scenarios. He finds that the stress testing procedures are well documented and no deviation is allowed from the procedure.

The bank's stress testing results produce multiple potential losses. In view of this, select the most appropriate option.

- A. The stress testing result must be actionable.
- B. The stress testing results must be integrated into decision-making, but only at the senior-most level of management.
- C. The stress testing result must accurately specify the exact amount of loss associated with a given variable.
- D. Stress testing produces potential losses and hence no action is required on the results.

The correct answer is **A**.

The stress testing result must be actionable. Stress testing is a simulation technique used in banking institutions to determine their ability to deal with an extremely stressed financial situation. The results of stress testing are not just for observation but should be actionable. This means that the results should feed into the decision-making process at the appropriate management level, including strategic business decisions of the board or senior management. The purpose of stress testing is to identify potential vulnerabilities in the bank's financial structure and to take corrective actions if necessary. Therefore, the results of stress testing should be used to make informed decisions about the bank's risk management strategies, capital allocation, and overall business operations. The actionable results of stress testing can help the bank to mitigate potential risks, enhance its financial stability, and ensure its long-term sustainability.

Choice B is incorrect. While it is important for senior management to consider stress testing results in their decision-making, it should not be limited to them only. All levels of management, especially those involved in risk management and strategic planning, should integrate these results into their decisions to ensure a comprehensive approach towards risk mitigation.

Choice C is incorrect. Stress testing does not necessarily specify the exact amount of loss associated with a given variable. Instead, it provides potential losses under various scenarios which can help in understanding the possible impact on capital under different conditions. The aim of stress testing is not precision but rather providing an insight into potential vulnerabilities and risks.

Choice D is incorrect. This statement contradicts the purpose of stress testing which aims at identifying potential risks and vulnerabilities that could impact an organization's capital or overall stability. Therefore, actions are indeed required based on the results of stress tests such as adjusting risk strategies or enhancing contingency plans.

Things to Remember

- Stress testing is a simulation technique used to assess a financial institution's ability to withstand adverse market conditions.
- The results of stress testing should be actionable, meaning they should inform decision-making processes at all levels of management.
- Stress testing helps identify potential vulnerabilities and risks in a bank's financial structure.
- Inputs from top economists can be valuable in modeling stress testing scenarios.
- Stress testing procedures should be well-documented and followed without deviation to ensure consistency and accuracy.

Q.1152 A bank has recently launched a fund for retail investors. The risk management team

carries out stress testing of the newly launched fund to determine the impact of the fund on the bank's overall capital. Jason Bloomberg, a newly recruited risk manager, observes that the bank has an independent risk management team. He notes that the entire risk assessment and identification process is carried exclusively through stress testing. While examining the stress testing result, John observes that the test produces multiple potential losses under various scenarios. Bloomberg also observes that the inputs from the bank's top economists were taken while modeling the stress testing scenarios. He finds that the stress testing procedures are well documented and no deviation is allowed from the procedure.

Inputs from the bank's top economist were considered while developing the scenario for stress testing. In view of this, select the most appropriate statement.

- A. Inputs from economists make the model more robust.
- B. Inputs from all stakeholders such as economists, business managers, fund managers, etc. must be taken into account.
- C. Only inputs from the risk management team in collaboration with the fund manager must be taken into account.
- D. The risk management team must independently design the models.

The correct answer is **B**.

The identification of relevant stress events requires the collaboration of all the stakeholders like traders, economists, fund managers, business managers, etc. Inputs from all stakeholders make the model more robust. This is because each stakeholder brings a unique perspective and understanding of the market, which can help in creating a more comprehensive and realistic stress testing scenario. Economists can provide insights on macroeconomic factors and trends, fund managers can provide information on the fund's strategy and potential risks, business managers can provide operational insights, and traders can provide market insights. By incorporating all these diverse inputs, the stress testing model becomes more robust and capable of capturing a wider range of potential stress events. This can help the bank in better managing its risks and ensuring the stability of its capital.

Choice A is incorrect. While inputs from economists can indeed contribute to the robustness of the model, they are not the only source of valuable input. Stress testing scenarios should incorporate a wide range of perspectives to ensure comprehensive risk assessment.

Choice C is incorrect. This statement is too restrictive and overlooks the potential contributions from other stakeholders such as economists and business managers. Their insights could provide additional dimensions to stress testing scenarios, enhancing their effectiveness in capturing potential risks.

Choice D is incorrect. Although it's important for the risk management team to maintain independence in its operations, this does not mean that they should be solely responsible for designing stress testing models. Inputs from various stakeholders can provide valuable insights that enhance model design and improve risk assessment accuracy.

Things to Remember

- Stress testing is a risk management technique used to evaluate the potential impact of financial risks on a bank or financial institution.
 - Inputs from different stakeholders such as economists, traders, fund managers, and business managers are crucial for creating comprehensive stress testing scenarios.
 - Collaboration among various departments within a bank, including risk management, finance, and operations, is essential for effective risk management.
 - Documentation of stress testing procedures is important to ensure transparency, consistency, and accountability in the risk assessment process.
 - Regular review and validation of stress testing models are necessary to ensure their accuracy and relevance in capturing potential risks.
-

Q.1153 Which of the following factors is most critical when calibrating a stressed Value at Risk (VaR) model for a financial institution's risk management framework?

- A. Identifying a historical period of significant financial stress that is relevant to the institution's current portfolio.
- B. Ensuring that the institution's trading activities are compliant with internal risk limits on a day-to-day basis.
- C. Evaluating the institution's ability to maintain regulatory capital ratios during normal market conditions.
- D. Assessing the performance of individual trading strategies and optimizing the allocation of capital across various business units.

The correct answer is **A**.

The calibration of a stressed Value at Risk (VaR) model requires the identification of a historical period of significant financial stress that is relevant to the institution's current portfolio. This period should be characterized by extreme market volatility and adverse economic conditions that could potentially impact the trading portfolio. The selection of a relevant stress period is crucial as it allows the financial institution to better understand its risk exposure and the potential for extreme losses during times of market stress. This information can be used to strengthen the institution's risk management framework and improve its resilience to future financial crises. Therefore, this factor is considered the most critical when calibrating a stressed VaR model.

Choice B is incorrect. While ensuring compliance with internal risk limits on a daily basis is important for overall risk management, it does not directly contribute to the calibration of a stressed VaR model. The calibration process primarily focuses on identifying periods of significant financial stress and adjusting the model accordingly.

Choice C is incorrect. Evaluating an institution's ability to maintain regulatory capital ratios during normal market conditions is crucial for assessing its financial stability, but it does not play a direct role in calibrating a stressed VaR model. The calibration process requires focusing on periods of significant financial stress rather than normal market conditions.

Choice D is incorrect. Assessing the performance of individual trading strategies and optimizing capital allocation across business units are important aspects of portfolio management, but they do not directly influence the calibration of a stressed VaR model. This process involves identifying relevant periods of significant financial stress and adjusting the model based on these observations.

Things to Remember

- Value at Risk (VaR) is a widely used risk management measure that quantifies the potential loss in value of a portfolio over a specific time horizon under normal market conditions at a given confidence level.
- Stressed VaR is a variation of VaR that incorporates extreme market conditions and financial stress scenarios to assess the potential losses during times of crisis or turmoil.
- Calibrating a stressed VaR model involves adjusting the model parameters to reflect the heightened risk environment during periods of significant financial stress.
- Historical data analysis is crucial in the calibration process to identify relevant stress periods and understand the impact of extreme market events on the institution's portfolio.
- Scenario analysis and stress testing are important tools in assessing the robustness of a stressed VaR model and evaluating the institution's resilience to adverse market conditions.

Q.1154 A bank has recently launched a fund for retail investors. The risk management team carries out stress testing of the newly launched fund to determine the impact of the fund on the bank's overall capital. Jason Bloomberg, a newly recruited risk manager, observes that the bank has an independent risk management team. He notes that the entire risk assessment and identification process is carried exclusively through stress testing. While examining the stress

testing result, John observes that the test produces multiple potential losses under various scenarios. Bloomberg also observes that the inputs from the bank's top economists were taken while modeling the stress testing scenarios. He finds that the stress testing procedures are well documented and no deviation is allowed from the procedure.

Select the most appropriate statement.

- A. Stress testing results must be confidential.
- B. Stress testing result must be used only internally.
- C. Stress testing results may be disclosed to outsiders with sufficient supporting information.
- D. Stress testing must be disclosed only to supervisors.

The correct answer is **C**.

Stress testing results may be disclosed to outsiders with sufficient supporting information. Stress testing is a critical tool used by banks to assess their vulnerability to adverse market conditions. The results of these tests can provide valuable insights into the bank's risk profile and its ability to withstand various stress scenarios. While the results of stress tests are often used internally for risk management and strategic planning purposes, they can also be disclosed to external parties, such as investors, regulators, and the public. However, when disclosing stress test results externally, it is important to provide sufficient supporting information. This includes details about the assumptions and methodologies used in the stress test, as well as the context in which the results should be interpreted. By providing this information, banks can help external parties understand the significance of the stress test results and make informed decisions based on them.

Choice A is incorrect. There is no rule that stress testing results must be confidential. While it's true that sensitive information should be handled with care, the disclosure of stress testing results can actually enhance transparency and trust among stakeholders, provided sufficient supporting information is given.

Choice B is incorrect. Stress testing results are not meant to be used only internally. They can provide valuable insights to external stakeholders such as investors, regulators, and rating agencies about the bank's risk profile and resilience under various scenarios.

Choice D is incorrect. While supervisors are indeed one of the key recipients of stress test disclosures, they are not the only ones who may have an interest in these results. Other stakeholders like investors or rating agencies may also find this information useful for their decision-making processes.

Things to Remember

- Stress testing is a risk management technique used by banks to assess their vulnerability to adverse market conditions.

- Stress testing helps banks understand their risk profile and ability to withstand various stress scenarios.
 - External parties such as investors, regulators, and the public may also be interested in the results of stress tests.
 - When disclosing stress test results externally, it is important to provide sufficient supporting information including details about assumptions and methodologies used.
 - Transparency and trust among stakeholders can be enhanced by disclosing stress testing results with proper supporting information.
-

Q.1155 XYZ Bank has multiple branches across the country. The bank has 10 verticals for each of its products headed by 10 Vice Presidents. The VPs report directly to the Chairman of the bank. The bank also has an independent risk management team that reports directly to the Chairman. Each vertical carries out its individual stress tests and submits the reports to the vertical head which then presents them to the Chairman. The stress test procedure indicates three scenarios which must be stress tested in each of the verticals, and all the VPs ensure that the procedure is fully complied with.

The current capital position of the bank indicates no material threat to the viability of the bank. In view of this, the bank's risk management team does not include scenarios that challenge the viability of the bank in the stress tests. The risk management team also suggests independent stress testing of market assets and the funding liquidity. The bank, in its investor presentation, proudly claims to stress test each component of the balance sheet.

Each vertical of the bank carries out stress tests independently. In view of this, select the most appropriate statement.

- A. Individual stress testing is desirable.
- B. Individual stress testing overestimates the risk.
- C. Risk arising due to linkages between the verticals must also be included in the stress testing.
- D. Individual stress testing makes the stress testing process straightforward.

The correct answer is **C**.

Stress tests should cover a wide range of risks, including those at the firm-wide level. The bank should be able to integrate effectively, in a meaningful fashion, across the range of its stress

testing activities and deliver a complete picture of firm-wide risk. The bank must be able to assess the risk arising due to the linkages between the different verticals. This is because the operations of the bank are not isolated, and the activities in one vertical can have implications for the others. For instance, a significant loss in one vertical could lead to a reduction in the overall capital of the bank, which could affect the operations of the other verticals. Similarly, a risk event in one vertical could lead to a loss of confidence in the bank, leading to a withdrawal of funds that could affect the liquidity position of the entire bank. Therefore, it is important for the stress tests to take into account the linkages between the different verticals to provide a comprehensive view of the risk profile of the bank.

Choice A is incorrect. While individual stress testing can provide insights into the risks associated with each vertical, it does not account for the interdependencies and correlations between different verticals. Therefore, relying solely on individual stress testing may not be desirable as it could lead to an underestimation of the overall risk profile of the bank.

Choice B is incorrect. Individual stress testing does not necessarily overestimate risk. In fact, it might underestimate risk if there are significant linkages or correlations between different verticals that are not captured in individual stress tests.

Choice D is incorrect. Although individual stress testing might simplify the process by breaking down complex systems into manageable parts, this approach overlooks potential interactions and correlations among different verticals which could amplify or mitigate risks under certain scenarios. Therefore, stating that individual stress testing makes the process straightforward oversimplifies its complexity and potential limitations.

Things to Remember

- Stress testing is a key tool used by banks to assess their resilience to adverse scenarios and shocks.
- Firm-wide stress testing involves evaluating the impact of stress scenarios on the entire organization, considering interdependencies and correlations.
- Linkages between different business units or verticals within a bank can amplify risks and impact the overall risk profile of the institution.
- Market assets and funding liquidity are critical components to include in stress testing to ensure comprehensive risk assessment.
- Balance sheet stress testing should cover all components to provide a holistic view of the bank's risk exposure.

Q.1156 XYZ Bank has multiple branches across the country. The bank has 10 verticals for each of its products headed by 10 Vice Presidents. The VPs report directly to the Chairman of the bank.

The bank also has an independent risk management team that reports directly to the Chairman. Each vertical carries out its individual stress tests and submits the reports to the vertical head which then presents them to the Chairman. The stress test procedure indicates three scenarios that must be stress tested in each of the verticals, and all the VPs ensure that the procedure is fully complied with.

The current capital position of the bank indicates no material threat to the viability of the bank. In view of this, the bank's risk management team does not include scenarios that challenge the viability of the bank in the stress tests. The risk management team also suggests independent stress testing of market assets and the funding liquidity. The bank, in its investor presentation, proudly claims to stress test each component of the balance sheet.

Select the most appropriate statement.

- A. Three scenarios, as indicated in the stress test procedure, is sufficient to assess the risk.
- B. Stress testing must include multiple scenarios.
- C. The stress testing procedure must be flexible and must include forward-looking scenarios.
- D. A minimum of 10 scenarios must be used to perform stress testing.

The correct answer is **C**.

An effective stress testing procedure should not only cover a spectrum of events and severity levels, but it should also be flexible and forward-looking. The stress testing procedure at XYZ Bank, as described, includes three scenarios for each vertical. However, this may not be sufficient to identify hidden vulnerabilities or anticipate future risks. The bank's current capital position does not pose a significant threat to its viability, which is why the risk management team does not include scenarios that challenge the bank's viability in the stress tests. However, this approach may overlook potential future risks that could threaten the bank's viability. Therefore, the stress testing procedure should be flexible enough to adapt to changing circumstances and include forward-looking scenarios that anticipate potential future risks. This approach would enhance the effectiveness of the stress testing procedure and provide a more comprehensive assessment of the bank's risk profile.

Choice A is incorrect. While the bank's stress test procedure does indicate three scenarios, it is not necessarily sufficient to assess all risks. Stress testing should be comprehensive and consider a variety of scenarios that could potentially impact the bank's financial health.

Choice B is incorrect. Although stress testing must indeed include multiple scenarios, this statement alone does not fully capture the requirements for effective stress testing. It lacks specificity about what kind of scenarios should be included and how they should be selected.

Choice D is incorrect. There isn't a fixed number of scenarios that must be used for stress testing. The number can vary depending on the complexity and risk profile of the institution being tested. Therefore, stating that a minimum of 10 scenarios must be used for stress testing may not always hold true.

Things to Remember

- Stress testing is a key tool used by financial institutions to assess their resilience to adverse events and shocks.
 - Effective stress testing should cover a spectrum of events and severity levels to provide a comprehensive assessment of risks.
 - Stress testing procedures should be flexible and forward-looking to anticipate potential future risks.
 - Scenarios used in stress testing should not only focus on current risks but also consider emerging risks and vulnerabilities.
 - Market assets and funding liquidity are important components to include in stress testing procedures.
-

Q.1157 XYZ Bank has multiple branches across the country. The bank has 10 verticals for each of its products headed by 10 Vice Presidents. The VPs report directly to the Chairman of the bank. The bank also has an independent risk management team that reports directly to the Chairman. Each vertical carries out its individual stress tests and submits the reports to the vertical head which then presents them to the Chairman. The stress test procedure indicates three scenarios that must be stress-tested in each of the verticals, and all the VPs ensure that the procedure is fully complied with.

The current capital position of the bank indicates no material threat to the viability of the bank. In view of this, the bank's risk management team does not include scenarios that challenge the viability of the bank in the stress tests. The risk management team also suggests independent stress testing of market assets and the funding liquidity. The bank, in its investor presentation, proudly claims to stress test each component of the balance sheet.

Select the most appropriate statement.

- A. In the case of bank XYZ, the risk management team is correct not to include scenarios which challenge the viability of the bank.
- B. In the case of bank XYZ, the risk management team must include scenarios with increased severity but must not challenge the viability of the bank.
- C. In the case of bank XYZ, the stress test must not include scenarios which challenge the viability of the bank.
- D. In the case of bank XYZ, the stress test must include scenarios which challenge the

viability of the bank.

The correct answer is **D**.

Stress tests should include scenarios that challenge the viability of the bank. This is because such stress tests can reveal hidden risks and interactions among risks. The global financial crisis has demonstrated the importance of such stress testing, as it allows for the development of a contingency plan in case such scenarios become reality. In the case of XYZ Bank, even though the current capital position does not indicate any significant threat to the bank's viability, it is still crucial to include such scenarios in the stress tests. This will ensure that the bank is prepared for any potential risks and can take appropriate action to mitigate these risks. Therefore, the statement that 'In the case of bank XYZ, the stress test must include scenarios which challenge the viability of the bank' is the most accurate.

Choice A is incorrect. The risk management team should include scenarios that challenge the viability of the bank in their stress tests. This is because stress tests are designed to assess how a bank would perform under severe economic conditions, and therefore, should include worst-case scenarios.

Choice B is incorrect. While it's true that the risk management team must include scenarios with increased severity, these scenarios should also challenge the viability of the bank. By excluding such scenarios, they may not be fully assessing all potential risks.

Choice C is incorrect. As explained above, stress tests need to include worst-case scenarios that challenge a bank's viability to ensure they are prepared for all possible outcomes.

Things to Remember

- Stress testing is a key tool used by banks to assess their resilience to adverse economic conditions and potential risks.
- Scenarios in stress tests should include a wide range of possibilities, including those that challenge the viability of the bank.
- Stress testing helps banks identify hidden risks and interactions among risks, allowing them to develop contingency plans.
- Market assets and funding liquidity are important components that should be independently stress tested by banks.
- Each component of a bank's balance sheet should be stress tested to ensure comprehensive risk assessment.

Q.1167 A fund manager examines the annual return generated by fund A for the last 10 years. The return generated by the fund is furnished in the table below:

Year	Return
2015	+9.45%
2014	−4.45%
2013	+5.34%
2012	−3.35%
2011	+2.45%
2010	−6.56%
2009	+7.41%
2008	−8.83%
2007	+2.33%
2006	+1.32%

The fund manager intends to stress test Fund A for various scenarios. Select the correct option with regards to the stress testing of Fund A.

- A. As the maximum loss during the last 10 year is 8.83%, the stress test scenario for maximum loss must not exceed 8.83%.
- B. As the average loss during the last 10 years is 5.80%, the stress test scenario for maximum loss must be equal to 5.80%.
- C. The stress test scenario for maximum loss must be more than the historical maximum loss posted by the fund.
- D. As the average return generated by the fund for the last 10 years is 0.51%, the stress test scenario for maximum loss must be -0.51%.

The correct answer is **C**.

Stress testing is a simulation technique used in the financial industry to evaluate the potential effects of severe but plausible events on portfolio values. It is a risk management tool that helps in understanding the risk exposure under extreme market conditions. The purpose of stress testing is not to predict outcomes, but to understand potential vulnerabilities in a portfolio. In this context, the stress test scenario for maximum loss should be more than the historical maximum loss posted by the fund. This is because stress testing is designed to assess the resilience of a fund to extreme market conditions that may not have been experienced in the past. Therefore, it is not sufficient to simply consider the maximum historical loss. The stress test should consider scenarios that are more severe than those experienced historically to ensure that the fund can withstand extreme market downturns. This is why choice C is correct.

Choice A is incorrect. While it's true that the maximum loss during the last 10 years was 8.83%, stress testing is not limited to historical losses. Stress tests are designed to evaluate how a fund would perform under extreme conditions, which may be worse than any historical scenario.

Choice B is incorrect. The average loss of 5.80% over the past ten years does not necessarily represent a worst-case scenario for stress testing purposes. Stress tests should consider potential losses that are significantly greater than average to ensure robustness under severe market conditions.

Choice D is incorrect. The average return of the fund over the last ten years has no direct relevance to determining a stress test scenario for maximum loss. Stress tests focus on potential future losses, not past returns.

Q.1170 A fund manager intends to carry out scenario analysis of his portfolio. The portfolio consists of 25% government bonds, 60% global equities, and 15% investment in gold ETFs. The fund manager lists the portfolio's risk factors. He intends to perform a scenario analysis by generating scenarios based on the adverse movement in the portfolio's identified risk factors. Such an approach to scenario analysis is referred to as:

- A. Event-driven scenario analysis.
- B. Portfolio-driven scenario analysis.
- C. Factor push method.
- D. Historical method.

The correct answer is **B**.

Scenario analysis is a process of evaluating possible future events by considering alternative possible outcomes. It is used to assess the impact of different factors on the value of an investment portfolio. There are two main types of scenario analysis: event-driven and portfolio-driven. In event-driven scenario analysis, scenarios are created based on plausible events that could cause changes in the risk factors. On the other hand, portfolio-driven scenario analysis involves identifying the risk factors of a portfolio and then creating scenarios based on adverse movements in these risk factors. In the given scenario, the fund manager first identifies the risk factors of his portfolio and then generates scenarios based on the adverse movement of these risk factors. This approach is characteristic of portfolio-driven scenario analysis. Therefore, the correct answer is Choice B: Portfolio-driven scenario analysis.

Choice A is incorrect. Event-driven scenario analysis is a type of scenario analysis that focuses on specific events, such as a financial crisis or natural disaster, and their potential impact on the portfolio. This does not match the approach described in the question where scenarios are generated based on adverse movement of identified risk factors.

Choice C is incorrect. The factor push method involves changing one factor at a time while keeping all other factors constant to see how it affects the portfolio. This differs from the approach described in the question where multiple risk factors are considered simultaneously.

Choice D is incorrect. The historical method involves creating scenarios based on historical data and trends, which isn't mentioned in this case. In contrast, this manager's approach

involves generating scenarios based on adverse movements of identified risk factors.

Things to Remember

- Scenario analysis helps in assessing the impact of different factors on the value of an investment portfolio.
 - Event-driven scenario analysis focuses on specific events and their potential impact on the portfolio.
 - The factor push method involves changing one factor at a time to see its effect on the portfolio.
 - The historical method involves creating scenarios based on historical data and trends.
 - Portfolio-driven scenario analysis involves identifying the risk factors of a portfolio and creating scenarios based on adverse movements in these risk factors.
-

Q.1172 The general belief that diversification leads to risk reduction was challenged during the Global Financial Crisis of 2007-2009. During the crisis, it was observed that the correlation between different assets increased due to which the concept of diversification failed.

Imagine yourself being a risk manager. In order to assess the likely impact of such events, the most appropriate tool is the:

- A. Scenario analysis.
- B. Sensitivity analysis.
- C. Historical simulation.
- D. Factor push analysis.

The correct answer is **A**.

Scenario analysis is a process of examining and evaluating possible events or scenarios that could take place in the future and predicting the various feasible outcomes. In the context of the Global Financial Crisis of 2007-2009, scenario analysis would be an appropriate tool for a risk manager. This is because the crisis can be considered an extreme event, and scenario analysis is particularly useful for examining such events. Different scenarios can be modeled based on the conditions during the crisis, and an analysis of the likely impact of these scenarios can be carried out. This would provide a comprehensive understanding of the potential risks and impacts associated with similar events in the future, thereby enabling effective risk management.

strategies.

Choice B is incorrect. Sensitivity analysis is a tool used to understand how different values of an independent variable impact a particular dependent variable under a given set of assumptions. However, it does not take into account the correlation between different assets or the potential impact of extreme events like the Global Financial Crisis.

Choice C is incorrect. Historical simulation involves re-running risk models based on historical data to assess potential future outcomes. While this can be useful for understanding how past events might influence future risks, it may not accurately capture the dynamics of extreme events like the Global Financial Crisis where correlations between assets increased significantly.

Choice D is incorrect. Factor push analysis involves changing one factor at a time while keeping all other factors constant to see how this impacts the output of a model. This method would not be suitable for assessing an event like the Global Financial Crisis where multiple factors and their interrelationships played significant roles.

Things to Remember

- Correlation between assets: Correlation measures the degree to which the prices of different assets move in relation to each other. High correlation between assets can reduce the benefits of diversification.
 - Systemic risk: Systemic risk refers to the risk of a widespread financial market collapse or crisis, where the failure of one institution or asset can trigger a chain reaction of failures throughout the financial system.
 - Extreme events: Extreme events, such as the Global Financial Crisis, are events that deviate significantly from the norm and can have a major impact on financial markets and portfolios.
 - Risk management strategies: Risk management strategies involve identifying, assessing, and prioritizing risks followed by coordinated and economical application of resources to minimize, monitor, and control the probability and/or impact of unfortunate events.
-

Q.2813 Which of the following is an advantage of reverse stress testing?

- A. By concentrating on different scenarios, reverse stress testing helps to identify the most profitable business lines.
- B. Reverse stress testing does not consider scenarios beyond its normal business

expectations and concentrates on issues that could affect business during the normal business operations.

C. Reverse stress testing does not consider scenarios beyond its normal business expectations and consequently does not require a comprehensive analysis.

D. By evaluating scenarios and circumstances that would render a business unviable, reverse stress testing identifies potential business vulnerabilities.

The correct answer is **D**.

Reverse stress testing is a risk management tool that starts from a hypothetical outcome of business failure and works backwards to identify the scenarios and circumstances that could lead to this outcome. This approach is particularly useful in identifying potential vulnerabilities in a business. By considering extreme scenarios that could render a business unviable, reverse stress testing allows businesses to proactively identify and address potential weaknesses, thereby enhancing their resilience and ability to withstand adverse conditions. This is the primary advantage of reverse stress testing.

Choice A is incorrect. Reverse stress testing does not focus on identifying the most profitable business lines. Instead, it focuses on identifying scenarios and circumstances that could potentially lead to the failure of a business.

Choice B is incorrect. This statement contradicts the purpose of reverse stress testing. Reverse stress testing specifically considers scenarios beyond normal business expectations to identify potential vulnerabilities that could lead to business failure.

Choice C is incorrect. Again, this statement contradicts the nature of reverse stress testing. It does consider scenarios beyond its normal business expectations and requires a comprehensive analysis to identify potential risks and vulnerabilities.

Things to Remember

- Reverse stress testing is a risk management tool that starts from a hypothetical outcome of business failure and works backwards to identify potential vulnerabilities.
- It helps in identifying extreme scenarios that could render a business unviable.
- Reverse stress testing allows businesses to proactively address potential weaknesses and enhance their resilience.
- It is important to consider a wide range of scenarios and circumstances during reverse stress testing to ensure comprehensive risk identification.
- Reverse stress testing is particularly useful in assessing the robustness of a business under severe adverse conditions.

Q.2814 Stress testing is an essential practice for assessing an organization's resilience against potential adverse economic scenarios. During which phase of the economic cycle is stress testing most important?

- A. During the beginning of an economic recession.
- B. In the middle of an economic recession.
- C. In the beginning of an economic expansion.
- D. After a long period of economic expansion.

The correct answer is **D**.

Stress testing is particularly important after a long period of economic expansion. During such periods, the economy is generally performing well, with high employment rates, increased consumer spending, and robust business growth. However, these conditions can also lead to complacency and the underpricing of risk. Financial institutions may become overly optimistic, underestimating the potential for economic downturns and the associated risks. Stress testing during these times helps institutions prepare for potential economic downturns by assessing their resilience to adverse scenarios. It allows them to identify vulnerabilities in their risk management strategies and make necessary adjustments to ensure their financial stability in the event of an economic downturn.

Choice A is incorrect. While stress testing is important during the beginning of an economic recession, it is not the most crucial stage. This phase typically involves a downturn in economic activity, and while stress tests can help identify vulnerabilities, they are more critical after a long period of economic expansion when risks may have built up unnoticed.

Choice B is incorrect. In the middle of an economic recession, financial institutions are already dealing with the impacts of the crisis. Stress testing at this stage can still be useful for managing ongoing risks but it's not as crucial as after a long period of expansion when potential vulnerabilities may have accumulated without detection.

Choice C is incorrect. At the beginning of an economic expansion, institutions are generally recovering from a previous downturn and hence less likely to face severe stress scenarios. Therefore, while stress testing remains important for risk management purposes, it becomes most crucial after prolonged periods of growth where complacency about risk can set in.

Things to Remember

- Stress testing is a risk management technique used to evaluate an organization's ability to withstand adverse economic conditions.
- It involves subjecting the organization to hypothetical scenarios that could potentially

impact its financial stability.

- Stress testing helps identify vulnerabilities in risk management strategies and allows for necessary adjustments to be made to improve resilience.
 - It is important for financial institutions to conduct stress testing regularly to ensure preparedness for unexpected economic downturns.
 - Stress testing is a key component of regulatory requirements for financial institutions to assess their capital adequacy and risk management practices.
-

Q.3445 In the context of regulatory compliance and effective risk management, stress testing programs are integral to ensuring the stability and resilience of banks. Which of the following parties bears the ultimate responsibility for stress testing programs in banks?

- A. The risk management function
- B. Senior management
- C. The board of directors
- D. Internal audit

The correct answer is **C**.

The board of directors is ultimately responsible for a bank's stress tests. This responsibility stems from the board's overarching role in setting the strategic direction of the bank and overseeing its operations. While the board members may not delve into the technical details of stress tests, they are expected to ensure that they remain sufficiently knowledgeable about the procedures involved in stress testing and the interpretation of results. Their engagement in this process is crucial for the effective operation of stress testing. They need to ensure that the stress testing program is robust, that it adequately captures the bank's risk profile, and that the results are used appropriately in decision-making. This includes understanding the assumptions and limitations of the stress tests, challenging the results where necessary, and ensuring that appropriate actions are taken in response to the findings of the stress tests.

Choice A is incorrect. While the risk management function plays a significant role in conducting stress tests and analyzing their results, they do not bear the ultimate responsibility for the implementation and oversight of these programs. Their role is more operational, focusing on executing the tests and interpreting the results.

Choice B is incorrect. Senior management also has an important role in stress testing, as they are responsible for ensuring that appropriate resources are allocated to this process and that it

aligns with the bank's overall strategic objectives. However, while they have a key role in implementing stress testing programs, ultimate oversight responsibility lies elsewhere.

Choice D is incorrect. The internal audit function provides independent assurance on the effectiveness of risk management processes including stress testing but does not have ultimate responsibility for its implementation or oversight. They review and assess whether all procedures are followed correctly but do not set up or oversee these procedures themselves.

Things to Remember

- Stress testing is a risk management tool used to evaluate the potential impact of adverse events or scenarios on a bank's financial condition.
 - Stress testing helps banks identify vulnerabilities, assess capital adequacy, and enhance their risk management practices.
 - The results of stress tests are used to inform decision-making processes, such as capital planning, risk mitigation strategies, and contingency planning.
 - Regulatory authorities often require banks to conduct stress tests to ensure they have sufficient capital to withstand severe economic conditions.
 - Stress testing programs typically involve scenario analysis, sensitivity analysis, and reverse stress testing to assess different aspects of a bank's risk exposure.
-

Q.3446 Which of the following options *most accurately* presents a key governance issue that played a critical role in the failure of banks in the lead up to the 2007/2009 financial crisis?

- A. Senior management played little or no role at all in the development and operation of stress testing.
- B. Stress testing reports would be passed up to the boards of directors without first being approved by senior management
- C. Stress testing did not appear to be sufficiently integrated into institutions' risk management frameworks, nor were test results taken into account during decision making
- D. Stress testing programs lacked clear, well-detailed policies meant to outline the procedure to follow from the start to the end, as well as describing the role played by various employees

The correct answer is **C**.

The financial crisis of 2007/2009 revealed that stress testing was not adequately integrated into the risk management frameworks of many banking institutions. This lack of integration meant that the results of stress tests were often not taken into account during decision-making processes. This was a significant governance issue as it meant that banks were not adequately prepared for the potential risks they faced. The few institutions that did show a commitment to stress testing often did not examine sufficiently severe scenarios, further exacerbating the issue. This lack of integration and consideration of stress test results in decision making was a key factor that contributed to the failure of many banks during the financial crisis.

Choice A is incorrect. While it's true that senior management involvement in stress testing was often lacking, this does not encapsulate the key governance issue. The main problem was not just the lack of involvement from senior management, but rather the insufficient integration of stress testing into overall risk management frameworks and decision-making processes.

Choice B is incorrect. Although there may have been instances where stress testing reports were passed to boards without approval from senior management, this was not a prevalent or significant contributing factor to bank failures during the crisis. The more critical issue was that these tests and their results were not adequately incorporated into risk management strategies or decision making.

Choice D is incorrect. While it's accurate that many institutions lacked clear policies for conducting stress tests, this alone does not capture the primary governance issue at hand during the financial crisis. The central problem lay in how these tests were used (or rather, ignored) within broader risk management frameworks and strategic decisions.

Things to Remember

- Stress testing is a key tool used by financial institutions to assess their resilience to adverse market conditions and potential risks.
- Senior management involvement in risk management processes is crucial for effective governance and decision-making.
- The 2007/2009 financial crisis highlighted the importance of robust risk management frameworks and the integration of stress testing results into decision-making processes.
- Clear policies and procedures for stress testing are essential for ensuring consistency and effectiveness in risk management practices.
- Regulatory oversight and compliance requirements regarding stress testing have increased significantly post the financial crisis to prevent similar failures in the future.

Q.4560 A bank carries out regular stress testing to determine its appropriate capital level. During a given year, the bank generates a scenario to assume that the GDP growth rate might decline by 3%. Under the impending circumstances, which type of scenario will the bank *most likely* internally generate?

- A. Ad hoc scenarios.
- B. Historical scenarios.
- C. Stressing key variables scenarios.
- D. Regulatory scenarios.

The correct answer is **C**.

In the context of stress testing, a scenario could be constructed by assuming that a significant change occurs in one or more key variables. In this case, the key variable is the GDP growth rate, which is assumed to decline by 3%. This type of scenario is known as 'stressing key variables scenario'. It is a common practice in stress testing to stress key variables and observe the potential impact on the bank's capital level. This helps the bank to prepare for adverse economic conditions and ensure that it has sufficient capital to withstand potential losses.

Choice A is incorrect. Ad hoc scenarios are typically generated in response to specific events or concerns, and not necessarily linked to key economic variables such as GDP growth rate. They are more event-specific and less systematic than what the question describes.

Choice B is incorrect. Historical scenarios are based on actual past events or periods of significant financial stress. While they may involve changes in GDP, they do not involve hypothetical changes in key variables like the scenario described in the question.

Choice D is incorrect. Regulatory scenarios are typically prescribed by regulatory authorities and are used to ensure that banks meet regulatory requirements.

Things to Remember

- Stress testing is a risk management technique used by banks to assess their ability to withstand adverse economic conditions.
- Key variables in stress testing scenarios can include interest rates, exchange rates, credit spreads, and other macroeconomic indicators.
- Ad hoc scenarios are typically used for specific events or concerns that may not be related to key economic variables.
- Historical scenarios are based on past events or periods of financial stress, but may not involve hypothetical changes in key variables.

- Regulatory scenarios are prescribed by regulatory authorities to ensure banks meet regulatory requirements.
-

Q.4564 While analyzing the stress testing results, analysts should consider the impacts of the stress testing scenarios and also knock-on effects. What is a knock-on effect?

- A. It is an effect due to the way a financial institution responds to an adverse condition.
- B. It is an effect on the way financial institutions implement stress testing.
- C. It is a negative influence from the staff conducting the stress test.
- D. It is an effect that arises due to the involvement of the Board and senior management in the stress testing process.

The correct answer is **A**.

A knock-on effect, in the context of stress testing in financial risk management, refers to the subsequent consequences that arise due to the way a financial institution responds to an adverse condition. This response could potentially exacerbate the adverse scenario, leading to further complications. For instance, if a bank responds to a stress scenario by selling off assets to maintain liquidity, this could potentially lead to a market downturn, causing a knock-on effect. Therefore, understanding and considering these knock-on effects is crucial for analysts when interpreting the results of stress tests.

Choice B is incorrect. The 'knock-on effect' does not pertain to the implementation process of stress testing by financial institutions. Rather, it refers to the subsequent effects that occur due to how a financial institution responds to adverse conditions.

Choice C is incorrect. The 'knock-on effect' has nothing to do with any negative influence from the staff conducting the stress test. It is about the cascading effects resulting from an institution's response to adverse situations, not about personnel conduct or performance during testing.

Choice D is incorrect. The involvement of Board and senior management in stress testing process does not define a 'knock-on effect'. While their involvement may influence decisions and responses, it doesn't directly relate to or cause a knock-on effect which specifically refers to subsequent impacts following an initial response.

Things to Remember

- Stress testing is a risk management technique used to evaluate the potential impact of adverse events on a financial institution's financial condition.
- There are different types of stress tests, including sensitivity analysis, scenario

analysis, and reverse stress testing.

- Knock-on effects can also be referred to as ripple effects or domino effects, highlighting the interconnectedness of risks in the financial system.
 - Analysts should also consider the interconnectedness of risks and the potential for systemic risk when analyzing stress testing results.
 - It is important for financial institutions to have robust risk management frameworks in place to address and mitigate the impacts of stress testing scenarios and knock-on effects.
-

Q.4566 In the context of regulatory stress testing in the United States, which of the following is **true**?

- A. Comprehensive Capital Analysis and Review (CCAR) is a stress test performed by the Federal Reserve on banks with consolidated assets of at least \$100 billion.
- B. Under the CCAR, banks are required to consider three scenarios: baseline, severe, and an internal scenario.
- C. Under the Dodd-Frank Act Stress Test (DFAST), banks are required to submit a capital plan.
- D. The Dodd-Frank Act Stress Test (DFAST) applies to banks with consolidated assets between USD 10 billion and USD 50 billion.

The correct answer is **D**.

The Dodd-Frank Act Stress Test (DFAST) indeed applies to banks with consolidated assets between USD 10 billion and USD 50 billion. This regulatory requirement is part of the Dodd-Frank Wall Street Reform and Consumer Protection Act, which was enacted in response to the financial crisis of 2008. The DFAST is designed to ensure that these banks have sufficient capital to absorb losses and support operations during adverse economic conditions. It requires banks to conduct stress tests to assess their potential losses from hypothetical adverse scenarios. The results of these tests are then used to inform the banks' capital planning decisions.

Choice A is incorrect. The Comprehensive Capital Analysis and Review (CCAR) is not performed on banks with consolidated assets of over USD 100 billion. Instead, it applies to bank holding companies with \$50 billion or more in total consolidated assets.

Choice B is incorrect. Under the CCAR, banks are required to consider three scenarios: baseline, adverse, and severely adverse. There's no requirement for an internal scenario as

stated in the option.

Choice C is incorrect. Under the Dodd-Frank Act Stress Test (DFAST), banks are not required to submit a capital plan. It's under CCAR that banks are required to submit a capital plan which includes their planned capital actions over a nine-quarter planning horizon.

Things to Remember

- Regulatory stress testing is a key tool used by regulators to assess the resilience of financial institutions to adverse economic conditions.
 - Stress tests help regulators ensure that banks have enough capital to withstand potential losses and continue operating in times of financial stress.
 - The scenarios used in stress tests typically include baseline, adverse, and severely adverse scenarios to cover a range of potential economic conditions.
 - Capital planning is an important component of regulatory stress testing, where banks outline their planned capital actions based on the stress test results.
 - The Dodd-Frank Act and the Comprehensive Capital Analysis and Review (CCAR) are key regulatory frameworks in the United States that govern stress testing for banks.
-

Reading 55: Pricing Conventions, Discounting, and Arbitrage

Q.1000 A 50-day US T-bill has a quoted price of 1.60. What is the cash price of the bill?

- A. 94.75
- B. 97.43
- C. 100.23
- D. 99.78

The correct answer is **D**.

Recall that the cash price of a US T-bill is given by:

$$C = 100 - Q \frac{n}{360}$$

where

C = Cash Price

Q = Quoted Price of the T-bill.

n = number of calendar days until the maturity of the Treasury bill

So in this case we have Q = 1.60 and n = 50 so that:

$$\begin{aligned} C &= 100 - 1.60 \times \frac{50}{360} \\ &= 99.77777 \approx 99.78 \end{aligned}$$

Q.1003 Ronam Ltd. invests in semi-annual US Treasury bonds with face values of USD 1,000 on 15 June 2017. A bond made a coupon payment of USD 40 on February 15, 2017. The next coupon is due on August 15, 2017. If the quoted price for the bond for delivery on June 15, 2017, is USD 1001-16, what is the bond's full price?

- A. USD 1,026.52
- B. USD 1,013.48
- C. USD 1,028.02
- D. USD 1,014.98

The correct answer is C.

We have 120 (13, 31, 30, 31, and 15 days in Feb, March, April, May, and June respectively) between the 1st coupon date (Feb 15, 2017) and the settlement date (June 15, 2017).

We have 61 days from the settlement date to the next coupon payment date (Aug 15, 2017).

Accrued interest using actual/actual day-count- convention:

Accrued interest = $\text{USD } 40 * (120/181) \text{ days} = \text{USD } 26.5193$

Full price = Quoted price + Accrued Interest = $1,001.50 + 26.5193 = \text{USD } 1028.0193$

Note: $1001-16 = 1,001 + 16/32 = 1,001.5$

Things to Remember

- US Treasury bonds are considered one of the safest investments as they are backed by the full faith and credit of the US government.
- Coupon payments on bonds are typically made semi-annually, although some bonds may have different payment frequencies.
- Accrued interest is the interest that has accumulated on a bond since the last coupon payment. It is typically calculated using the actual/actual day-count convention.
- The quoted price of a bond is typically expressed as a percentage of the face value, with fractions of 1/32 being common in the bond market.
- Settlement date is the date on which the buyer must pay for the bond and take ownership, while the next coupon payment date is when the next interest payment is due to the bondholder.

Q.1004 Suppose that a U.S. Corporate bond has a face value of \$1,000 and is redeemed at par. The annual coupon rate is 10%, payable semiannually. Assume the last coupon was paid 100 days ago and there are 30 days in each month. The accrued interest is closest to:

- A. \$55
- B. \$28
- C. \$25
- D. \$30

The correct answer is **B**.

$$AI = \text{Coupon} \times \frac{\text{Number of days from last coupon to the settlement date}}{\text{Number of days in coupon period}}$$

Thus,

$$AI = \$50 \times \frac{100}{180} = \$27.7778 \approx 28$$

Q.2774 A fixed-income trader summarizes in the table below the prices of Treasury Bonds with semiannual coupon payment. The data is as of 01/01/17.

	Maturity	Coupon~Rate	Price (per \$100 face value)
Tranche 1	30/06/2017	3.5%	99 – 00
Tranche 2	31/12/2017	4%	100 – 16
Tranche 3	30/06/2018	5%	101 – 04

What are the discount factors for 0.5, 1 and 1.5 years?

A. $d(0.5) = 0.9730$; $d(1) = 0.9662$; $d(1.5) = 0.9393$

B. $d(0.5) = 0.9551$; $d(1) = 0.9422$; $d(1.5) = 0.9102$

C. $d(0.5) = 0.9633$; $d(1) = 0.9523$; $d(1.5) = 0.9085$

D. $d(0.5) = 0.98990$; $d(1) = 0.9782$; $d(1.5) = 0.8787$

The correct answer is **A**.

The cash flows are as follow:

	T(0.5)	T(1.0)	T(1.5)
Tranche 1	101.75	–	–
Tranche 2	2	102	–
Tranche 3	2.5	2.5	102.5

$$99.00 = 101.75 * d(0.5)$$

$$100.50 = 2 * d(0.5) + 102 * d(1)$$

$$101.125 = 2.5 * d(0.5) + 2.5 * d(1) + 102.5 * d(1.5)$$

$$d(0.5) = 0.9730$$

$$d(1) = 0.9662$$

$$d(1.5) = 0.9393$$

Q.2775 Consider a 2-year Treasury Bond that is currently trading on the market at a price of 97.75. The bond has a coupon rate of 5%, which is paid out semiannually.

d(0.5)	0.9777
d(1)	0.9471
d(1.5)	?
d(2)	0.8845

Given the discount factor structure shown in the table above, d(1.5) is closest to:

- A. 0.9385
- B. 0.9228
- C. 0.9205
- D. 0.9107

The correct answer is **D**.

The cash flows are as follow:

T(0.5)	T(1.0)	T(1.5)	T(2.0)
2.5	2.5	2.5	102.5

$$P_0 = \frac{c_1}{1 + \frac{y}{2}} + \frac{c_2}{1 + \frac{y}{2}} + \frac{F + c_n}{1 + \frac{y}{2}}$$

Where

P_0 = Price of the bond

$c_1, c_2, \dots, c_n$ are semi-annual coupon payments

F is the face value

y is the yield on the bond.

Note this question has given us discount factors:

$$\frac{1}{1 + \frac{y_1}{2}}, \frac{1}{1 + \frac{y_2}{2}} \text{ and } \frac{1}{1 + \frac{y_4}{2}}$$

We need to find the discount factor in the third coupon payment. Also note that the Treasury Bond is quoted per 100 face value. Thus

$$\begin{aligned} 97.75 &= 2.5 \times d(0.5) + 2.5 \times d(1) + 2.5 \times d(1.5) + 102.5 \times d(2) \\ &= 2.5 \times 0.9777 + 2.5 \times 0.9471 + 2.5 \times d(1.5) + 102.5 \times 0.8845 \\ \Rightarrow d(1.5) &= 0.9107 \end{aligned}$$

Q.2776 A risk manager is concerned with the pricing of one of the bank's treasury bonds that pays 3% semiannual interest and matures in two years. The discount factors for different maturities are as per the following table:

	Discount factor
d(0.5)	0.9950
d(1.0)	0.9727
d(1.5)	0.9327
d(2)	0.9045

What is the price of the bond?

- A. 93.577
- B. 94.187
- C. 95.847
- D. 96.157

The correct answer is **D**.

The price of the bond is given by:

$$1.5 * d(0.5) + 1.5 * d(1) + 1.5 * d(1.5) + 101.5 * d(2) = 96.157$$

Q.2777 An institutional investor is analyzing U.S. Treasury securities. They are considering investing in either a 10-year Treasury note or creating a portfolio of Treasury STRIPS with a similar maturity. Which of the following statements correctly describes a key difference between investing in the 10-year Treasury note and a portfolio of Treasury STRIPS?

- A. A portfolio of Treasury STRIPS eliminates reinvestment risk associated with coupon payments, whereas the 10-year Treasury note exposes the investor to such risk.
- B. A portfolio of Treasury STRIPS provides a more stable yield compared to a 10-year Treasury note.
- C. Treasury STRIPS are priced at a premium to their face value due to their zero-coupon nature, whereas 10-year Treasury notes are typically priced closer to par.

D. A portfolio of Treasury STRIPS provides lower total returns compared to a 10-year Treasury note

The correct answer is **A**.

Treasury STRIPS (Separate Trading of Registered Interest and Principal Securities) are created by separating the coupon payments and principal repayment of a Treasury bond into individual zero-coupon securities. By investing in STRIPS, an investor receives a single payment at maturity, eliminating the need to reinvest coupon payments and thus avoiding reinvestment risk. In contrast, a traditional Treasury note pays coupons semi-annually, requiring the investor to reinvest these payments at potentially fluctuating interest rates, thereby exposing them to reinvestment risk.

B is incorrect. While STRIPS eliminate reinvestment risk, they are more price-sensitive to interest rate changes due to their higher duration. Their yield is not inherently more stable because their value fluctuates significantly with interest rates compared to coupon-bearing notes.

C is incorrect. Treasury STRIPS are priced at a discount to their face value because they do not pay periodic coupons, and their value accrues solely through appreciation to par at maturity. Suggesting they are priced at a premium is factually incorrect.

D is incorrect. The absence of periodic coupon payments does not inherently result in lower total returns. STRIPS and Treasury notes with the same maturity offer similar yields to maturity, reflecting the time value of money. The difference lies in cash flow timing and reinvestment risk, not total return potential.

Things to Remember

- Treasury STRIPS are zero-coupon securities created by separating the coupon and principal payments of U.S. Treasury securities.
- There are two types of STRIPS: P-STRIPS (Principal STRIPS) represent the principal repayment at maturity, while C-STRIPS (Coupon STRIPS) represent individual coupon payments.
- Because STRIPS are zero-coupon bonds, their yield to maturity is the discount rate that equates the present value of the single future payment (either principal or coupon) to the current price.
- A key advantage of STRIPS is the elimination of reinvestment risk, as there are no interim coupon payments to reinvest.
- A common misunderstanding is that STRIPS are less sensitive to interest rate changes.

The opposite is true: STRIPS are more sensitive to interest rate fluctuations than coupon-bearing bonds of the same maturity. This higher sensitivity is due to the fact that the entire return is received at maturity.

- A practical example: An investor saving for a specific future liability, like a child's college education in 10 years, might use P-STRIPS maturing in 10 years to lock in a known future value and avoid reinvestment risk.

Q.2778 In fixed-income markets, certain instruments often trade at a premium due to their unique features or favorable market conditions. Which of the below presented fixed-income instruments typically trade rich?

- A. Long-term P-STRIPS.
- B. AAA-rated Corporate bonds.
- C. Short-term C-STRIPS.
- D. Mortgage-backed securities.

The correct answer is **C**.

Separate Trading of Registered Interest and Principal of Securities (STRIPS) are debt securities issued by the U.S. Treasury. They are sold at a significant discount to face value and offer no interest payments because they mature at par. STRIPS can be either principal-only (P-STRIPS) or interest-only (C-STRIPS), and their market prices are influenced by factors such as interest rates, demand, and the time to maturity.

Short-term C-STRIPS often trade at higher prices ('rich') compared to similar issues. This is partly due to the high demand for short-term securities, which provides a financing advantage. Specifically, these securities can be used as collateral to borrow money at rates lower than general collateral (GC) rates. This financing advantage makes short-term C-STRIPS attractive to investors, leading to higher demand and consequently, higher prices.

Choice A is incorrect. Long-term P-STRIPS (Principal STRIPS) are not typically traded at a premium. They are zero-coupon bonds that pay no interest until maturity, which makes them less attractive to investors seeking regular income, and hence they do not usually trade rich.

Choice B is incorrect. AAA-rated Corporate bonds may trade at a premium due to their high credit rating and the perceived low risk associated with them. However, they do not typically trade rich as compared to other fixed income securities because the yield on these bonds is generally lower due to their high credit quality.

Choice D is incorrect. Mortgage-backed securities (MBS) also do not typically trade rich

because of their inherent prepayment risk and the complexity involved in pricing these instruments. The uncertainty regarding cash flows from MBS can make them less attractive to certain investors, thereby reducing demand and preventing them from trading at a premium.

Things to Remember

- Fixed-income instruments can trade at a premium due to unique features or favorable market conditions.
 - Market prices of fixed-income instruments are influenced by factors such as interest rates, demand, and time to maturity.
 - Short-term C-STRIPS often trade at higher prices ('rich') compared to similar issues due to high demand and financing advantages.
 - Long-term P-STRIPS are zero-coupon bonds that do not typically trade at a premium.
 - AAA-rated Corporate bonds may trade at a premium due to high credit rating but do not typically trade rich compared to other fixed income securities.
 - Mortgage-backed securities (MBS) do not typically trade rich due to prepayment risk and pricing complexity.
-

Q.2779 A trader prepares a presentation to the investment committee of a bank with a suggestion to invest \$10,000,000 in U.S. Treasury 2 and 3/8s of December 31, 2019, tranche with semiannual coupon payment frequency. The trader wants to invest in this tranche, as the quoted market price is much lower than his estimate of a fair price. To prove the point, the trader includes a table below with market prices of C-STRIPS as of June 30, 2017 (valuation date).

	Maturity	Market Price (per 100 face value)
Tranche 1	31/12/2017	\$99.753
Tranche 2	30/06/2018	\$97.257
Tranche 3	31/12/2018	\$95.012
Tranche 4	30/06/2019	\$94.332
Tranche 5	31/12/2019	\$93.805

What is closest to the price of the Treasury bond?

A. \$99.507

- B. \$105.209
- C. \$106.235
- D. \$98.079

The correct answer is **A**.

$$\begin{aligned}
 \text{Price of the bond} &= 1.1875 * \frac{99.753}{100} + 1.1875 * \frac{97.257}{100} + 1.1875 * \frac{95.012}{100} \\
 &\quad + 1.1875 * \frac{94.332}{100} + 101.1875 * \frac{93.805}{100} \\
 &= 99.507
 \end{aligned}$$

Further explanation:

The price of a bond is simply equal to the present value of all its future payments (coupons plus face value/redemption value). This is the logic applied here.

We wish to find the price of US 2 3/8s of June 30, 2017. A 2 3/8 bond implies it pays an annual coupon of 2% + 3/8% = 2.375%. We always compute the price of bonds per \$100 face value.

We are told that the bond pays semiannual coupons. That implies a coupon of 2.375%/2 = 1.1875% every six months. Per \$100 face value, that's \$1.1875. At redemption, the investor will receive the face value plus the coupon of the last six-month period, i.e. 100 + 1.1875

Now, we need to discount all these to the present, and that means we need to establish the relevant discount factors. Luckily, we have the prices of C-STRIPS maturing at each of those dates. A C-STRIP is priced at a discount to face value, just like T-bills.

$$d(t) \times \text{face value} = \text{price}; \text{ therefore, } d(t) = \frac{\text{price}}{\text{face value}}$$

Q.3418 The following table presents the characteristics of three different bonds with semiannual coupons and different times to maturity:

Maturity	Coupon	Price
6 months	6.0%	102 – 20
12 months	12%	104 – 08+
18 months	7.5%	98 – 24

If the principal repayment for each bond is \$100, which of the following is closest to the discount factor for 1.5 years?

- A. 0.9964
- B. 0.8823

C. 0.8865

D. 0.992

The correct answer is **B**.

The 6-month bond has cash flows only at maturity. It makes its interest payment of $\$3(\frac{6\%}{2} \times \$100)$ plus the principal repayment of \$100 at $t = 0.5$. To find $d(0.5)$ you should equate the present value (price) of the bond to the cash flows. 102-20 is equivalent to 102 full USD plus 20 fractions of a dollar where total no. of fractions is 32. Hence,

$$102 - 20 = 102 + 20/32 = \$102.63$$

$$\$102.63 = \$103d(0.5)$$

$$d(0.5) = 0.9964$$

The 12-month bond makes payments at $t = 0.5$ and at $t = 1$:

$$104 - 08+ = 104 + 8.5/32 = 104.27$$

Note: A “+” sign at the end of a quote represents half a tick.

$$104.27 = \$6d(0.5) + \$106d(1)$$

$$104.27 = \$6 \times 0.9964 + \$106d(1)$$

$$d(1) = \frac{104.27 - 5.98}{106} = 0.9273$$

Similarly, for the 18-month bond,

$$98.75 = \$3.75d(0.5) + \$3.75d(1) + 103.75d(1.5)$$

$$d(1.5) = \frac{98.75 - 3.74 - 3.48}{103.75} = 0.8823$$

Q.3419 As the chief investment manager of one of your corporate clients, you determine that the use of STRIPS (separate trading of registered interest and principal securities) issued by the U.S. Treasury would help match assets with liabilities at various points in the future. Which of the following statements regarding STRIPS is correct?

- A. Shorter-term STRIPS tend to trade cheap while longer-term STRIPS tend to trade rich
- B. Shorter-term STRIPS tend to trade rich while longer-term STRIPS tend to trade cheap
- C. Shorter-term C-STRIPS tend to trade at a discount
- D. Longer-term C-STRIPS tend to trade at a premium

The correct answer is **B**.

Shorter-term STRIPS tend to trade rich while longer-term STRIPS tend to trade cheap. This is because shorter-term STRIPS are in higher demand due to their lower risk and higher liquidity. Investors are willing to pay a premium for these securities, causing them to trade rich. On the other hand, longer-term STRIPS are riskier and less liquid, causing them to trade cheap. Investors demand a higher yield for holding these securities, which lowers their price.

Choice A is incorrect. Shorter-term STRIPS do not tend to trade cheap. In fact, the opposite is true. Shorter-term STRIPS usually trade rich due to their lower duration and less interest rate risk compared to longer-term STRIPS.

Choice C is incorrect. The question does not specify anything about C-STRIPS, which are a specific type of STRIP issued by corporations rather than the U.S Treasury. Therefore, this choice does not accurately reflect the trading tendencies of U.S Treasury STRIPS.

Choice D is incorrect. Similar to Choice C, this option refers to C-STRIPS and not U.S Treasury STRIPS as specified in the question prompt. Furthermore, it's also important to note that longer-term securities generally do not trade at a premium due to their higher interest rate risk.

Things to Remember

- STRIPS (Separate Trading of Registered Interest and Principal Securities) are fixed-income securities that separate the interest and principal components of a bond and trade them individually.
- Investors can purchase either the interest component (known as a coupon STRIP) or the principal component (known as a principal STRIP) of a bond.
- Shorter-term STRIPS typically have lower duration and less interest rate risk compared to longer-term STRIPS.
- Investors are willing to pay a premium for shorter-term STRIPS due to their lower risk and higher liquidity.
- Longer-term STRIPS are riskier and less liquid, causing them to trade at a discount.

- C-STRIPS are a specific type of STRIP issued by corporations rather than the U.S. Treasury.
-

Q.4578 Suppose that the cash price of a US Treasury bill is 90 per 100 of face value. If the bill has 60 days to maturity, what is the quoted price of the Treasury bill?

- A. 60
- B. 61
- C. 54.75
- D. 58.93

The correct answer is **A**.

We find the quoted price by the formula,

$$Q = \frac{360}{n}(100 - C)$$

Where

Q = Quoted price of the Treasury bill.

C = Cash price of the Treasury bill.

n = number of calendar days until maturity of the Treasury bill.

$$Q = \frac{360}{60}(100 - 90) = 6(10) = \$60$$

Q.4579 Company ABC wishes to invest in a 184-day Treasury bill from the US government. The bill is currently issued at the cash price of 98.50. What is the quoted price of the bill?

A. 2.86

B. 2.08

C. 2.93

D. 2.75

The correct answer is **C**.

The quoted price of the US T-bill is given by:

$$Q = \frac{360}{n}(100 - C)$$

Where

Q = Quoted price of the Bill

C = Cash price of the bill

n = number of calendar days until the maturity of the Treasury bill

Then we have:

$$Q = \frac{360}{184}(100 - 98.50) = 2.93$$

Q.4580 Suppose a bond with a par value of 1000 has coupon payments of 10% per annum and a yield to maturity of 5%. If the bond has 4 years to maturity, what is the price of the bond?

- A. 841.51
- B. 1,177.30
- C. 1,259.57
- D. 3,628.22

The correct answer is **B**.

Price of the bond is given by.

$$PV = \sum \left(\frac{C}{(1+y)^n} \right) + \frac{C + FV}{(1+y)^n}$$

Where.

C= coupon payments

y= yield to maturity

n= number of years to maturity of the bond.

FV= face value of the bond

Coupon payments=10% of 1000=\$100

$$PV = \frac{100}{1.05} + \frac{100}{1.05^2} + \frac{100}{1.05^3} + \frac{100}{1.05^4} + \frac{1000}{1.05^4} = \$1,177.297 \approx 1,177.30$$

Q.4581 A 20-days US treasury bill has a quoted price of 1.50. What is the cash price?

- A. 99.84
- B. 99.92
- C. 98.5
- D. 99.98

The correct answer is **B**.

We can get the cash price of the bill using the following formula.

$$Q = \frac{360}{n}(100 - C)$$

Where.

n= the number of calendar days until maturity of the treasury bill.

Q= the quote price

C= the cash price

$$1.50 = \frac{360}{20}(100 - C)$$

$$C = 100 - 1.50 \times \frac{20}{360} = 99.9166$$

Q.6294 A USD 250,000 par value U.S. government bond pays 4.25% coupon with a semiannual frequency. As of October 14, 2024, the last coupon was paid on August 23, 2024, and the next coupon is due on February 23, 2025. U.S. government bonds use an actual/actual day count convention. The accrued interest as of October 14, 2024, is closest to:

- A. USD 1,534
- B. USD 1,501
- C. USD 3,002
- D. USD 1,472

The correct answer is **B**.

Let's solve this step-by-step:

1) Calculate the coupon payment:

Annual coupon rate = 4.25%

Semiannual coupon rate = $4.25\% / 2 = 2.125\%$

Coupon payment = $\$250,000 * 2.125\% = \$5,312.50$

2) Determine the number of days for which interest has accrued:

Last coupon date: August 23, 2024

Date of calculation: October 14, 2024

Days between:

- August (31 - 23) = 8 days

- September = 30 days

- October = 14 days

Total = $8 + 30 + 14 = 52$ days

3) Calculate the total number of days in this coupon period:

August 23, 2024 to February 23, 2025

- August (31 - 23) = 8 days

- September to January = $30 + 31 + 30 + 31 + 31 = 153$ days

- February = 23 days

Total = $8 + 153 + 23 = 184$ days

4) Calculate the accrued interest:

Accrued Interest = (Coupon Payment * Days accrued) / Days in coupon period

= $(\$5,312.50 * 52) / 184$

= \$1,501.36

Therefore, the accrued interest as of October 14, 2024, is closest to \$1,501.

A is incorrect. It results from incorrectly using 180 days in the denominator.

C is incorrect. It results from using the annual coupon instead of the semiannual coupon.

D is incorrect. It results from using 51 days instead of 52 days.

Things to Remember

- In bonds using the actual/actual day count convention, the precise number of days between coupon periods and accrual periods must be used, which may vary each year due to leap years or different month lengths.
- Semiannual bonds distribute coupon payments twice per year, meaning the annual coupon rate must be halved when calculating each payment.

- Accrued interest is calculated by prorating the coupon payment based on the fraction of the current coupon period that has elapsed since the last payment.
 - For U.S. government bonds, the bondholder is entitled to interest accrued from the last coupon date up to, but not including, the settlement date.
-

Reading 56: Interest Rates

Q.1005 Grant Bank pays an interest of 8% with interest getting compounded quarterly. The effective annual rate is equal to:

- A. 8%
- B. 8.2%
- C. 8.4%
- D. 8.1%

The correct answer is **B**.

Let's assume that a sum of \$1000 is invested initially and the annual rate is assumed to be x.

Sum at the end of 1 year = $\$1000 * (1 + \frac{x}{100})^1$

As the interest paid by the bank is 8% with quarterly compounding, the aggregate value at the end of one year is given by:

$$\$1000 * (1 + \frac{0.08}{4})^4 = \$1000 * 1.082$$

Hence,

$$\begin{aligned} \$1000 * (1 + \frac{x}{100}) &= \$1000 * 1.082 \\ \Rightarrow 1 + \frac{x}{100} &= 1.082 \\ \Rightarrow \frac{x}{100} &= (1.082 - 1) = 0.082 \\ x &= 8.20\% \end{aligned}$$

Therefore, the effective annual rate = 8.20%

Q.1006 Royal Bank extends a loan of \$1000 to a customer for 2 years. The bank charges interest with half-yearly compounding frequency. If the spot rate for a 2-year loan is 10% per annum, then the amount that the customer needs to pay after 2 years is closest to:

- A. \$1,200
- B. \$1,216
- C. \$1,210
- D. \$1,222

The correct answer is **B**.

$$\text{Amount (after T years)} = \text{Principal} * \left(1 + \frac{(\text{Spot Rate})^{(T * \text{Compounding Frequency})}}{(\text{Compounding Frequency})}\right)$$

Where:

Principal = \$1000

Spot rate = 10%

Compounding Frequency = Half yearly = 2

T = 2 Years

$$\text{Amount (After 2 years)} = \$1000 * \left(1 + \frac{0.10^{2*2}}{2}\right) = \$1,215.51$$

Q.1007 The spot rate for 1 year and 2 years are 10% and 12% respectively. The forward rate for a loan to be given in 1 year for a term of 1 year is:

- A. 11%
- B. 13%
- C. 14%
- D. 12%

The correct answer is **C**.

The relation between spot rates and forward rate can be indicated by the expression below:

$$(1 + \text{Spot rate}_{\text{for } T})^T = (1 + \text{Spot rate}_{\text{for } t})^t * (1 + \text{Forward rate}_{t \text{ to } T})$$

Where

Spot rate for 1 year = 10%

Spot rate for 2 years = 12%

Therefore,

$$\begin{aligned}(1 + 0.12)^2 &= (1 + 0.10)^1 * (1 + f_{1,1}) \\ 1 + f_{1,1} &= \frac{(1 + 0.12)^2}{(1 + 0.10)^1} = 1.14 \\ f_{1,1} &= 0.14 = 14\%\end{aligned}$$

Q.1008 John Marauder observes that two zero-coupon bonds issued by ACC Limited are currently trading at prices given in the table below:

Residual maturity	Price	Face value
1 year	\$91.74	\$100
2 year	\$82.64	\$100

The 1-year and 2-year spot rates are:

- A. 9% and 10% respectively.
- B. 10% and 11% respectively.
- C. 8% and 9% respectively.
- D. 11% and 12% respectively.

The correct answer is **A**.

Zero-coupon bonds are bonds that do not carry coupons. Zero coupons bonds trade at a discount to face value, the discount rates used are the spot rates. Hence the price of a zero-coupon bond can be calculated by using the expression below:

$$\text{Price (zero-coupon bond of maturity T years)} = FV * \frac{1}{(1 + S_T)^T}$$

Therefore, using the above expression the spot rates are:

$$\begin{aligned} \$91.74 &= \$100 * \left(\frac{1}{(1 + S_1)} \right)^1 \\ 1 + S_1 &= \frac{100}{91.74} = 1.09 \\ S_1 &= 9\% \\ \$82.64 &= \$100 * \left(\frac{1}{(1 + S_2)} \right)^2 \\ (1 + S_2)^2 &= \frac{100}{82.64} = 1.21 \\ S_2 &= 10\% \end{aligned}$$

Q.1009 John Marauder observes that two zero-coupon bonds issued by ACC Limited are currently trading at prices given in the table below:

Residual maturity	Price	Face value
1 year	\$91.74	\$100
2 year	\$82.64	\$100

ABC Traders Private Limited managers are planning to issue zero-coupon bonds with a maturity of 1 year next year. As per the present market conditions, the price of the zero-coupon bond will most likely be:

- A. \$85.09
- B. \$95.09
- C. \$80.09
- D. \$90.09

The correct answer is **D**.

Zero-coupon bonds are bonds that do not carry coupons. Zero coupons bonds trade at a discount to face value, the discount rates used are the spot rates. Hence the price of a zero-coupon bond can be calculated by using the expression below:

$$\text{Price (zero-coupon bond of maturity T years)} = FV * \frac{1}{(1 + S_T)^T}$$

Therefore, using the above expression the spot rates are:

$$\$91.74 = \$100 * \left(\frac{1}{(1 + S_1)^1} \right)$$

$$1 + S_1 = \frac{100}{91.74} = 1.09$$

$$S_1 = 9\%$$

$$\$82.64 = \$100 * \left(\frac{1}{(1 + S_2)^2} \right)$$

$$(1 + S_2)^2 = \frac{100}{82.64} = 1.21$$

$$S_2 = 10\%$$

The forward rate can be computed using the following expression:

$$(1 + S_{\text{for } T})^T = (1 + S_{\text{for } t})^t * (1 + f_{t \text{ to } T})$$

Where

1 year spot rate = 9%

2 year spot rate = 10%

$$f_{1,1} = f$$

Therefore, the 1 year forward rate is equal to:

$$\begin{aligned}(1 + 0.10)^2 &= (1 + 0.09) * (1 + f_{1 \text{ year rate after 1 year}}) \\ 1 + f_{1 \text{ year rate after 1 year}} &= \frac{(1.1)^2}{1.09} = 1.11 \\ f_{1 \text{ year rate after 1 year}} &= 0.11 = 11\% \\ \text{The 1 year forward rate after 1 year} &= 11\%\end{aligned}$$

Hence the price of the zero-coupon bond issued for 1 year after 1 year will be:

$$\begin{aligned}P_{\text{Zero-coupon bond of maturity T years}} &= FV * \frac{1}{(1 + S_T)^T} \\ P &= \frac{\$100}{1.11} = \$90.09\end{aligned}$$

Q.1010 John Marauder observes that two zero-coupon bonds issued by ACC Limited are currently trading at prices given in the table below:

Residual maturity	Price	Face value
1 year	\$91.74	\$100
2 year	\$82.64	\$100

Select the most appropriate statement.

- A. The market expects the interest rates to fall, therefore the yield curve is downward sloping.
- B. The market expects the interest rate to rise, therefore the yield curve is upward sloping.
- C. The market expects the interest rate to remain constant, therefore the yield curve is flat.
- D. The yield curve is always downward sloping.

The correct answer is **B**.

The market expects the interest rate to rise, therefore the yield curve is upward sloping. The yield curve is a graphical representation of the interest rates on debt for a range of maturities. It shows the relationship between the interest rate (or cost of borrowing) and the time to maturity of the debt for a given borrower in a given currency. In this case, the prices of the bonds suggest that the yield for the 2-year bond is higher than the yield for the 1-year bond. This is indicative of an upward sloping yield curve, which typically occurs when the market expects interest rates to increase in the future. The expectation of rising interest rates is inferred from the forward rates calculated from the bond prices. The current 1-year spot rate is 9%, and the 1-year forward rate after 1 year is 11%. This suggests that the market expects the interest rate to rise from 9% to 11% after one year. Therefore, the yield curve is upward sloping, and the market expects the interest rate to rise.

Choice A is incorrect. The market does not expect the interest rates to fall. If that were the case, the price of a bond with longer maturity would be higher than that of a shorter maturity bond, which is not observed in this scenario.

Choice C is incorrect. The market does not expect the interest rate to remain constant. If it did, then both bonds would have similar prices regardless of their maturities, which contradicts our given data.

Choice D is incorrect. The yield curve is not always downward sloping. It can be upward sloping (as in this case), flat or even humped depending on various factors such as market expectations about future interest rates and risk premiums required by investors for holding long-term bonds.

Q.1011 If the term structure of spot interest rates is flat, then the term structure of forward interest rates must be:

- A. Upward sloping
- B. Downward sloping
- C. Flat
- D. Humped shaped

The correct answer is C.

The term structure of forward rates will also be flat if the term structure of spot rates is flat. This is because the forward rate is derived from the spot rates for different maturities. If the spot rates are the same for all maturities, then the forward rates will also be the same for all maturities. To illustrate this, let's assume that the spot rates for 1 year and 2 years are equal to r . The one-year forward rate starting in one year can be calculated as follows:

$$(1 + r)^2 = (1 + r)^1 * (1 + \text{extForwardrate}_{\text{ext1yearto1year}}) \quad 1 + \text{extForwardrate}_{\text{ext1yearto1year}} = 1 + r$$

As shown in the calculation above, the forward rate is equal to the spot rate r . Therefore, if the term structure of spot rates is flat, the term structure of forward rates will also be flat.

Choice A is incorrect. An upward sloping term structure of forward interest rates would imply that future short-term interest rates are expected to be higher than current short-term rates. This is not consistent with a flat term structure of spot interest rates, which implies that all future short-term rates are expected to be the same as the current rate.

Choice B is incorrect. A downward sloping term structure of forward interest rates would suggest that future short-term interest rates are expected to be lower than current short-term rates. This contradicts a flat term structure of spot interest rates, where all future short-term rates are anticipated to remain at the present level.

Choice D is incorrect. A humped shaped term structure of forward interest rate suggests that the forward rate for some future period will be higher than both the current and more distant future periods' spot rate, which contradicts with a flat term structure where all maturities have same yield.

Q.1013 Jack Mangers observes that the spot rates for 2 years and 3 years are 6% and 8% respectively while the 2y1y (1-year forward rate beginning after 2 years) rate is 12.50%. Select the most appropriate option.

- A. As there is no interest rate discrepancy, no transaction can generate riskless profit.
- B. Borrowing funds for 3 years, lending the funds for 2 years and an agreement to lend

funds for 1 year after 2 years will generate a riskless profit of \$0.43 per \$100.

C. Lending funds for 3 years, borrowing the funds for 2 years and an agreement to lend funds for 1 year after 2 years will generate riskless profit of \$0.50 per \$100.

D. Borrowing funds for 3 years, lending the funds for 2 years and an agreement to lend funds for 1 year after 2 years will generate riskless profit of \$0.70 per \$100.

The correct answer is **B**.

Borrowing funds for 3 years, lending the funds for 2 years and an agreement to lend funds for 1 year after 2 years will generate a riskless profit of \$0.43 per \$100. This is because the interest rates for borrowing and lending are different. The strategy involves borrowing funds for 3 years at an interest rate of 8%, then lending these funds for 2 years at a lower interest rate of 6%. After 2 years, the repaid money is then lent for 1 year as per the forward rate agreement at 12.50%. The difference in interest rates allows for a riskless profit to be made. The cash flow at the end of 3 years is \$126.41 (from the repayment of the lent money with 1 year worth of interest) minus \$125.97 (the repayment of the initially borrowed \$100 for 3 years), resulting in a profit of \$0.43.

Choice A is incorrect. There is indeed an interest rate discrepancy in the market. The forward rate for a 1-year period beginning after 2 years (2y1y) is higher than the spot rates for both a 2-year and a 3-year period, which indicates an arbitrage opportunity.

Choice C is incorrect. This strategy involves lending funds for longer duration and borrowing for shorter duration, which would not generate riskless profit due to the higher forward rate compared to spot rates. The cost of borrowing would exceed the returns from lending, resulting in a loss rather than profit.

Choice D is incorrect. While this strategy involves similar transactions as option B, it incorrectly estimates the potential riskless profit at \$0.70 per \$100 borrowed/lent instead of \$0.43 per \$100 borrowed/lent as calculated using interest rate parity condition.

Things to Remember

- Interest rate parity (IRP) is a fundamental concept in finance that links interest rates, spot exchange rates, and forward exchange rates in a way that eliminates the potential for riskless profit.
- Spot rates represent the current market interest rates for a specific period, while forward rates are the expected future interest rates for the same period.
- Arbitrage opportunities arise when there is a discrepancy between spot rates, forward rates, and interest rates, allowing investors to exploit market inefficiencies for riskless profit.

- Understanding the relationship between spot rates, forward rates, and interest rates is crucial for identifying profitable investment opportunities in the financial markets.
-

Q.1014 The details of a bond currently trading is given below:

Face value: \$1,000

Coupon: 6%

YTM: 6%

Tenure: 10 years

The price of the bond is:

- A. \$1,100
- B. 990
- C. \$1,000
- D. 980

The correct answer is **C**.

Yield to maturity (YTM) is the discount rate which is used to discount the cash flows of the bond to arrive at its price.

The price of the bond is equal to its face value if and only if the YTM is equal to the coupon rate.

The rate at which the price of the bond is equal to its face value is known as *Par rates*.

Therefore, in the above case, the price of the bond is equal to \$1,000.

Things to Remember

- Bond Price and Yield have an inverse relationship -as yield increases, bond price decreases and vice versa.
- Yield to Maturity (YTM) is the total return anticipated on a bond if it is held until it matures.
- Duration of a bond measures its price sensitivity to interest rate changes.
- Bond prices are affected by factors such as interest rates, credit quality, and time to maturity.
- Callable bonds give the issuer the right to redeem the bond before it matures.

Q.1015 A rate of 5% is quoted with continuous compounding. What is the equivalent rate, quoted with monthly compounding?

- A. 5.01%
- B. 6.03%
- C. 6.02%
- D. 5.59%

The correct answer is **A**.

If the rates are equivalent, then the following equation must hold:

$$e^R = \left(1 + \frac{R_m}{m}\right)^m$$

Where

R = Continuously compounded rate of interest

R_m = interest rate compounded m-times.

Then in this case we have:

$$\begin{aligned} e^{0.05} &= \left(1 + \frac{R_{12}}{12}\right)^{12} \\ \Rightarrow R_{12} &= 12 \left[e^{\frac{0.05}{12}} - 1\right] \\ &\approx 5.01\% \end{aligned}$$

Q.1016 The 5-year and 6-year continuously compounded interest rates are 6.75% and 7.25%, respectively. What is the forward rate of interest between year five and year six?

- A. 9.75%
- B. 8.25%
- C. 7.50%
- D. 8.56%

The correct answer is **A**.

When the rates are quoted with continuous compound, the forwards rate between times T_1 and T_2 is given by:

$$F = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1}$$

Where

R_1 = Spot rate for maturity T_1

R_2 = Spot rate for maturity T_2

Thus for this case we have:

$$F = \frac{0.0725 \times 6 - 0.0675 \times 5}{6 - 5} = 9.75\%$$

Alternatively, note that the following equation must hold:

$$\begin{aligned} e^{0.0675 \times 5} e^{F \times 1} &= e^{0.0725 \times 6} \\ \Rightarrow F &= (0.0725 \times 6) - (0.0675 \times 5) \\ &= 0.0975 = 9.75\% \end{aligned}$$

Q.2781 In the table below, an analyst has summarized the current swap rates prevailing on the market.

Term in Years	Swap Rate
0.5	2.00%
1.0	2.80%
1.5	3.20%

What is the 1.5-year discount factor?

- A. 0.9533
- B. 0.9995
- C. 0.9725
- D. 0.9999

The correct answer is **A**.

Consider an interest rate swap. If we assumed that the notional amount is exchanged, the fixed leg of the swap would resemble a fixed coupon-paying bond, with fixed leg payments acting like semi-annual, fixed coupons, and the notional amount acting like the principal payment. We can therefore write an equation for each “bond” that equates the present value of its cash flows to its price of par.

$$\begin{aligned} (100 + \frac{2}{2}) * d(0.5) &= 100 \Rightarrow d(0.5) = 0.9901 \\ \frac{2.8}{2} * d(0.5) + (100 + \frac{2.8}{2}) * d(1.0) &= 100 \Rightarrow d(1.0) = 0.9725 \\ \frac{3.2}{2} * d(0.5) + \frac{3.2}{2} * d(1.0) + (100 + \frac{3.2}{2}) * d(1.5) &= 100 \Rightarrow d(1.5) = 0.9533 \end{aligned}$$

Q.2783 After a recent FED’s announcement, a trader observed significant changes across the whole spot rates curve. The correct spot rates are as follows:

Year	Spot rate
0.5	2.0%
1.0	2.1%
1.5	2.3%
2.0	2.2%
2.5	2.5%

What is the 6-month forward rate in 2 years?

- A. 0.30%
- B. 1.85%
- C. 2.21%
- D. 3.71%

The correct answer is **D**.

To calculate the 6-month forward rate starting in 2 years, we use the given spot rates and the formula for forward rates. The forward rate between two points in time is derived from the idea that investing at the spot rate for a certain term should be equivalent to investing at the spot rate for a shorter term and then reinvesting the proceeds for the remaining term at the forward rate.

We use the following relation between spot rates and forward rates:

$$(1 + S_t)^t \times (1 + F_{t,t+\tau})^\tau = (1 + S_{t+\tau})^{t+\tau}$$

Where:

- S_t is the spot rate for time t .
- $S_{t+\tau}$ is the spot rate for time $t + \tau$.
- $F_{t,t+\tau}$ is the forward rate from time t to $t + \tau$.

We need the 6-month forward rate starting from 2 years (2.0 years to 2.5 years). Using the annualized spot rates for $t = 2.0$ years and $t + \tau = 2.5$ years, we convert these percentages to decimals:

$$\begin{aligned} S_{2.0} &= 2.2\% = 0.022 \\ S_{2.5} &= 2.5\% = 0.025 \end{aligned}$$

The formula rearranges to solve for $F_{2,2.5}$:

$$(1 + 0.022)^2 \times (1 + F_{2,2.5})^{0.5} = (1 + 0.025)^{2.5}$$

First, calculate the left-hand side part:

$$(1 + 0.022)^2 = 1.044484$$

And the right-hand side part:

$$(1 + 0.025)^{2.5} = 1.063677$$

Now, solving for $F_{2,2.5}$:

$$1.044484 \times (1 + F_{2,2.5})^{0.5} = 1.063677$$

$$(1 + F_{2,2.5})^{0.5} = \frac{1.063677}{1.044484} \approx 1.018376$$

$$1 + F_{2,2.5} = (1.018376)^2 \approx 1.037089$$

$$F_{2,2.5} = 1.037089 - 1 = 0.037089$$

Thus, the 6-month forward rate starting in 2 years is approximately **3.71%**.

Q.2785 Par rates prevailing on the market, are:

Term in years	Par Rates
0.5	1.60%
1.0	2.00%
1.5	2.60%
2.0	3.20%

What is the two-year discount factor? (Assume semiannual coupons.)

- A. 0.9871
- B. 0.9619
- C. 0.9421
- D. 0.938

The correct answer is **D**.

$(\frac{C_T}{2}) * A_T + d(T) = 1$, where C_T is the T year par rate and A_T is the annuity factor.

Term in years	Discount factor	Comments
0.5	0.9921	$(\frac{C_{0.5}}{2}) * d(0.5) + d(0.5) = 1$
1.0	0.9803	$(\frac{C_{1.0}}{2}) * (d(0.5) + d(1.0)) + d(1.0) = 1$
1.5	0.9619	$(\frac{C_{1.5}}{2}) * (d(0.5) + d(1.0) + d(1.5)) + d(1.5) = 1$
2.0	0.9380	$(\frac{C_{2.0}}{2}) * (d(0.5) + d(1.0) + d(1.5) + d(2.0)) + d(2.0) = 1$

Expanded Approach

Consider an interest rate swap. If we assumed that the notional amount is exchanged, the fixed leg of the swap would resemble a fixed coupon-paying bond, with fixed leg payments acting like semi-annual, fixed coupons, and the notional amount acting like the principal payment. We can therefore write an equation for each "bond" that equates the present value of its cash flows to its price of par.

To work out the discount factors, our swap rates become our "coupon rates"; then we determine the discounted value of each cash flow and equate the total to 100 (par amount)

To get $d(0.5)$, we assume we have a six-month 1.6% coupon bond priced at par. That means we have two cashflows -- $1.6\%/2 * 100$ and 100. The discount factor at 0.5 years, i.e., $d(0.5)$, is worked out using the following equation:

$$1.6/2 * d(0.5) + 100 * d(0.5) = 100$$

$$\rightarrow (1.6/2 + 100)d(0.5) = 100$$

$$\rightarrow d(0.5) = 100/100.8 = 0.9921$$

To get $d(1.0)$, we assume we have a one-year 2.00% coupon bond priced at par. That means we have three cashflows -- $2.0\%/2 * 100$, $2.0\%/2 * 100$, and 100. The discount factor at 1 year, i.e., $d(1.0)$, is worked out using the following equation:

$$1 * d(0.5) + 1 * d(1.0) + 100 * d(1.0) = 100$$

We already know the value of $d(0.5)$

$$\rightarrow 0.9921 + (101)d(1.0) = 100$$

$$\rightarrow d(1.0) = (100 - 0.9921)/101 = 0.9803$$

To get $d(1.5)$, we assume we have a one-and-a-half-year 2.60% coupon bond priced at par. That means we have four cashflows -- $2.6\%/2 * 100$, $2.6\%/2 * 100$, $2.6\%/2 * 100$, and 100. The discount factor at 1.5 years, i.e., $d(1.5)$, is worked out using the following equation:

$$1.3d(0.5) + 1.3d(1.0) + 1.3d(1.5) + 100d(1.5) = 100$$

We already have $d(0.5)$ and $d(1.0)$

$$\rightarrow 1.28973 + 1.27439 + 101.3d(1.5) = 100$$

$$\rightarrow d(1.5) = (100 - 1.28973 - 1.27439)/101.3 = 0.9619$$

To get $d(2.0)$, we assume we have a two-year 3.20% coupon bond priced at par. That means we have five cashflows -- $3.2\%/2 * 100$, $3.2\%/2 * 100$, $3.2\%/2 * 100$, $3.2\%/2 * 100$, and 100. The discount factor at 2 years, i.e., $d(2.0)$, is worked out using the following equation:

$$1.6d(0.5) + 1.6d(1.0) + 1.6d(1.5) + 1.6d(2.0) + 100d(2.0) = 100$$

We already have $d(0.5)$, $d(1.0)$, and $d(1.5)$

$$\rightarrow 1.5874 + 1.56848 + 1.53904 + 101.6d(2.0) = 100$$

$$\rightarrow d(2.0) = (100 - 1.5874 - 1.56848 - 1.53904)/101.6 = 0.9380$$

Q.3421 The price of a five-year zero-coupon government bond is \$72.25. The price of a similar six-year bond is \$67.34. The one-year implied forward rate from year 5 to year 6 is closest to:

- A. 7.29%
- B. 7.00%
- C. 6.79%
- D. 6.24%

The correct answer is **A**.

First, we need to find the 6-year and 5-year spot rates, assuming the price is given per \$100 par value. 5-year spot, S_5 :

$$72.25 = 100/(1 + S_5)^5$$

$$S_5 = 6.717\%$$

6-year spot, S_6 :

$$67.34 = 100/(1 + S_6)^6$$

$$S_6 = 6.812\%$$

Then, recall that:

$$(1 + 5 \text{ year spot})^5 * (1 + 1\text{-year forward rate}) = (1 + 6\text{-year spot})^6 \text{ Thus,}$$

$$(1 + 1\text{-year forward rate}) = (1 + 6\text{-year spot})^6 / (1 + 5\text{-year spot})^5$$

$$1\text{-year forward rate} = 1.484978899 / 1.38410182 - 1 = 7.288\%$$

Things to Remember

- Spot rate is the interest rate applicable to a zero-coupon bond.
 - Forward rates are implied future interest rates that can be calculated from current spot rates.
 - Implied forward rates are used to calculate expected future interest rates based on current market conditions.
 - The relationship between spot rates and forward rates is important in understanding the term structure of interest rates.
-

Q.3422 The term structure of swap rates is:

n – year	Swap rate
1 – year	3.5%
2 – year	4%
3 – year	4.5%
4 – year	5%
5 – year	5.5%

The two-year forward swap rate starting in three years is closest to:

- A. 0.5%
- B. 4%
- C. 7.02%
- D. 6.22%

The correct answer is **C**.

First, we compute the accrual of a dollar over three and five (3+2) years:

For $T = 3$, this is $(1 + 0.045)^3 = 1.1412$

For $T = 5$, this is $(1 + 0.055)^5 = 1.3070$

$$(1 + R_T)^T = (1 + R_n)^n (1 + F_{n,T})^{T-n}$$

Where $n < T$

$$1.3070 = 1.1412 \times (1 + F_{3,5})^2$$

$$F_{3,5} = \sqrt{\frac{1.3070}{1.1412}} - 1 = 7.02\%$$

Q.3423 Below is the term structure for swap rates:

Maturity in Years	Swap Rate
1	4.0%
2	4.5%
3	5.0%
4	5.5%
5	6.0%

The 1-year forward swap rate starting in four years is closest to:

- A. 4.0%
- B. 6.0%
- C. 7.0%
- D. 8.0%

The correct answer is **D**.

$$(1 + R_T)^T = (1 + R_n)(1 + F_{n,T})^{T-n}$$

Where $n < T$

First, we compute the accrual of a dollar over four and five years:

For $T = 4$, this is $(1 + 0.055)^4 = 1.2388$

For $T = 5$, this is $(1 + 0.06)^5 = 1.3382$

$$1.3382 = 1.2388 \times (1 + F_{4,5})^1$$

$$F_{4,5} = \frac{1.3382}{1.2388} - 1 = 8.0\%$$

Q.4572 A trader invests \$100 million in a savings account. After two years, the total amount in his account is \$150 million. What is the rate of interest compounded semi-annually?

- A. 20.32%
- B. 21.34%
- C. 22.47%
- D. 25.32%

The correct answer is **B**.

From the FV formula, we have:

$$FV = PV(1 + \frac{i_m}{m})^{mn}$$

Where

FV = future value

PV = present value

m = number of compounds per year

n = number of years

From the formula above, we have:

$$150 = 100(1 + \frac{i_2}{2})^4$$

Solving for i_2 , we have:

$$i_2 = 2 \left\{ \left(\frac{150}{100} \right)^{\frac{1}{4}} - 1 \right\} = 0.2134 = 21.34\%$$

An **alternative** to solving this question is by using the financial calculator. The variables are as follows: N = 4, PV = -100, PMT = 0 and FV = 150 so that CPT $\Rightarrow$ I/Y = 10.67%.

Therefore, the annual rate of interest is $10.67\% \times 2 = 21.34\%$. This is the **easiest** way!

Q.4573 If the one-year spot rate is 5% and the two years spot rate is 8%, what is the one-year forward rate, one year from now?

A. 0.0555

B. 0.111

C. 0.067

D. 0.134

The correct answer is **B**.

The spot rate and the forward rate have the following relationship:

$$(1 + y_2)^2 = (1 + y_1) \times (1 + f_{[1,1]})$$

Where

y_1 = one-year spot rate

y_2 = two-year spot rate and,

$f_{[1,1]}$ = one-year forward rate, one year from now

$$(1.08)^2 = (1.05)^1 \times (1 + f_{[1,1]})$$
$$f_{[1,1]} = \frac{1.08^2}{1.05} - 1 = 0.111 = 11.1\%$$

Q.4574 1000 is invested in an account that pays an annual nominal interest of 8% compounded quarterly per year. What is the value of the amount in the account after three years?

A. 1259.71

B. 1268.24

C. 1061.21

D. 2518.17

The correct answer is **B**.

$$FV = PV \left(1 + \frac{i_m}{m}\right)^{mn}$$

Where:

FV= the future value

PV=the present value

m= number of compounds in a year

n= number of years

The value of the amount in the account after three years is given by:

$$FV = 1000 \left(1 + \frac{0.08}{4}\right)^{12} = 1268.24$$

Q.4575 Suppose a company is obliged to make annual payments of \$5,000 for the premises it occupies. Payments are due on 1st January 2001, 1st January 2002, and 1st January 2003. If the company wishes to cover these payments by investing a single sum in its bank account that pays 8% per annum compound, what sum must be invested on 1st January 2000?

A. 12,885.48

B. 14,853.44

C. 13,916.32

D. 11,298.56

The correct answer is **A**.

An annuity is a series of annual payments of PMT until the final time T. The value of an ordinary annuity is given by:

$$PV_{\text{annuity}} = PMT \frac{1 - (1 + r)^{-T}}{r}$$

Where:

r = discount rate

This is an annuity ordinary because the first payment would happen a year after January 2000.

The present value of the annuity on 1st January 2000 is given by:

$$5000 \left(\frac{1 - 1.08^{-3}}{0.08} \right) = 12,885.48$$

Q.4576 Spot rates with semi-annual compounding are as below:

Maturity (years)	Spot rates (%)
1.0	2.0
1.5	2.5
2.0	3.0

What is the forward rate for the period between time 1.0 and 1.5, expressed annually?

A. 0.035

B. 0.0175

C. 0.07

D. 0.005

The correct answer is **A**.

The forward rate for the period between time 1.0 and 1.5 is given by:

$$\left(1 + \frac{0.025}{2}\right)^3 = \left(1 + \frac{0.02}{2}\right)^2 \left(1 + \frac{f_1}{2}\right)$$

Solving for f_1 , we have

$$\frac{f_1}{2} = \frac{\left(1 + \frac{0.025}{2}\right)^3}{\left(1 + \frac{0.02}{2}\right)^2} - 1 = 0.0175$$

And therefore, $f_1 = 0.0175 \times 2 = 0.035 = 3.5\%$

Q.4577 Spot rates with semi-annual compounding are as below:

Maturity (years)	Spot rates (%)
1.0	2.0
1.5	2.5
2.0	3.0

What is the forward rate for the semi-annual period between time 1.5 and 2, expressed annually?

- A. 0.045
- B. 0.0225
- C. 0.09
- D. 0.0977

The correct answer is **A**.

The forward rate between time 1.5 and 2.0 will be:

$$\frac{f_2}{2} = \frac{(1 + \frac{0.030}{2})^4}{(1 + \frac{0.025}{2})^3} - 1 = 0.0225$$
$$f_2 = 0.0225 \times 2 = 0.045 = 4.5\%$$

Reading 57: Bond Yields and Return Calculations

Q.1021 Understanding the relationship between corporate bonds and government bonds is fundamental in fixed-income investing. Corporate bonds trade at a positive spread to government bonds because:

- A. Corporate bonds are more liquid than government bonds.
- B. Corporate bonds have higher credit risks than government bonds.
- C. Corporate bonds generate higher returns than government bonds.
- D. Corporate bonds pay less coupon than government bonds.

The correct answer is **B**.

Corporate bonds have higher credit risks than government bonds. Credit risk refers to the risk that a borrower will default on any type of debt by failing to make required payments. In the context of bonds, it is the risk that the issuer will not be able to make the required interest payments or to repay the principal upon maturity. Corporate bonds are issued by companies, which are subject to business risks and therefore have a higher credit risk compared to government bonds. Government bonds are considered to be virtually risk-free because they are backed by the full faith and credit of the government. Therefore, to compensate for the higher credit risk, corporate bonds offer a higher yield than government bonds, which results in a positive spread.

Choice A is incorrect. Corporate bonds are not necessarily more liquid than government bonds. In fact, government bonds are often considered to be more liquid due to their high demand and the large volume of trade in the market.

Choice C is incorrect. While it's true that corporate bonds can generate higher returns, this isn't the primary reason for their positive spread compared to government bonds. The higher return on corporate bonds is a compensation for the additional risks (including credit risk) associated with them, which leads us back to option B as being correct.

Choice D is incorrect. The coupon payment of a bond does not directly affect its trading spread. While it's true that lower coupon payments might make a bond less attractive and therefore could indirectly lead to wider spreads, this isn't typically the primary factor affecting spreads between corporate and government bonds.

Things to Remember

- Government bonds are considered risk-free because they are backed by the full faith and credit of the government.
- Credit risk is the risk that a borrower will default on debt payments, and it is higher for corporate bonds compared to government bonds.

- The spread between corporate bonds and government bonds is a reflection of the compensation for the higher credit risk associated with corporate bonds.
 - Liquidity in the bond market refers to how easily a bond can be bought or sold without significantly affecting its price.
 - Higher returns on corporate bonds are a result of the additional risks they carry, such as credit risk.
-

Q.1022 Consider the following details with respect to a bond:

Face value: \$1000

Coupon: 10%

Frequency: Semi-annually

Coupon payment dates: January 1st and July 1st

An investor buys the bond on January 22nd at a price of \$990.23, and sells it on August 3rd at \$1030.34. The gross realized return on the bond investment is:

- A. 4.05%
- B. 9.1%
- C. 8.8%
- D. 9.4%

The correct answer is **B**.

January 22nd: -990.23

July 1st: +50 (Coupon payment)

August 3rd: +1030.34

$$\text{Gross Realized Return} = \frac{(\$1030.34 + \$50 - \$990.23)}{\$990.23} = 9.10\%$$

Q.1023 Consider the following details with respect to a bond:

Face value: \$1000

Coupon: 10%

Frequency: Semi-annually

Coupon payment dates: January 1st and July 1st

An investor buys this bond at \$1043.43 on January 1st, 2016 and sells it on January 1st, 2017 at \$995.23. The coupon received on July 1st, 2016 is reinvested for six months at a semi-annually compounded rate of 9%. The realized gross holding period return is:

- A. 0.39%
- B. 5.18%
- C. 4.99%
- D. 5.6%

The correct answer is **B**.

The realized gross holding period return (HPR) for a bond can be calculated with the following formula:

$$\text{HPR} = \frac{\text{Ending Value}}{\text{Beginning Value}} - 1$$

In this case, the beginning value of the bond is the purchase price, and the ending value is the sale price plus the coupon payments received (including any interest earned on reinvested coupons).

Here's how you can calculate each component:

1. **Beginning Value.** This is the price the investor paid for the bond, which is \$1043.43.
2. **Ending Value.** This is the sale price of the bond plus the reinvested coupon. The bond was sold for \$995.23. The coupon payment is 10% of the face value, paid semi-annually, so each payment is $\$1000 * 10\% / 2 = \50 . The July 1st coupon was reinvested for six months at a semi-annually compounded rate of 9%, so it grew to $\$50 * (1 + 9\%/2) = \51.25 . So, the ending value is $\$995.23$ (sale price) + $\$50$ (January 1st, 2017 coupon) + $\$52.25$ (reinvested July 1st, 2016 coupon) = $\$1097.48$.

Using these values in the HPR formula:

$$\text{HPR} = \frac{\$1097.48}{\$1043.43} - 1$$

This results in an HPR of approximately 0.0518 or 5.18%.

Things to Remember

- Holding period return (HPR) is a measure of the total return received from holding an asset or portfolio over a period of time.
- HPR takes into account not only the capital gains or losses from the asset's price appreciation or depreciation but also any income generated, such as dividends or coupon payments.
- Reinvesting coupons or dividends can have a significant impact on the overall return of an investment, especially when compounded over time.
- Semi-annual compounding means that interest is calculated and added to the investment twice a year, which can lead to slightly higher returns compared to annual compounding.
- Face value of a bond is the amount that the issuer agrees to repay the bondholder at maturity. It is also known as par value.
- Coupon rate is the fixed interest rate that the issuer pays to the bondholders. It is usually expressed as a percentage of the bond's face value.

Q.1024 A corporate bond has a residual maturity of 2 years and pays a 10% coupon annually. Two zero-coupon bonds are currently trading at the price mentioned below:

Price	Residual maturity
USD 92.38	1 year
USD 84.17	2 years

The price of the corporate bond is closest to:

- A. USD 101.01
- B. USD 117.66
- C. USD 101.82
- D. USD 93.41

The correct answer is **C**.

A bond is priced using the spot rates as:

$$\text{Price}_{\text{Bond}} = \frac{C}{(1 + \text{Spot rate}_1)^1} + \frac{(FV + C)}{(1 + \text{Spot rate}_2)^2}$$

$$\text{Spot Rate}_{1 \text{ year}} = \left(\frac{100}{92.38}\right) - 1 = 0.0825 = 8.25\%$$

$$\text{Spot Rate}_{2 \text{ year}} = \left(\frac{100}{84.17}\right)^{\frac{1}{2}} - 1 = 0.0900 = 9.00\%$$

The price of the bond can be calculated as:

$$\begin{aligned} \text{Bond Price} &= \frac{10}{(1 + 0.0825)^1} + \frac{100 + 10}{(1 + 0.09)^2} \\ &= 9.24 + 8.42 + 84.17 = 101.82 \end{aligned}$$

Q.1025 A bond with a residual maturity of 2 years that pays a 10% coupon annually is currently trading at \$102.10. It is also observed that two zero-coupon bonds are currently trading at the price mentioned below:

Price	Residual maturity
\$92.38	1 year
\$84.17	2 years

The transaction which will generate riskless profit is:

- A. Sell the bond and purchase the zero-coupon bonds.
- B. Borrow funds and purchase the zero-coupon bonds.
- C. Riskless profit cannot be generated.
- D. Buy the bond and sell the zero-coupon bonds.

The correct answer is **A**.

The bond is priced using the spot rates as:

$$\text{Price}_{\text{Bond}} = \frac{C}{(1 + \text{Spot rate}_1)^1} + \frac{(FV + C)}{(1 + \text{Spot rate}_2)^2}$$

$$\text{Spot Rate}_{1 \text{ year}} = \left(\frac{100}{92.38}\right) - 1 = 0.0825 = 8.25\%$$

$$\text{Spot Rate}_{2 \text{ year}} = \left(\frac{100}{84.17}\right)^{\frac{1}{2}} - 1 = 0.0900 = 9.00\%$$

The price of the bond can thus be calculated as:

$$\begin{aligned}\text{Price}_{\text{Bond}} &= \frac{10}{(1 + 0.0825)^1} + \frac{(100 + 10)}{(1 + 0.0900)^2} \\ &= 101.82\end{aligned}$$

As can be observed from the above, the price of the bond is \$100.82. Hence, the bond is overvalued if its market price is greater than \$100.82. As observed, if the bond is overvalued, the investor must sell the overvalued bond. The funds obtained by selling the bond must be invested in the zero-coupon bonds. By the above transaction, the investor can make a profit if the market price is greater than \$101.82.

Choice B is incorrect. Borrowing funds to purchase the zero-coupon bonds would not result in a riskless profit. This is because borrowing costs would need to be considered, and these could potentially exceed the returns from the zero-coupon bonds, leading to a net loss rather than a profit.

Choice C is incorrect. A riskless profit can indeed be generated in this scenario by selling the bond and purchasing the zero-coupon bonds (as explained in choice A). Therefore, stating that a riskless profit cannot be generated is incorrect.

Choice D is incorrect. Buying the bond and selling the zero-coupon bonds would not result in a riskless profit either. The cost of buying the bond (\$102.10) exceeds the proceeds from selling both zero-coupon bonds (\$92.38 + \$84.17 = \$176.55), which means there would be an initial cash outflow without any guaranteed return, hence no riskless profit can be achieved through this strategy.

Q.1027 The bonds of country A is trading at a spread of x with respect to US Treasury bonds in country B and at a spread of y with respect to US Treasury bonds in country C. A relative value trader wants to generate returns by trading in bonds of country A. If $x > y$, then the trade must be:

- A. To buy the bond in country B and sell it in country C.
- B. To buy the bond in country C and sell it in country B.
- C. To sell the bond in country B and C.
- D. To buy the bond in country B and C.

The correct answer is **A**.

The bond of country A trades at a higher spread in country B as compared to country C. The spread of a bond is the difference between its yield and the yield of a risk-free bond, in this case, the US Treasury bonds. A higher spread indicates a higher yield, which in turn implies a lower price for the bond due to the inverse relationship between bond prices and yields. Therefore, the

bond of country A is cheaper in country B than in country C. The relative value trader can take advantage of this price difference by buying the bond in country B, where it is cheaper, and selling it in country C, where it is more expensive. This strategy, known as arbitrage, allows the trader to make a profit without taking on any risk. The profit from this trade would be the difference in the spreads, i.e., $(x-y)\%$, assuming no transaction costs or other market frictions.

Choice B is incorrect. Buying the bond in country C and selling it in country B would not be a profitable strategy because the spread in country C is less than that of country B. This means that the bond's yield relative to US Treasury bonds is lower in country C, making it less attractive for purchase compared to the bond in country B.

Choice C is incorrect. Selling the bond in both countries would not take advantage of the difference in spreads between countries B and C. The trader's goal should be to exploit this discrepancy by buying where yields are higher (country B) and selling where they are lower (country C).

Choice D is incorrect. Buying bonds from both countries does not capitalize on the observed spread difference between them, as there would be no arbitrage opportunity exploited here.

Things to Remember

- Spread: The difference between the yield of a bond and the yield of a risk-free bond, such as US Treasury bonds.
 - Arbitrage: The practice of exploiting price differences in different markets to make a profit without taking on any risk.
 - Inverse relationship between bond prices and yields: When bond prices go up, yields go down, and vice versa.
 - Relative value trading: A strategy that involves comparing the value of different securities to identify mispricings and profit from them.
 - Transaction costs: Additional costs incurred when buying and selling securities, which can impact the profitability of trades.
-

Q.2786 Aram Bauer is considering an investment in fixed income instruments. He is interested in a U.S. Treasury tranche as of December 31, 2019. The tranche pays coupons of 4.5% per year compounded semiannually. The price of this tranche as of December 31, 2016, is 97.124.

What is the yield to maturity of the tranche?

- A. 2.78%
- B. 3.22%
- C. 5.55%
- D. 6.44%

The correct answer is **C**.

To compute a bond's yield to maturity, we use the following formula:

$$P = \frac{C_1}{(1+y)^1} + \frac{C_2}{(1+y)^2} + \frac{C_3}{(1+y)^3} \cdots + \frac{F + C_N}{(1+y)^N}$$

Where:

P = price of the bond

C_t = annual cash flow in year t

N = time to maturity in years

y = annual yield (YTM to maturity)

F = face value

$$\frac{2.25}{(1 + \frac{y}{2})^1} + \frac{2.25}{(1 + \frac{y}{2})^2} + \frac{2.25}{(1 + \frac{y}{2})^3} + \frac{2.25}{(1 + \frac{y}{2})^4} + \frac{2.25}{(1 + \frac{y}{2})^5} + \frac{102.25}{(1 + \frac{y}{2})^6} = 97.124$$

Using trial and error, we get $y=5.554\%$

Alternatively, using a financial calculator with the following inputs:

N = 6; PV = -97.124; PMT = 2.25 (=4.5/2); FV = 100

We get, I/Y = 2.777%

$$\Rightarrow YTM = 2 \times 2.777 = 5.554\%$$

Q.2787 On Jan 1 2017, Commercial Bank of India issued a six-year bond paying an annual coupon of 6% at a price reflecting a yield to maturity of 4%. As of 31 Dec 2017, interest rates remain unchanged. Holding all other factors constant, and assuming a flat term structure of interest rates, how was the bond's price affected? The price:

- A. Remained constant.
- B. Decreased.
- C. Increased.
- D. Increased, but only in the second half of the year.

The correct answer is **B**.

From the data given, it's clear that the bond's coupon is greater than the yield. As such, the bond must have traded at a premium – implying the price must have been greater than the face value. Provided the yield doesn't change, a bond's price will always converge to its face value. Since the price starts higher, it must decrease.

Things to Remember

- A bond's price and yield have an inverse relationship -as the yield decreases, the price of the bond increases, and vice versa.
 - Yield to maturity (YTM) is the total return anticipated on a bond if it is held until it matures.
 - A bond trading at a premium means its price is higher than its face value, while a bond trading at a discount means its price is lower than its face value.
 - Flat term structure of interest rates implies that interest rates are constant across all maturities.
 - Duration of a bond is a measure of its price sensitivity to interest rate changes.
-

Q.2788 A trader borrows \$3,000,000 with a term of two years at a simple annual interest rate of 2% from his broker. He purchases at par a bond with a 5% coupon paid annually. The bond matures exactly in 10 years. Two years later, the trader sells the bond at the price of \$101 and repays the loan on an annual basis. Assuming that all of the coupons received are reinvested at the rate of 1.5%, for a period of 1 year, what is the trader's net realized return on the transaction described above?

- A. +7.0000%
- B. +7.0750%
- C. +11.0000%
- D. +11.0750%

The correct answer is **B**.

The net realized return is the return after financing costs have been subtracted

$$\begin{aligned}
 \text{Proceeds from the sale of the bond} &= \$3,000,000 * \frac{101}{100} = \$3,030,000.00 \\
 \text{Coupons received from the bond} &= \$150,000.00 * 1.015 + \$150,000.00 = \$302,250.00 \\
 \text{Interest paid to the broker} &= -\$60,000.00 - \$60,000.00 = -\$120,000 \\
 \text{Net proceeds} &= \$3,030,000.00 + \$302,250.00 - \$120,000 = \$3,212,250.00 \\
 \text{Net realized return} &= \frac{(\$3,212,250.00 - \$3,000,000)}{\$3,000,000} = 7.0750\%
 \end{aligned}$$

Q.2789 Alice Tuck invested her annual bonus in a bond with a face value of \$55,000. The bond pays a 5% coupon semiannually and matures in 10 years. At the purchase date, the bond had a yield to maturity of 7%. Six months later Alice received the coupon and sold the bond at the market yield of 6.5%. What is the net realized return on Alice's transaction?

- A. +7.0000%
- B. +7.1250%
- C. +7.2312%
- D. +7.5475%

The correct answer is **C**.

Investment in the bond = \$47,183.18

(N = 20; I/Y = 3.5%; PMT = 1,375 (=55,000 *5% / 2); FV = 55,000
CPT => PV = - 47,183.18)

Proceeds from the sale of the bond = \$49,220.10

(N = 19; I/Y = 3.25%; PMT = 1,375 (=55,000 *5% / 2); FV = 55,000
CPT => PV = - 49,220.10)

Coupons received from the bond = \$1,375

$$\begin{aligned} \text{Total proceeds} &= \$49,220.10 + \$1,375 = \$50,595.10 \\ \text{Net realized return} &= \frac{(\$50,595.10 - \$47,183.18)}{\$47,183.18} = 7.2312\% \end{aligned}$$

Q.3425 Bank A and Bank B both have a credit rating of BBB. Bank A issues a fixed-rate bond with a 10-year term to maturity, while Bank B issues a similar bond with a 5-year term to maturity. Holding all other factors constant, which of the following statements is *most likely* true?

- A. Bond A has a higher interest rate risk than bond B
- B. Bond B has a higher interest rate than bond A
- C. Bond A has a lower coupon rate than bond B
- D. Bond B has a higher coupon rate than bond A

The correct answer is **A**.

Bond A has a higher interest rate risk than bond B. The term 'interest rate risk' refers to the risk that the value of a bond or other fixed-income investment will suffer as the result of unexpected fluctuations in interest rates. In general, the longer the term to maturity of a bond, the greater the interest rate risk. This is because the bond's value could be impacted by changing interest rates over a longer period before it matures. Therefore, a bond with a longer maturity, such as Bond A in this case, generally has a higher interest rate risk than a similar bond with a shorter maturity, such as Bond B. To compensate investors for taking on this additional risk, long-term bonds typically offer higher coupon rates than short-term bonds of the same credit quality. Therefore, it is most likely that Bond A has a higher interest rate risk than Bond B.

Choice B is incorrect. The interest rate of a bond is not determined by its term to maturity but rather by the creditworthiness of the issuer and prevailing market conditions. Since both banks have the same credit rating, it's unlikely that Bond B would have a higher interest rate than Bond A solely based on its shorter maturity period.

Choice C is incorrect. The coupon rate of a bond, like its interest rate, is determined by factors such as the issuer's creditworthiness and market conditions, not its term to maturity. Therefore, it's not accurate to say that Bond A would necessarily have a lower coupon rate than Bond B just because it has a longer term to maturity.

Choice D is incorrect. As explained above in Choice C explanation, the coupon rate of a bond isn't determined by its term to maturity but rather other factors such as issuer's creditworthiness and prevailing market conditions. Hence, there isn't enough information provided in this scenario to conclude that Bond B would have higher coupon rates than Bond A.

Things to Remember

- Interest rate risk is the risk that the value of a bond or fixed-income investment will decrease due to unexpected fluctuations in interest rates.
- Longer-term bonds generally have higher interest rate risk compared to shorter-term bonds.
- Coupon rate of a bond is determined by factors such as the issuer's creditworthiness and prevailing market conditions, not just the term to maturity.
- Credit rating is an important factor in determining the interest rate of a bond, as it reflects the issuer's ability to repay the bond.
- Market conditions, such as overall interest rate levels, also play a significant role in determining bond interest rates.

Q.3426 On Jan 1 2017, a 5-year corporate bond, paying an annual coupon of 8%, was selling at a discount. As of 31 Dec 2017, interest rates remain unchanged. Holding all other factors constant,

which of the following relationships holds true? (P_0 represents the price of the bond and YTM is the bond's yield to maturity.)

- A. $P_0 < \text{par}$ and $\text{YTM} < 8\%$
- B. $P_0 < \text{par}$ and $\text{YTM} > 8\%$
- C. $P_0 > \text{par}$ and $\text{YTM} > 8\%$
- D. $P_0 > \text{par}$ and $\text{YTM} < 8\%$

The correct answer is **B**.

The bond is trading at a discount, which means that the price (P_0) is lower than its face value (par). This situation typically arises when the bond's coupon rate is less than its yield to maturity (YTM). Therefore, the relationship $P_0 < \text{par}$ and $\text{YTM} > 8\%$ holds true. As the bond approaches its maturity date, its price will gradually 'pull to par', meaning it will rise to its face value. This is because the market price of a bond tends to move towards its face value as it nears maturity, assuming that the market interest rates remain constant.

Choice A is incorrect. The bond was purchased at a discount, which means $P_0 < \text{par}$. However, when a bond is bought at a discount, the yield to maturity (YTM) is greater than the coupon rate. Therefore, YTM cannot be less than 8%.

Choice C is incorrect. Given that the bond was purchased at a discount, it implies that P_0 cannot be greater than par value. Hence this option contradicts with the given scenario.

Choice D is incorrect. This choice suggests that $P_0 > \text{par}$ and $\text{YTM} < 8\%$. However, as per the question's premise where it states that the bond was bought at a discount (P_0

Things to Remember

- Yield to Maturity (YTM) is the total return anticipated on a bond if it is held until it matures.
- A bond is said to be trading at a discount when its market price is below its face value.
- The relationship between bond price, coupon rate, and YTM is crucial in understanding bond valuation.
- As a bond approaches maturity, its price tends to move towards its face value assuming constant market interest rates.
- The relationship between bond price and YTM is inversely related -as YTM increases, bond price decreases and vice versa.

Q.4582 A zero-coupon bond with three years to maturity has a face value of 100. If the current market price of the bond is 88, what is the yield to maturity of the bond?

- A. 12%
- B. 13.6%
- C. 4.35%
- D. 3.79%

The correct answer is **C**.

The yield to maturity of a zero-coupon bond is given by:

$$\text{YTM} = \left(\frac{\text{Face value}}{\text{Current price of the bond}} \right)^{\frac{1}{\text{year to maturity}}} - 1$$

In this case,

$$\text{YTM} = \left(\frac{100}{88} \right)^{\frac{1}{3}} - 1 = 4.35\%$$

Q.4583 An investor buys a two-year 100 par-value bond at \$95 per \$100 face value. The bond pays semiannual coupons at a rate of 5% per annum. Suppose that after six months, the coupon is invested and earns 2% for the next 6 months. One year after the bond was purchased, the investor decides to cash out and sell the bond at \$97. What is the gross realized return for the investor?

- A. 7.37%
- B. 7.42%
- C. 4.22%
- D. 10.05%

The correct answer is **B**.

$$\text{Gross Realised Return} = \frac{\text{Ending Value} + \text{Coupon} - \text{Beginning Value}}{\text{Beginning Value}}$$

Given that the coupon rate is 5% paid semiannually, then the bond pays coupons of \$2.5 after every 6-months, and since the coupon is invested for the next 6 months at 2%, then we have:

$$\text{Gross Realised Return} = \frac{97 + 2.5 * 1.02 + 2.5 - 95}{95} \approx 7.42\%$$

Q.4584 Suppose that James is offered a bond that pays \$40 per annum in perpetuity. If the discount rate is 6%, what is the bond's expected price?

- A. 600
- B. 666.67
- C. 667.77
- D. 666

The correct answer is **B**.

The price of a perpetual bond is given by,

$$PV_{\text{perpetuity}} = \frac{PMT}{r}$$

Where;

r = the discount rate; and

PMT = the periodic payment.

In this case,

$$PV_{\text{perpetuity}} = \frac{40}{0.06} = \$666.67$$

Q.4585 What is the present value of an annuity that pays \$100 per year at the end of each year for the next five years at an effective rate of 5% per annum?

- A. 435
- B. 432.95
- C. 495
- D. 487.98

The correct answer is **B**.

We can find the present value of the annuity using the formula;

$$PV_{\text{annuity}} = PMT \left(\frac{1 - (1 + r)^{-T}}{r} \right)$$

Where,

PV_{annuity} = present value of the annuity

PMT = periodic premium payment

r = effective discount rate

In this case,

$$PV_{\text{annuity}} = 100 \left(\frac{1 - (1.05)^{-5}}{0.05} \right) = \$432.95$$

Reading 58: Applying Duration, Convexity, and DV01

Q.1175 The price of a bond at various rates is given in the table below:

Spot rate	Price
3.45%	95.8680
3.40%	96.0780

The DV01 of the bond is:

- A. 11.04
- B. 1.1
- C. 0.042
- D. 0.906

The correct answer is **C**.

$$\begin{aligned} \text{DV01} &= -\frac{\Delta P}{10,000 * \Delta y} \\ \Delta P &= 95.8680 - 96.0780 = -0.21 \\ \Delta y &= 3.45\% - 3.40\% = 0.05\% \\ \text{DV01} &= -\frac{-0.21}{10,000 * 0.05\%} = 0.042 \end{aligned}$$

Q.1176 The price of a bond at various rates is given in the table below:

Par rates	price
3.45%	95.8680
3.40%	96.0780
3.35%	96.3210

The effective duration of the bond is:

- A. 2.3575
- B. 4.7149
- C. -2.3575
- D. 0.04719

The correct answer is **B**.

The effective duration of the bond can be calculated using the formula:

$$\text{Effective Duration} = \frac{P_+ - P_-}{2 \times P_0 \times \Delta y}$$

Where:

- P_+ is the bond price when the yield decreases (in this case, at 3.35%).
- P_- is the bond price when the yield increases (in this case, at 3.45%).
- P_0 is the bond price at the initial yield (at 3.40%).
- Δy is the change in yield in decimal form.

Substituting the given values into the formula:

$$\begin{aligned}\text{Effective Duration} &= \frac{96.3210 - 95.8680}{2 \times 96.0780 \times 0.0005} \\ &= \frac{0.4530}{0.096078} \\ &= 4.71491\end{aligned}$$

Q.1181 Details of portfolio X is given below: Note: Portfolio X consists of bonds A, B, C and D, and the value of each bond is given in the above table. The duration of portfolio X is:

Bond	Value	Duration	Convexity
Bond A	\$120,000	5.453	230.453
Bond B	\$100,000	7.213	350.361
Bond C	\$150,000	2.348	120.714
Bond D	\$130,000	8.190	480.341

A. 5.26

B. 5.59

C. 5.1

D. 5.69

The correct answer is **B**.

$$\begin{aligned}\text{Total value of the portfolio} &= \$120,000 + \$100,000 + \$150,000 + \$130,000 = \$500,000 \\ \text{Duration of the portfolio} &= \frac{\$120,000 * 5.453 + \$100,000 * 7.213 + \$150,000 * 2.348 + \$130,000 * 8.190}{\$500,000} \\ &= 5.59\end{aligned}$$

Q.1182 Details of portfolio X is given below:

Bond	Value	Duration	Convexity
Bond A	\$120,000	5.453	230.453
Bond B	\$100,000	7.213	350.361
Bond C	\$150,000	2.348	120.714
Bond D	\$130,000	8.190	480.341

Note: Portfolio X consists of bonds A, B, C and D, and the value of each bond is furnished in the above table.

The convexity of portfolio X is:

A. 360.426

B. 386.484

C. 200

D. 286.484

The correct answer is **D**.

$$\begin{aligned}\text{Total value of the portfolio} &= \$120,000 + \$100,000 + \$150,000 + \$130,000 = \$500,000 \\ \text{Convexity of the portfolio} &= \frac{(\$120,000 * 230.453 + \$100,000 * 350.361 + \$150,000 * 120.714 + \$130,000 * 480.341)}{(\$500,000)} \\ &= 286.484\end{aligned}$$

Q.1184 Data on three bonds are given below. Assume the current date is March 31, 2015.

Bond	Maturity	Price	Yield	Duration	Convexity
A	March 31, 2020	110.321	2.32%	4.321	34.09
B	March 31, 2030	109.320	3.23%	9.102	78.32
C	March 31, 2045	103.211	4.11%	18.112	323.11

The fund manager is considering purchasing \$10 million (face value) of bonds B at the cost of \$10.932 million. The fund manager expects the interest volatility to increase and hence wants to maximize his returns. However, another fund manager makes the suggestion that instead of investing in bond B, the fund manager should invest in a combination of bonds A and C. The fund manager has a surplus of \$10.932 million and wants the duration of the portfolio to be equal to that of bond B. Given that the fund manager expects increased interest volatility, the fund manager should invest in:

- A. \$10.932 million of Bond B.
- B. \$10.932 million of Bond C.
- C. \$5.12 million of bond A and \$5.81 of bond C.
- D. \$7.14 million of bond A and \$3.79 of bond C.

The correct answer is **D**.

The fund manager must invest in such a bond/portfolio which has the highest convexity, as higher convexity will help the fund manager to maximize return due to interest rate volatility.

Let the investments in A and C be x and y respectively. As per the constraint of having a duration of the portfolio equal to that of bond B (9.102 years):

$$x + y = 10.932$$

$$\left(\frac{1}{10.932}\right) * (x * 4.321 + y * 18.112) = 9.102$$

Solving for x and y:

$$\begin{aligned} x &= \$7.14 \\ y &= \$3.79 \end{aligned}$$

$$\text{Portfolio convexity} = \left(\frac{7.14}{10.932}\right) * 34.09 + \left(\frac{3.79}{10.932}\right) * 323.11 = 134.29$$

As the convexity of the portfolio is higher than the convexity of bond B and the portfolio meets the duration constraints (as it is equal to bond B), the preferred investment is in a portfolio consisting of bond A and C.

Q.1185 Data on three bonds are given below. Assume the current date is March 31, 2015.

Bond	Maturity	Price	Yield	Duration	Convexity
A	March 31, 2020	110.321	2.32%	4.321	34.09
B	March 31, 2030	109.320	3.23%	9.102	78.32
C	March 31, 2045	103.211	4.11%	18.112	323.11

The return generated by a portfolio consisting of \$7.14 million of bond A and \$3.79 million of bond C in the case the interest rate remains constant is:

A. 0.0232

B. 0.0411

C. 0.0294

D. 0.0151

The correct answer is **C**.

$$\begin{aligned}\text{The return generated in case the rate remains constant} &= \left(\frac{7.14}{10.932}\right) * 2.32\% + \left(\frac{3.79}{10.932}\right) * 4.11\% \\ &= 2.94\%\end{aligned}$$

Q.2791 Suppose the yield on a zero-coupon bond declines from 5.00% to 4.99%, and the price of the bond increases from \$50.0 to \$51.5. Compute the DV01.

A. \$0.0512

B. \$1.5

C. \$2.5

D. \$0.1

The correct answer is **B**.

$$DV01 = -\frac{\Delta BV}{10,000 \times \Delta y}$$

Where:

ΔBV = change in bond value

Δy = change in yield

$$DV01 = -\frac{51.5 - 50.0}{10,000 \times -0.0001} = 1.5$$

Q.2792 Ted Oster wants to calculate the DV01 of a new position in a bond with a face value of \$1,000,000. The bond was bought today for \$84.102 for \$100 face value. Oster knows that the Macaulay duration is 5.25.

What is the DV01 of the position if the bond has a yield to maturity of 10% per annum compounded annually?

- A. \$401.39
- B. \$405.40
- C. \$483.47
- D. \$492.73

The correct answer is **A**.

$$\begin{aligned}\text{Modified duration} &= \frac{5.25}{(1 + 10\%)} = 4.7727 \\ \text{DV 01} &= \text{Mod dur} \times 0.0001 \times \text{Bond value} \\ &= 4.7727 \times 0.0001 \times \left(\frac{84.102}{100}\right) \times \$1,000,000 \\ &= \$401.39\end{aligned}$$

Note that the reason why we multiply by 0.0001 is to get the dollar change for a 1% change in yield in terms of basis points.

Q.3315 Peter Drury, a risk manager at Capital Bank, is evaluating the price sensitivity of an investment-grade callable bond using the bank's valuation system. The table below gives a breakdown of the bond and the embedded option. The current interest rate environment is flat at 5%.

Interest rate level	Bond value per 100 USD face value	Call option value per 100 USD face value
4.95%	98.2520	1.7480
5.0%	98.0000	1.5000
5.05%	97.7500	1.3225

The DV01 of a comparable bond with no embedded options and with the same maturity and coupon rate as the callable bond is closest to:

- A. 0.09275
- B. 0.08015

C. 0.1285

D. 0.075

The correct answer is **A**.

The call option negatively affects the price of a bond because investors lose future coupon payments if the call option is exercised by the issuer.

The value of a plain-vanilla bond can be given by:

$$\text{Price (plain-vanilla bond)} = \text{price (callable bond)} + \text{price (call option)}$$

Thus, the price of the plain-vanilla bond with no embedded options at a rate of 5.0% would be 99.5000, the price at a rate of 4.95% would be 100.0000, and the price at a rate of 5.05% would be 99.0725.

DV01 is the dollar value change in price (value) of a fixed income instrument, such as a bond, in response to a change in the yield of the instrument. It is given by:

$$DV01 = -\frac{\Delta P}{10,000 \times \Delta Y}$$

where:

ΔP = change in price, and

ΔY = change in yield

Thus

$$DV01 = -\frac{99.0725 - 100}{10,000 \times (0.0505 - 0.0495)} = -\frac{-0.9275}{10} = 0.09275$$

Q.3319 The current price of a bond is 950. The duration of the bond is 8.83 and the convexity of the bond is 6.43. What is the change in the price of the bond for a one percentage point increase in the interest rate?

- A. -38.0
- B. 52.25
- C. 22.8
- D. -83.58

The correct answer is **D**.

For a one basis point of change in the interest rate, the formula for the bond price change is:

$$\begin{aligned}\text{Change in price} &= [-\text{Modified Duration} \times \text{Bond Price} \times \text{Change in yield}] + \left[\frac{1}{2} \times \text{Convexity} \times \text{Bond Price} \times (\text{Change in yield})^2\right] \\ &= -8.83 \times 0.01 \times 950 + \frac{6.43}{2} \times 950 \times 0.01^2 = -83.5796\end{aligned}$$

Q.3427 Given the following portfolio of bonds:

Bond	Price	Par Amount Held (USD Million)	Modified Duration
A	101.22	4	2.45
B	85.53	6	4.25
C	115.50	9	7.61

What is the value of the portfolio's DV01 (dollar value of a basis point)?

- A. \$10,960
- B. \$11,000
- C. \$11,060
- D. \$12,600

The correct answer is **C**.

$$\text{Portfolio DV01} = \text{Portfolio Modified duration} \times \text{Portfolio Market Value} \times 0.0001$$

But we have to compute the portfolio modified duration first:

Modified duration of the portfolio = Weighted Average of Modified Duration of Individual Bonds is

$$= w_1 D_1 + w_2 D_2 + \dots + w_k D_k$$

where :

$$w_i = \frac{\text{Market value of bond } i}{\text{Portfolio market value}}$$

D_i = Modified duration of bond i

k = total no. of bonds in the portfolio

Based on the table above, these are the values for bonds A, B, and C:

Bond	Value	Weight in the Portfolio
A	$= 101.22 \times \frac{4,000,000}{100} = 4,048,800$	$= \frac{4,048,800}{19,575,600} = 0.21$
B	$= 85.53 \times \frac{6,000,000}{100} = 5,131,800$	$= \frac{5,131,800}{19,575,600} = 0.26$
C	$= 115.5 \times \frac{9,000,000}{100} = 10,395,000$	$= \frac{10,395,000}{19,575,600} = 0.53$
Portfolio	$= 4,048,800 + 5,131,800 + 10,395,000$ $= 19,575,600$	

$$\text{Portfolio Modified Duration} = 0.21 \times 2.45 + 0.26 \times 4.25 + 0.53 \times 7.61 = 5.65$$

Therefore,

$$\text{Portfolio DV01} = 5.65 \times 19,575,600 \times 0.0001 = \$11,060.2$$

Q.3428 The current interest rate environment in a certain developing economy is flat at 5%. A risk manager has compiled the following data regarding a callable bond in two other interest rate environments (all values in USD per USD 100 face value):

Level of interest rate	Callable bond	Call option
4.96%	102.00	2.0893
5.00%	102.4465	2.0255
5.04%	101.2111	2.0021

The convexity of the callable bond is closest to:

- A. -102,608
- B. -51,304
- C. -100,020
- D. -103,000

The correct answer is **A**.

Convexity is the second derivative of the formula for change in bond prices with a change in interest rates.

$$\text{Convexity} = \frac{1}{P} \times \frac{d^2P}{dy^2}$$

This approach can be quite rigorous in terms of computation, so you need to estimate convexity using the formula:

$$\text{Convexity} = \frac{P_{-\Delta y} + P_{+\Delta y} - 2P_0}{P_0 \times \Delta y^2}$$

Where:

$P_{-\Delta y}$ =price estimate if yield decreases by a given amount, Δy

$P_{+\Delta y}$ =price estimate if yield increases by a given amount, Δy

P_0 =initially observed bond price (at the flat rate)

Δy =change in yield in one step, expressed in decimal form (0.04% in this case)

$$\begin{aligned} \text{Convexity} &= \frac{102.00 + 101.2111 - 2 \times 102.4465}{102.4465 \times 0.0004^2} \\ &= -\frac{1.6819}{0.00001639} = -102,608 \end{aligned}$$

Q.4611 A bank has a position of USD 2 million with a duration of 10 years. To completely hedge its position, the bank takes a short position of USD 1.4 million in bond B. What is the duration of bond B?

- A. 10
- B. 11
- C. 13
- D. 14

The correct answer is **D**.

The position required in the bond to hedge a position is given by:

$$P = -\frac{V D_V}{D_B}$$

Where

V = a value of the position

D_V = duration of the position

D_B = duration of the bond

So in this case,

$$\begin{aligned} P &= -\frac{V D_V}{D_B} \\ \Rightarrow -1.4 &= -\frac{2 \times 10}{D_B} \\ \therefore D_B &= \frac{20}{1.4} \approx 14 \end{aligned}$$

Q.4613 Suppose the yield on a zero-coupon bond declines from 7.00% to 6.95% and the price of the zero increases from \$202.45 to \$203.87. What is the value of DV01?

A. 0.342

B. 0.324

C. 0.284

D. 0.242

The correct answer is **C**.

DV01 is defined as:

$$DV01 = -\frac{\Delta P}{\Delta r}$$

Where

Δr = the size of a parallel shift in the interest rate term structure measured in **basis points**

ΔP = resultant change in the value of the position being considered

$$\Rightarrow DV01 = -\frac{203.87 - 202.45}{(-0.0005) \times 10,000} = \frac{1.42}{5} = 0.284$$

Q.4614 The Macaulay duration of a coupon bond is 10.25. If the yield on the bond is 8% compounded semi-annually, what is the corresponding modified duration for the bond?

- A. 10.25
- B. 9.86
- C. 11.45
- D. 9.54

The correct answer is **B**.

Denoting the bond yield by y , modified duration is calculated by dividing Macaulay duration by $(1 + \frac{y}{2})$. So in this case,

$$\text{Modified duration} = \frac{10.25}{(1 + \frac{0.08}{2})} = 9.86$$

Q.4616 A bond portfolio consists of three bonds:

- Bond A worth 9 million with a duration of 4;
- Bond B worth 5 million with a duration of 6; and
- Bond C worth 6 million with a duration of 7.

What is the duration of the portfolio?

- A. 7.8
- B. 6.6
- C. 5.4
- D. 5.7

The correct answer is **C**.

Recall that the portfolio duration is calculated as the **weighted sum** of the individual durations where the weight attached to each security is equal to its value as a percentage of total portfolio value.

The total value of the portfolio is $9 + 5 + 6 = \text{USD } 20$ million. The duration of the bond is given by:

$$\frac{9}{20} \times 4 + \frac{5}{20} \times 6 + \frac{6}{20} \times 7 = 5.4$$

Q.4617 A position worth USD 3 million has a duration of 4 and a convexity of 5. What is the estimated change in the position for a five-basis-point increase in all rates?

- A. Increase by USD 5,998.125
- B. Decrease by USD 5,998.125
- C. Increase by USD 5,546.670
- D. Decrease by USD 5,546.670

The correct answer is **B**.

By using convexity and duration, the price change estimation is given by:

$$\begin{aligned}\Delta P &= -DP \Delta r + \frac{1}{2} CP (\Delta r)^2 \\ &= -4 \times 3 \times 0.0005 + \frac{1}{2} \times 5 \times 3 \times (0.0005)^2 \\ &= -0.005998125 = -\text{USD } 5,998.125\end{aligned}$$

Reading 59: Modeling and Hedging Non-Parallel Term Structure Shifts

Q.1148 All the following are assumptions of Key Rate Shifts, EXCEPT:

- A. Rates can be determined as a function of a relatively small number of key rates.
- B. There is a parallel shift of rates across the key rates.
- C. There is a linear shift of rates across the term structure.
- D. The rate of a given term is not affected by its neighboring key rates.

The correct answer is **D**.

The statement that 'The rate of a given term is not affected by its neighboring key rates' is incorrect in the context of Key Rate Shifts. The fundamental premise of Key Rate Shifts is that rates can be determined as a function of a relatively small number of key rates. This implies that the rate of a given term is influenced by its neighboring key rates. For example, the 5-year rate could be a function of the 2-year and 3-year rates. Therefore, the assertion that a given term's rate is unaffected by its neighboring key rates contradicts the basic assumptions of Key Rate Shifts.

Choice A is incorrect. The assumption that rates can be determined as a function of a relatively small number of key rates is indeed one of the fundamental principles behind Key Rate Shifts. This concept allows for the simplification and modeling of complex interest rate structures.

Choice B is incorrect. The assumption that there is a parallel shift of rates across the key rates aligns with the assumptions of Key Rate Shifts. In this model, all key rates are assumed to move in unison, which simplifies analysis and prediction.

Choice C is incorrect. The assumption that there is a linear shift of rates across the term structure also aligns with Key Rate Shifts' assumptions. This implies that changes in short-term interest rates have an equal impact on long-term interest rates, which aids in creating simplified models for risk management purposes.

Things to Remember

- Key Rate Shifts are a method used to analyze and model changes in interest rates across different maturities.
- Key Rate Shifts assume that rates can be determined as a function of a relatively small number of key rates, simplifying the analysis of interest rate movements.
- Parallel shift of rates across key rates implies that all key rates move in the same direction by the same amount, simplifying the modeling process.

- Linear shift of rates across the term structure means that changes in short-term rates have an equal impact on long-term rates, aiding in risk management modeling.
- Key Rate Shifts are commonly used in the analysis of yield curves and interest rate risk management.

Q.2606 A risk manager prepares a presentation on the interest rate risk of the bank's bond portfolio. The table below shows the value of the portfolio in case of shifts in key rates by one basis point and corresponding key rate duration.

	Value	Key Rate Duration
Initial Curve	500.425	
2-year Shift	500.227	3.957
5-year Shift	500.201	4.476
10-year Shift	?	28.476
30-year Shift	499.500	18.484
Total		55.393

What is the value of the portfolio in the case of a 10-year shift?

- A. 501.85
- B. 500.043
- C. 499.5
- D. 499.0

The correct answer is **D**.

$$\text{Key rate duration} = -\frac{1}{p} \frac{\partial p}{\partial y}$$

Where:

P = initial bond value/price

∂p = change in bond price

∂y = change in yield

Let the value of the bond portfolio in the case of a 10-year shift be "x"

Thus, the key rate duration with respect to the 10-year shift is calculated as:

$$\begin{aligned}
\text{Key rate duration} &= -\frac{1}{500.425} \frac{x - 500.425}{0.01\%} \\
\Rightarrow 28.476 &= -\frac{1}{500.425} \times \frac{x - 500.425}{0.01\%} \\
\Rightarrow 28.476 \times 500.425 &= -\frac{x - 500.425}{0.01\%} \\
\Rightarrow 28.476 \times 500.425 \times 0.01\% &= -(x - 500.425) \\
\Rightarrow 28.476 \times 500.425 \times 0.01\% &= (500.425 - x) \\
\Rightarrow 500.425 - 28.476 \times 500.425 \times 0.01\% &= x \\
\Rightarrow x &= 499.000
\end{aligned}$$

Q.2607 Frank Capper wants to estimate the impact of key rate changes on the value of C-STRIPs. Capper uses 2-year, 5-year, 10-year, and 30-year key rates in his analysis. He also wants to incorporate an unexpected 50 basis point shock of the 10-year rate in the model.

Which of the following rates will be affected by the change of the 10-year rate key rate?

- A. 2-year and 5-year
- B. 30-year rate
- C. 5-year and 30-year rates
- D. All of the rates will be affected

The correct answer is **C**.

The 5-year and 30-year rates will be affected by the change of the 10-year rate key rate. This is because each key rate influences the yields from the term of the previous key rate (or zero) to the term of the next key rate. In this case, the 10-year key rate is sandwiched between the 5-year and 30-year key rates. Therefore, any change in the 10-year key rate will directly impact the 5-year and 30-year rates. This is a fundamental concept in the analysis of key rate changes and their impact on the value of financial instruments such as C-STRIPs.

Choice A is incorrect. The 2-year and 5-year rates are not directly impacted by a change in the 10-year key rate. Key rates are specific points on the yield curve, and changes in one key rate do not necessarily affect other key rates, especially those with shorter durations.

Choice B is incorrect. While it's true that a change in the 10-year key rate might have some influence on the 30-year rate due to their relative closeness on the yield curve, this choice ignores the potential impact on the 5-year rate which can also be affected.

Choice D is incorrect. Although all rates could potentially be influenced by market dynamics and shifts in economic conditions, a shock to a specific key rate (in this case, the 10-year) does not automatically translate into changes across all other durations. In this context, only those

durations that are close to or longer than the shocked duration would typically be affected.

Things to Remember

- Key rate changes can have a significant impact on the value of fixed income securities and derivatives.
 - Understanding the relationship between different key rates and their influence on each other is crucial for risk management and valuation.
 - Duration and convexity are important concepts to consider when analyzing the impact of interest rate changes on bond prices.
 - Yield curves provide valuable information about the term structure of interest rates and can help in making investment decisions.
-

Q.2610 A trader wants to hedge the 2-year and 5-year rates exposure of a portfolio. To perform the hedge, the trader can use Bond 1 and Bond 2 presented below.

	Key Rate '01 (per 100 face amount)	
Hedging Bonds	<u>2-year</u>	<u>5-year</u>
Bond 1	0.0080	—
Bond 2	0.0099	0.0160

	Key Rate '01(\$)	
	<u>2-year</u>	<u>5-year</u>
Fixed Income Portfolio	250	320

What is the face value of Bond 1 required to perform the hedge?

- A. \$200,000
- B. \$650,000
- C. \$950,000
- D. \$1,250,000

The correct answer is **B**.

Let's denote $F(i)$ as the face amount of hedging bond i that we need to sell.

We have the following equations:

$$(1): F(2) * \frac{0.0099}{100} + F(1) * \frac{0.0080}{100} = \$250$$

$$(2): F(2) * \frac{0.0160}{100} = \$320$$

$$\text{From (2)} \rightarrow F(2) = \$2,000,000$$

$$\text{From (1) and (2)} \rightarrow F(1) = \$650,000$$

Q.2611 The risk manager of a regional bank is concerned with possible shocks in short-term rates. He wants to find a transaction that will completely eliminate the 2-year exposure and decrease by half the current 5-year exposure. A trader proposes the following two bonds as hedging instruments:

	Key Rate '01 (per 100 face amount)	
Hedging Bonds	<u>2-year</u>	<u>5-year</u>
Bond 1	0.0010	0.0050
Bond 2	0.0015	0.0025

	Key Rate '01(\$)	
	<u>2-year</u>	<u>5-year</u>
Fixed Income Portfolio	1,000	4,000

What is the trader's hedging transaction?

- A. Sell \$60,000,000 of bond 1; Sell \$10,000,000 of bond 2.
- B. Sell \$10,000,000 of bond 1; Sell \$60,000,000 of bond 2.
- C. Sell \$10,000,000 of bond 1; Buy \$60,000,000 of bond 2.
- D. Buy \$60,000,000 of bond 1; Sell \$60,000,000 of bond 2.

The correct answer is **B**.

Let's denote $F(i)$ as the face amount of hedging bond i that we need to sell.

To completely hedge the 2-year exposure, $F(1)$ and $F(2)$ should solve the following equation:

$$F(2) * \frac{0.0015}{100} + F(1) * \frac{0.0010}{100} = \$1000 \dots (1)$$

To decrease by half the 5-year exposure of \$4,000, $F(1)$ and $F(2)$ should solve the following equation:

$$F(2) * \frac{0.0025}{100} + F(1) * \frac{0.0050}{100} = \$2,000 \dots (2)$$

Solving equation (1) and (2), simultaneously, we get,
 $F(2) = \$60,000,000$ and $F(1) = \$10,000,000$

Q.2612 The head of the trading department of a bank suggests speculating on the interest rate curve by eliminating the exposure of the bond portfolio to long-term rates (10-year and 30-year rates) and increasing the exposure to medium-term rates. The table below represents the key rate '01s of the current portfolio and corresponding hedging instruments.

	Key Rate '01 (per 100 face amount)		
Hedging Bonds	<u>5-year</u>	<u>10-year</u>	<u>30-year</u>
Bond 1	0.0050	–	–
Bond 2	0.0170	0.0250	–
Bond 3	0.0100	0.0300	0.0350

	Key Rate '01(\$)		
	<u>5-year</u>	<u>10-year</u>	<u>30-year</u>
Fixed Income Portfolio	1,000	2,000	1,500

What will be the portfolio's 5-year exposure after hedging 10-year and 30-year exposures?

- A. \$85.70
- B. \$171.43
- C. \$8,570.00
- D. \$17,143.00

The correct answer is **A**.

Let's denote $F(i)$ as the face amount of hedging bond i that we need to sell.

We have the following equations:

$$(1): F(3) * \frac{0.0300}{100} + F(2) * \frac{0.025}{100} = \$2,000$$

$$(2): F(3) * \frac{0.0350}{100} = \$1,500$$

$$\text{From (3)} \rightarrow F(3) = \$4,285,714$$

$$\text{From (2) and (3)} \rightarrow F(2) = \$2,857,143$$

$$\text{Finally, the portfolio's 5-year exposure} = \$1000 - F(2) * \frac{0.0170}{100} - F(3) * \frac{0.0100}{100} = \$85.70$$

Note. In this scenario, the aim is to neutralize the interest rate risk of the bond portfolio associated with the 10-year and 30-year key rates, while enhancing the exposure to the 5-year key rate. The key rate '01' for each bond signifies the change in bond price for a 1 basis point (0.01%) shift in the respective yield curve segment. By selling appropriate face amounts of

hedging bonds, we can counteract the portfolio's sensitivity to movements in specific sections of the yield curve. The equations are set up to match the portfolio's exposures (in terms of dollar value) to those of the hedging instruments. By solving these equations, we determine the face amounts of the hedging bonds to be sold. The resultant 5-year exposure is then calculated by adjusting the portfolio's original 5-year exposure for the impact of these hedging bonds. This approach allows for effective control the bond portfolio's exposure to specific interest rate movements, based on the market outlook.

Q.2613 The risk manager at a regional bank is trying to interpret the results of an interest rate curve shocks simulation. The table below represents the key rate '01s for the fixed income portfolio of the bank.

	Value	Key Rate '01
Initial Curve	2,000.000	
2-year Shift	1,998.500	1.500
5-year Shift	1,998.300	1.700
10-year Shift	1,999.000	1.000
30-year Shift	1,998.000	2.000
Total		6.200

What is the approximate value of the portfolio in the case of a 5 basis point increase of the 2-year rate and a 10 basis point increase of the 30-year rate?

- A. 1,972.500
- B. 1,980.000
- C. 1,992.500
- D. 1,996.500

The correct answer is **A**.

To determine the approximate value of the portfolio after a 5 basis point increase in the 2-year rate and a 10 basis point increase in the 30-year rate, we use the Key Rate '01s provided. The Key Rate '01s measure the change in the portfolio value for a 1 basis point shift in the interest rate at specific maturities.

The impact of a 5 basis point increase in the 2-year rate is calculated as:

$$\text{Change in value} = 1.500 \times 5 = 7.5$$

The impact of a 10 basis point increase in the 30-year rate is calculated as:

$$\text{Change in value} = 2.000 \times 10 = 20.0$$

The total change in portfolio value is the sum of these impacts:

$$\text{Total change} = 7.5 + 20.0 = 27.5$$

Given the initial portfolio value of GBP 2,000.000, the new portfolio value is:

$$\text{New portfolio value} = 2,000.000 - 27.5 = 1,972.500$$

The approximate value of the portfolio in the case of a 5 basis point increase in the 2-year rate and a 10 basis point increase in the 30-year rate is: 1,972.500}

Q.2614 Anna White, head of the risk management department of a regional bank, asks one of the interns to analyze the volatility of the bank's medium-term fixed-income portfolio with a value of \$10,000. For the analysis of the portfolio, the intern uses 2-year rates (annual volatility of 20%) and 5-year rates (annual volatility of 30%). He calculates that the key rate '01s for 2-year and 5-year shifts are 2 per \$ 100 and 5 per \$ 100 respectively.

What is the volatility of the portfolio if the correlation between 2-year and 5-year rates is 0.45?

- A. \$155.24
- B. \$171.76
- C. \$22,570.00
- D. \$29,500

The correct answer is **B**.

$$\begin{aligned} \text{Portfolio variance} &= \left(\frac{2}{100} \times 0.20 \times 10000\right)^2 + \left(\frac{5}{100} \times 0.30 \times 10000\right)^2 \\ &\quad + 2 \times \frac{2}{100} \times \frac{5}{100} \times 0.20 \times 10000 \times 0.30 \times 10000 \times 0.45 \\ &= 29,500 \\ \text{Portfolio volatility} &= \$29,500^{0.5} = \$171.76. \end{aligned}$$

Q.2615 Initially, a fixed income portfolio of an investment bank had the following key rate '01s:

	Value	Key Rate
Initial Curve	10,000	
2-year Shift	9,990	10
5-year Shift	9,985	15
10-year Shift	9,981	19
30-year Shift	9,984	16
Total		60

After a recommendation from the risk management department, a trader completely hedged the 30-year exposure with a bond that had the following characteristics:

Hedging Bond	Key Rate '01 (per USD 100 face value)
Initial Curve	
2-year Shift	—
5-year Shift	—
10-year Shift	3
30-year Shift	4

How much will the bank save, if immediately after the hedge, the interest rate curve experienced a 20 basis point upward parallel shift?

- A. 320
- B. 360
- C. 480
- D. 560

The correct answer is **D**.

To hedge the 30-year exposure, the trader will need to sell $\$400 (= \$\frac{16}{4} * 100)$ face value of hedging bond. The sale of hedging bond will also impact the 10-year exposure by $-\$12 (= -\frac{3}{100} * \$400)$.

The portfolio key rate '01s after hedge will become:

Hedging Bond	Key Rate '01
Initial Curve	
2-year Shift	10
5-year Shift	15
10-year Shift	7
30-year Shift	—
Total	32

The portfolio value drop without hedging transaction = $20 * 60 = 1,200$

The portfolio value drop with hedging transaction = $20 * 32 = 640$

The hedging transaction will save the bank $560 (= 1,200 - 640)$.

Q.3430 Kelvin Mertens, FRM, regularly participates in bond trading in the US. He is using key rate analysis to assess the effect of yield changes on bond prices. He finds that the 20-year yield has increased by 10 basis points. Moreover, this shock decreases linearly to zero for the 30-year yield. What is the effect of this shock on the 26-year yield?

- A. Increase of zero basis points
- B. Increase of six basis points
- C. Increase of ten basis points
- D. Increase of four basis points

The correct answer is **D**.

To determine the effect of the 20-year yield increasing by 10 basis points on the 26-year yield, you can use key rate analysis. In this case, the increase in yield for the 20-year bond is linearly reduced to zero for the 30-year bond.

This means that for each year between the 20-year and 30-year maturities, the increase in yield decreases by an equal amount. Since there are 10 years between the 20-year and 30-year maturities, you can calculate the decrease in yield increase per year as 10 basis points divided by 10 years, which is 1 basis point per year.

So, for a 26-year bond (which is 6 years away from the 20-year bond), the effect of the shock on its yield would be 6 basis points. However, since the shock is an increase, you would subtract 6 basis points from the 26-year yield.

Therefore, the effect of this shock on the 26-year yield is:

26-year yield increase = 10 basis points (from the 20-year yield) - 6 basis points (adjustment for the 6-year difference) = 4 basis points.

Things to Remember

- Key rate analysis is a technique used to assess the impact of changes in interest rates on bond prices across different maturities.
- Yield curve is a graphical representation of the yields on bonds of similar credit quality but different maturities.
- Duration is a measure of the sensitivity of the price of a bond to a change in interest rates.
- Convexity is a measure of the curvature of the price-yield relationship of a bond and helps to refine the estimate of the bond's price change given a change in yield.

Q.3431 The following table provides the initial price of a C-STRIP and its present value after the application of a one basis point shift in four key rates.

	Value
Initial value	26.14485
2-year shift	26.14582
5-year shift	26.14885
10-year shift	26.14885
30-year shift	26.02192

The key rate '01 with respect to the 10-year shift is closest to:

- A. -0.004
- B. -0.04
- C. -4
- D. -0.4

The correct answer is **A**.

$$\text{Key rate '01} = -\frac{\Delta BV}{10,000 \times \Delta y}$$

Where:

ΔBV =change in bond value

Δy =change in yield (0.01%)

The change in bond value here is measured in reference to the initial bond value.

$$= -\frac{26.14885 - 26.14485}{10,000 \times 0.01\%} = -0.004$$

Q.3432 The following table provides the initial price of a C-STRIP and its present value after the application of a one basis point shift in four key rates.

	Value
Initial value	26.11485
2-year shift	26.11582
5-year shift	26.11885
10-year shift	26.13885
30-year shift	26.01192

The key rate duration with respect to the 30-year shift is closest to:

- A. 39
- B. 51
- C. 38
- D. 10

The correct answer is **A**.

First, determine the corresponding key rate '01:

$$\text{Key rate '01} = -\frac{\Delta BV}{10,000 \times \Delta y}$$

Where:

ΔBV =change in bond value

Δy =change in yield (0.01%)

The change in bond value here is measured in reference to the initial bond value.

$$= -\frac{26.01192 - 26.11485}{10,000 \times 0.01\%} = 0.1029$$

Next, you can now compute the key rate duration using the formula:

$$DV01 = \text{Duration} \times 0.0001 \times \text{Bond value}$$

Thus,

$$\begin{aligned} \text{Duration} &= \frac{DV01}{0.0001 \times \text{Bond value}} \\ &= \frac{0.1029}{0.0001 \times 26.11485} \\ &= 39.41 \end{aligned}$$

Alternatively, recall that:

$$\text{Key rate duration} = -\frac{1}{P} \left(\frac{\partial P}{\partial y} \right)$$

Thus, the key rate duration with respect to the 30-year shift can be calculated as:

$$\text{Key rate duration} = -\left(\frac{1}{26.11485} \right) \times \left(\frac{26.01192 - 26.11485}{0.01\%} \right) = 39.41$$

Reading 60: Binomial Trees

Q.1205 Australian Financial Associates is holding the non-dividend paying stock of Neevan Holdings which is trading at USD 10. The continuously compounded risk-free rate is 5 percent per annum, and the annual standard deviation of the stock is 20 percent. What is the value of a 2-year European call option with a strike price of USD 10 using a two-period binomial model?

- A. USD 1.64
- B. USD 1.48
- C. USD 1.58
- D. USD 1.69

The correct answer is **B**.

To price options in the binomial model, we need:

U = size of the up move factor $= e^{\sigma\sqrt{\Delta t}}$

D = size of the down move factor $= e^{-\sigma\sqrt{\Delta t}} = \frac{1}{U}$

σ is the annual volatility of the underlying asset's returns and t is the length of the step in the binomial model ($t = 1$ in this case).

π_u = probability of an up move $= \frac{e^{r\Delta t} - D}{U - D}$

π_d = probability of a down move $= 1 - \pi_u$

Working with the data provided,

$U = 1.2214$ and $D = 0.8187$

$$\pi_u = 0.5775 \text{ and } \pi_d = 0.4225$$

Let S represent the price of the stock and f represent the value of the call:

Stock Price	Value of the Call
$S_{uu} = \$10 * 1.2214 * 1.2214 = \14.92	$f_{uu} = \max(\$14.92 - \$10, \$0) = \4.92
$S_{ud} = \$10 * 1.2214 * 0.8187 = \10	$f_{ud} = \max(\$10 - \$10, \$0) = \0
$S_{du} = \$10 * 0.8187 * 1.2214 = \10	$f_{du} = \max(\$10 - \$10, \$0) = \0
$S_{dd} = \$10 * 0.8187 * 0.8187 = \6.70	$f_{dd} = \max(\$6.70 - \$10, \$0) = \0

As the table shows, the value of the call will be positive only when the stock moves upward twice.

The expected value of the call 2 years from now is given by:

$$\begin{aligned}
&0.5775 \times 0.5775 \times \$4.92 + 0.5775 \times 0.4225 \times \$0 \\
&+ 0.4225 \times 0.5775 \times \$0 + 0.4225 \times 0.4225 \times \$0 \\
&= \$1.64
\end{aligned}$$

$$\text{Value of the option today} = \frac{1.64}{e^{0.05 \times 2}} = \$1.48$$

Q.1210 David Yung, an analyst working at the New Zealand Bank, uses Girsanov's Theorem to study portfolios in a risk-neutral world and in the real world. The Girsanov's Theorem states that:

- A. When we move from the risk-neutral world to the real world, the expected return from the stock price changes, but its volatility remains the same.
- B. When we move from the risk-neutral world to the real world, the expected return from the stock price remains the same, but its volatility changes.
- C. When we move from the risk-neutral world to the real world, both the expected return from the stock price and its volatility remains the same.
- D. When we move from the risk-neutral world to the real world, both the expected return from the stock price and its volatility change.

The correct answer is **A**.

Girsanov's Theorem, a fundamental concept in financial mathematics, states that when we transition from a risk-neutral world to the real world, the expected return from the stock price changes, but its volatility remains the same. In a risk-neutral world, all assets are assumed to grow at the risk-free rate, and risk preferences do not affect asset prices. However, in the real world, investors demand a risk premium for holding risky assets, which results in a higher expected return. Despite this change in expected return, the volatility of the stock price, which is a measure of the degree of variation of a trading price series over time, remains unchanged. This is because volatility is a function of the underlying uncertainties and risks inherent in the asset, which are not affected by the transition from a risk-neutral to a real-world scenario.

Choice B is incorrect. According to Girsanov's Theorem, when transitioning from a risk-neutral world to the real world, the expected return of a stock price changes due to the introduction of risk premiums. However, its volatility remains constant as it is not affected by this transition. Therefore, it is incorrect to say that only the volatility changes while the expected return remains the same.

Choice C is incorrect. This choice contradicts Girsanov's Theorem which states that in moving from a risk-neutral environment to a real-world one, there will be an alteration in expected returns due to risk premiums but no change in volatility. Hence, it's not accurate to assert that both expected return and volatility remain unchanged.

Choice D is incorrect. As explained above in choices B and C explanations, Girsanov's Theorem

implies that only the expected return changes when transitioning from a risk-neutral world to the real world while its volatility stays constant. Thus stating both parameters change isn't correct.

Things to Remember

- Risk-neutral world: In a risk-neutral world, all assets are assumed to grow at the risk-free rate, and risk preferences do not affect asset prices.
 - Real world: In the real world, investors demand a risk premium for holding risky assets, resulting in a higher expected return compared to the risk-neutral world.
 - Volatility: Volatility is a measure of the degree of variation of a trading price series over time and is a function of the underlying uncertainties and risks inherent in the asset.
 - Risk premium: The additional return that investors require for holding a risky asset compared to a risk-free asset.
 - Girsanov's Theorem: A fundamental concept in financial mathematics that explains the relationship between expected return and volatility when transitioning from a risk-neutral world to the real world.
-

Q.1211 Australian Financial Associates is holding the non-dividend paying stock of Neevan holdings which is trading at USD 10. The continuously compounded risk-free rate is 5 percent per year, and the annual standard deviation of the stock is 20%. What is the value of a 2-year European put option with a strike price of USD 10 using a two-period binomial model?

- A. USD 0.5325
- B. USD 2.4356
- C. USD 0.6884
- D. USD 2.3456

The correct answer is **A**.

The up-move and down-move factors are given by:

$$u = e^{\sigma\sqrt{\Delta t}} = e^{0.20 \times \sqrt{1}} = 1.2214$$

$$d = \frac{1}{u} = \frac{1}{1.2214} = 0.8187$$

Now, we can calculate the risk-neutral probability of an up-move:

$$p = \frac{e^{r\Delta t} - d}{u - d} = \frac{e^{0.05 \times 1} - 0.8187}{1.2214 - 0.8187} = 0.5775$$

Year 0	Year 1	Year 2
		$S_{uu} = \$12.214 \times 1.2214 = \14.918
	$S_u = \$10 \times 1.2214 = \12.214	$S_{ud} = \$12.214 \times 0.8187 = \10
$S_0 = \$10$	$S_d = \$10 \times 0.8187 = \8.187	$S_{dd} = \$8.187 \times 0.8187 = \6.703

The put option payoff at the 2-year node is given by:

$$f_{uu} = \max(K - S, 0) = \max(10 - 14.918, 0) = \$0$$

$$f_{ud} = \max(10 - 10, 0) = \$0$$

$$f_{dd} = \max(10 - 6.703, 0) = \$3.297$$

The put option payoff at the 1-year node is given by:

$$\begin{aligned} f_u &= e^{-r\Delta T} [p * f_{uu} + (1 - p) * f_{ud}] \\ &= e^{-0.05 \times 1} [0.5775 \times 0 + (1 - 0.5775) * 0] = 0 \end{aligned}$$

$$\begin{aligned} f_d &= e^{-r\Delta T} [p * f_{ud} + (1 - p) * f_{dd}] \\ &= e^{-0.05 \times 1} [0.5775 \times 0 + (1 - 0.5775) \times 3.297] = 1.3250 \end{aligned}$$

Now, the value of the option today is given by:

$$\begin{aligned} f_0 &= e^{-r\Delta T} [p * f_u + (1 - p) * f_d] \\ &= e^{-0.05 \times 1} [0.5775 \times 0 + (1 - 0.5775) \times 1.3250] = \$0.5325 \end{aligned}$$

Q.1213 Rose Associates is holding stocks of Xerox limited. The current value of the stock is USD 100 and the current continuously compounded risk-free rate is 3 percent. The stock pays a dividend at a continuous dividend yield of 2 percent. The annual standard deviation of the stock is 9 percent. What is the risk-neutral probability of an up-move and down-move for a 1-year European call option on the stock?

- A. 0.68 and 0.32
- B. 0.78 and 0.22
- C. 1.09 and 0.91
- D. 0.53 and 0.47

The correct answer is **D**.

$$U = \text{size of the up-move factor} = e^{\sigma\sqrt{t}} = e^{0.09\sqrt{1}} = 1.094$$

$$D = \text{size of the down-move factor} = \frac{1}{U} = 0.9139$$

The risk-neutral probabilities of upward and downward movements:

$$\begin{aligned}\pi_u &= \text{Risk-neutral probability of an up-move} \\ &= \frac{(e^{(r-q)t} - D)}{(U - D)} \\ &= \frac{(e^{(0.03-0.02)*1} - 0.9139)}{(1.094 - 0.9139)} = 0.53\end{aligned}$$

$$\begin{aligned}\pi_d &= \text{Risk-neutral probability of a down-move} \\ &= 1 - \pi_u = 1 - 0.53 \\ &= 0.47\end{aligned}$$

Where r is the risk-free rate and q is the dividend rate

Q.3400 Willy Smith, FRM, has a two-year European put with $K = \$41$. The current price of the underlying is $\$40$. Over the past year, the stock has exhibited a standard deviation of 20%. The risk-free rate of return is 5%. Compute the value of the put today using a two-step Binomial.

- A. \$2.71
- B. \$3
- C. \$0
- D. \$16.1

The correct answer is **A**.

$$u = e^{\sigma\sqrt{\Delta t}} = e^{0.2 \times \sqrt{1}} = 1.22$$

$$d = e^{-\sigma\sqrt{\Delta t}} = e^{-0.2 \times \sqrt{1}} = 0.82$$

$$\pi_u = \frac{e^{rt} - d}{u - d} = \frac{e^{0.05 \times 1} - 0.82}{1.22 - 0.82} = 0.5782,$$

$$\pi_d = 1 - 0.5782 = 0.4218$$

Let S represent the price of the stock and f represent the value of the put

Stock Price	Option Payoff
$S_{uu} = \$40 \times 1.22 \times 1.22 = \59.54	$f_{uu} = \max(\$41 - \$59.54, 0) = \$0$
$S_{ud} = \$40 \times 1.22 \times 0.82 = \40	$f_{ud} = \max(\$41 - \$40, 0) = \$1$
$S_{du} = \$40 \times 0.82 \times 1.22 = \40	$f_{du} = \max(\$41 - \$40, 0) = \$1$
$S_{dd} = \$40 \times 0.82 \times 0.82 = \26.90	$f_{dd} = \max(\$41 - \$26.90, 0) = \$14.10$

The expected value of the put 2 years from now is given by:

$$\begin{aligned} &0.5782 \times 0.5782 \times \$0 + 0.5782 \times 0.4218 \times \$1 \\ &+ 0.4218 \times 0.5782 \times \$1 + 0.4218 \times 0.4218 \times \$14.10 \\ &= \$3 \end{aligned}$$

$$\text{Value of the put today} = \frac{\$3}{e^{0.05 \times 2}} = \$2.71$$

Q.3402 XYZ stock is a non-dividend-paying stock currently priced at $\$108$. According to analysis, the annual standard deviation of returns on XYZ stock is 8% and the risk-free rate on interest,

compounded continuously, is 5.5%. Using a two-period binomial model, compute the value of a 6-month American call option on XYZ stock with a strike price of \$110.

- A. \$2.43
- B. \$7.04
- C. \$2.98
- D. \$4.58

The correct answer is **C**.

up move factor = $u = e^{\sigma\sqrt{t}} = e^{0.08 \times \sqrt{0.25}} = 1.041$

down move factor = $d = \frac{1}{1.041} = 0.961$

probability of an up move = $\pi_u = \frac{e^{0.055 \times 0.25} - 0.961}{1.041 - 0.961} = 0.66$

probability of a down move = $1 - \pi_u = 0.34$

Let S represent the price of the stock and f represent the value of the call

Stock Price	Option Payoff
$S_u = \$108 \times 1.041 = \112.43	$f_u = \max(112.43 - 110, 0) = 2.43$
$S_d = \$108 \times 0.961 = \103.79	$f_d = \max(103.79 - 110, 0) = 0$
$S_{uu} = \$108 \times 1.041^2 = \117.04	$f_{uu} = \max(117.04 - 110, 0) = 7.04$
$S_{ud} = \$108 \times 1.041 \times 0.961 = \108.04	$f_{ud} = \max(108.04 - 110, 0) = 0$
$S_{du} = \$108 \times 0.961 \times 1.041 = \108.04	$f_{du} = \max(108.04 - 110, 0) = 0$
$S_{dd} = \$108 \times 0.961^2 = \99.74	$f_{dd} = \max(99.74 - 110, 0) = 0$

The \$110 call option is in the money when the stock price finishes at \$117.04 at which time the call has a value of \$7.04. At the end of 3 months(3-month node), the expected payoff on the option in the next 3 months, given an up move up to that point, is:

$$\frac{\$7.04 \times 0.66 + \$0 \times 0.34}{e^{0.055 \times 0.25}} = \$4.58$$

In this case, if the holder of the option chose to exercise early, they'd receive a maximum of $\max(112.43 - 110, 0) = 2.43$. Since $\$2.43 < \4.58 , it would not be optimal to exercise the option early. At the end of 3 months(3-month node), the expected payoff on the option in the next 3 months, given a down move up to that point, is:

$$\frac{\$0 \times 0.66 + \$0 \times 0.34}{e^{0.055 \times 0.25}} = \$0$$

If the holder of the option chose to exercise early (3 months following a down move), they'd receive a maximum of $\max(\$103.79 - 110, 0) = \0 . Again, it would not be optimal to exercise the option early. The value of the option today is:

$$\frac{\$4.58 \times 0.66 + \$0 \times 0.34}{e^{0.055 \times 0.25}} = \$2.98$$

Q.3403 You have been provided the following information for a call option on the stock of VeloMedia:

- Current stock price = \$100
- Strike price = \$100
- Time to maturity = 1 year
- Exponential compounding interest rate = 10%
- Annual standard deviation = 30%

What is the value of a European call option using a two-period binomial tree with two distinct intervals of 6 months?

- A. \$15.38
- B. \$10.21
- C. \$0
- D. \$5.86

The correct answer is **A**.

Binomial parameters:

$$u = e^{\sigma\sqrt{\Delta t}} = e^{0.30\sqrt{0.5}} = 1.2363, \quad d = \frac{1}{u} = 0.8089$$

Risk-neutral probability:

$$p = \frac{e^{r \times \Delta t} - d}{u - d} = \frac{e^{0.10 \times 0.5} - 0.8089}{1.2363 - 0.8089} = 0.5671, \quad 1 - p = 0.4329$$

$$X = \$100$$

$$C_{uu} = 52.85, \quad C_{du} = C_{ud} = C_{dd} = 0$$

$$C_u = \frac{p \times C_{uu} + (1 - p) \times C_{ud}}{e^{r \times \Delta t}} = \frac{0.5671 \times 52.85 + (1 - 0.4329) \times 0}{e^{0.10 \times 0.5}} = 28.51, \quad C_d = 0$$

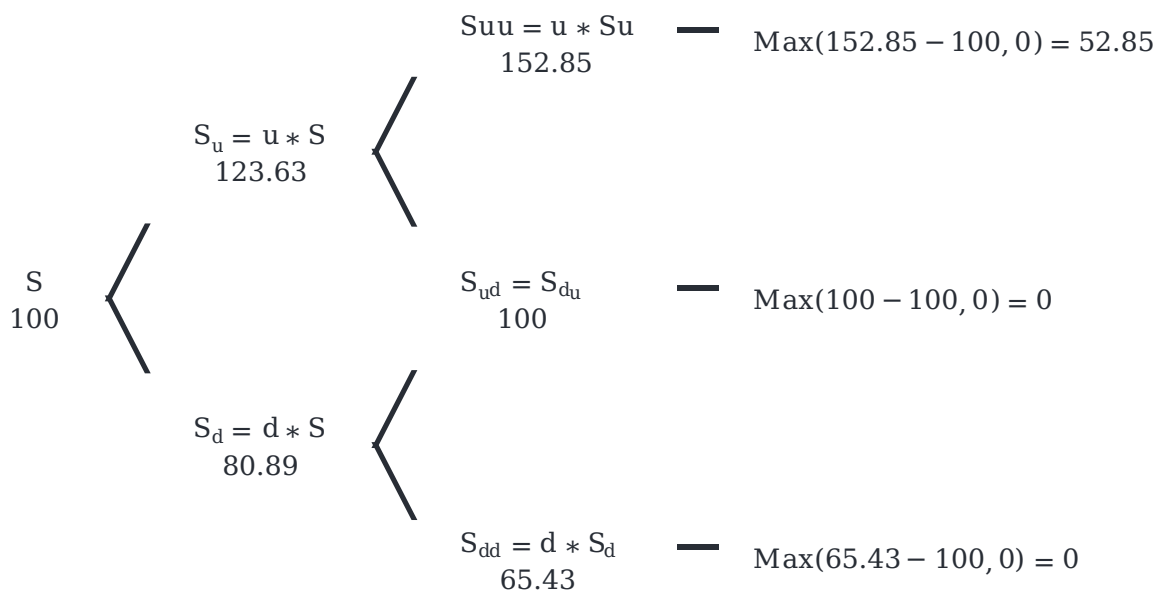
$$C = \frac{0.5671 \times 28.51 + (1 - 0.4329) \times 0}{e^{0.10 \times 0.5}} = 15.38$$

$$u = 1.2363$$

$$d = 0.8089$$

$$p = 0.5671$$

$$1 - p = 0.4329$$



Q.3404 You have been provided the following information for a European call option on the stock of VeloMedia:

- Current stock price = \$100
- Strike price = \$100

- Time to maturity = 1 year
- Exponential compounding interest rate = 10%
- Annual standard deviation = 30%

What is the call option delta at the current date? Use a two-period binomial tree with two distinct intervals of 6 months.

- A. 0
- B. 0.6669
- C. 1
- D. 0.5

The correct answer is **B**.

Binomial parameters:

$$u = e^{\sigma\sqrt{\Delta t}} = e^{0.30\sqrt{0.5}} = 1.2363, \quad d = \frac{1}{u} = 0.8089$$

Risk-neutral probability:

$$p = \frac{e^{r \times \Delta t} - d}{u - d} = \frac{e^{0.10 \times 0.5} - 0.8089}{1.2363 - 0.8089} = 0.5671, \quad 1 - p = 0.4329$$

$$X = \$100$$

$$C_{uu} = 52.85, \quad C_{du} = C_{ud} = C_{dd} = 0$$

$$C_u = \frac{p \times C_{uu} + (1 - p) \times C_{ud}}{e^{r \times \Delta t}} = \frac{0.5671 \times 52.85 + (1 - 0.4329) \times 0}{e^{0.10 \times 0.5}} = 28.51, \quad C_d = 0$$

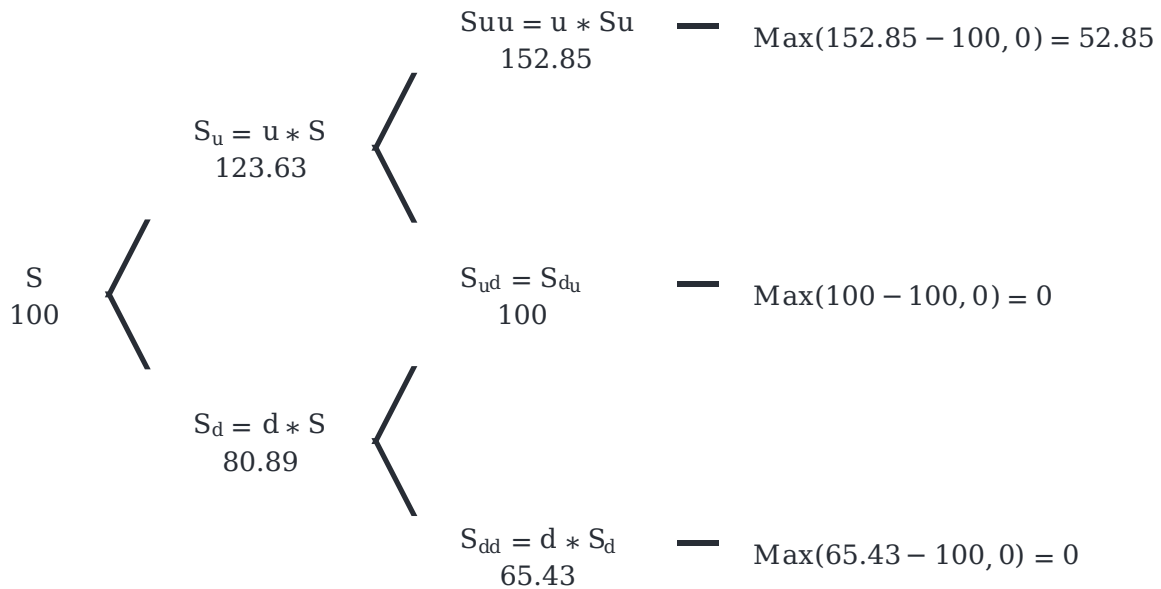
$$C = \frac{0.5671 \times 28.51 + (1 - 0.4329) \times 0}{e^{0.10 \times 0.5}} = 15.38$$

$$u = 1.2363$$

$$d = 0.8089$$

$$p = 0.5671$$

$$1 - p = 0.4329$$



Now, the call option delta at the current date is:

$$\Delta = \frac{\Delta C}{\Delta S} = \frac{C_u - C_d}{S_u - S_d} = \frac{28.51 - 0}{123.63 - 80.89} = 0.6669$$

Q.3405 You have been provided the following information for a put option on the stock of VeloMedia:

- Current stock price = \$100
- Strike price = \$100
- Time to maturity = 1 year
- Exponential compounding interest rate = 10%
- Annual standard deviation = 30%

Compute the value of the European put option using a two-period binomial tree with two distinct intervals of 6 months.

A. \$15.38

- B. \$5.86
- C. \$0
- D. \$10

The correct answer is **B**.

The up-move and down-move factors are:

$$u = e^{\sigma\sqrt{\delta t}} = e^{0.30 \times \sqrt{0.5}} \approx 1.2363$$

$$d = \frac{1}{u} = \frac{1}{1.2363} \approx 0.8089$$

The probability of an up-move is thus,

$$p = \frac{e^{r\delta t} - d}{u - d} = \frac{e^{0.10 \times 0.5} - 0.8089}{1.2363 - 0.8089} \approx 0.5671$$

$$p_{uu} = p_{du} = p_{ud} = 0, \quad p_{dd} = \max(X - S_{dd}, 0) = \max(100 - 100 * 0.8089 * 0.8089, 0) = \max(100 - 6$$

$$P = PV[p^2 p_{uu} + 2p(1-p)p_{ud} + (1-p)^2 p_{dd}] = \frac{p^2 p_{uu} + 2p(1-p)p_{ud} + (1-p)^2 p_{dd}}{R^2}$$

$$P = \frac{0.5671^2 \times 0 + 2 \times 0.5671 \times (1 - 0.5671) \times 0 + (1 - 0.5671)^2 \times 34.87}{e^{0.10 \times 0.5 \times 2}}$$

$$P = \frac{(0.4329)^2 \times 34.57}{e^{0.10}} = 5.86$$

Q.4697 The current price of a stock is \$40. Its volatility is 10% per annum and the risk-free rate is 5% per annum with continuous compounding. Using a two-step binomial, what is the value of a six-month European call option on the stock with a strike price of \$40?

- A. 1.54
- B. 2.0
- C. 1.58
- D. 1.83

The correct answer is **A**.

In this case,

$$\begin{aligned}u &= e^{\sigma\sqrt{t}} = e^{0.10\sqrt{0.25}} = 1.0513 \\d &= \frac{1}{1.0513} = 0.9512 \\p &= \frac{e^{rt} - d}{u - d} = \frac{e^{0.05 \times 0.25} - 0.9512}{1.0513 - 0.9512} = 0.6132 \\1 - p &= 1 - 0.6132 = 0.3868\end{aligned}$$

Where:

u= size of the up-move factor

d= size of the down move factor

σ = volatility of the stock

p= probability of an up move

1 – p= probability of a down move

Using a two-step binomial,

$$S_{uu} = 44.2093$$

$$S_{ud} = 40$$

$$S_{du} = 40$$

$$S_{dd} = 36.1913$$

Notice that the only time the option is in the money is when two upward price movements lead to an ending price of \$44.2093 and a call value of \$4.2093. The expected value of the option at the end of the second period is the value of the option in each state multiplied by the probability of that state occurring:

$$\text{Expected call value in two periods} = 0.6132 * 0.6132 * 4.2093 = \$1.5828$$

The value of the call option today is the expected value in six months discounted at the risk-free rate of 5%:

$$\text{Call} = \frac{\$1.5828}{e^{0.05(0.5)}} = \$1.5437$$

Q.4698 A 6-month stock currently trading at \$40 pays a continuous dividend of 2%, and the current continuously compounded risk-free rate is 3%. Assuming an annual standard deviation of 3%, and a strike price of 40, what is the value of the put today?

- A. 0.32
- B. 0.28
- C. 0.51
- D. 0.52

The correct answer is **A**.

In this case,

$$u = e^{\sigma\sqrt{t}} = e^{0.03\sqrt{0.5}} = 1.02144$$

$$d = \frac{1}{1.02144} = 0.97901$$

$$p = \frac{(e^{(r-q)t}) - d}{u - d}$$

Where:

u= size of the up-move factor

d= size of the down move factor

σ = volatility of the stock

p= probability of an up move

1 – p= probability of a down move

$$p = \frac{e^{(0.03-0.02)0.5} - 0.97901}{1.02144 - 0.97901} = 0.613$$

$$\text{Probability of a down move} = 1 - 0.613 = 0.387$$

Let S represent the stock price, and f represent the value of the put option.

$$S_u = 40.84$$

$$S_d = 39.16$$

The payoffs at the final node are:

$$f_u = \max(40 - 40.84, 0) = 0$$

$$f_d = \max(40 - 39.16, 0) = 0.84$$

The value of the put today is given by:

$$\begin{aligned}
 f &= (f_u p + f_d (1 - p)) e^{-rt} \\
 &= (0 \times 0.613 + 0.8396 \times 0.387) e^{-0.03 \times 0.5} = 0.32008
 \end{aligned}$$

Reading 61: The Black-Scholes-Merton Model

Q.993 Stephen Hawking, a trader working at Orange Securities, collects the following data of a 1-year European put and call options on the stock Mango Apparel. The current stock price is USD 120, and the strike price of the option is USD 125. The risk-free rate is 10 percent. If the prices of a European call and European put are USD 10 and USD 15, respectively, then what is the implied dividend yield of the stock Mango Apparel?

- A. 0.11439
- B. 0.045323
- C. 0.10439
- D. 0.055323

The correct answer is **C**.

Put-call parity:

$$\text{Call option} - \text{Put option} = S_0 e^{-qxT} - K e^{-rT}$$

Where

S_0 = Initial stock price;

q = implied dividend yield;

r = risk-free rate; and

T = time

$$\begin{aligned} 10 - 15 &= 120e^{-qx1} - 125e^{-0.10 \times 1} \\ e^{-q} &= 0.9009 \\ \ln(0.9009) &= -q \end{aligned}$$

Implied dividend yield (q) = 10.439%

Q.994 Chinese International Bank is analyzing the stock of Jatoka International. One-year European call and put options are written on the stock of Jatoka International which is a non-dividend paying stock. The initial stock price is Yuan 100 and the risk-free rate is 5%. The time to maturity is 1 year, and the strike price is Yuan 125. Furthermore, $N(d_1) = 0.6925$, $N(d_2) = 0.5435$. What are the values of European put and call options (approx.) using the Black-Scholes differential equation?

- A. Call option value is USD 4.626 and Put option value is USD 23.52.
- B. Call option value is USD 23.52 and put option value is USD 4.626.
- C. Call option value is USD 2.626 and Put option value is USD 13.52.
- D. Call option value is USD 13.52 and put option value is USD 2.626.

The correct answer is **A**.

$$\begin{aligned}
 \text{Call value} &= S_0 N(d_1) - X e^{-R_f \times T} N(d_2) \\
 &= [100 \times 0.6925] - \{125 e^{(-0.05)} \times 0.5435\} \\
 &= \text{USD } 4.626 \\
 \text{Put value} &= [X e^{-R_f \times T} \times (1 - N(d_2))] - [S_0 \times (1 - N(d_1))] \\
 &= [125 e^{-0.05} \times (1 - 0.5435)] - [100 \times (1 - 0.6925)] \\
 &= \text{USD } 23.52
 \end{aligned}$$

Q.995 Raheja Financials is holding the stock of Duckworth Limited which is trading at USD 50. A European call option that expires in 3 months with a strike price of USD 51 is available for trading. The annualized standard deviation is 20 percent, and the risk-free rate of interest is 4 percent. What is the value of the European call option using the Black-Scholes-Merton model expiring in 3 months if $N(d_1) = 0.48085$ and $N(d_2) = 0.44116$?

- A. USD 2.422
- B. USD 2.224
- C. USD 1.767
- D. USD 1.259

The correct answer is **C**.

Value of European call option as per the Black-Scholes-Merton Model:

$$\begin{aligned}\text{Call value} &= S_0 N(d_1) - Xe^{-R_f \cdot T} N(d_2) \\ &= [50 \times 0.48085] - \{51e^{(-0.04 \cdot 0.25)} \cdot 0.44116\} \\ &= \text{USD } 1.767\end{aligned}$$

Q.996 Michael Wong is holding the stock of Duckworth limited which is trading at USD 50. A European put option that expires in 3 months with a strike price of USD 51 is available for trading. The annualized standard deviation is 20 percent, and the continuously compounded risk-free rate is 4 percent. What is the value of the European put option using the Black-Scholes-Merton model expiring in 3 months if $N(d_1) = 0.48085$ and $N(d_2) = 0.44116$?

- A. USD 1.77
- B. USD 1.26
- C. USD 2.19
- D. USD 2.26

The correct answer is **D**.

The value of European put option as per the Black-Scholes-Merton Model is calculated as follows:

$$\begin{aligned}\text{Put value} &= [Xe^{-R_f \cdot T} * (1 - N(d_2))] - [S_0 * (1 - N(d_1))] \\ &= [51e^{-0.04 * 0.25} * (1 - 0.44116)] - [50 * (1 - 0.48085)] \\ &= \text{USD } 2.26\end{aligned}$$

Q.3407 A stock price has an expected return of 10% and a volatility of 30%. The current price is \$30. What is the probability that a European call option on the stock with an exercise price of \$32 and a maturity date in six months will be exercised? Please click here if you want to use the Standard Normal table

A. 0.5032

B. 0.247

C. 0.4309

D. 0.008

The correct answer is C.

The required probability is the probability of the stock price being above \$32 in six months' time since a call option is only exercised if the stock price is higher than the strike price. Suppose that the stock price in six months is S_T . The probability distribution of $\ln S_T$ is

$$\ln S_T \sim N\left(\ln 30 + \left(0.10 - \frac{0.3^2}{2}\right)0.5, 0.3^2 \times 0.5\right) \sim N(3.429, 0.045)$$

The required probability is given by:

$$P(\ln S_T > \ln 32) = 1 - P(\ln S_T \leq \ln 32) = 1 - N\left(\frac{3.466 - 3.429}{\sqrt{0.045}}\right) = 1 - N(0.1746)$$

Note that: $\ln 32 = 3.466$.

From the standard normal distribution table, $N(0.1746) = 0.5691$

Therefore, the required probability is $1 - 0.5691 = 0.4309$

Alternatively,

$$\text{Probability of exercise} = N(d_2)$$

From the Black Scholes formula,

$$\begin{aligned} d_1 &= \frac{\ln\left(\frac{30}{32}\right) + \left(0.10 + \frac{0.30^2}{2}\right)0.5}{0.30\sqrt{0.5}} - 0.30\sqrt{0.5} = 0.03754 \\ \Rightarrow d_2 &= 0.03753 - 0.30\sqrt{0.5} = -0.1746 \\ \Rightarrow N(d_2) &= 0.4309 \end{aligned}$$

Q.3408 What is the price of a European call option on a non-dividend-paying stock when the stock price is \$68, the strike price is \$65, the risk-free interest rate is 16% per annum, the volatility is 39% per annum, and the time to maturity is three months?

- A. 5.35
- B. 4.85
- C. 8.31
- D. 0.536

The correct answer is **C**.

Recall that the price of European call option is given by:

$$c = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

Where:

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

In this case,

$$S_0 = 68$$

$$K = 65$$

$$r = 0.16$$

$$\sigma = 0.39 \text{ and,}$$

$$T = 0.25$$

$$d_1 = \frac{\left(\ln\frac{68}{65} + \left(0.16 + \frac{0.39^2}{2}\right)0.25\right)}{0.39\sqrt{0.25}} = \frac{0.1041}{0.195} = 0.5338$$
$$d_2 = d_1 - 0.39\sqrt{0.25} = 0.3388$$

The price of the European call is

$$\begin{aligned}c &= 68 \times N(0.5338) - 65e^{-0.16 \times 0.25} N(0.3388) \\&= 68 \times 0.7032 - 65 \times 0.9608 \times 0.6326 \\&= 8.31\end{aligned}$$

Q.3409 Consider a European call option when the stock price is \$20, the exercise price is \$22, the time to maturity is six months, the volatility is 20% per annum, and the risk-free interest rate is 15% per annum. Two equal dividends of \$1 are expected during the life of the option, with ex-dividend dates at the end of two months and five months. What is the value of the option?

- A. \$0.30
- B. \$0.40
- C. \$0.25
- D. \$0.26

The correct answer is **A**.

Note that S_0 is reduced to S by the present value of the dividends payable, but all other variables remain the same

$$S = S_0 - PV$$

Where

$$PV = D_1 e^{-(r)\frac{\Delta t_1}{m}} + D_2 e^{-(r)\frac{\Delta t_2}{m}} = 1e^{-(0.15)\frac{2}{12}} + 1e^{-(0.15)\frac{5}{12}} = 1.9147$$

$$S = \$20 - \$1.9147 = \$18.09$$

$$K = 22,$$

$$T = 0.5,$$

$$\sigma = 0.2$$

$$d_1 = \frac{\ln \frac{S}{K} + [R_f^C + (\frac{1}{2} \times \sigma^2)] T}{\sigma \sqrt{T}}$$

$$= \frac{\ln \frac{18.09}{22} + [0.15 + (\frac{1}{2} \times 0.2^2)] 0.5}{0.2 \sqrt{0.5}} = -0.7826$$

$$d_2 = d_1 - (\sigma \sqrt{T}) = -0.7826 - 0.2 \sqrt{0.5} = -0.9241$$

From statistical tables,

$$N(d_1) = N(-0.7826) = 1 - N(0.7826) = 1 - 0.7831 = 0.2169$$

And

$$N(d_2) = N(-0.9241) = 1 - N(0.9241) = 1 - 0.8223 = 0.1777$$

$$C_0 = [S \times N(d_1)] - [K \times e^{-R_f^C \times T} \times N(d_2)]$$

$$= 18.09 \times 0.2169 - 22e^{-0.15 \times 0.5} \times 0.1777$$

$$= 3.9237 - 3.6269 = 0.2968 \approx 0.30$$

Q.3410 Consider a European option on a non-dividend paying stock with the following characteristics:

- Current stock price = \$50
- Exercise price = 50
- Continuous compounding interest rate = 8%
- Standard deviation = 34%
- Time to expiration = 2 years

Calculate the price of the call option and its delta using the Black-Scholes-Merton model. Please [click here](#) if you want to use the standard normal table.

- A. Call = \$12.97; $\Delta = 0.7167$
- B. Call = \$5.57; $\Delta = 0.7167$
- C. Call = \$5.57; $\Delta = -0.2832$
- D. Call = \$12.97; $\Delta = -0.2832$

The correct answer is **A**.

The price of a call option on a non-dividend-paying stock:

$$C = S \times N(d_1) - PV(X) \times N(d_2)$$

$$d_1 = \frac{\ln\left[\frac{S}{PV(X)}\right] + \frac{\sigma\sqrt{T}}{2}}{\sigma\sqrt{T}} = \frac{\ln\left[\frac{S}{X}\right] + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} \quad \text{and} \quad d_2 = d_1 - \sigma\sqrt{T}$$

$$d_1 = \frac{\ln\left(\frac{50}{50}\right) + \left(0.08 + \frac{0.34^2}{2}\right)2}{0.34\sqrt{2}} = 0.5732$$

$$d_2 = 0.5732 - 0.34\sqrt{2} = 0.0923$$

From the standard normal table $N(d_1) = N(0.5732) = 0.7167$ and $N(d_2) = N(0.0923) = 0.5367$

$$\begin{aligned} \text{Value of the call} &= S * N(d_1) - PV(X) * N(d_2) \\ &= 50 * 0.7167 - 50e^{-0.08 \times 2} \times 0.5367 \\ &= 12.9677 \end{aligned}$$

Using put-call parity relation,

$$\text{Value of the put} = C + PV(X) - S = 12.9677 + 50e^{-0.08 \times 2} - 50 = 5.5749$$

$$\text{Delta of call} = N(d_1) = 0.7167$$

$$\text{Delta of put} = -N(-d_1) = -N(-0.5732) = -(1 - 0.7167) = -0.2833$$

Where N is the normal distribution.

Things to Remember

The delta of a call option is the amount by which the price of the option will change for a given change in the price of the underlying asset. The delta is always positive for calls, meaning that as the underlying asset increases in price, the call will also increase in price. The size of the delta will depend on how much time is left until expiration and how far the underlying asset is from the strike price. At-the-money and near-the-money options will have the largest deltas, while deep in-the-money and deep out-of-the-money options will have smaller deltas. As expiration approaches, all options will tend to have deltas closer to 1.0.

As you can see, the delta of the call option here is high because it is at the money.

Delta of a call option = $N(d_1)$, where $N(x)$ denotes the cumulative standard normal distribution function.

Delta of a put option = $-N(-d1)$

Finally, because the standard normal distribution is symmetric and centered at zero, the standard normal cumulative distribution function has a very useful property:

$$N(-x) = 1 - N(x)$$

$$-N(-x) = N(x) - 1$$

$$-N(-d1) = N(d1) - 1$$

With these in mind, you don't need to calculate the delta of a put once you get the delta of the corresponding call:

Using the last equation, put delta = call delta - 1 = $0.7167 - 1 = -0.2833$

Q.3411 Consider a European option on a non-dividend paying stock with the following characteristics:

- Current stock price = \$50
- Exercise price = 50
- Continuous compounding interest rate = 8%
- Standard deviation = 34%
- Time to expiration = 2 years

Calculate the price of a put option and its delta using the Black-Scholes-Merton model.

- A. Put = \$12.97; $\Delta = 0.7167$
- B. Put = \$5.57; $\Delta = 0.7167$
- C. Put = \$5.57; $\Delta = -0.2832$
- D. Put = \$12.97; $\Delta = -0.2832$

The correct answer is **C**.

The price of a put option on a non-dividend-paying stock can be calculated as:

$$P + S = C + PV(X)$$

$$P = S [N(d_1) - 1] - PV(X)[N(d_2) - 1]$$

$$N(d) + N(-d) = 1$$

$$P = -S \times N(-d_1) + PV(X) \times N(-d_2)$$

$$d_1 = \frac{\ln\left[\frac{S}{X}\right] + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} \quad \text{and} \quad d_2 = d_1 - \sigma\sqrt{T}$$

The price of a call option on a non-dividend-paying stock:

$$C = S \times N(d_1) - PV(X) \times N(d_2)$$

$$d_1 = \frac{\ln\left[\frac{S}{PV(X)}\right] + \frac{\sigma\sqrt{T}}{2}}{\sigma\sqrt{T}} = \frac{\ln\left[\frac{S}{X}\right] + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} \quad \text{and} \quad d_2 = d_1 - \sigma\sqrt{T}$$

$$d_1 = \frac{\ln\left(\frac{50}{50}\right) + \left(0.08 + \frac{0.34^2}{2}\right)2}{0.34\sqrt{2}} = 0.5732$$

$$d_2 = 0.5732 - 0.34\sqrt{2} = 0.0923$$

From the standard normal table $N(d_1) = N(0.5732) = 0.7167$ and $N(d_2) = N(0.0923) = 0.5367$

$$\begin{aligned} \text{Value of the call} &= S * N(d_1) - PV(X) * N(d_2) \\ &= 50 * 0.7167 - 50e^{-0.08 \times 2} \times 0.5367 \\ &= 12.9677 \end{aligned}$$

Using put-call parity relation,

$$\text{Value of the put} = C + PV(X) - S = 12.9677 + 50e^{-0.08 \times 2} - 50 = 5.5749$$

$$\text{Delta of call} = N(d_1) = 0.7167$$

$$\text{Delta of put} = -N(-d_1) = -N(-0.5732) = -(1 - 0.7167) = -0.2833$$

Where, N is the normal distribution.

Q.3412 Consider a company with N million shares outstanding, each worth S_0 , that is contemplating issuing M warrants. Each warrant would grant the holder the right to purchase one share with a strike price of X in a year. Assuming the value of a corresponding 1-year European call option is worth C, the cost of issuing the warrants would take which of the following forms?

- A. $\frac{C}{N+M}$
- B. $\frac{MC}{N+M}$
- C. $\frac{MNC}{N+M}$
- D. $\frac{NC}{N+M}$

The correct answer is **C**.

It can be shown the cost of issuing each warrant is given by:

$$\frac{N}{N+M} \times \text{Price of the Warrant}$$

Where

N = Number of existing shares

M = Number of warrants issued

Note that a warrant is defined as options issued by a company on its own stock. So, in this case, the cost of issuing M warrants is given by:

$$M \times \frac{N}{N+M} \times C = \frac{MNC}{N+M}$$

Q.3414 Dividends play a significant role in the pricing of options, as they impact the underlying stock's price and, consequently, the option's value. Understanding this effect is crucial for options traders and investors. What is the effect of dividends on option prices?

- A. Call option prices increase; Put option prices increase
- B. Call option prices increase; Put option prices decrease
- C. Call option prices decrease; Put option prices increase
- D. Call option prices decrease; Put option prices decrease

The correct answer is **C**.

Dividends have a direct impact on the price of the underlying stock. When a company announces a dividend, the stock price decreases by the present value of the dividend on the ex-dividend date. This is because the cash paid out as dividends is no longer part of the company's assets, reducing the company's overall value. This decrease in stock price has a corresponding effect on the prices of call and put options. Call options, which give the holder the right to buy the stock at a predetermined price, become less valuable when the stock price decreases. This is because the holder of the call option would prefer to buy the stock at the lower market price rather than at the higher strike price of the option. Therefore, the price of the call option decreases. On the other hand, put options, which give the holder the right to sell the stock at a predetermined price, become more valuable when the stock price decreases. This is because the holder of the put option can sell the stock at the higher strike price of the option rather than at the lower market price. Therefore, the price of the put option increases. Hence, when dividends are paid, call option prices decrease and put option prices increase.

Choice A is incorrect because it suggests that both call and put option prices increase when dividends are paid. This is not accurate. When dividends are paid, the stock price decreases, which has a negative impact on call option prices and a positive impact on put option prices. Call options become less valuable because the holder would prefer to buy the stock at the lower market price rather than at the higher strike price of the option. Conversely, put options become more valuable because the holder can sell the stock at the higher strike price of the option rather than at the lower market price. Therefore, the price of the call option decreases and the price of the put option increases when dividends are paid.

Choice B is incorrect because it suggests that call option prices increase and put option prices decrease when dividends are paid. This is not accurate. When dividends are paid, the stock price decreases, which has a negative impact on call option prices and a positive impact on put option prices. Call options become less valuable because the holder would prefer to buy the stock at the lower market price rather than at the higher strike price of the option. Conversely, put options become more valuable because the holder can sell the stock at the higher strike price of the option rather than at the lower market price. Therefore, the price of the call option decreases and the price of the put option increases when dividends are paid.

Choice D is incorrect because it suggests that both call and put option prices decrease when dividends are paid. This is not accurate. When dividends are paid, the stock price decreases, which has a negative impact on call option

Things to Remember

- Dividends are cash payments made by a company to its shareholders out of its profits or reserves.
- Ex-dividend date is the date on which the seller, rather than the buyer, of a stock will be entitled to a recently announced dividend.
- Call options give the holder the right to buy the underlying asset at a specified price within a specific time frame.
- Put options give the holder the right to sell the underlying asset at a specified price

within a specific time frame.

- Option pricing models, such as the Black-Scholes model, take into account factors like dividends, interest rates, volatility, and time to expiration.
-

Q.4618 A stock has an initial price of \$50, an expected annual return of 20%, and annual volatility of 25%. What is the 95% confidence interval for the stock price at the end of 3 years?

- A. $\$22.83 < S_T < \301.27
- B. $\$44.18 < S_T < \57.33
- C. $\$31.28 < S_T < \57.08
- D. $\$35.49 < S_T < \193.77

The correct answer is **D**.

In this case, $S_0=50$, $\mu=0.20$, $\sigma=0.25$, and $T=3$

The logarithm of the stock follows a normal distribution with the following parameters:

$$\ln S_T \sim N \left(\ln S_0 + \left(\mu - \frac{\sigma^2}{2} \right) T, \sigma \sqrt{T} \right)$$

Where:

μ = the expected annual return of the stock.

σ = the annual volatility for the stock.

T = time

S_T = The stock price at time T .

$$\begin{aligned} \ln S_T &\sim N \left(\ln S_0 + \left(\mu - \frac{\sigma^2}{2} \right) T, \sigma \sqrt{T} \right) \\ &= N \left[\ln 50 + \left(0.20 - \frac{0.25^2}{2} \right) 3, 0.25 \times \sqrt{3} \right] \\ \ln S_T &\sim N(4.418, 0.433^2) \end{aligned}$$

The distribution has a mean of 4.418 and standard deviation of 0.433

To obtain the 95% confidence interval for stock price using the above data, we will proceed as follows:

$$\begin{aligned}\ln S_T &\sim N(4.418, 0.433^2) \\ \ln S_T &= \mu \pm Z_\alpha \times \sigma \\ (\text{In this case, we have } \sigma &= 0.433) \\ \text{and } Z_\alpha &= 1.96 \\ 4.418 - 1.96 \times 0.433 &< \ln S_T < 4.418 + 1.96 \times 0.433 \\ e^{3.56932} < S_T &< e^{5.26668} \\ \$35.49 < S_T &< \$193.77\end{aligned}$$

Q.4619 Suppose the current exchange rate for a currency is 1.25, and the exchange rate volatility is 15%. Calculate the value of a call option to buy 1000 units of the currency in 5 years at an exchange rate of 2.50. The domestic and foreign risk-free interest rates are 1% and 2%, respectively. Please click here to view the standard normal table

- A. 2.9
- B. 3.8
- C. 43.5
- D. 1249.9

The correct answer is **A**.

In this case $S_0=1.25$, $K=2.50$, $r=0.01$, $r_f=0.02$, $s=0.15$, and $T=5$

From BSM pricing formula,

$$C_0 = S_0 e^{-r_f T} \times N(d_1) - K e^{-r T} \times N(d_2)$$

Where:

T = time to maturity

S_0 = current stock price

K = strike price

r = domestic risk-free rate

r_f = foreign risk-free rate

s = volatility of the stock price

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + [r - r_f + \left(\frac{\sigma^2}{2}\right)]T}{\sigma\sqrt{T}} = \frac{\ln\frac{1.25}{2.50} + [0.01 - 0.02 + \frac{0.15^2}{2}]5}{0.15\sqrt{5}} = -2.05$$

$$d_2 = d_1 - \sigma\sqrt{T} = -2.05 - 0.15\sqrt{5} = -2.39$$

From the standard normal tables,

$$N(d_1) = N(-2.05) = 1 - 0.9798 = 0.0202$$

$$N(d_2) = N(-2.39) = 1 - 0.9916 = 0.0084$$

The value of the call is therefore given by:

$$C_0 = S_0 e^{-r_f T} \times N(d_1) - K e^{-r T} \times N(d_2)$$

$$= 1.25 e^{-0.02 \times 5} \times 0.0202 - 2.50 e^{-0.01 \times 5} \times 0.0084 = 0.0029$$

This is the value of the option to buy one unit of the currency. The value of an option to buy 1000 units is:

$$0.0029 \times 1000 = 2.9$$

Q.4620 The futures price of an asset is USD 40, and the annual volatility of the futures price is 20%. If the risk-free rate is 5%, what is the value of a put option to sell futures in 6 months for USD 45?

A. USD 2.75

B. USD 5.52

C. USD 2.68

D. USD 4.82

The correct answer is **B**.

In this case,

$$F_0 = 40, K = 45, r = 0.05, s = 0.20, T = 0.5$$

The following formula gives the value of the put option:

$$P_0 = K e^{-r T} \times N(-d_2) - F_0 e^{-r T} \times N(-d_1)$$

Where:

P_0 = value of the put option

K = strike price

s = volatility of the futures price

r = risk-free rate

T = time

Now define:

F_0 = current futures price

Then we have:

$$d_1 = \frac{\ln\left(\frac{F_0}{K}\right) + T * \frac{\delta^2}{2}}{\delta\sqrt{T}} = \frac{\ln\left(\frac{40}{45}\right) + 0.5 * \frac{0.20^2}{2}}{0.20\sqrt{0.5}} = -0.76214$$

Thus,

$$d_2 = d_1 - \delta\sqrt{T} = -0.76214 - 0.20\sqrt{0.5} = -0.90356$$

The value of the put option is given by:

$$\begin{aligned} P_0 &= Ke^{-rT} \times N(-d_2) - F_0e^{-rT} \times N(-d_1) \\ &= 45e^{-0.05 \times 0.5} \times 0.8159 - 40e^{-0.05 \times 0.5} \times 0.7764 = 5.5197 \end{aligned}$$

Q.4621 The current price of a stock is USD 50. If this price grows to USD 74.59 in two years, what is the realized return on the stock per annum?

A. 0.2

B. 0.49

C. 0.34

D. 0.22

The correct answer is **A**.

We can derive the formula for calculating the realized return on a stock from the formula for calculating the expected price of a stock, i.e.,

$$E(S_T) = S_0 e^{\mu T}$$
$$\mu = \frac{1}{T} \ln \frac{S_T}{S_0}$$

Where:

μ = Realized return on a stock

T = time to maturity

S_0 = initial stock price

S_T = stock price at time T

$$\text{Realized return} = \frac{1}{2} \ln \frac{74.59}{50} = 0.20 = 20\%$$

Q.4622 The following are monthly stock prices in EUR: 21, 35, 40, and 28. From this data, what is the estimated volatility of the log-returns per month?

- A. 0.435
- B. 0.189
- C. 0.402
- D. 0.355

The correct answer is **A**.

Month	Stock price, S_i	$\frac{S_i}{S_{i-1}}$	$X_i = \ln \frac{S_i}{S_{i-1}}$	X_i^2
0	21			
1	35	1.6667	0.5108	0.2609
2	40	1.1429	0.1336	0.0178
3	28	0.7000	-0.3567	0.1272

We calculate volatility of returns per month using the variance formula below:

$$\text{Variance} = \frac{1}{n-1} \left\{ \sum X_i^2 - n \left(\frac{\sum X_i}{n} \right)^2 \right\}$$

In this case:

$$\sum X_i = 0.2877, \sum X_i^2 = 0.4059, n=3, \frac{\sum X_i}{n} = \frac{0.2877}{3} = 0.0959$$

Therefore,

$$\text{Variance} = \frac{1}{2} \{ 0.4059 - 3 \times 0.0959^2 \} = 0.1892$$

The volatility per month is, therefore $\sqrt{0.1892} = 0.435 = 43.5\%$

Reading 62: Option Sensitivity Measures: The “Greeks”

Q.1186 Fintech Company Inc. is planning to purchase a call option on European Airlines. The continuous dividend yield is 2 percent and the time to maturity is 2 years. If the continuous risk-free rate is 5 percent and $N(d1)$ is 0.45, what is the Delta of the call option?

- A. 0.432
- B. -0.432
- C. 0.864
- D. -0.864

The correct answer is **A**.

The Delta of a call option with a continuous dividend yield is calculated using the formula below:

$$\text{Delta} = N(d1) e^{(-qT)}$$

Where

Continuous dividend yield (q) = 2%

Time to maturity = 2 years

$$\text{Delta} = 0.45 * e^{(-0.02 * 2)} = 0.432$$

Things to Remember

- Delta measures the sensitivity of the option price to changes in the price of the underlying asset.
- $N(d1)$ is the probability that the call option will be exercised.
- Continuous dividend yield is the rate of return an investor would expect to receive from holding the stock.
- Risk-free rate is the theoretical rate of return of an investment with no risk of financial loss.
- Time to maturity is the time remaining until the option contract expires.
- Call options give the holder the right to buy the underlying asset at a specified price within a specific time frame.

Q.1187 John Augustus, an equity analyst at Fintech Inc., is evaluating a portfolio of American Airlines stock and options on the same stock. The portfolio is currently Delta neutral but has a positive Gamma. If Augustus would like to make the portfolio both Delta and Gamma neutral, then Johnson will:

- A. Sell stock of American Airlines and sell Put options on stock of American Airlines.
- B. Buy stock of American Airlines and buy Put options on stock of American Airlines.
- C. Buy stock of American Airlines and sell Put options on stock of American Airlines.
- D. Sell stock of American Airlines and buy Put options on stock of American Airlines.

The correct answer is **A**.

To achieve both Delta and Gamma neutrality in a portfolio, one must understand the fundamental concepts of Delta and Gamma in options trading. Delta measures the rate of change of an option's price with respect to changes in the price of the underlying asset, while Gamma measures the rate of change in Delta with respect to changes in the underlying asset's price. In this scenario, the portfolio is Delta neutral but has a positive Gamma, indicating that the portfolio's Delta will increase if the underlying asset's price increases, and decrease if the underlying asset's price decreases. To make the portfolio Gamma neutral, options on the stock of American Airlines need to be sold, as selling options results in negative Gamma. However, selling put options on American Airlines will make the portfolio Delta positive, as put options have a negative Delta and selling them results in positive Delta. To counteract this and make the portfolio Delta neutral, the stock of American Airlines needs to be sold. Selling the stock results in negative Delta, which will offset the positive Delta from selling the put options. Therefore, to make the portfolio both Delta and Gamma neutral, Augustus should sell stock of American Airlines and sell put options on the same stock, as stated in choice A.

Choice B is incorrect. Buying both the stock and put options on the same stock would increase the portfolio's Delta, not neutralize it. Additionally, buying more put options would further increase the portfolio's positive Gamma, which contradicts Augustus' intention to achieve Gamma neutrality.

Choice C is incorrect. While buying the stock and selling put options could potentially neutralize Delta, it would not achieve Gamma neutrality. Selling put options decreases Gamma; however, since the portfolio already has a positive Gamma, this action alone will not be sufficient to neutralize it.

Choice D is incorrect. Selling both American Airlines' stocks and buying Put options on these stocks will decrease Delta but will increase Gamma due to long position in Put Options which contradicts with Augustus' intention of achieving both Delta and Gamma neutrality.

Things to Remember

- Delta measures the rate of change of an option's price with respect to changes in the price of the underlying asset.

- Gamma measures the rate of change in Delta with respect to changes in the underlying asset's price.
 - To achieve Delta neutrality, the sum of the Deltas of all positions in the portfolio should be zero.
 - To achieve Gamma neutrality, the sum of the Gammas of all positions in the portfolio should be zero.
 - Selling options results in negative Gamma, while buying options results in positive Gamma.
 - Put options have a negative Delta, while call options have a positive Delta.
-

Q.1188 Trenor Johnson, a portfolio manager, working at Sterile Finances Limited, is analyzing the delta of a portfolio. Which of the following statements is (are) true about the delta of a portfolio?

- A. Call options have negative deltas while put options have positive Deltas.
- B. Delta on options lie between -1 and +1.
- C. The delta of the underlying asset is always zero.
- D. The delta of the portfolio is always the sum of individual deltas.

The correct answer is **B**.

The delta of an option is a measure of how much the price of an option is expected to change per \$1 change in the price of the underlying asset. It is essentially a measure of the option's sensitivity to changes in the price of the underlying asset. For call options, the delta ranges from 0 to 1. This is because as the price of the underlying asset increases, the price of the call option also increases. Conversely, for put options, the delta ranges from -1 to 0. This is because as the price of the underlying asset increases, the price of the put option decreases. Therefore, the delta of an option, whether it's a call or a put, will always lie between -1 and +1. This is a fundamental concept in options trading and portfolio management, and understanding it is crucial for making informed investment decisions.

Choice A is incorrect. This statement is not accurate. In the context of options trading, call options have positive deltas because they increase in value as the price of the underlying asset increases. Conversely, put options have negative deltas because their value decreases as the price of the underlying asset increases.

Choice C is incorrect. The delta of an underlying asset is not always zero. In fact, it's typically one for a long position and negative one for a short position in an asset because a \$1 increase in the price of an underlying asset will result in approximately \$1 change in the value of that position.

Choice D is incorrect. The delta of a portfolio is typically calculated as the weighted sum of the deltas of the individual assets within the portfolio.

Things to Remember

- Delta measures the sensitivity of an option's price to changes in the price of the underlying asset.
 - For call options, delta ranges from 0 to 1, while for put options, delta ranges from -1 to 0.
 - Delta can be used to hedge against price movements in the underlying asset.
 - Delta is just one of the Greeks used in options trading, along with gamma, theta, vega, and rho.
 - Delta hedging involves adjusting the position in the underlying asset to offset changes in the option's price.
-

Q.1194 Consider a call option on a non-dividend paying stock where the stock price is \$95, the risk-free rate is 5%, the time to maturity is 40 weeks (= 0.7692 years) and $N'(d_1) = 0.398185$. A 1% increase in the volatility will increase the value of the option by approximately:

- A. -33.176
- B. -0.33176
- C. 33.176
- D. 0.33176

The correct answer is **D**.

$$\begin{aligned}\text{The Vega of a call option} &= S_0 \sqrt{T} N'(d_1) \\ &= \$95 * \sqrt{0.7692} * 0.398185 \\ &= 33.176\end{aligned}$$

Thus, a 1% (0.01) increase in the volatility increases the value of the option by approximately $0.01 * 33.176 = 0.33176$

Note that:

Both call and put options have the same value of vega and this value is a positive number

Note. We multiply by 0.01 (divide by 100) because of how volatility is expressed in the vega formula. The formula for vega assumes that volatility is expressed as a decimal rather than a percentage. Therefore, when we calculate vega, this represents the change in option value for a change in volatility of 1.0 in decimal terms (or 100%). Since we want the change for just 1% (0.01 in decimal), we need to multiply our vega by 0.01. $33.15 \times 0.01 = \$0.3315 = \text{aprox. } \0.33 . Remember, volatility is required when calculating the value of d_1 , but luckily, we have already been given the value of d_1 in this case.

Q.1195 Which of the following statement is true with regard to Rho?

- A. The Rho of a portfolio of options is the rate of change of the value of the portfolio with respect to the interest rate.
- B. The Rho of a portfolio of derivatives is the rate of change of the value of the portfolio with respect to the volatility of the underlying asset.
- C. The Rho of a portfolio of options on an underlying asset is the rate of change of the portfolio's Delta.
- D. The Rho of a portfolio of options is the rate of change of the value of the portfolio with respect to the passage of time with all else remaining the same.

The correct answer is **A**.

Rho is a measure of the sensitivity of an option or options portfolio to changes in the interest rate. It is part of the group of measures known as the Greeks, which are used to identify the different types of risks that can affect the price of derivatives. Rho measures the expected change in an option's price for a 1% change in interest rates. This is particularly important for options traders because changes in interest rates can have a significant impact on the price of options. For example, if interest rates increase, the value of call options generally increases and the value of put options generally decreases. Therefore, a positive Rho indicates that the price of the option will increase with an increase in interest rates, while a negative Rho indicates that the price of the option will decrease with an increase in interest rates.

Choice B is incorrect. The rate of change of the value of a portfolio with respect to the volatility of the underlying asset is not Rho, but Vega. Vega measures sensitivity to volatility.

Choice C is incorrect. The rate of change of a portfolio's Delta is not described by Rho, but by Gamma. Gamma measures the rate at which Delta changes for each unit change in the price of the underlying asset.

Choice D is incorrect. The rate of change in an option's price given a one-unit (day) change in time, all else remaining constant, is referred to as Theta and not Rho.

Things to Remember

- Rho is one of the Greeks used to measure the sensitivity of options to changes in interest rates.
 - Rho measures the expected change in an option's price for a 1% change in interest rates.
 - Changes in interest rates can have a significant impact on the price of options.
 - Vega measures sensitivity to volatility, not Rho.
 - Gamma measures the rate at which Delta changes for each unit change in the price of the underlying asset, not Rho.
 - Theta measures the rate of change in an option's price given a one-unit change in time, not Rho.
-

Q.1196 A stock is currently trading at \$25. The delta of the call option is 0.482. A fund manager buys 100,000 call options with a strike price of \$26.50 on the stock. To maintain a delta neutral position, the fund manager must:

- A. Buy 4,820,000 shares
- B. Sell 4,820,000 shares
- C. Buy 1,000,000 shares
- D. Sell 1,000,000 shares

The correct answer is **B**.

As the fund manager buys 100,000 call options, the delta of the option position is:

$$\delta \text{ of option position} = 0.482 * 100,000 = +48,200$$

To get a delta neutral position, the fund manager needs to sell 4,820,000(= 48,200 × 100) shares since 1 call option typically gives the holder the right to buy 100 shares at strike price.

Q.1197 A stock is currently trading at \$25. The delta of the call option is 0.482. A fund manager buys 100,000 call option contracts on the stock with a strike price of \$29. What action is most likely to be taken by the fund manager to maintain a delta neutral position?

- A. Sell 4,820,000 shares.
- B. Buy 4,820,000 shares.
- C. Buy 48,200 call options.
- D. Sell 48,200 shares.

The correct answer is **A**.

The fund manager should sell 4,820,000 shares to maintain a delta-neutral position. This is because the delta of the call option is 0.482, which means that for every \$1 increase in the price of the underlying stock, the price of the option will increase by \$0.482. Since the fund manager has bought 100,000 options, the portfolio delta is $100,000 \times 0.482 = 48,200$. However, since each option contract represents 100 shares, the fund manager should sell 4,820,000 shares ($48,200 \times 100$) of the underlying stock. This way, any gain or loss on the option will be offset by a corresponding loss or gain on the shares, thereby maintaining a delta-neutral position.

Choice B is incorrect. Buying 4,820,000 shares would increase the delta of the position rather

than neutralizing it. The fund manager needs to offset the positive delta of the call options by taking a position with a negative delta. Therefore, buying shares which have a positive delta would not achieve this.

Choice C is incorrect. Buying additional call options would also increase the overall delta of the position as call options have a positive delta. This action would move the portfolio further away from being delta neutral.

Choice D is incorrect. Selling only 48,200 shares will not be sufficient to offset the positive delta from 100,000 call option contracts and achieve a delta-neutral position. The number of shares sold must match with total deltas from all option contracts in order to maintain neutrality.

Things to Remember

- Delta measures the sensitivity of the option price to changes in the price of the underlying asset.
 - To maintain a delta-neutral position, an investor must offset the positive delta of call options by taking a position with a negative delta, and vice versa for put options.
 - Delta hedging involves adjusting the position in the underlying asset to offset the delta of the options held in the portfolio.
 - Delta-neutral strategies are used to reduce directional risk and focus on profiting from other factors such as volatility or time decay.
 - Gamma is another option Greek that measures the rate of change of an option's delta in response to changes in the price of the underlying asset.
-

Q.1198 A portfolio manager buys 100 APR 45 call option selling for \$3.58 that have a delta of 0.4 and a gamma of 0.1. If the underlying trades downwards by \$1, then the delta of the overall position will now be:

- A. 0.3
- B. 30
- C. 27
- D. 0.27

The correct answer is **B**.

Gamma reflects the change in delta in response to a one-point movement of the underlying stock price.

Our gamma here is 0.1. For every one-point move in the price of the underlying, delta of the corresponding option will change by 0.1. A one point (1 USD) increase in price will prompt a 0.1 increase in delta. Similarly, a one point decrease in price will prompt a 0.1 decrease in delta.

In our case, delta will decrease from 0.4 to 0.3.

But that's delta for just one call option. For 100 options, $\text{delta} = 0.3 * 100 = 30$.

Note that gamma is always a positive number regardless of whether you are buying calls or puts but is effectively negative when you write options.

Note that:

100 APR 45 call option:

100 here represents the number of call options.

45 here means that an investor can exercise the right to buy the stock at 45 per share.

Things to Remember

- Delta measures the rate of change of the option price with respect to the price of the underlying asset.
 - Gamma measures the rate of change of delta with respect to the price of the underlying asset.
 - Theta measures the rate of decline in the value of an option over time.
 - Vega measures the sensitivity of the option price to changes in implied volatility.
 - Options can be used for speculation, hedging, or generating income.
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Q.1199 A portfolio of derivatives on a stock has a delta of 2400 and a gamma of -100. Also available for trading is an option on the stock with a delta of 0.5 and a gamma of 0.04. To make the portfolio gamma neutral, the portfolio manager should:

- A. Buy 2,500 options.
- B. Sell 2,500 options.
- C. Buy 1,200 options.
- D. Sell 1,200 options.

The correct answer is **A**.

To create a gamma-neutral position (sometimes called gamma-neutral hedging), the manager must add the appropriate number of options that equals the existing portfolio gamma position. In this case, the existing gamma position is -100, and an available option exhibits a gamma of 0.04, which translates into buying approximately 2,500 options ($100 / 0.04$).

Things to Remember

Gamma-neutral hedging is the construction of options trading positions that are hedged such that the total gamma value of the position is zero or near zero. The goal is to take options combinations that will make the overall gamma value as close to zero as possible. This results in the delta value of the positions remaining stagnant no matter how strongly the underlying stock moves.

Q.1201 Consider a European put option with the following parameters:

- Stock price: USD 50
- Strike price: USD 55
- Risk-free rate: 5%
- Volatility: 16%
- Time to maturity: 6 months

What is the rho of the put option?

- A. -20.15
- B. 6.07
- C. -13.35
- D. -18.31

The correct answer is **A**.

The rho of the put option can be calculated as follows:

$$\text{rho(put)} = -KTe^{-rT}N(-d_2)$$

But we know that, from the Black Scholes formula,

$$\begin{aligned}d_1 &= \frac{\ln(\frac{S_0}{K}) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}} \\&= \frac{\ln(\frac{50}{55}) + (0.05 + \frac{0.16^2}{2}) \cdot 0.5}{0.16\sqrt{0.5}} = -0.5649\end{aligned}$$

$$\Rightarrow d_2 = -0.5649 - 0.16\sqrt{0.5} = -0.6780$$

$$\begin{aligned}\therefore \text{rho(put)} &= -KTe^{-rT}N(-d_2) \\&= -55 \times 0.5e^{-0.05 \times 0.5}N(-(-0.6780)) \\&= -55 \times 0.5e^{-0.05 \times 0.5} \times 0.7511 = -20.14527 \approx -20.15\end{aligned}$$

Q.1203 A fund manager sells 200,000 call options on stock A, a non-dividend paying stock. The delta of the stock option is 0.45, and the risk-free rate is 6%.
Select the most appropriate statement.

- A. The position can be made delta neutral by selling 90,000 shares of the underlying asset.
- B. The position can be made delta neutral by going short 9,000,000 shares on the underlying asset.
- C. The position can be made delta-neutral by buying 9,000,000 shares of the underlying asset.
- D. The position can be made delta neutral by going short on call options.

The correct answer is C.

To determine the correct answer, we need to calculate the total delta of the options sold by the fund manager and figure out how to offset that delta to achieve a delta-neutral position:

Calculate the total delta of the options position. We know that one option controls 100 shares. The delta of one option is 0.45. Since the fund manager sold 200,000 call options, the total delta from the options position is:

$$0.45 \times 200,000 \times 100 = 9,000,000$$

Selling call options results in a negative position in delta (because you are taking a position that benefits if the stock price goes down), thus the total delta of the position is: -9,000,000.

Offsetting the delta to achieve neutrality. To neutralize this delta, the fund manager would need to offset this negative delta with a positive delta of equal magnitude. The most straightforward way to achieve this is by buying shares of the underlying stock, since each share has a delta of 1. To neutralize a delta of -9,000,000, the manager would need to purchase 9,000,000 shares of stock A (since each share has a delta of +1).

A is incorrect as selling 90,000 shares would drastically overshoot the necessary delta adjustment.

B is incorrect as going short 9,000,000 shares would also overshoot the necessary delta adjustment.

D is incorrect since being short on more call options would increase the negative delta, making the position further from neutral, not balancing it.

Things to Remember

- Delta measures the sensitivity of the option price to changes in the price of the underlying asset.

- Delta-neutral means that the overall position has a delta of zero, resulting in the portfolio being less sensitive to changes in the underlying asset's price.
 - Delta hedging involves adjusting the position in the underlying asset to offset the delta of the options position.
 - Delta of an option can range from -1 to 1, depending on whether it is a call option or a put option.
 - Delta hedging is a risk management strategy used by options traders to reduce directional risk in their portfolios.
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Q.3415 Which one of the following statements is true regarding option Greeks?

- A. Gamma is greatest for in-the-money options with long maturities
- B. Delta of deep-in-the-money put options tends toward +1
- C. Vega is greatest for at-the-money options with long maturities
- D. When buying at-the-money options, theta tends to be positive

The correct answer is **C**.

Vega is a measure of an option's price sensitivity to changes in the volatility of the underlying asset. It represents the amount that an option contract's price changes in reaction to a 1% change in the volatility of the underlying asset. Vega is greatest for at-the-money options with long maturities because these options have the highest sensitivity to changes in volatility. This is because at-the-money options have the greatest uncertainty regarding whether they will end up in-the-money at expiration. Therefore, any change in the underlying asset's volatility significantly impacts the option's price. Long maturity options also have higher vegas because the longer the time to expiration, the more chance there is for the underlying asset's price to move, which increases the option's price sensitivity to changes in volatility.

Choice A is incorrect. Gamma is not greatest for in-the-money options with long maturities. In fact, gamma is highest for at-the-money options and decreases as the option moves further into or out of the money. Additionally, gamma decreases as maturity increases.

Choice B is incorrect. The delta of deep-in-the-money put options does not tend toward +1; it tends towards -1 instead. This means that if the underlying asset's price increases by \$1, the price of a deep-in-the-money put option would decrease by approximately \$1.

Choice D is incorrect. When buying at-the-money options, theta does not tend to be positive; it

tends to be negative instead because time decay works against buyers of at-the-money options.

Things to Remember

- Gamma is highest for at-the-money options and decreases as the option moves further into or out of the money.
 - Delta of deep-in-the-money put options tends towards -1, not +1.
 - Theta tends to be negative for options buyers as time decay works against them.
 - Vega is a measure of an option's price sensitivity to changes in the volatility of the underlying asset.
 - Options with longer maturities have higher vegas due to increased price sensitivity to changes in volatility over time.
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Q.3417 A trader has a short option position that's delta-neutral but has a gamma of -800. In the market, there's a tradeable option with a delta of 0.8 and a gamma of 2. To maintain the position gamma-neutral and delta-neutral, the most appropriate strategy is to:

- A. Sell 320 options and buy 400 shares of the underlying
- B. Buy 400 options and sell 320 shares of the underlying
- C. Buy 400 options and buy 320 shares of the underlying
- D. Sell 320 options and buy 320 shares of the underlying

The correct answer is **B**.

The trader needs to buy 400 options and sell 320 shares of the underlying. The current position of the trader is gamma-negative, which means the trader needs to increase the gamma back to zero. This can be achieved by buying call options. The number of options that need to be added to the existing portfolio to create a gamma-neutral position is calculated by the formula: $-\frac{\{\text{Gamma}\}_{\text{p}}}{\{\text{Gamma}\}_{\text{T}}}$, where $\{\text{Gamma}\}_{\text{p}}$ is the gamma of the existing portfolio position and $\{\text{Gamma}\}_{\text{T}}$ is the gamma of a traded option that can be added. In this case, it is $-(-800/2)$ which equals 400. Therefore, the trader needs to buy 400 call options. However, buying 400 call options increases the delta from zero to 320 (calculated as 400×0.8). To restore the delta back to zero, the trader needs to sell 320 shares. Positions in shares always have zero gamma, hence this strategy will ensure that the position remains both gamma-neutral and delta-neutral.

Choice A is incorrect. Selling 320 options would increase the negative gamma of the position,

making it further from gamma-neutral. Buying 400 shares of the underlying would not affect the gamma of the position as shares have a gamma of zero.

Choice C is incorrect. Buying both options and shares would make the position delta-positive, which contradicts with maintaining a delta-neutral position.

Choice D is incorrect. Selling 320 options and buying 320 shares will not neutralize both delta and gamma at once because selling more options will increase negative gamma while buying stocks has no effect on it as stocks have zero gamma.

Things to Remember

- Delta-neutral position: A position where the overall delta of the portfolio is zero, meaning the portfolio is not affected by small changes in the price of the underlying asset.
 - Gamma: Gamma measures the rate of change of an option's delta relative to a change in the price of the underlying asset. It shows how sensitive the option's delta is to changes in the underlying price.
 - Gamma-neutral position: A position where the overall gamma of the portfolio is zero, meaning the portfolio is not affected by small changes in the price of the underlying asset's volatility.
 - Options trading strategies: Various combinations of buying and selling options and the underlying asset to achieve specific risk profiles such as delta-neutral, gamma-neutral, etc.
 - Calculating position adjustments: Formulas can be used to calculate the number of options or shares needed to adjust the position to maintain delta-neutral or gamma-neutral status.
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